ST77922

Single-Chip TFT Controller/Driver/Touch

Datasheet

Version 1.2

2024/07

Sitronix Technology Corporation

LIST OF CONTENT

1 GENERAL DESCRIPTION

The ST77922 is a System-on-Chip (SoC) driver LSI designed for TFT LCD controller with a build-in touch panel controller and suitable for small to medium size portable devices. ST77922 can support up to 160,000 pixels in resolution with RAM for dual gate, 560RGBx640 Ram-less for dual gate and support 16M color. The 840-channel source driver has true 6-bit resolution, which generates 64 Gamma-corrected values by an internal D/A converter. The ST77922 is capable of connecting directly to an external microprocessor, and provides 8-bits parallel interface, 8-bit RGB Interface, MIPI interface, 3/4-line serial peripheral interface (3/4 SPI), and Quad serial peripheral interface (QSPI). The ST77922 touch protocol via standard integrated circuit bus(I2C) or serial peripheral interface(SPI). Display data can be stored in the on-chip display data RAM . It can perform display data RAM read/write operation with no external operation clock to minimize power consumption. In addition, because of the integrated power supply circuit necessary to drive liquid crystal; it is possible to make a display system with fewest components.

2 FEATURES

2.1 Display

- Single chip TFT-LCD Controller/Driver with On-chip Display 1/2RAM(200x400*24bits)
- Display Resolution: Dual Gate
  - 400*RGB (H) *400(V) ~ 120*RGB (H) *120(V)
  - 400RGB~560*RGB (H) *640(V) , Ram-less
  - 560RGB (H) *960(V) , Ram-less (Note)
- LCD Driver Output Circuits
  - Source Outputs: 280 RGB Channels
  - Support gate control signals to gate driver in the panel
- Display Colors (Color Mode)
  - Full Color: 16M, RGB=(888) max., Idle Mode Off
  - Color Reduce: 8-color, RGB=(111), Idle Mode On
- Programmable Pixel Color Format (Color Depth) for Various Display Data Input Format
  - 16-bit/pixel: RGB=(565) 65K color
  - 18-bit/pixel: RGB=(666) 262K color
  - 24-bit/pixel: RGB=(888) 16M color
- Interface
  - Parallel 8080-series MCU Interface (8-bit)
  - 8-bit RGB Interface (VSYNCX, HSYNCX, DOTCLK, ENABLE, DB[7:0])
  - Serial Peripheral Interface (SPI Interface) and 2 data lane SPI
  - Quad Serial Peripheral Interface (QSPI Interface)
  - MIPI Display Serial Interface (DSI V1.01 r11 and D-PHY V1.0, 1 clock and 1 data lane pairs)
- Display Features
  - CDC Function
  - CABC Function
  - Sunlight Readability Enhancement (SRE)
- On Chip Build-In Circuits
  - DC/DC Converter
  - Non-Volatile (NV) Memory
  - Adjustable VCOM Generation
  - Timing Controller
  - Internal VPP for NV Memory
- Build-In NV Memory for LCD Initial Register Setting
  - OTP to store VCOM & ID1~ID3
- Driving Algorithm
  - 1/2/4-dot Inversion

- Column Inversion

- Wide Supply Voltage Range
  - I/O Voltage (VDDI to GND): 1.65V ~ 3.6V (VDDI≦VDD)
  - Voltage for Analog Circuit (VDD to GND): 2.65V ~ 3.6V
- On-Chip Power System
  - VCOM level: GND
  - Gamma(+) voltage range: 3.8V~6.4V
  - Gamma(-) voltage range: -4.4V~-2V
  - VGH voltage range: 7.5V~15.5V
  - VGL voltage range: -6.5V~-12V
  - Adjustable voltage range for feed through compensation: 0.1V~2.2V
- Power saving modes
  - Sleep in mode
  - Deep Standby mode
  - Low frame mode 1~50Hz
- Others
  - Zero-Cap
  - GIP + Dual-Gate driving
  - Dual-Gate: pixel number must be a multiple of 4
- Optimized layout for COG Assembly
- Operate temperature range: −30ºC to +85 ºC
- Lower Power Consumption

Note: Max 960 Gate channels varied by vertical resolution & frame rate

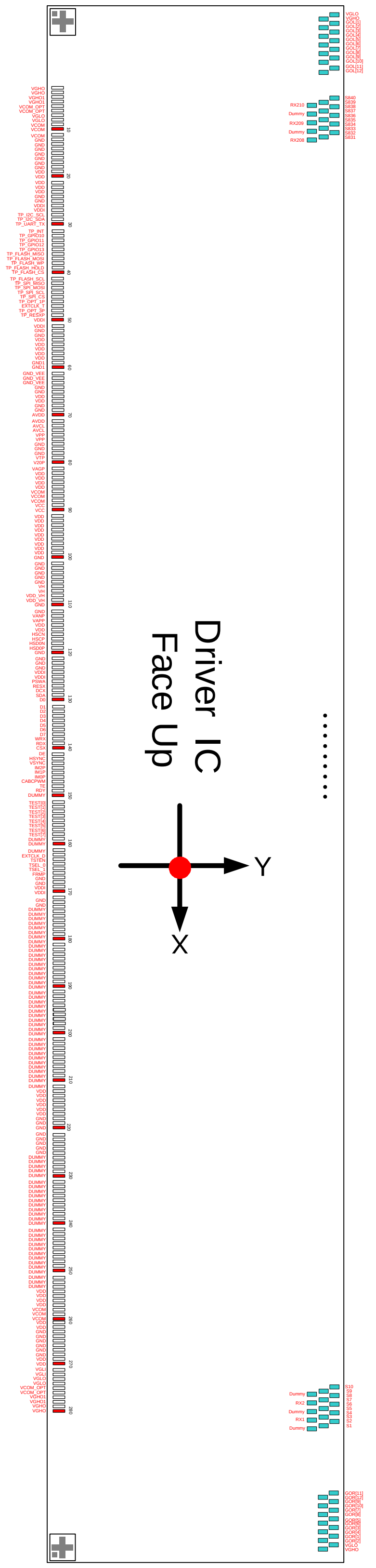
2.2 Touch

- 16-bits MCU optimized for capacitive sensing and other human interactions
- 5-point coordinate sensing without ghost point
- 210 RX Channel
- TP Sensor Pitch : 4mm x 4mm or more
- High signal to noise ratio(SNR)
- I2C & SPI protocol for communication with the host
- Support Long-V sensing mode
- Support noise detection and automatic frequency hopping
- Support Flush function
- Wake Up gesture
- Self-Capacitance detection
- Power saving for Finger Detection
- Touch mapping flexible
- Touch interface access Display register
- Touch FW Host/Flash download

3 PAD ARRANGEMENT

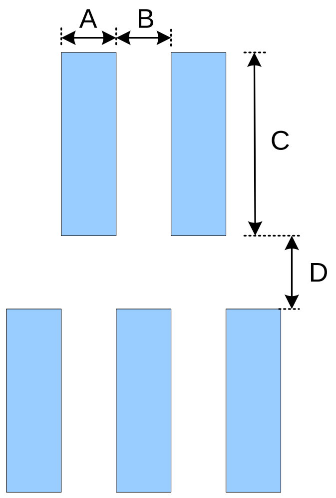
3.1 Output Bump Dimension

Au bump height	9um
Au bump size	**13µm x 70µm :** GIP : VGHO、VGL、GOR1~GOR12、GOL1~GOL12 Source : S1~S840 RX1~210
	30µm x 73µm : Pad1~Pad280



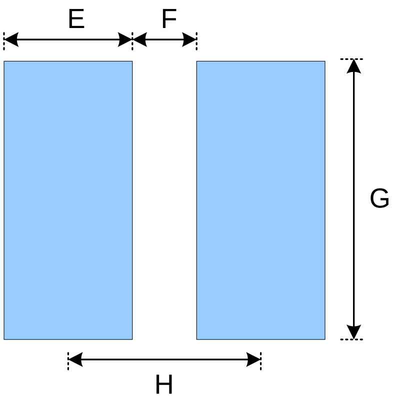
3.2 Bump Dimension

●Output Pads



Symbol	Item	Size
A	Bump Width	13 um
B	Bump Gap 1 (Horizontal)	15 um
C	Bump Height	70 um
D	Bump Gap 2 (Vertical)	20 um

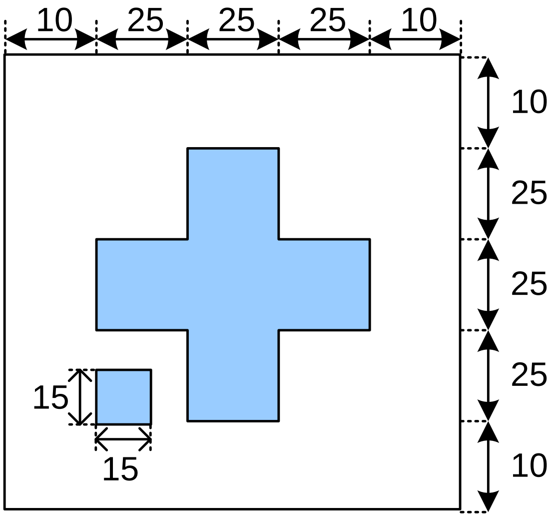
●Input Pads



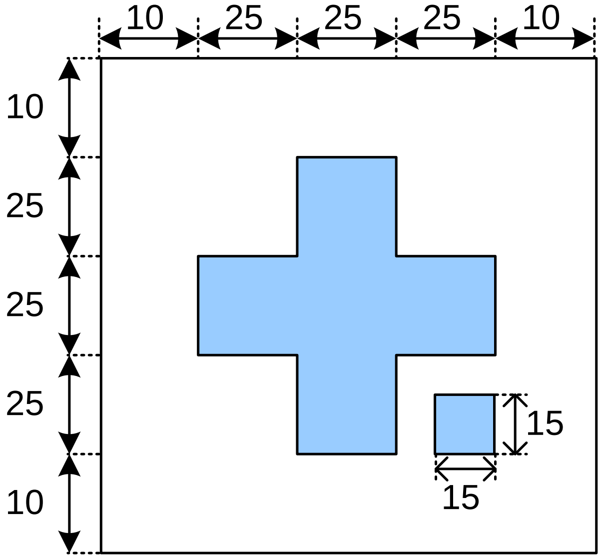
Symbol	Item	Size
E	Bump Width	30 um
F	Bump Gap	15 um
G	Bump Height	73 um
H	Bump Pitch	45 or 49 um

3.3 Alignment Mark Dimension

● **Alignment Mark Left: L(X,Y)=(-6676.8, -431.954)**



● **Alignment Mark Right: R(X,Y)= (6677.201, -431.954)**

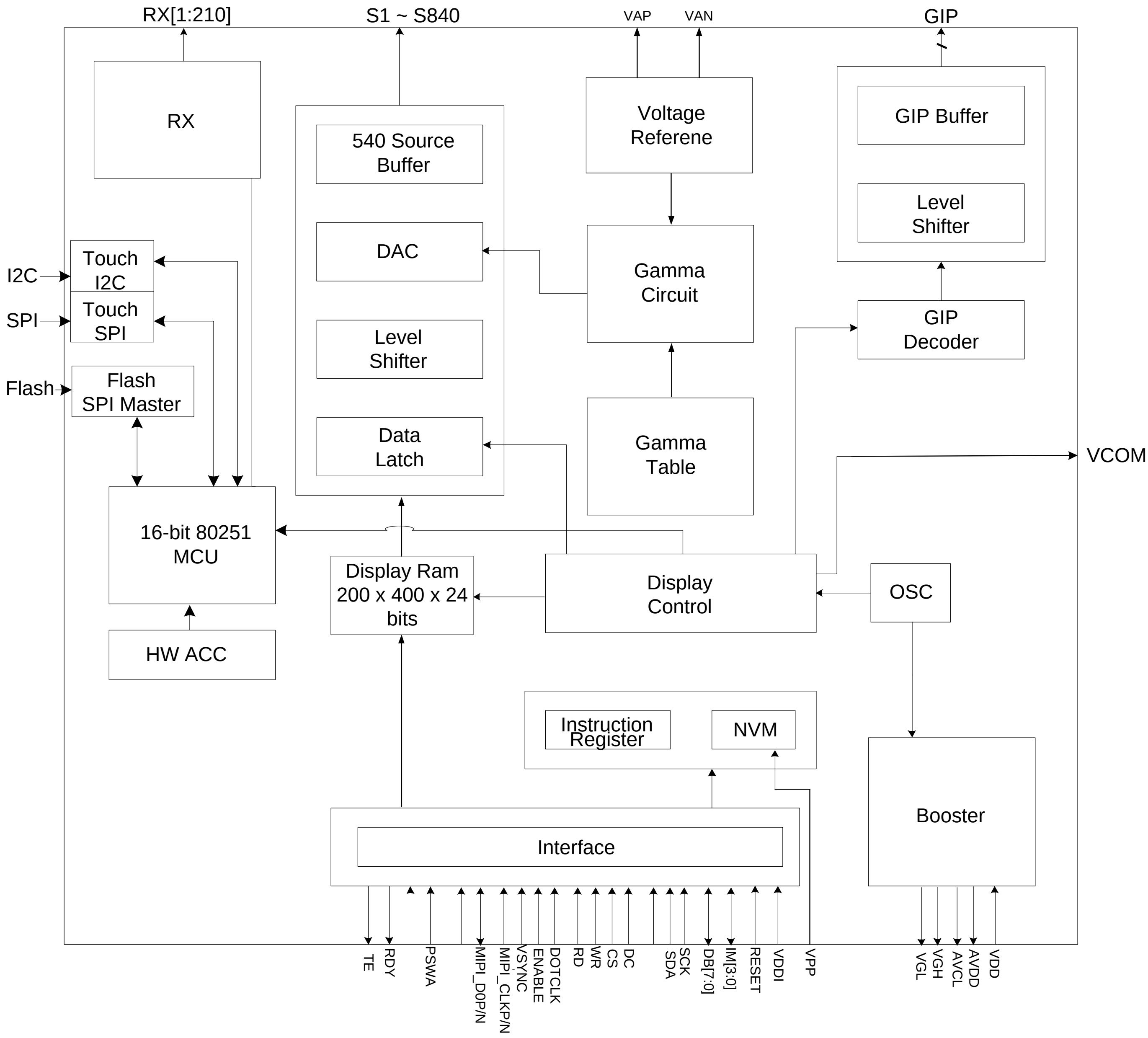


3.4 Chip Information

Chip size	13460 um x 970 um
Chip thickness	200 um
Pad Location	Pad center
Coordinate Origin	Chip center

Note: The chip size is not include the scribe line.u

4 BLOCK DIAGRAM



5 PIN DESCRIPTION

5.1 Power Supply Pins

Name	I/O	Description	Connect Pin
VDD	I	- Power Supply for Analog, Digital System and Booster Circuit.	VDD
VDD_VH	I	- Power Supply for internal Circuit.	VDD
VDDI	I	- Power Supply for I/O System. - VDDI must be lower than or equal to VDD.	VDDI
GND, GND1, GND_VEE	I	- System Ground for Analog System, Digital System, I/O System and Booster Circuit.	GND
VPP	I	- Power Supply for Internal NVM. - When programming NVM, It needs external power supply voltage (8.25V). - The current of Ivpp must be more than 10mA. **- If select internal power then leaves these pins open when not on use.**	External Power

5.2 Interface Logic Pins

<table>
<tr><th>Name</th><th>I/O</th><th>Description</th><th>Connect Pin</th></tr>
<tr><td>IM2P
IM1P
IM0P</td><td>I</td><td>-The System interface mode select.
<table>
<tr><th>IM2</th><th>IM1</th><th>IM0</th><th>MPU Interface Mode</th><th>Data pin</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>3-line 9bit serial I/F</td><td>SDA: in/out</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>4-line 8bit serial I/F</td><td>SDA: in/out</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>2 data lane serial I/F</td><td>SDA0: in/out、SDA1: in</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>QSPI I/F</td><td>SDA[3:0]: in/out</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>RGB_3-line 9bit serial I/F</td><td>SDA: in/out、DB[7:0]</td></tr>
<tr><td>1</td><td>0</td><td>1</td><td>RGB_4-line 8bit serial I/F</td><td>SDA: in/out、DB[7:0]</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>MIPI I/F</td><td>DP/DN</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>80-8bit parallel I/F</td><td>D[7:0]</td></tr>
</table>
</td><td>GND/VDDI</td></tr>
<tr><td>RESX</td><td>I</td><td>-Global reset signal</td><td>MCU</td></tr>
<tr><td>TE</td><td>O</td><td>-Tearing effect output pin to synchronies MCU to frame writing, activated by S/W command. When this pin is not activated (TE function OFF), this pin is GND level.
-If not used, please let this pin open.</td><td>MCU</td></tr>
</table>

Name	I/O	Description	Connect Pin
D[7:0]	I/O	-D[7:0] are used as MCU parallel interface data bus, RGB interface data bus. -SDA2(D0),SDA3(D1) are used as QSPI interface data bus. (SDA2、SDA3) -If not used, please fix this pin at GND.	MCU
SDA	I/O	-Data pin - SDA signal in 3-line serial ,4-line serial , RGB 3-line ,RGB 4-line - SDA0 signal in 2 data lane serial ,QSPI (SDA0) -If not used, please fix this pin at GND.	MCU
CSX	I	-Chip select pin. CSX='0' : Low enable. CSX='1' : High disable -If not used, please fix this pin at VDDI.	MCU
DCX	I	-Display data/command selection pin in parallel interface. DCX='1': display data or parameter. DCX='0': command data. -Display data/command selection pin in 4-line serial interface. (A0) DCX='1': display data or parameter. DCX='0': command data. -SDA1 signal in 2 data lane serial ,QSPI. (SDA1) -If not used, please fix this pin at VDDI.	MCU
WRX	I	-Write enable in MCU parallel interface. - Dot clock signal in RGB interface. (DOTCLK) -If not used, please fix this pin at VDDI	MCU
RDX	I	-Read enable in MCU parallel interface. -SCL signal in 3-line serial ,4-line serial , RGB 3-line ,RGB 4-line, 2 data lane serial ,QSPI (SCL) -If not used, please fix this pin at VDDI.	MCU
DE	I	-Data enable signal in RGB interface -If not used, please fix this pin at VDDI.	MCU
HSYNC	I	-Horizontal (Line) synchronizing input signal in RGB interface. -If not used, please fix to the VDDI.	MCU
VSYNC	I	-Vertical (Frame) synchronizing input signal in RGB interface. -If not used, please fix to the VDDI.	MCU
PSWA	I	-Differential clock polarity swap in MIPI-DSI interface. -If not used, please fix to the GND. -	GND/VDDI

<table>
<tr><th>Name</th><th>I/O</th><th>Description</th><th>Connect Pin</th></tr>
<tr><td></td><td></td><td><table>
<tr><th rowspan="2">PSWA</th><th rowspan="2">IM2</th><th rowspan="2">IM1</th><th rowspan="2">IM0</th><th colspan="4">MIPI I/F</th></tr>
<tr><th>HSCP</th><th>HSCN</th><th>HSD0P</th><th>HSD0N</th></tr>
<tr><td>0</td><td rowspan="2">1</td><td rowspan="2">1</td><td rowspan="2">0</td><td>HSCP</td><td>HSCN</td><td>HSD0P</td><td>HSD0N</td></tr>
<tr><td>1</td><td>HSCN</td><td>HSCP</td><td>HSD0N</td><td>HSD0P</td></tr>
</table></td><td></td></tr>
<tr><td>HSCP</td><td>I</td><td>-MIPI-DSI clock lane positive-end input pin.
-If not used, please fix this pin at GND.</td><td>MCU</td></tr>
<tr><td>HSCN</td><td>I</td><td>-MIPI-DSI clock lane negative-end input pin.
-If not used, please fix this pin at GND.</td><td>MCU</td></tr>
<tr><td>HSD0P</td><td>I/O</td><td>-MIPI-DSI data lane positive-end input pin.
-If not used, please fix this pin at GND.</td><td>MCU</td></tr>
<tr><td>HSD0N</td><td>I/O</td><td>-MIPI-DSI data lane negative-end input pin.
-If not used, please fix this pin at GND.</td><td>MCU</td></tr>
<tr><td>RDY</td><td>O</td><td>-Compressed processing busy flag is used to notice Host that compressed processing has completed.
-If not used, please let this pin open.</td><td>OPEN</td></tr>
</table>

5.3 Driver Output Pins

Name	I/O	Description	Connect pin
S[840:1]	O	-Source output voltage signals applied to liquid crystal.	LCD
GOR[12:1] GOL[12:1]	O	-Gate control signals and the swing voltage level is VGH to VGL.	LCD
VGHO	O	-Power output (Positive) pin for gate driver.	LCD
VGLO	O	-Power output (Negative) pin for gate driver.	LCD
VGHO1	O	-Power output (Positive) pin for IGZO case. -Short VGHO for A-Si case	OPEN
VCOM	O	-A power supply for the TFT-LCD common electrode.	LCD
VCOM_OPT	O	-VCOM with Display scan , modulation signal with touch scan	LCD
CABCPWM	O	-PWM output signal to driving LED. -If not used, please let this pin open.	CABC

5.4 Touch Related Pin

Name	I/O	Description	Connect pin
TP_RESXP	I	-External reset for TP.	MCU
TP_OPT_1P	I	-Boot From Flash Pin , TP_OPT_1P ='0' : Host download. TP_OPT_1P ='1' : Flash Boot.	GND/VDDI
TP_OPT3P	I	-Touch test pin. -If not used, please let this pin GND.	GND
TP_UART_TX	O	-UART TX pad. -If not used, please let this pin open.	OPEN
TP_INT	O	-Touch screen interrupt. -If not used, please let this pin open.	MCU
TP_I2C_SCL	I/O	-I2C interface data pin. -If not used, please let this pin open.	MCU
TP_I2C_SDA	I/O	-I2C interface clock pin. -If not used, please let this pin open.	MCU
TP_SPI_MISO	O	-Slave input data pin in SPI interface. -If not used, please let this pin open.	MCU
TP_SPI_MOSI	I	-Slave output data pin in SPI interface. -If not used, please let this pin VDDI or Open.	MCU
TP_SPI_SCL	I	-Slave clock pin in SPI interface. -If not used, please let this pin VDDI or Open.	MCU
TP_SPI_CS	I	-Slave chip select pin in SPI interface. -If not used, please let this pin VDDI.	MCU
TP_FLASH_ HOLD	I/O	- Master hold signal in SPI interface. -If not used, please let this pin open.	FLASH
TP_FLASH_WP	I/O	-Master write protect in SPI interface. -If not used, please let this pin open.	FLASH
TP_FLASH_CS	O	-Master chip select in SPI interface. -If not used, please let this pin open.	FLASH
TP_FLASH_ MISO	I/O	-Master input data in SPI interface. -If not used, please let this pin open.	FLASH
TP_FLASH_ MOSI	I/O	-Master output data in SPI interface. -If not used, please let this pin open.	FLASH
TP_FLASH_ SCL	O	-Master clock signal in SPI interface. -If not used, please let this pin open.	FLASH
VAGP	O	-Modulation signal	OPEN

RX[210:1]	O	-TP Sensor Pins.	LCD

5.5 Test and Other Pins

Name	I/O	Description	Connect pin
TP_GPIO10~13	I/O	-Touch test pin. -Please leave it open	OPEN
EXTCLK_T	I	-Touch test pin. -If not used, please let this pin VDDI.	VDDI
EXTCLK_D	I	-Driver test pin. -If not used, please let this pin VDDI.	VDDI
TSTEN	I	-Driver test pin. -Please leave it open	OPEN
TSEL_0	I	-Driver test pin. -Please leave it open	OPEN
TSEL_1	I	-Driver test pin. -Please leave it open	OPEN
FRMP	I	-Driver test pin. -If not used, please let this pin GND.	GND
VGLI	O	-Used for monitoring -Please leave it open	OPEN
VCC	O	-Used for monitoring. -Please leave it open	OPEN
AVDD	O	-Power Pad for analog Circuit. -Please leave it open	OPEN
AVCL	O	-Power Pad for analog Circuit. -Please leave it open	OPEN
VAPP	O	-A power output of grayscale voltage. -Please leave it open	OPEN
VANP	O	-A power output (negative) of gray scale voltage. -Please leave it open	OPEN
VH	O	-Used for monitoring. -Please leave it open	OPEN
VTP	O	-Used for monitoring. -Please leave it open	OPEN
V20P	O	-Used for monitoring. -Please leave it open	OPEN

TEST[7:0]	O	-This pin is for testing -Please leave it open.	OPEN
DUMMY	-	-These pins are dummy. -Leave the pin open.	OPEN

6 DRIVER ELECTRICAL CHARACTERISTICS

6.1 Absolute Operation Range

Item	Symbol	Range	Unit
Supply Voltage (Analog)	VDD	- 0.3 ~ +4.6	V
Supply Voltage (I/O)	VDDI	- 0.3 ~ +4.6	V
Supply Voltage (Logic)	VCC	-0.3 ~ +2	V
Driver Supply Voltage	VGH-VGL	-0.3 ~ +30.0	V
Logic Input Voltage Range	VIN	0.5 ~ VDDI + 0.5	V
Logic Output Voltage Range	VO	0.5 ~ VDDI + 0.5	V
Operating Temperature Range	TOPR	-30 ~ +85	℃
Storage Temperature Range	TSTG	-40 ~ +125	℃

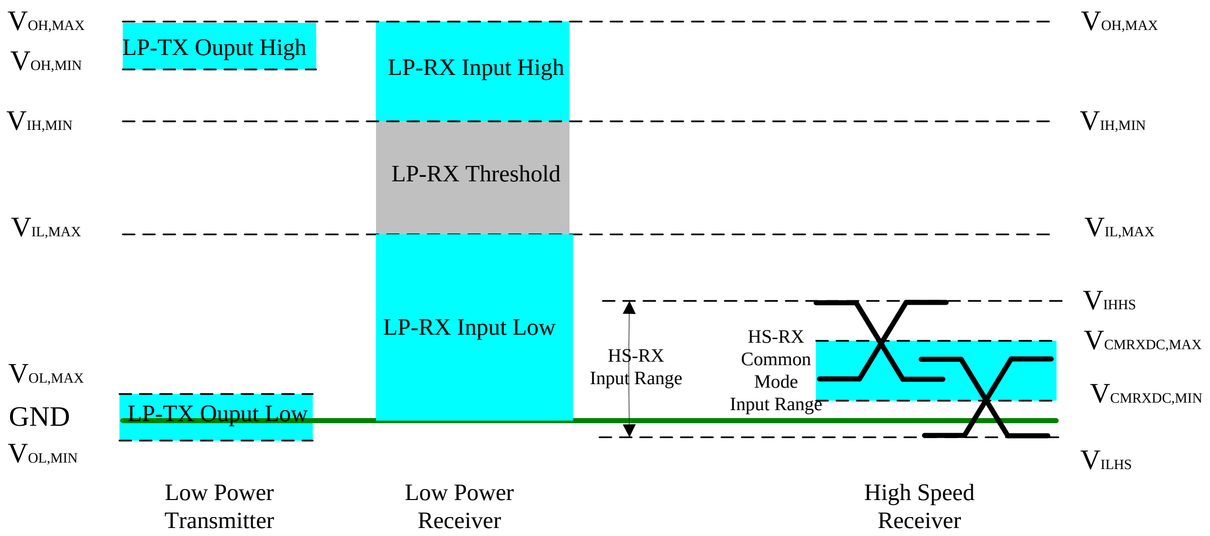
Absolute Operation Range

Note: If one of the above items is exceeded its maximum limitation momentarily, the quality of the product may be degraded. Absolute maximum limitation, therefore, specify the values exceeding which the product may be physically damaged. Be sure to use the product within the recommend range.

6.2 DC Characteristics

6.2.1 DC characteristics for MIPI DSI

● MIPI Signaling Voltage Levels



● MIPI DC characteristics

Parameter	Symbol	Specification			Unit
		MIN	TYP	MAX	
Operation Voltage for MIPI Receiver					
Low power mode operating voltage	V_{LPH}	1.1	1.2	1.3	V
MIPI Characteristics for High Speed Receiver					
Single-ended input low voltage	V_{ILHS}	-40	-	-	mV
Single-ended input high voltage	V_{IHHS}	-	-	460	mV
Common-mode voltage	V_{CMRXDC}	70	-	330	mV
Differential input impedance	Z_{ID}	80	100	125	ohm
MIPI Characteristics for Low Power Mode					
Pad signal voltage range	V_I	-50	-	1350	mV
Logic 0 input threshold	V_{IL}	0	-	550	mV
Logic 1 input threshold	V_{IH}	880	-	1350	mV
Output low level	V_{OL}	-50	-	50	mV
Output high level	V_{OH}	1.1	1.2	1.3	V

6.2.2 DC characteristics for Panel Driving

Parameter	Symbol	Condition	Specification			Unit	Related Pins
			MIN.	TYP.	MAX.		
Power & Operation Voltage							
System Voltage	VDD	Operating voltage	2.65	2.8	3.6	V	-
Interface Operation Voltage	VDDI	I/O Supply Voltage	1.65	1.8	3.6	V	-
Gate Driver High Voltage	VGH	-	7.5	-	15.5	V	-
Gate Driver Low Voltage	VGL	-	-12	-	-6.5	V	-
Gate Driver Supply Voltage	-	| VGH-VGL |	-	-	27.5	V	-
Input / Output							
Logic-High Input Voltage	VIH	-	0.7VDDI	-	VDDI	V	Note 1
Logic-Low Input Voltage	VIL	-	GND	-	0.3VDDI	V	Note 1
Logic-High Output Voltage	VOH	IOH = -1.0mA	0.8VDDI	-	VDDI	V	Note 1
Logic-Low Output Voltage	VOL	IOL = +1.0mA	GND	-	0.2VDDI	V	Note 1
Logic-High Input Current	IIH	VIN = VDDI	-	-	1	uA	Note 1
Logic-Low Input Current	IIL	VIN = GND	-1	-	-	uA	Note 1
Input Leakage Current	ILI	IOH = -1.0mA	-0.1	-	+0.1	uA	Note 1
VCOM Voltage							
VCOM Voltage	VCOM	-	-	GND	-	V	-
Source Driver							
Gamma Reference Voltage(Positive)	VAP	-	3.8	-	6.4	V	-
Gamma Reference Voltage(Negative)	VAN	-	-4.4	-	-2	V	-
Source Output Settling Time	Tr	Below with 99% precision	-	-	20	us	Note 2

Basic DC Characteristics

Notes:

1. The max. value is between measured point of source output and gamma setting value.

6.3 Power Consumption

Ta=25℃, Registers setting are IC default setting.

Operation Mode	Current Consumption	
	Typical	
	IDDI(mA)	IDD(mA)
Sleep-in Mode	0	6
Deep Standby Mode	0	0

Notes:

1. The Current Consumption is DC characteristics of ST77922.

2. Sleep in Mode (TP Flash Boot mode Enable, TP Not Power Down mode)

6.4 AC Characteristics

6.4.1 8080 Series MCU Parallel Interface Characteristics: 8-bit Bus

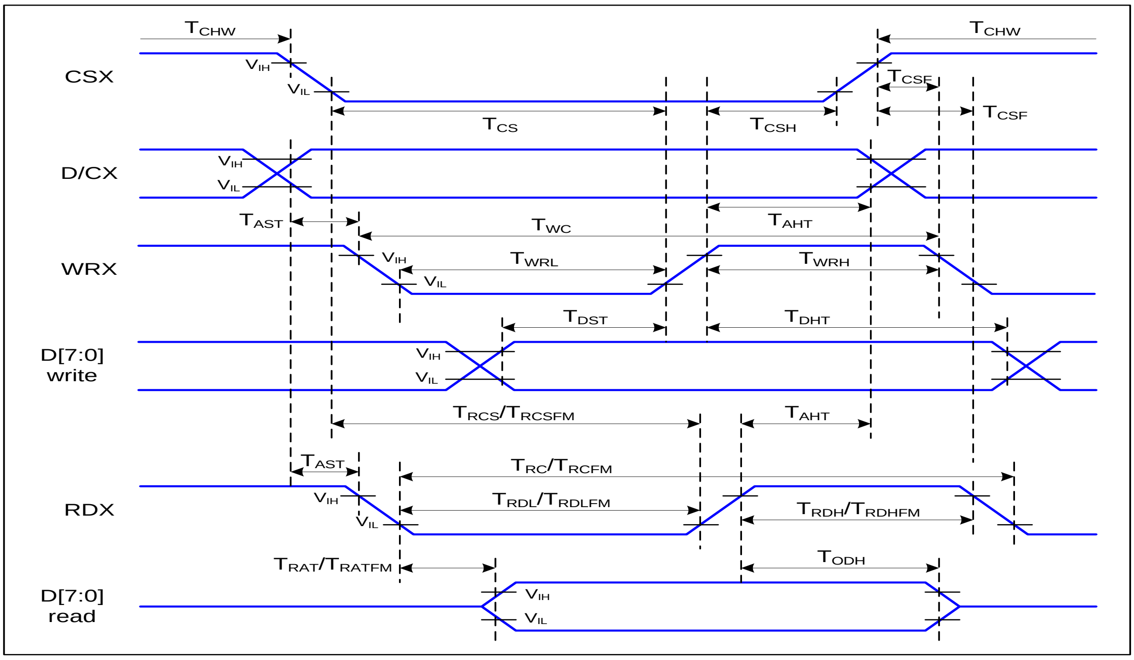


Figure 1 Parallel Interface Timing Characteristics (8080-Series MCU Interface)

VDDI=1.8V, VDD=2.8V, Ta=25℃

Signal	Symbol	Parameter	Min	Max	Unit	Description
D/CX	T_{AST}	Address setup time	0		ns	-
	T_{AHT}	Address hold time (Write/Read)	10		ns	
CSX	T_{CHW}	Chip select “H” pulse width	0		ns	-
	T_{CS}	Chip select setup time (Write)	15		ns	
	T_{RCS}	Chip select setup time (Read ID)	45		ns	
	T_{RCSFM}	Chip select setup time (Read FM)	355		ns	
	T_{CSF}	Chip select wait time (Write/Read)	10		ns	
	T_{CSH}	Chip select hold time	10		ns	
WRX	T_{WC}	Write cycle	26		ns	
	T_{WRH}	Control pulse “H” duration	13		ns	
	T_{WRL}	Control pulse “L” duration	13		ns	
RDX (ID)	T_{RC}	Read cycle (ID)	160		ns	When read ID data
	T_{RDH}	Control pulse “H” duration (ID)	90		ns	
	T_{RDL}	Control pulse “L” duration (ID)	45		ns	
RDX (FM)	T_{RCFM}	Read cycle (FM)	450		ns	When read from frame memory
	T_{RDHFM}	Control pulse “H” duration (FM)	90		ns	

	T_{RDLFM}	Control pulse “L” duration (FM)	355		ns	
D[7:0]	T_{DST}	Data setup time	10		ns	For CL=30pF
	T_{DHT}	Data hold time	10		ns	
	T_{RAT}	Read access time (ID)		40	ns	
	T_{RATFM}	Read access time (FM)		340	ns	
	T_{ODH}	Output disable time	20	80	ns	

Table 1 8080 Parallel Interface Characteristics

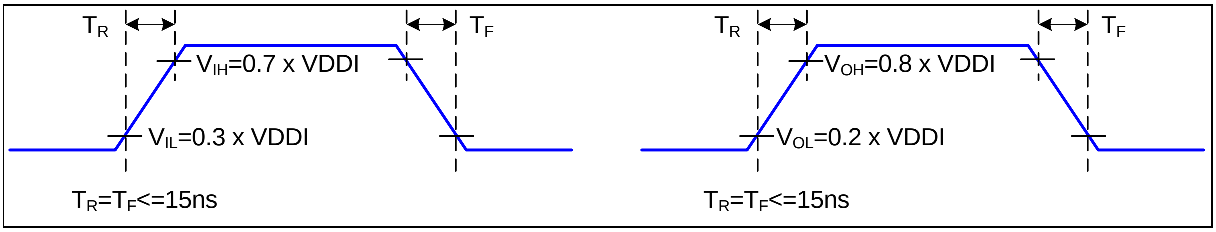


Figure 2 Rising and Falling Timing for I/O Signal

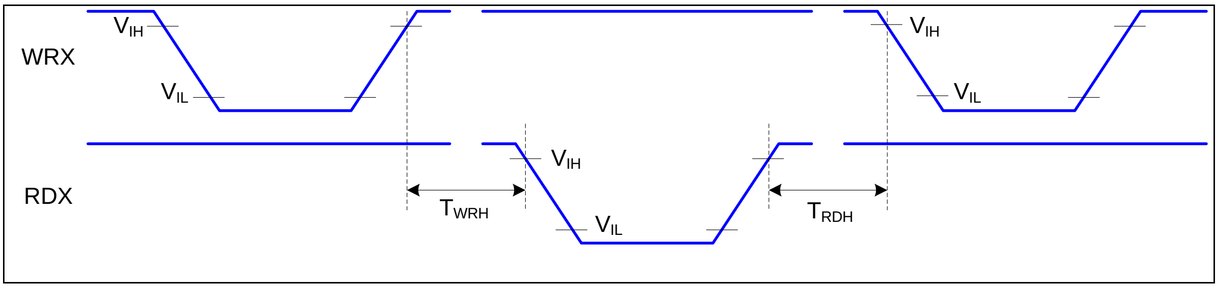


Figure 3 Write-to-Read and Read-to-Write Timing

Note: The rising time and falling time (Tr, Tf) of input signal and fall time are specified at 15 ns or less. Logic high and low levels are specified as 30% and 70% of VDDI for Input signals.

6.4.2 Serial Interface Characteristics (3-line serial):

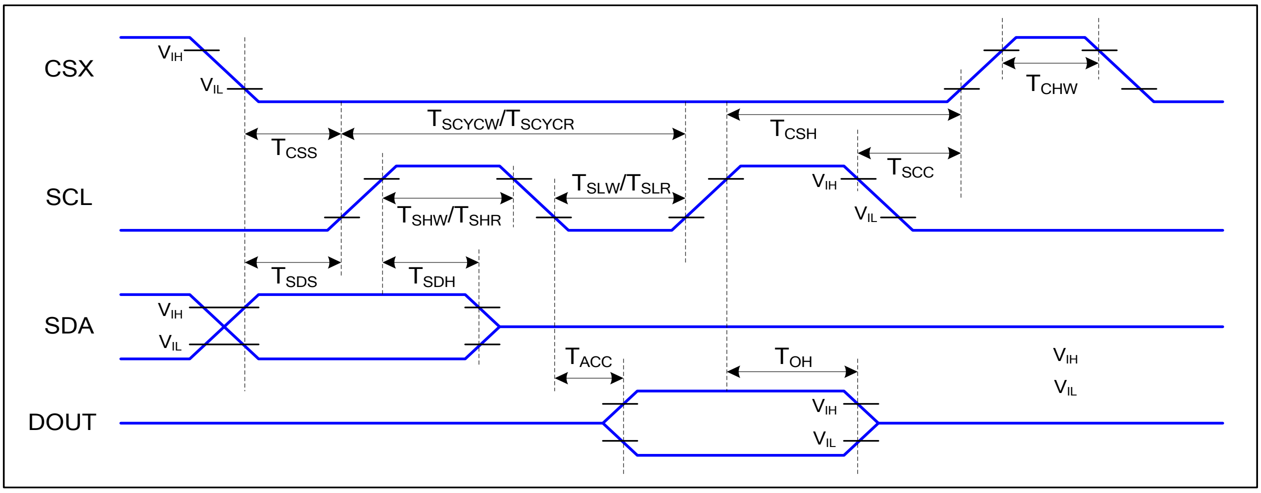


Figure 4 3-line serial Interface Timing Characteristics

VDDI=1.8V, VDD=2.8V, Ta=25℃

Signal	Symbol	Parameter	Min	Max	Unit	Description
CSX	T_{CSS}	Chip select setup time (write)	15		ns	
	T_{CSH}	Chip select hold time (write)	15		ns	
	T_{CSS}	Chip select setup time (read)	60		ns	
	T_{SCC}	Chip select hold time (read)	65		ns	
	T_{CHW}	Chip select “H” pulse width	40		ns	
SCL	T_{SCYCW}	Serial clock cycle (Write)	16		ns	
	T_{SHW}	SCL “H” pulse width (Write)	7		ns	
	T_{SLW}	SCL “L” pulse width (Write)	7		ns	
	T_{SCYCR}	Serial clock cycle (Read)	150		ns	
	T_{SHR}	SCL “H” pulse width (Read)	60		ns	
	T_{SLR}	SCL “L” pulse width (Read)	60		ns	
SDA (DIN)	T_{SDS}	Data setup time	7		ns	
	T_{SDH}	Data hold time	7		ns	
DOUT	T_{ACC}	Access time	10	50	ns	For maximum CL=30pF
	T_{OH}	Output disable time	15	50	ns	For minimum CL=8pF

Table 2 3-line serial Interface Characteristics

Note : The rising time and falling time (Tr, Tf) of input signal are specified at 15 ns or less. Logic high and low levels are specified as 30% and 70% of VDDI for Input signals.

6.4.3 Serial Interface Characteristics (4-line serial):

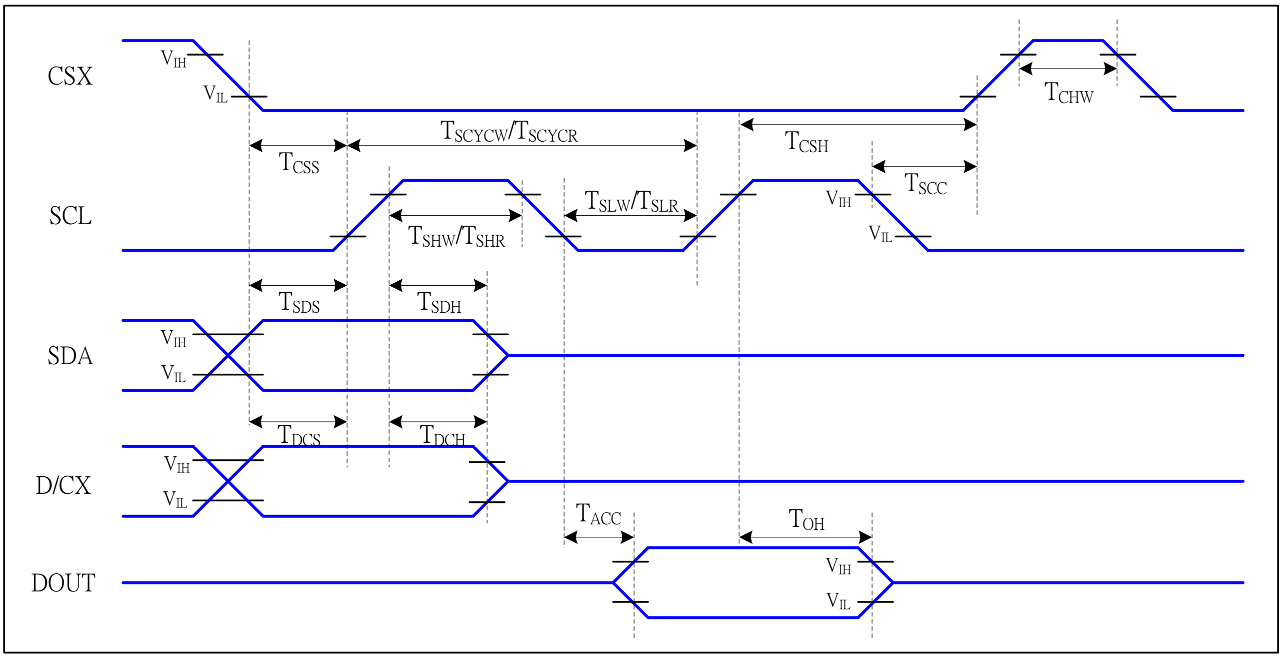


Figure 5 4-line serial Interface Timing Characteristics

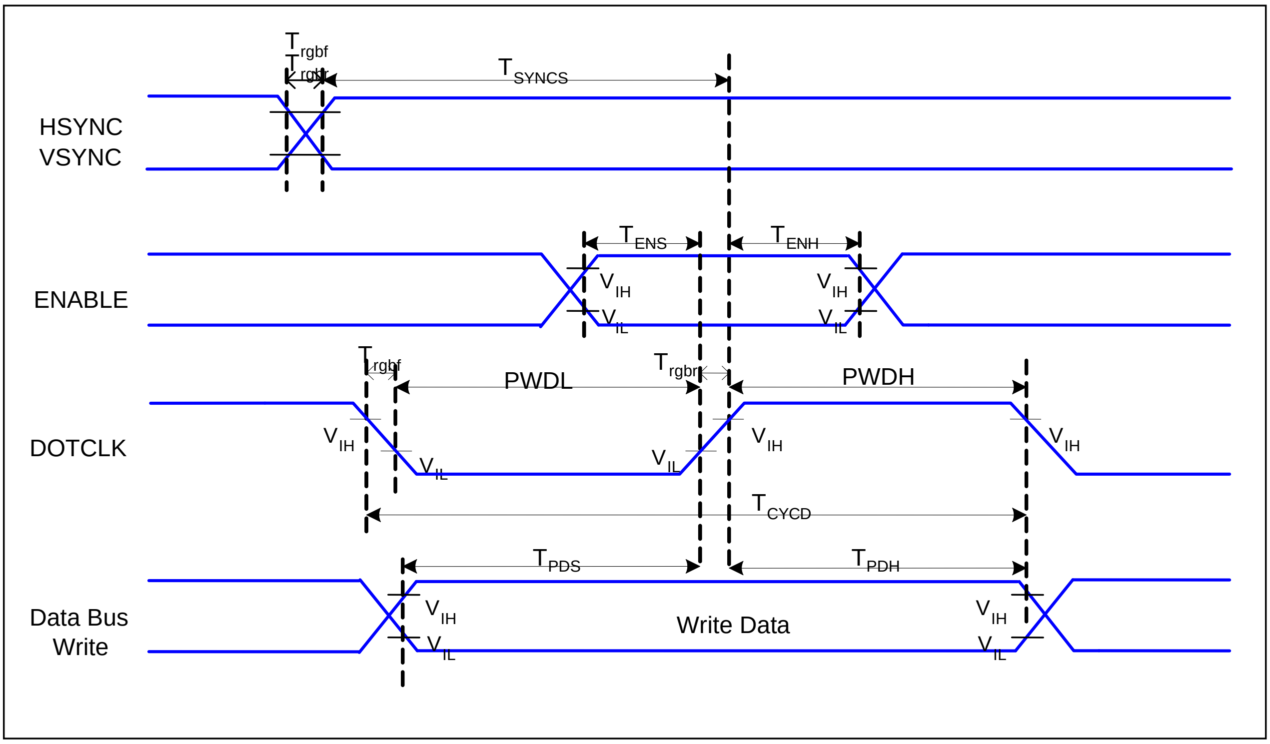
VDDI=1.8V, VDD=2.8V, Ta=25℃

Signal	Symbol	Parameter	MIN	MAX	Unit	Description
CSX	T_{CSS}	Chip select setup time (write)	15		ns	
	T_{CSH}	Chip select hold time (write)	15		ns	
	T_{CSS}	Chip select setup time (read)	60		ns	
	T_{SCC}	Chip select hold time (read)	65		ns	
	T_{CHW}	Chip select “H” pulse width	40		ns	
SCL	T_{SCYCW}	Serial clock cycle (Write)	16		ns	-write command & data ram
	T_{SHW}	SCL “H” pulse width (Write)	7		ns	
	T_{SLW}	SCL “L” pulse width (Write)	7		ns	
	T_{SCYCR}	Serial clock cycle (Read)	150		ns	-read command & data ram
	T_{SHR}	SCL “H” pulse width (Read)	60		ns	
	T_{SLR}	SCL “L” pulse width (Read)	60		ns	
D/CX	T_{DCS}	D/CX setup time	10		ns	
	T_{DCH}	D/CX hold time	10		ns	
SDA (DIN)	T_{SDS}	Data setup time	7		ns	
	T_{SDH}	Data hold time	7		ns	
DOUT	T_{ACC}	Access time	10	50	ns	For maximum CL=30pF
	T_{OH}	Output disable time	15	50	ns	For minimum CL=8pF

Table 3 4-line serial Interface Characteristics

Note : The rising time and falling time (Tr, Tf) of input signal are specified at 15 ns or less. Logic high and low levels are specified as 30% and 70% of VDDI for Input signals.

6.4.4 RGB Interface Characteristics:



VDDI=1.8V, VDD=2.8V, Ta=25℃

Signal	Symbol	Parameter	MIN	MAX	Unit	Description
HSYNC, VSYNC	T_{SYNCS}	VSYNC, HSYNC Setup Time	15	-	ns	
ENABLE	T_{ENS}	Enable Setup Time	15	-	ns	
	T_{ENH}	Enable Hold Time	15	-	ns	
DOTCLK	PWDH	DOTCLK High-level Pulse Width	15	-	ns	
	PWDL	DOTCLK Low-level Pulse Width	15	-	ns	
	T_{CYCD}	DOTCLK Cycle Time	30	-	ns	
	Trghr, Trghf	DOTCLK Rise/Fall time	-	10	ns	
DB	T_{PDS}	PD Data Setup Time	15	-	ns	
	T_{PDH}	PD Data Hold Time	15	-	ns	

Table 4 6 Bits RGB Interface Timing Characteristics

6.4.5 QSPI Interface Characteristics:

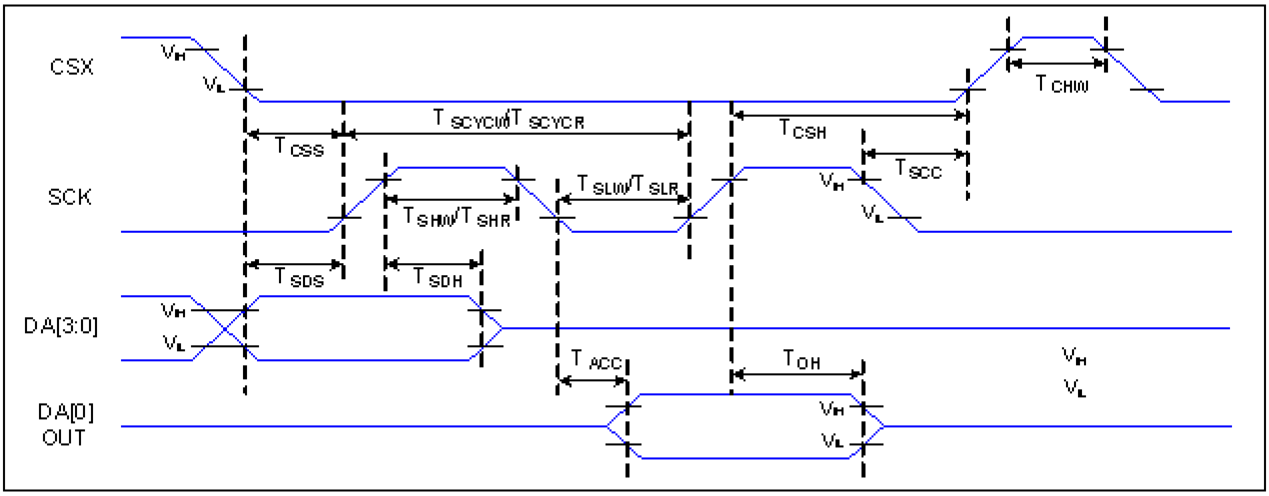


Figure 7 QSPI Interface Timing Characteristics

VDDI=1.8V, VDD=2.8V, Ta=25℃

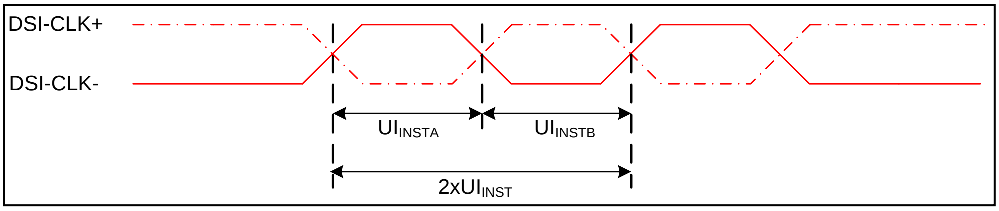
Signal	Symbol	Parameter	Min	Max	Unit	Description
CSX	T_{CSS}	Chip select setup time (write)	15		ns	
	T_{CSH}	Chip select hold time (write)	15		ns	
	T_{CSS}	Chip select setup time (read)	60		ns	
	T_{SCC}	Chip select hold time (read)	65		ns	
	T_{CHW}	Chip select “H” pulse width	40		ns	
			200		ns	*Note1*
SCL	T_{SCYCW}	Serial clock cycle (Write)	16		ns	
	T_{SHW}	SCL “H” pulse width (Write)	7		ns	
	T_{SLW}	SCL “L” pulse width (Write)	7		ns	
	T_{SCYCR}	Serial clock cycle (Read)	150		ns	
	T_{SHR}	SCL “H” pulse width (Read)	60		ns	
	T_{SLR}	SCL “L” pulse width (Read)	60		ns	
SDA (DIN)	T_{SDS}	Data setup time	7		ns	
	T_{SDH}	Data hold time	7		ns	
DOUT	T_{ACC}	Access time	10	50	ns	For maximum CL=30pF
	T_{OH}	Output disable time	15	50	ns	For minimum CL=8pF

Table 5 QSPI Interface Characteristics

Note : The rising time and falling time (Tr, Tf) of input signal are specified at 15 ns or less. Logic high and low levels are specified as 30% and 70% of VDDI for Input signals.

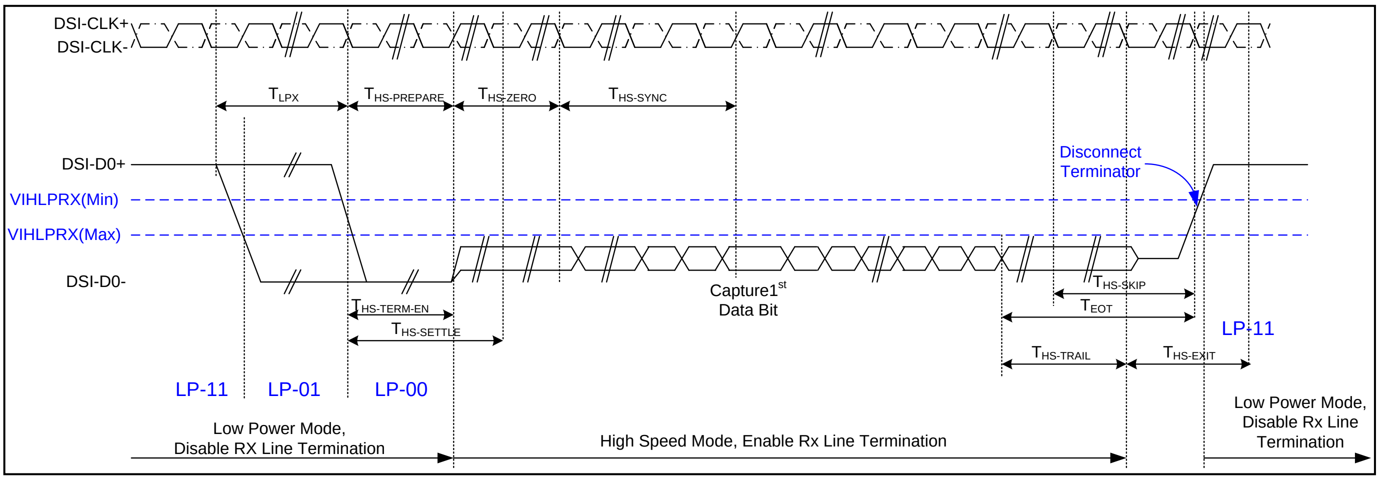
6.4.6 MIPI Interface Characteristics

High Speed Mode – Clock Channel Timing



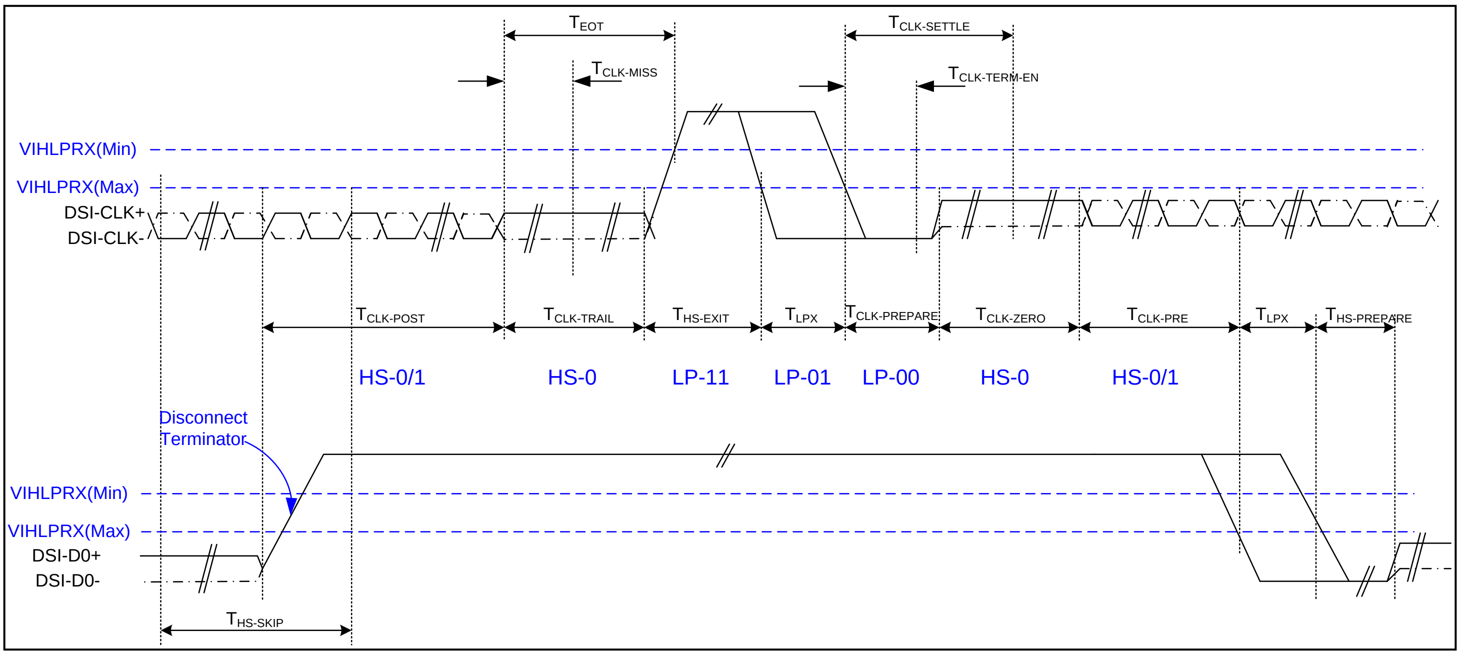
Signal	Symbol	Parameter	MIN	MAX	Unit	Description
DSI-DATA_P/N	2xUI INST	Double UI instantaneous	2.66	25	ns	
DSI-DATA_P/N	UI INSTA ,UI INSTB	UI instantaneous Half	1.33	12.5	ns	

High-Speed Data Transmission



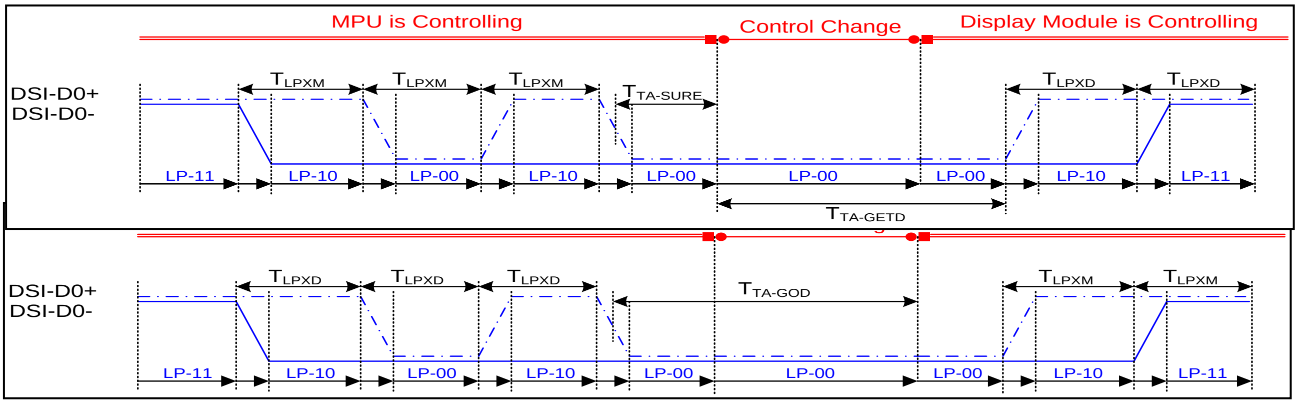
Parameter	Symbol	MIN	TYP	MAX	Unit
Time to drive LP-00 to prepare for HS transmission	$T_{HS\text{-}PREPARE}$	40+4UI		85+6UI	ns
Time from start of t HS-TRAIL or t CLK-TRAIL period to start of LP-11 state	T_{EOT}			105+12UI	ns
Time to enable data receiver line termination measured from when Dn crosses VILMAX	$T_{HS\text{-}TERM\text{-}EN}$			35+4UI	ns
Time to drive flipped differential state after last payload data bit of a HS transmission	$T_{HS\text{-}TRAIL}$	60+4UI			ns
Time-out at RX to ignore transition period of EoT	$T_{HS\text{-}SKIP}$	40		55+4UI	ns
Time to drive LP-11 after HS burst	$T_{HS\text{-}EXIT}$	100			ns
Length of any Low-Power state period	T_{LPX}	50			ns
Sync sequence period	$T_{HS\text{-}SYNC}$		8UI		ns
Minimum lead HS-0 drive period before the Sync sequence	$T_{HS\text{-}ZERO}$	105+6UI			ns

Switching the Clock Lane between Clock Transmission and Low-Power Mode



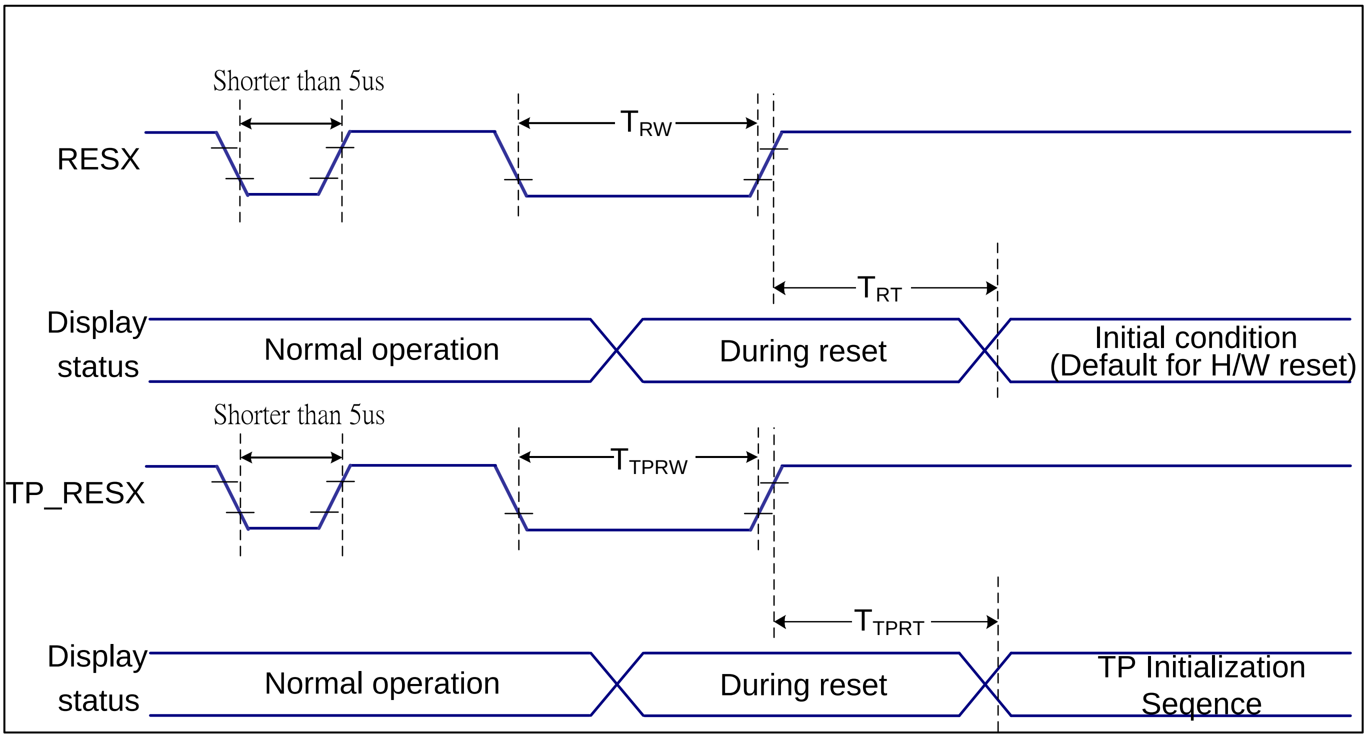
Parameter	Symbol	MIN	TYP	MAX	Unit
Time that the transmitter shall continue sending HS clock after the last associated Data Lane has transitioned to LP mode	$T_{CLK-POST}$	60+52UI			ns
Detection time that the clock has stopped toggling	$T_{CLK-MISS}$			60	ns
Time to drive LP-00 to prepare for HS clock transmission	$T_{CLK-PREPARE}$	38		95	ns
Minimum lead HS-0 drive period before starting Clock	$T_{CLK-PREPARE}$ $+T_{CLK-ZERO}$	300			ns
Time to enable Clock Lane receiver line termination measured from when Dn cross VIL,MAX	$T_{HS-TERM-EN}$			38	ns
Minimum time that the HS clock must be set prior to any associated date lane beginning the transmission from LP to HS mode	$T_{CLK-PRE}$	8			UI
Time to drive HS differential state after last payload clock bit of a HS transmission burst	$T_{CLK-TRAIL}$	60			ns

Bus Turnaround Procedure



Parameter	Symbol	MIN	TYP	MAX	Unit
Length of any Low-Power state period : Master side	T_{LPX}	50		75	ns
Length of any Low-Power state period : Slave side	T_{LPX}	47.5	50	52.5	ns
Ratio of T_{LPX} (MASTER)/ T_{LPX} (SLAVE) between Master and Slave side	Ratio T_{LPX}	2/3		3/2	
Time-out before new TX side start driving	$T_{TA\text{-}SURE}$	T_{LPX}		2 T_{LPX}	ns
Time to drive LP-00 by new TX	$T_{TA\text{-}GET}$		5 T_{LPX}		Ns
Time to drive LP-00 after Turnaround Request	$T_{TA\text{-}GO}$		4 T_{LPX}		ns

6.4.7 Reset Timing



VDDI=1.8V,VDD=2.8V, Ta=25℃

Related Pins	Symbol	Parameter	MIN	MAX	Unit
RESX	TRW	Reset pulse duration	10	-	us
	TRT	Reset cancel	-	5 (Note 1, 5)	ms
			-	120 (Note 1, 6, 7)	ms
TP_RESX	TTPRW	Reset pulse duration	10	-	us
	TTPRT	Reset cancel	-	100	ms

Notes:

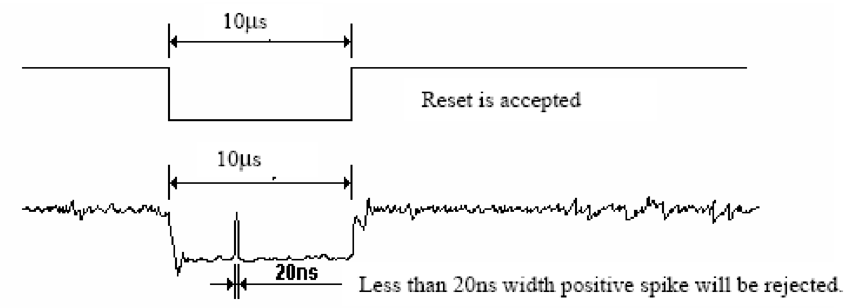
1. The reset cancel includes also required time for loading ID bytes, VCOM setting and other settings from NVM (or similar device) to registers. This loading is done every time when there is HW reset cancel time (tRT) within 5 ms after a rising edge of RESX.

2. Spike due to an electrostatic discharge on RESX line does not cause irregular system reset according to the table below:

RESX Pulse	Action
Shorter than 5us	Reset Rejected
Longer than 9us	Reset
Between 5us and 9us	Reset starts

3. During the Resetting period, the display will be blanked (The display is entering blanking sequence, which maximum time is 120 *ms, when Reset Starts in Sleep Out –mode. The display remains the blank state in Sleep In –mode.) and then return to Default condition for Hardware Reset.*

4. Spike Rejection also applies during a valid reset pulse as shown below:

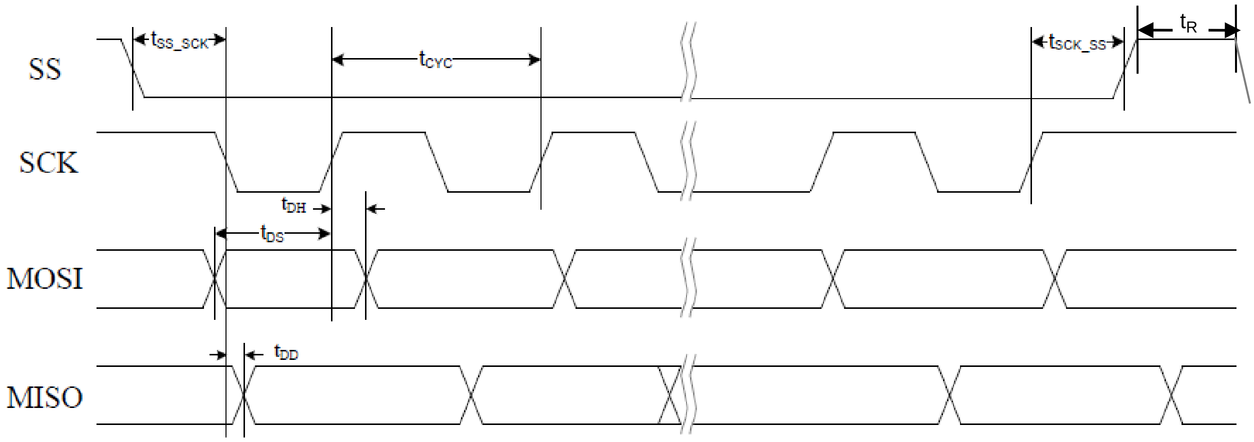


5. When Reset applied during Sleep In Mode.

6. When Reset applied during Sleep Out Mode.

7. It is necessary to wait 5msec after releasing RESX before sending commands. Also Sleep Out command cannot be sent for 120msec.

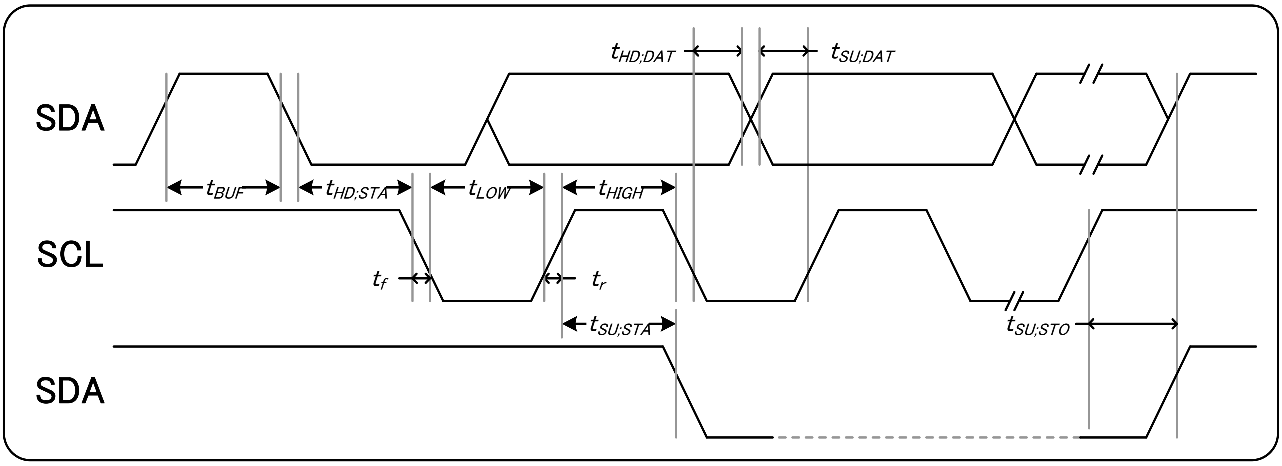
6.4.8 Touch SPI Timing



VDDI=1.8V,VDD=2.8V, Ta=25℃

Symbol	Parameter	Rating			Unit
		Min.	Typ.	Max.	
f_{SCK}	SCK frequency	-	-	12	Mhz
t_{CYC}	SCK cycle time	83	-	-	ns
t_{DS}	Data setup time prior SCK rising	25	-	-	ns
t_{DH}	Data hold time after SCK rising	25	-	-	ns
t_{DD}	MISO data output delay from SCK falling	-	-	50	ns
t_{SS_SCK}	SS falling to 1st SCK falling	1000	-	-	ns
t_{SCK_SS}	SCK rising to SS rising	1000	-	-	ns
t_R	CS recovery time	28	-	-	us

6.4.9 Touch I2C Timing



VDDI=1.8V,VDD=2.8V, Ta=25℃

Item	Signal	Symbol	Rating			Unit
			Min.	Typ.	Max.	
SCL clock frequency	SCL	f_{SCL}			400	khz
SCL clock low period		t_{LOW}	1300			ns
SCL clock high period		t_{HIGH}	600			ns
Data set-up time	SDA	$t_{SU;DAT}$	100			ns
Data hold time		$t_{HD;DAT}$	0			ns
Setup time for a repeated START condition	SDA	$t_{SU;STA}$	600			ns
Start condition hold time		$t_{HD;STA}$	600			ns
Setup time for STOP condition		$t_{SU;STO}$	600			ns
Bus free time between a STOP and START		t_{BUF}	1300			ns

7 INTERFACE

7.1 MPU Interface Type Selection

ST77922 supports 8 bit parallel data bus for 8080 series CPU, RGB serial interfaces, SPI interface, QSPI interface, MIPI interface. Selection of these interfaces are set by IM[2:0] pins as shown below.

IM2	IM1	IM0	MPU Interface Mode	Data pin
0	0	0	3-line 9bit serial I/F	SDA: in/out
0	0	1	4-line 8bit serial I/F	SDA: in/out
0	1	0	2 data lane serial I/F	SDA0: in/out、SDA1: in
0	1	1	QSPI I/F	SDA[3:0]: in/out
1	0	0	RGB_3-line 9bit serial I/F	SDA: in/out、DB[7:0]
1	0	1	RGB_4-line 8bit serial I/F	SDA: in/out、DB[7:0]
1	1	0	MIPI I/F	DP/DN
1	1	1	80-8bit parallel I/F	DB[7:0]

Table 6 Interface Type Selection

7.2 8080-Ⅰ Series MCU Parallel Interface

The MCU can use 8080-8bits parallel interface: 11-lines with 8-data parallel interface. The chip-select CSX (active low) enables/disables the parallel interface. RESX (active low) is an external reset signal. WRX is the parallel data write enable, RDX is the parallel data read enable and D[7:0] is parallel data bus.

The LCD driver reads the data at the rising edge of WRX signal. The D/CX is the data/command flag. When D/CX='1', D[7:0] bits is either display data or command parameter. When D/C='0', D[7:0] bits is command. The interface functions of 8080-8bits parallel interface are given in following table.

IM2	IM1	IM0	Interface	D/CX	RDX	WRX	Read back selection
1	1	1	8-bit parallel	0	1	↑	Write 8-bit command (D7 to D0)
				1	1	↑	Write 8-bit display data or 8-bit parameter (D7 to D0)
				1	↑	1	Read 8-bit display data (D7 to D0)
				1	↑	1	Read 8-bit parameter or status (D7 to D0)

Table 7 the function of 8080 - 8 bits parallel interface

7.2.1 Write cycle sequence

The write cycle means that the host writes information (command / data) to the display via the interface. Each write cycle (WRX high-low-high sequence) consists of 3 control signals (DCX, RDX, WRX) and data signals (DB[7:0]). DCX bit is a control signal, which tells if the data is a command or a data. The data signals are the command if the control signal is low (='0') and vice versa it is data (='1').

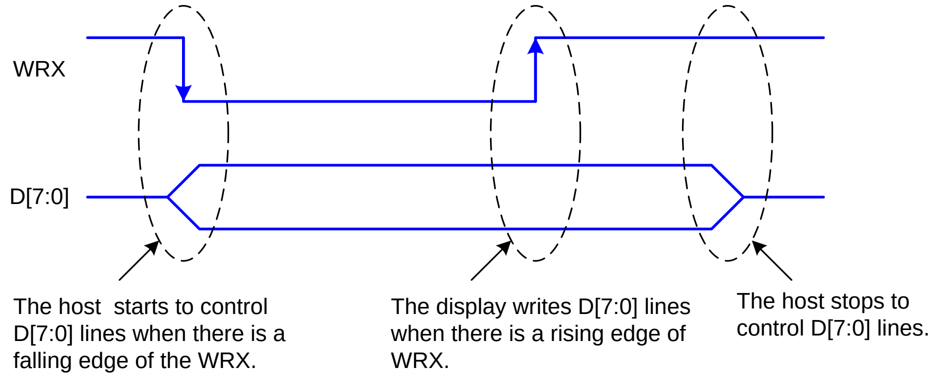


Figure 8 8080 - 8 bits WRX Protocol

Note: WRX is an unsynchronized signal (It can be stopped).

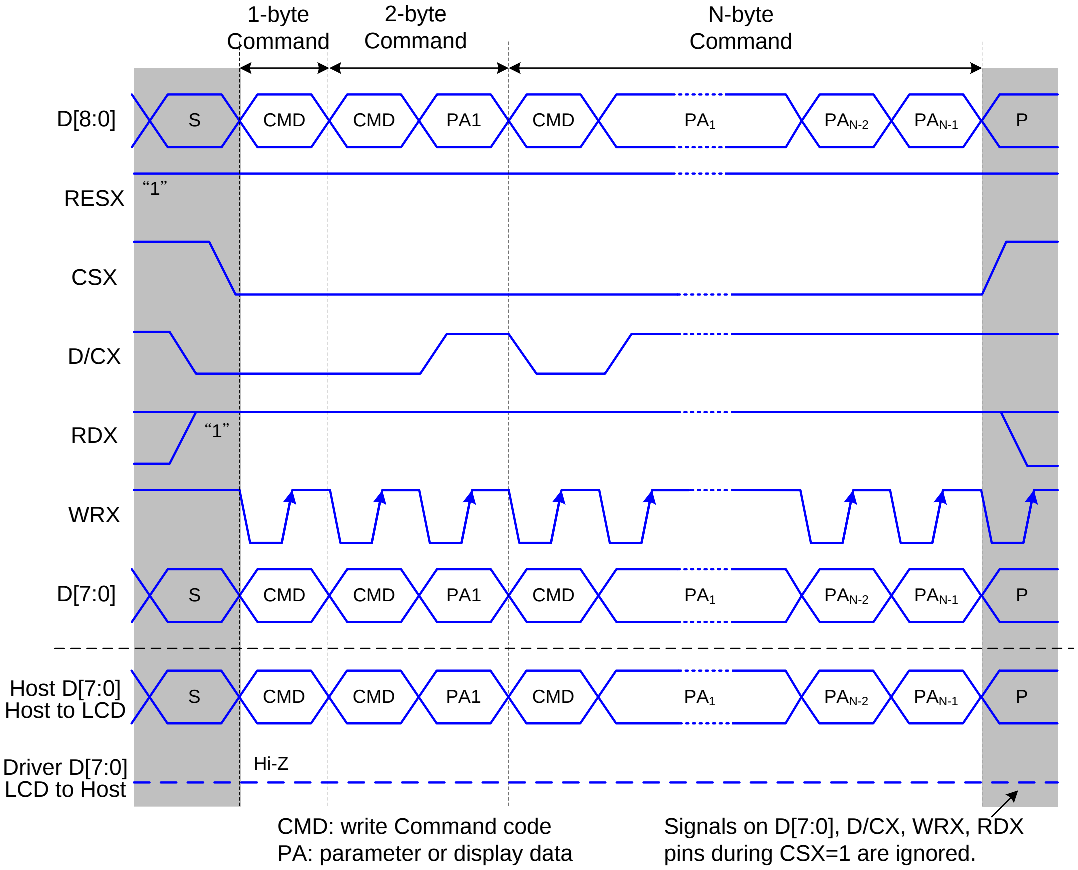


Figure 9 8080 - 8 bits Parallel Bus Protocol, Write to Register or Display RAM

7.2.2 Read cycle sequence

The read cycle (RDX high-low-high sequence) means that the host reads information from LCD driver via interface. The driver sends data (D[7:0]) to the host when there is a falling edge of RDX and the host reads data when there is a rising edge of RDX.

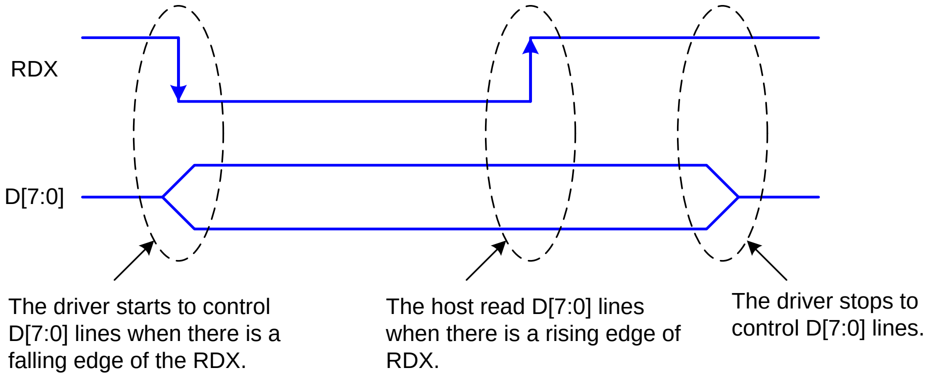


Figure 10 8080 – 8 bits RDX protocol

Note: RDX is an unsynchronized signal (It can be stopped).

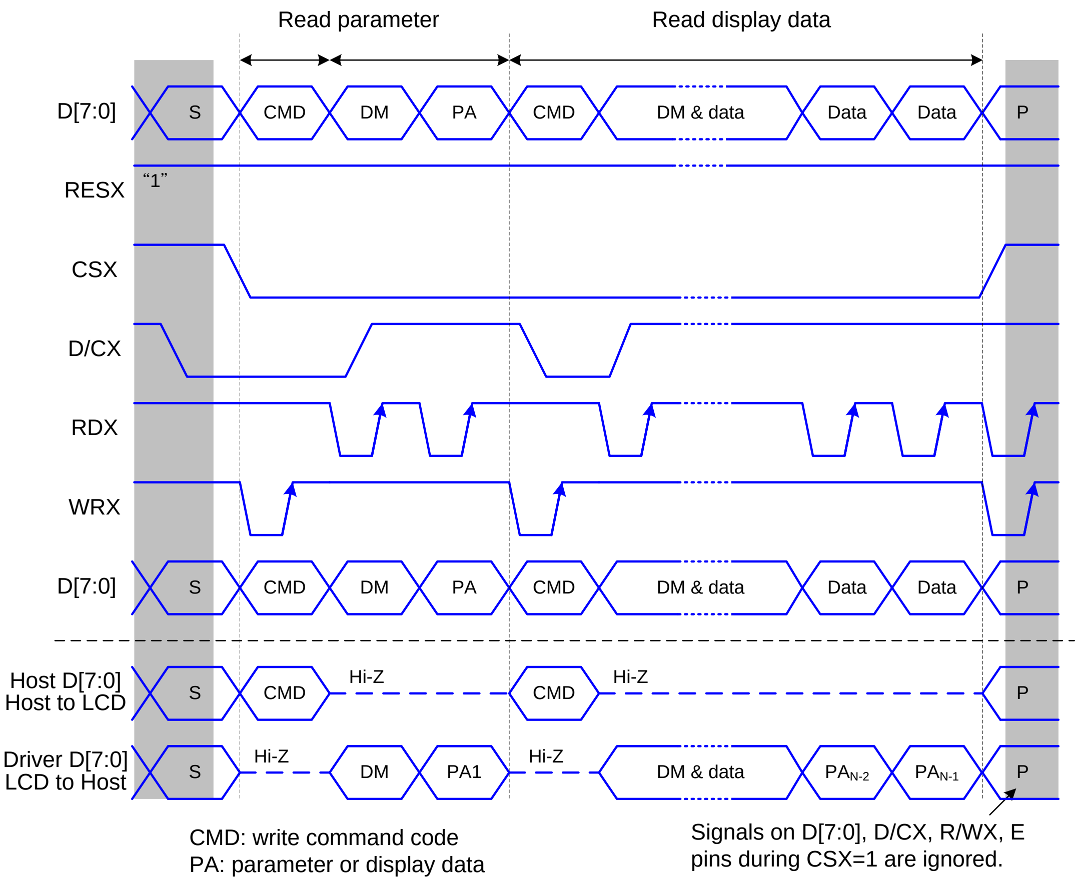


Figure 11 8080 - 8 bits parallel bus protocol, read data from register or display RAM

7.3 Serial Interface

IM2	IM1	IM0	Interface	Read back selection
0	0	0	3-line serial interface Ⅰ	Via the read instruction (8-bit, 24-bit and 32-bit read parameter)
0	0	1	4-line serial interface Ⅰ	

Table 8 Selection of serial interface

The serial interface is either 3-lines/9-bits or 4-lines/8-bits bi-directional interface for communication between the micro controller and the LCD driver. The 3-lines serial interface use: CSX (chip enable), SCL (serial clock) and SDA (serial data input/output), and the 4-lines serial interface use: CSX (chip enable), D/CX (data/ command flag), SCL (serial clock) and SDA (serial data input/output). Serial clock (SCL) is used for interface with MCU only, so it can be stopped when no communication is necessary.

7.3.1 Pin description

3-line serial interface Ⅰ

Pin Name	Description
CSXP	Chip selection signal
RDXP (SCL)	Clock signal
SDA	Serial input/output data

4-line serial interface Ⅰ

Pin Name	Description
CSXP	Chip selection signal
DCXP (A0)	Data is regarded as a command when DCXP is low Data is regarded as a parameter or data when DCXP is high
RDXP (SCL)	Clock signal
SDA	Serial input/output data

Table 9 pin description of serial interface

7.3.2 Command write mode

The write mode of the interface means the micro controller writes commands and data to the LCD driver. 3-lines serial data packet contains a control bit D/CX and a transmission byte. In 4-lines serial interface, data packet contains just transmission byte and control bit D/CX is transferred by the D/CX pin. If D/CX is “low”, the transmission byte is interpreted as a command byte. If D/CX is “high”, the transmission byte is stored in the display data RAM (memory write command), or command register as parameter.

Any instruction can be sent in any order to the driver. The MSB is transmitted first. The serial interface is initialized when CSX is high. In this state, SCL clock pulse or SDA data have no effect. A falling edge on CSX enables the serial interface and indicates the start of data transmission.

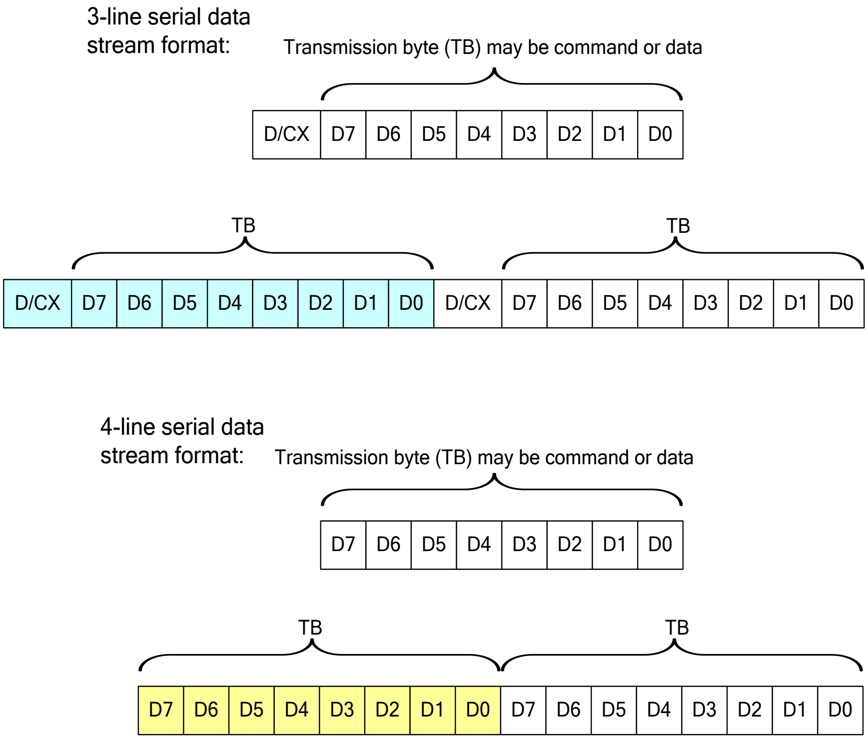


Figure 12 Serial interface data stream format

When CSX is "high", SCL clock is ignored. During the high period of CSX the serial interface is initialized. At the falling edge of CSX, SCL can be high or low. SDA is sampled at the rising edge of SCL. D/CX indicates whether the byte is command (D/CX='0') or parameter/RAM data (D/CX='1'). D/CX is sampled when first rising edge of SCL (3-line serial interface) or 8th rising edge of SCL (4-line serial interface). If CSX stays low after the last bit of command/data byte, the serial interface expects the D/CX bit (3-line serial interface) or D7 (4-line serial interface) of the next byte at the next rising edge of SCL..

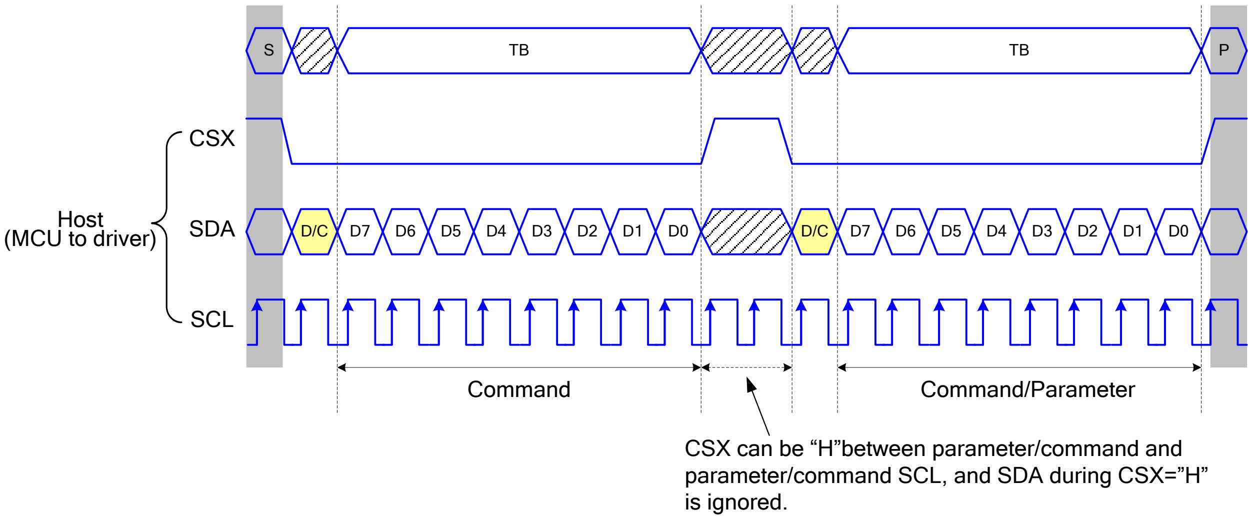


Figure 13 3-line serial interface write protocol (write to register with control bit in transmission)

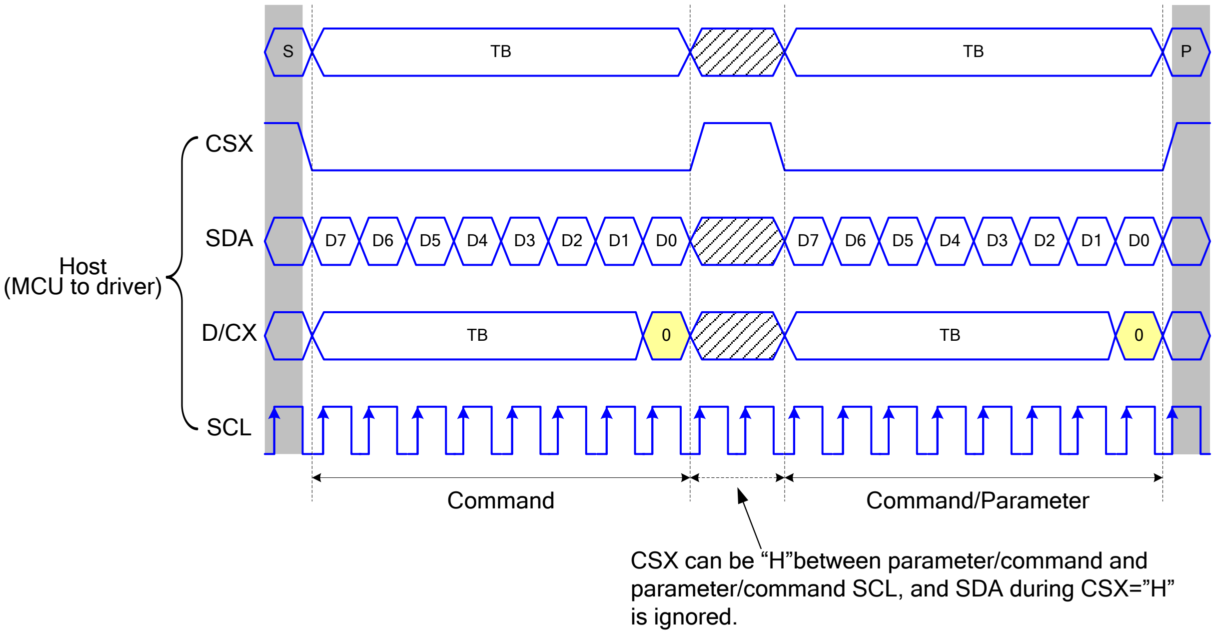


Figure 14 4-line serial interface write protocol (write to register with control bit in transmission)

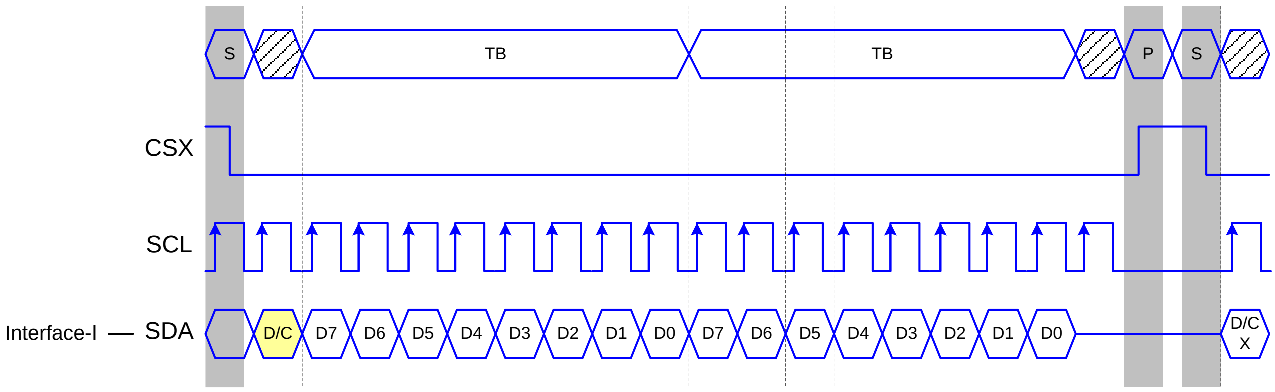
7.3.3 Read function

The read mode of the interface means that the micro controller reads register value from the driver. To achieve read function, the micro controller first has to send a command (read ID or register command) and then the following byte is transmitted in the opposite direction. After that CSX is required to go to high before a new command is send (see the below figure). The driver samples the SDA (input data) at rising edge of SCL, but shifts SDA (output data) at the falling edge of SCL. Thus the micro controller is supported to read at the rising edge of SCL.

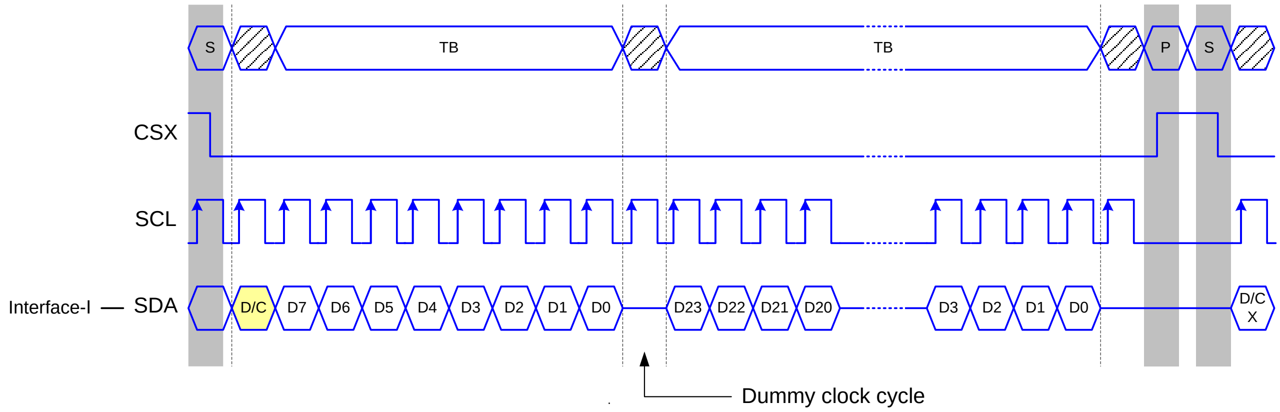
After the read status command has been sent, the SDA line must be set to tri-state no later than at the falling edge of SCL of the last bit.

7.3.4 3-line serial interface Ⅰ protocol

3-line serial protocol (for RDID1/RDID2/RDID3/0Ah/0Bh/0Ch/0Dh/0Eh/0Fh command: 8-bit read):



3-line serial protocol (for RDDID command: 24-bit read):



3-line Serial Protocol (for RDDST command: 32-bit read):

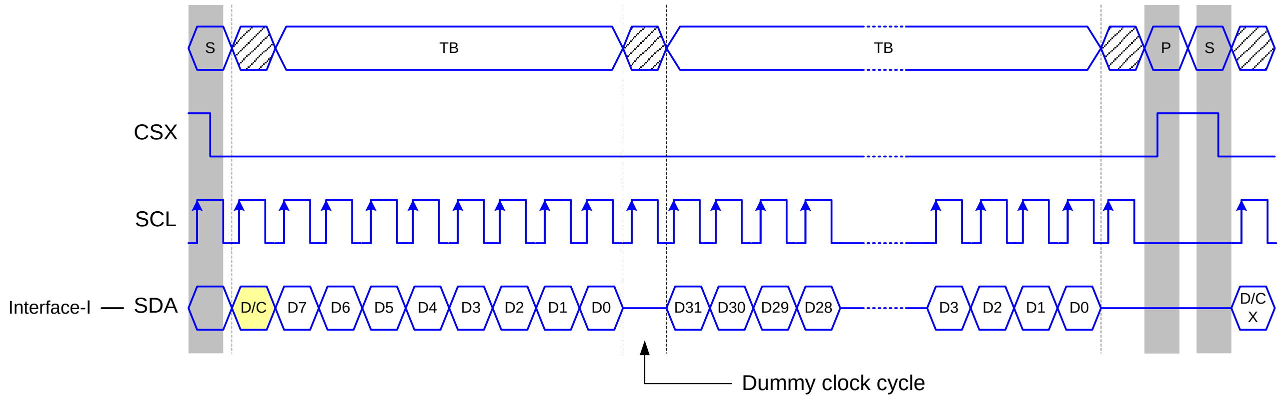
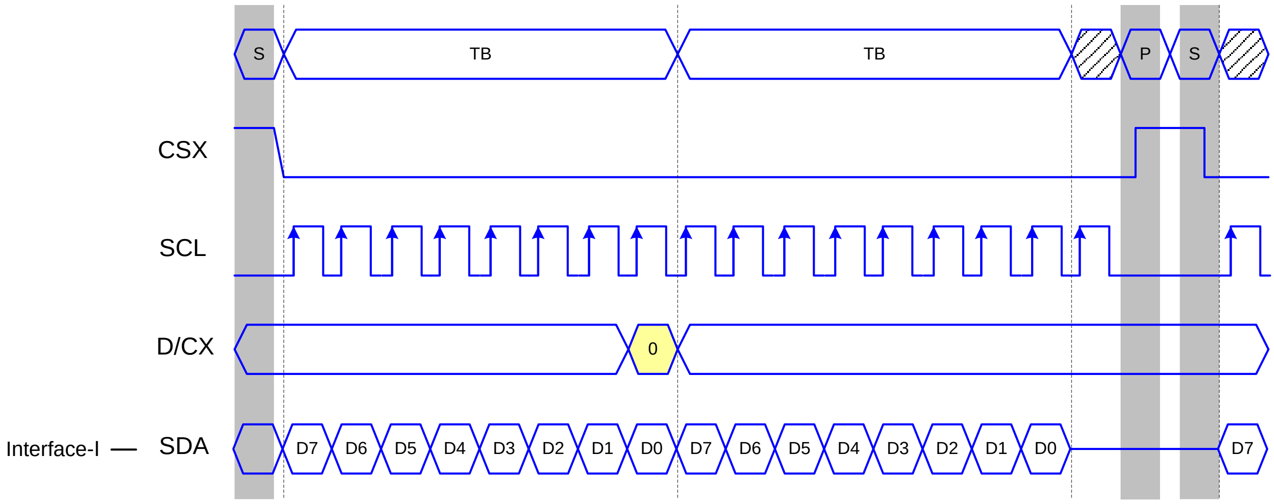


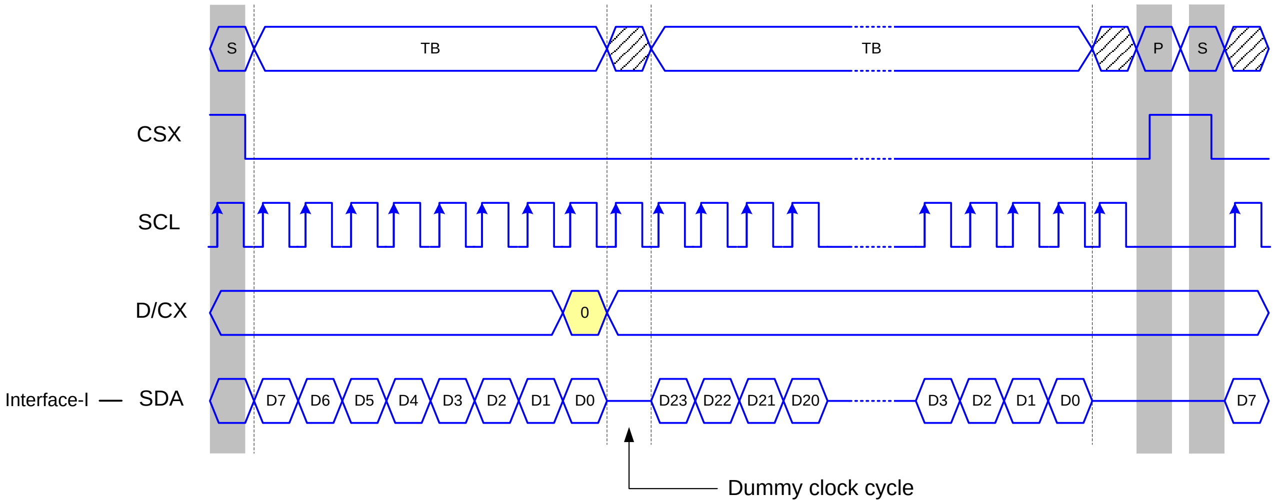
Figure 15 3-line serial interface read protocol

7.3.5 4-line serial protocol

4-line serial protocol (for RDID1/RDID2/RDID3/0Ah/0Bh/0Ch/0Dh/0Eh/0Fh command: 8-bit read):



4-line serial protocol (for RDDID command: 24-bit read)



4-line Serial Protocol (for RDDST command: 32-bit read)

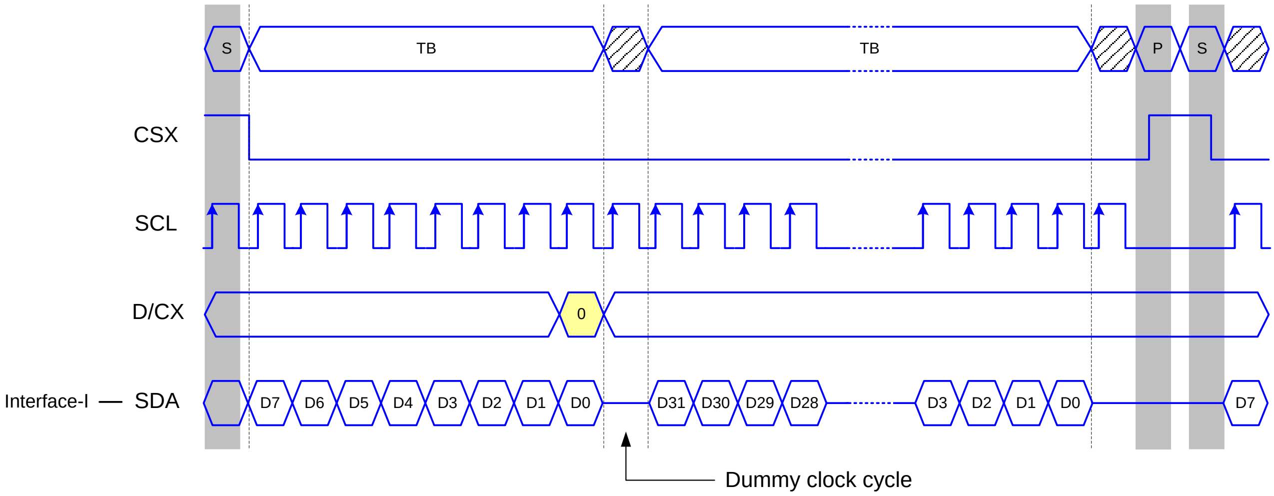


Figure 16 4-line serial interface read protocol

7.4 2 data lane serial Interface

Interface selection:

IM2	IM1	IM0	Interface	Read back selection
0	1	0	2 data lane serial interface	Via the read instruction (8-bit, 24-bit and 32-bit read)

Table 10 IM pin selection

2-wire data lane serial interface use: CSX (chip enable), SCL (serial clock) and SDA1 (serial data input/output 1), and SDA2 (serial data input 2).

2 data lane hardware suggestion and Pin description:

2 data lane serial interface, IM[2:0]=010

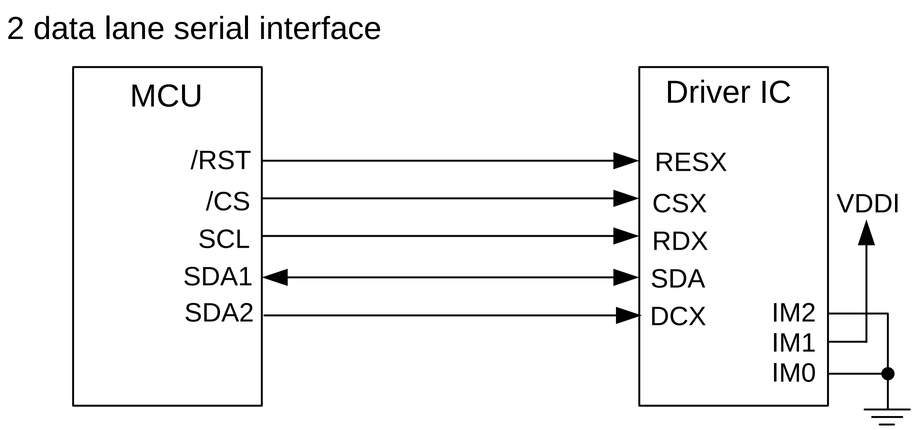


Figure 17 Hardware suggestion of 2 data lane serial interface

Pin Name	Description
CSX	Chip selection signal
RDX (SCL)	Clock signal
SDA (SDA1)	Serial data input/output1
DCX(SDA2)	Serial data input2

Table 11 Pin description of 2 data lane serial interface

Command write mode:

The command write protocol of 2-wire data lane serial interface is the same with the 3-line serial interface, so users can ignore the input data of DCX.

Any instruction can be sent in any order to the driver. The MSB is transmitted first. The serial interface is initialized when CSX is high. In this state, SCL clock pulse or SDA data have no effect. A falling edge on CSX enables the serial interface and indicates the start of data transmission.

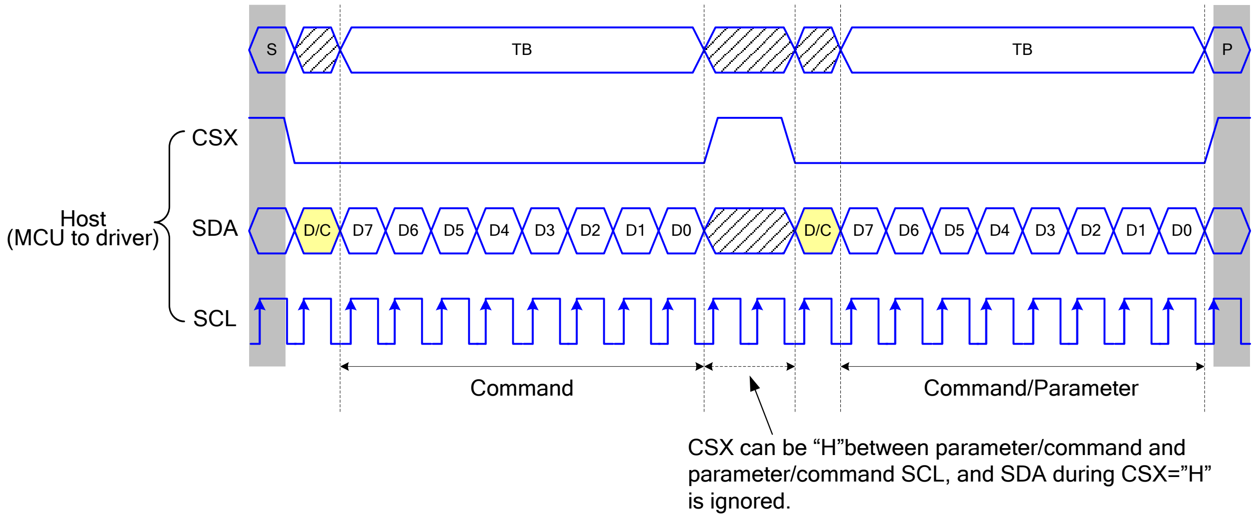


Figure 18 3-line serial interface write protocol (write to register with control bit in transmission)

SRAM write mode:

The SRAM write mode of 2-wire data line serial interface need use SDA pin and DCXP pin to be data input pins.

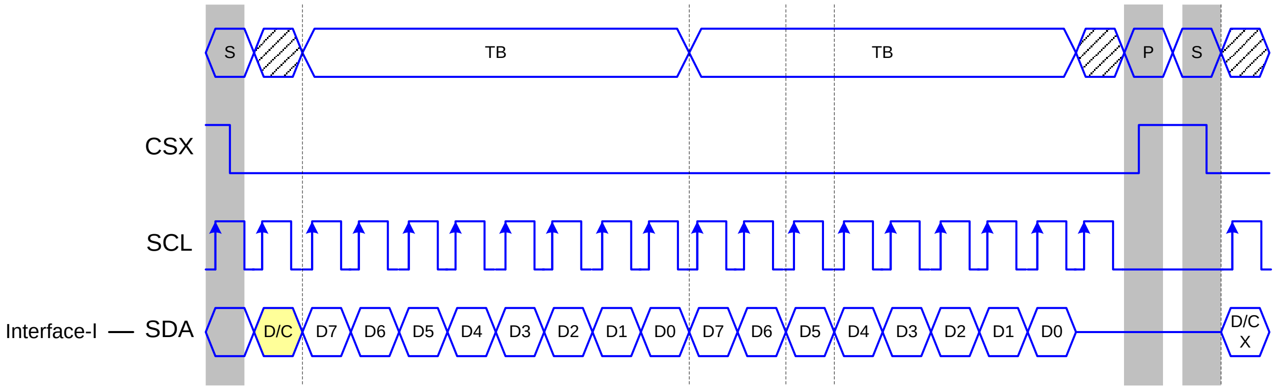
Read function:

The read mode of 2-wire data lane serial interface is the same with the 3-line serial interface and DCXP pin can be ignored. To achieve read function, the micro controller first has to send a command (read ID or register command) and then the following byte is transmitted in the opposite direction. After that CSX is required to go to high before a new command is send (see the below figure). The driver samples the SDA (input data) at rising edge of SCL, but shifts SDA (output data) at the falling edge of SCL. Thus the micro controller is supported to read at the rising edge of SCL.

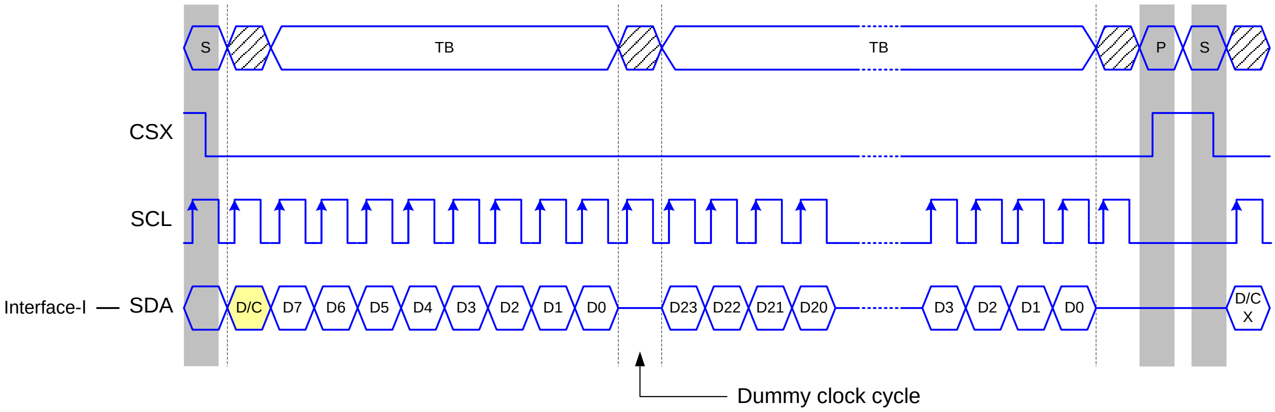
After the read status command has been sent, the SDA line must be set to tri-state no later than at the falling edge of SCL of the last bit.

3-line serial interface Ⅰ protocol:

3-line serial protocol (for RDID1/RDID2/RDID3/0Ah/0Bh/0Ch/0Dh/0Eh/0Fh command: 8-bit read):



3-line serial protocol (for RDDID command: 24-bit read)



3-line Serial Protocol (for RDDST command: 32-bit read)

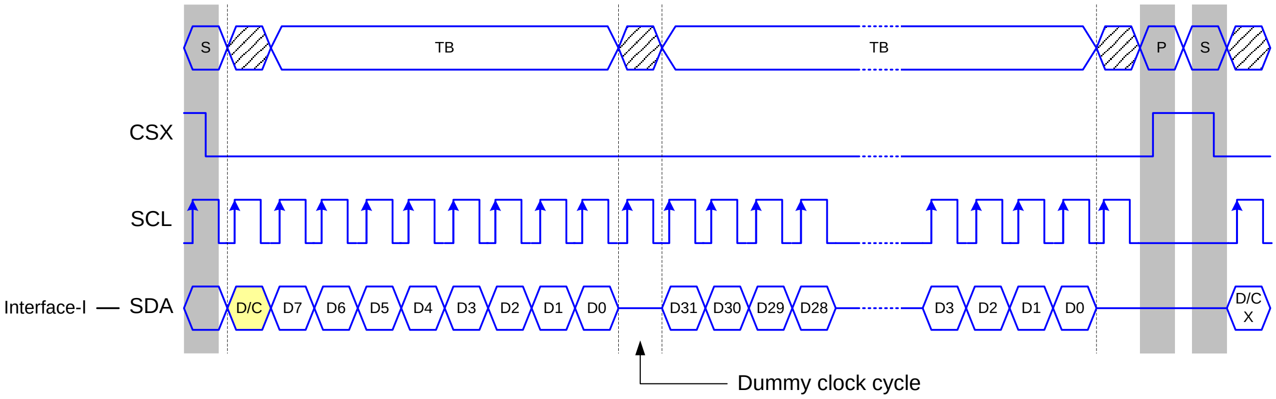


Figure 19 3-line serial interface read protocol

7.5 Data Transfer Break and Recovery

If there is a break in data transmission by RESX pulse, while transferring a command or frame memory data or multiple parameter command data, before Bit D0 of the byte has been completed, then driver will reject the previous bits and have reset the interface such that it will be ready to receive command data again when the chip select line (CSX) is next activated after RESX have been HIGH state.

If there is a break in data transmission by CSX pulse, while transferring a command or frame memory data or multiple parameter command data, before Bit D0 of the byte has been completed, then driver will reject the previous bits and have reset the interface such that it will be ready to receive the same byte re-transmitted when the chip select line (CSX) is next activated.

If 1, 2 or more parameter commands are being sent and a break occurs while sending any parameter before the last one and if the host then sends a new command rather than re-transmitting the parameter that was interrupted, then the parameters that were successfully sent are stored and the parameter where the break occurred is rejected. The interface is ready to receive next byte as shown below.

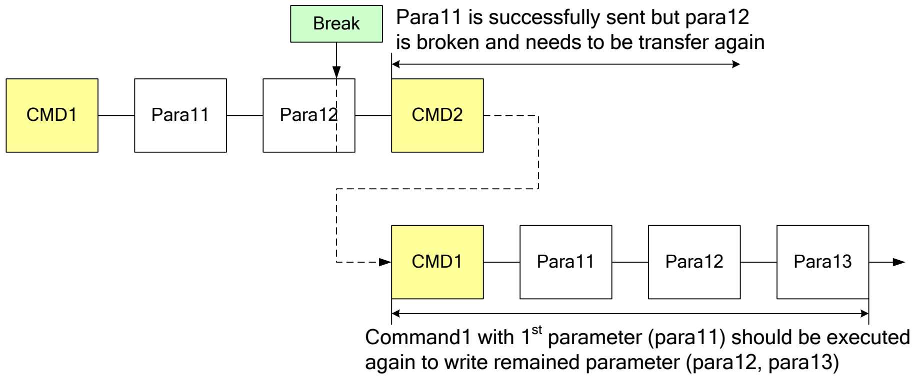


Figure 20 Write interrupts recovery (serial interface)

If a 2 or more parameter commands are being sent and a break occurs by the other command before the last one is sent, then the parameters that were successfully sent are stored and the other parameter of that command remains previous value.

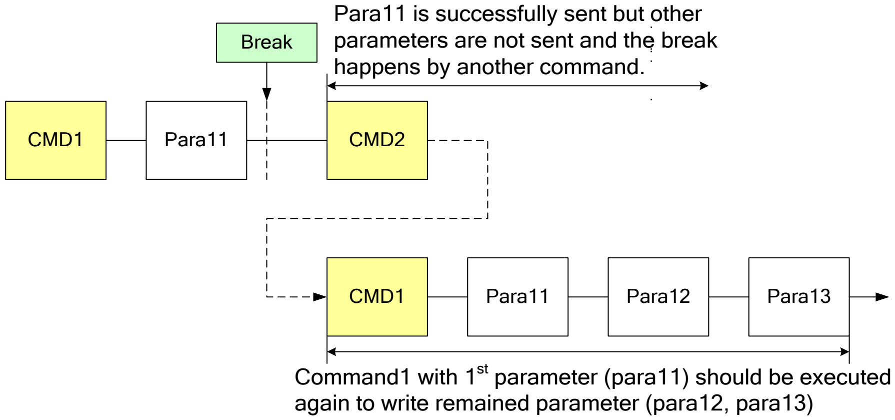


Figure 21 Write interrupts recovery (both serial and parallel Interface)

7.6 Data Transfer Pause

It will be possible when transferring a command, frame memory data or multiple parameter data to invoke a pause in the data transmission. If the Chip Select line is released after a whole byte of a frame memory data or multiple parameter data has been completed, then driver will wait and continue the frame memory data or parameter data transmission from the point where it was paused. If the Chip Select line is released after a whole byte of a command has been completed, then the display module will receive either the command's parameters (if appropriate) or a new command when the Chip Select line is next enabled as shown below.

This applies to the following 4 conditions:

1) Command-Pause-Command
2) Command-Pause-Parameter
3) Parameter-Pause-Command
4) Parameter-Pause-Parameter

7.6.1 Parallel interface pause

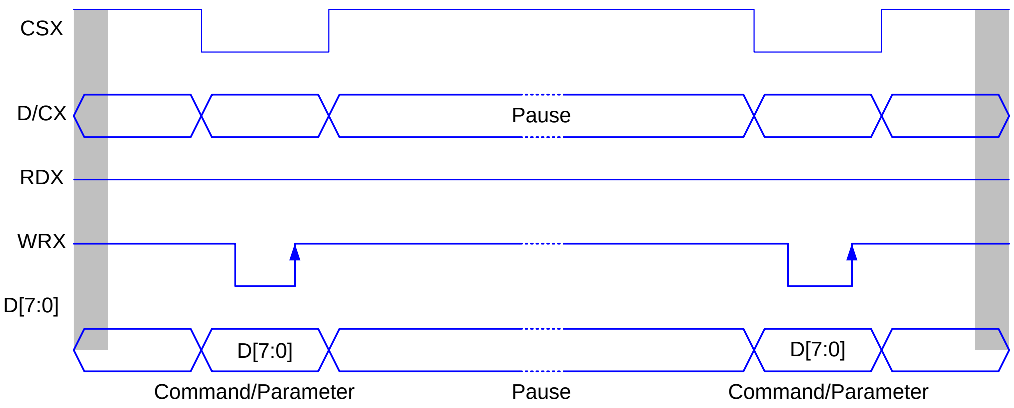


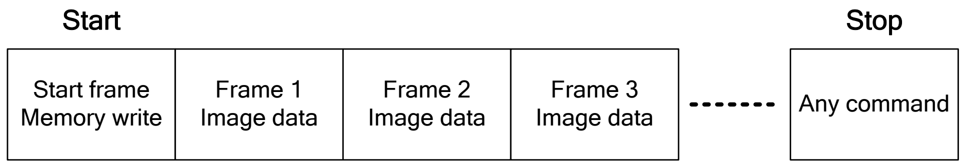
Figure 22 Parallel bus pause protocol (paused by CSX)

7.7 Data Transfer Mode

The module has two kinds color modes for transferring data to the display RAM. These are 16-bit color per pixel and 18-bit color per pixel. The data format is described for each interface. Data can be downloaded to the frame memory by 2 methods.

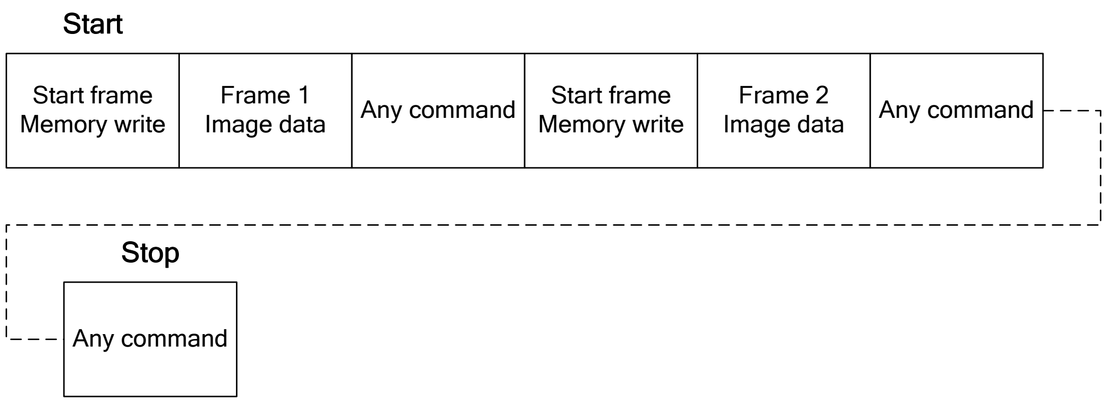
7.7.1 Method 1

The image data is sent to the frame memory in successive frame writes, each time the frame memory is filled, the frame memory pointer is reset to the start point and the next frame is written.



7.7.2 Method 2

The image data is sent and at the end of each frame memory download, a command is sent to stop frame memory write. Then start memory write command is sent, and a new frame is downloaded.



Note 1: These apply to all data transfer Color modes on both serial and parallel interfaces.

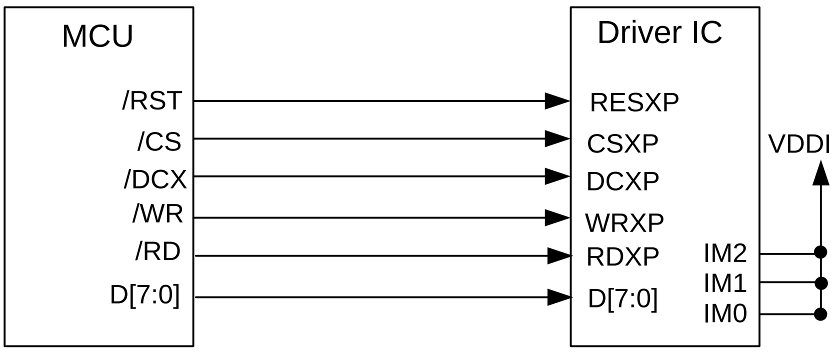
Note 2: The frame memory can contain both odd and even number of pixels for both methods. Only complete pixel data will be stored in the frame memory.

7.8 Data Color Coding

7.8.1 8080 Series 8-bit Parallel Interface

The 8080 series 8-bit parallel interface of ST77922 can be used by setting IM[2:0]="111b". Different display data formats are available for two Colors depth supported by listed below.

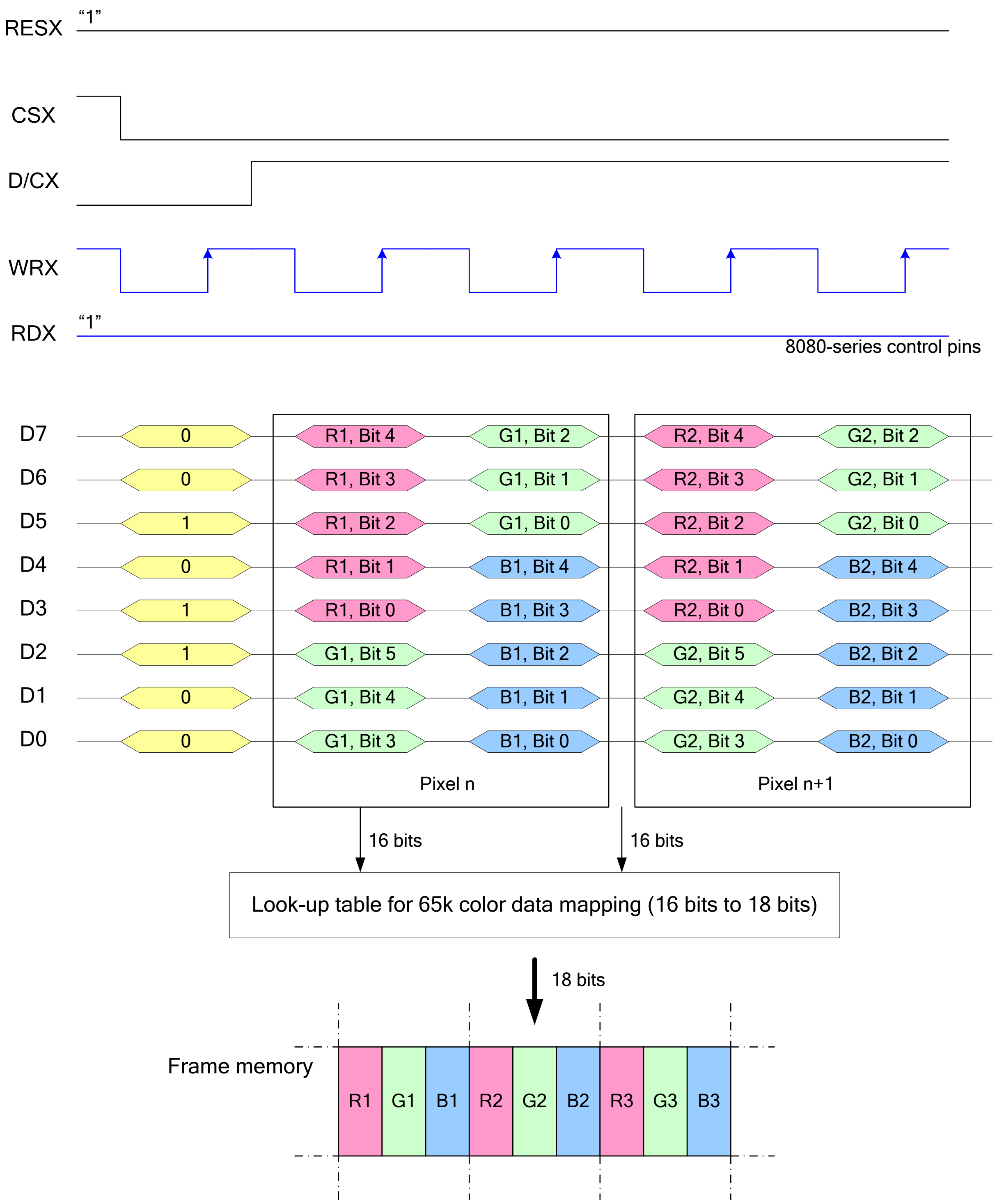
- 65k colors, RGB 5,6,5-bit input.
- 262k colors, RGB 6,6,6-bit input.
- 16M colors, RGB 8,8,8-bit input.



8080 Series 8-bit Interface Connection

7.8.1.1 8-bit data bus for 16-bit/pixel (RGB 5-6-5-bit input), 65K-Colors, 3Ah=”01h”

There is 1pixel (3 sub-pixels) per 2-byte



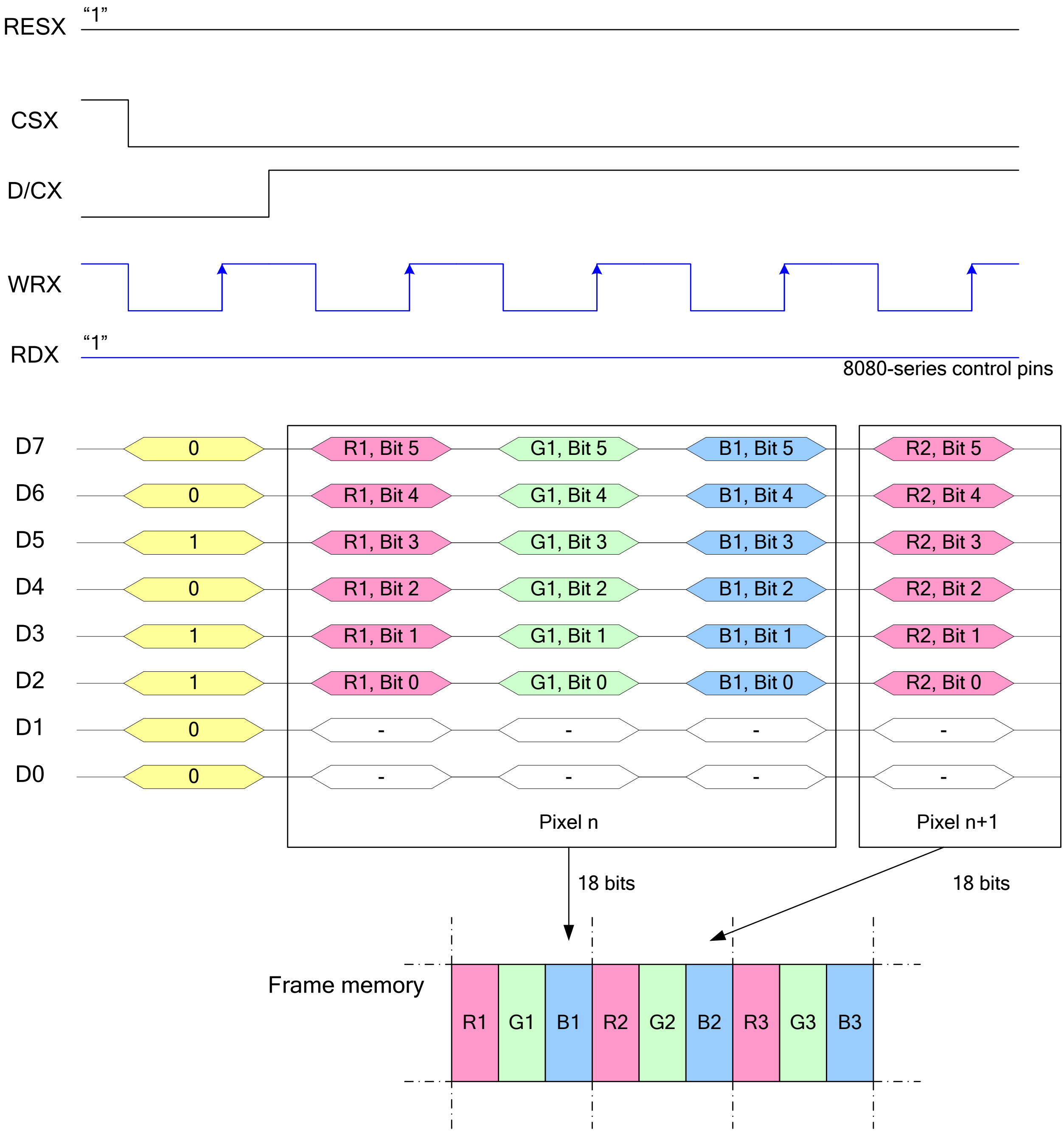
Note 1: The data order is as follows, MSB=D7, LSB=D0 and picture data is MSB=Bit 5, LSB=Bit 0 for Green, and MSB=Bit 4, LSB=Bit 0 for Red and Blue data.

Note 2: 2-times transfer is used to transmit 1 pixel data with the 16-bit color depth information.

Note 3: ‘-‘ = Don’t care – Can be set to ‘0’ or ‘1’

7.8.1.2 8-bit data bus for 18-bit/pixel (RGB-6-6-6-bit input), 262K-Colors, 3Ah=”02h”

There is 1pixel (3 sub-pixels) per 3-bytes.



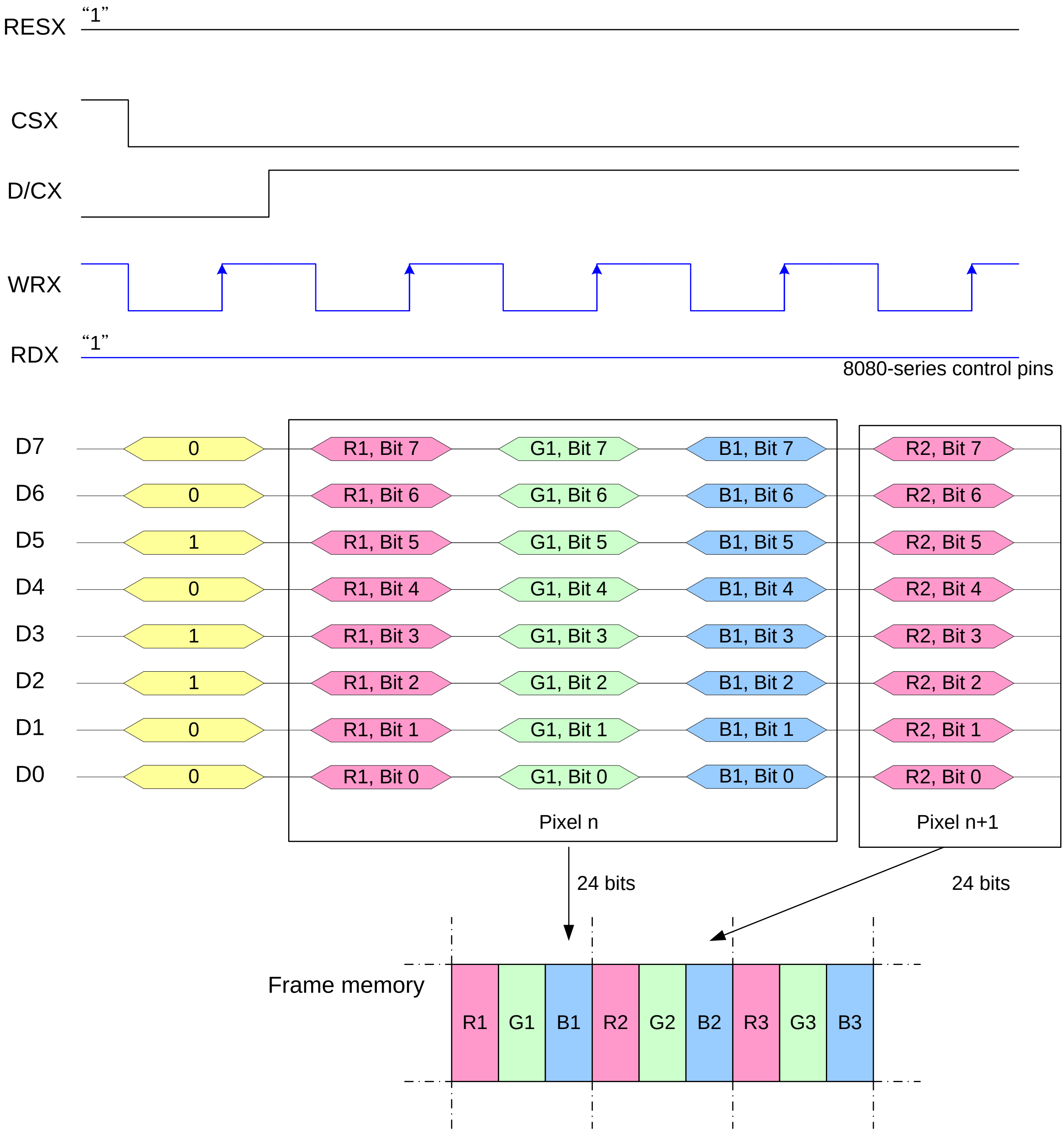
Note 1: The data order is as follows, MSB=D7, LSB=D0 and picture data is MSB=Bit 5, LSB=Bit 0 for Red, Green and Blue data.

Note 2: 3-times transfer is used to transmit 1 pixel data with the 18-bit color depth information.

Note 3: ‘-‘ = Don’t care – Can be set to ‘0’ or ‘1’

7.8.1.3 8-bit data bus for 24-bit/pixel (RGB-8-8-8-bit input), 16M-Colors, 3Ah=”03h”

There is 1pixel (3 sub-pixels) per 3-bytes.



Note 1: The data order is as follows, MSB=D7, LSB=D0 and picture data is MSB=Bit 7, LSB=Bit 0 for Red, Green and Blue data.

Note 2: 3-times transfer is used to transmit 1 pixel data with the 24-bit color depth information.

Note 3: ‘-‘ = Don’t care – Can be set to ‘0’ or ‘1’

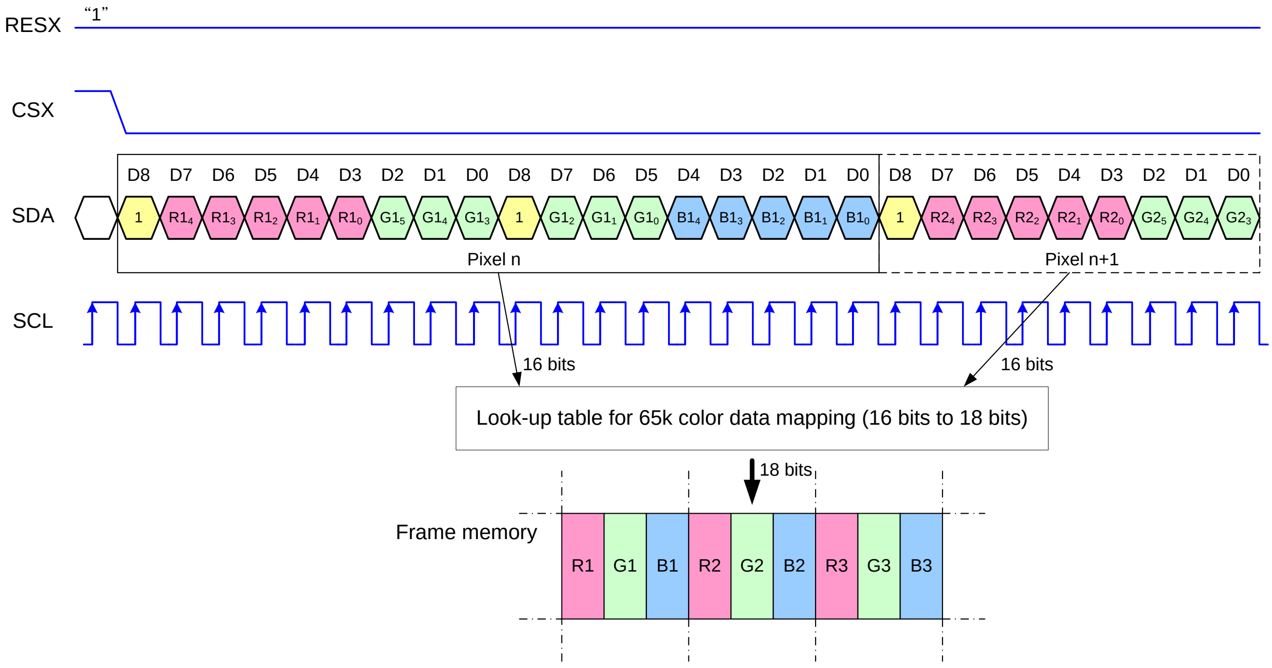
7.8.2 3-Line Serial Interface

Different display data formats are available for two colors depth supported by the LCM listed below.

65k colors, RGB 5-6-5-bit input

262k colors, RGB 6-6-6-bit input

7.8.2.1 Write data for 16-bit/pixel (RGB 5-6-5-bit input), 65K-Colors, 3Ah=”01h”

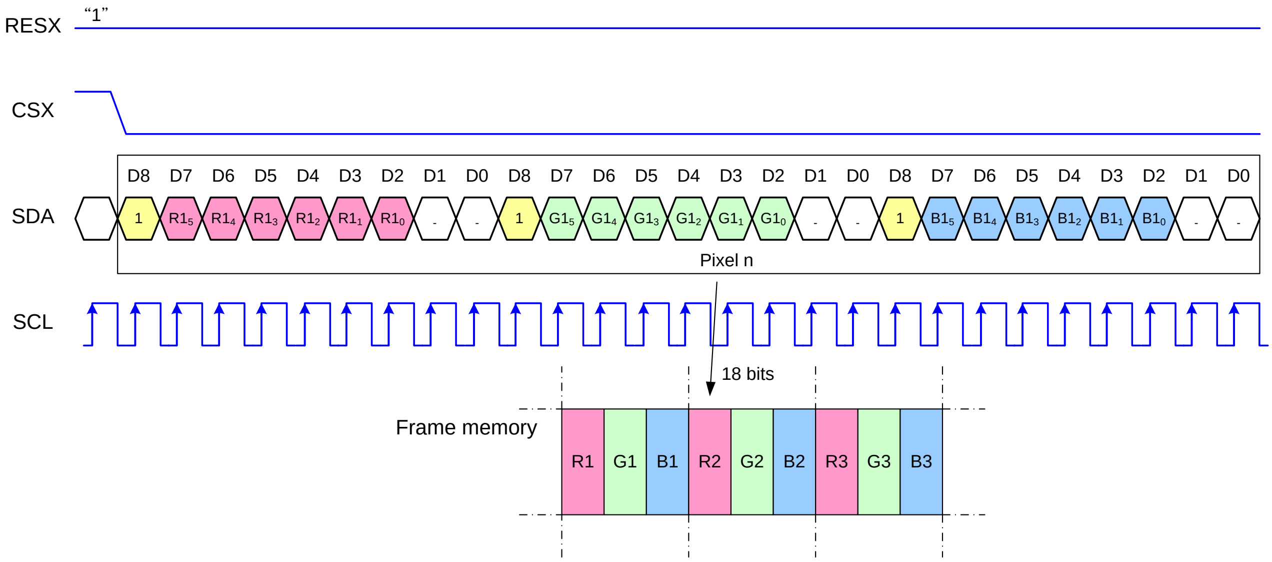


Note 1: Pixel data with the 16-bit color depth information

Note 2: The most significant bits are: Rx4, Gx5 and Bx4

Note 3: The least significant bits are: Rx0, Gx0 and Bx0

7.8.2.2 Write data for 18-bit/pixel (RGB-6-6-6-bit input), 262K-Colors, 3Ah=”02h”

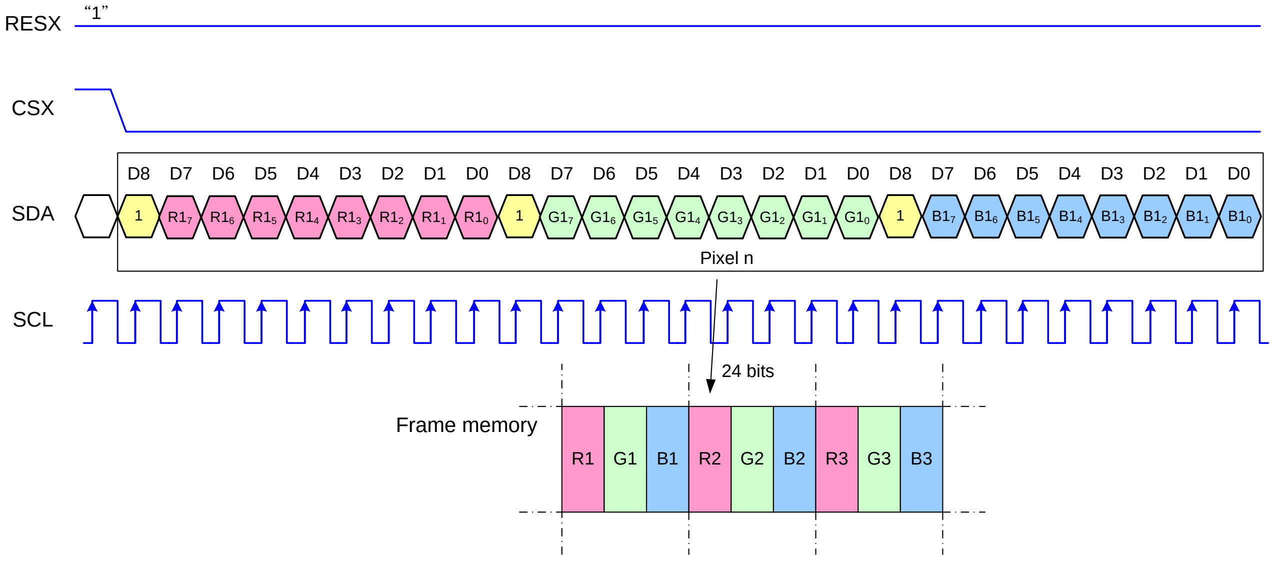


Note 1: Pixel data with the 18-bit color depth information

Note 2: The most significant bits are: Rx5, Gx5 and Bx5

Note 3: The least significant bits are: Rx0, Gx0 and Bx0

7.8.2.3 Write data for 24-bit/pixel (RGB-8-8-8-bit input), 16M-Colors, 3Ah=”03h”



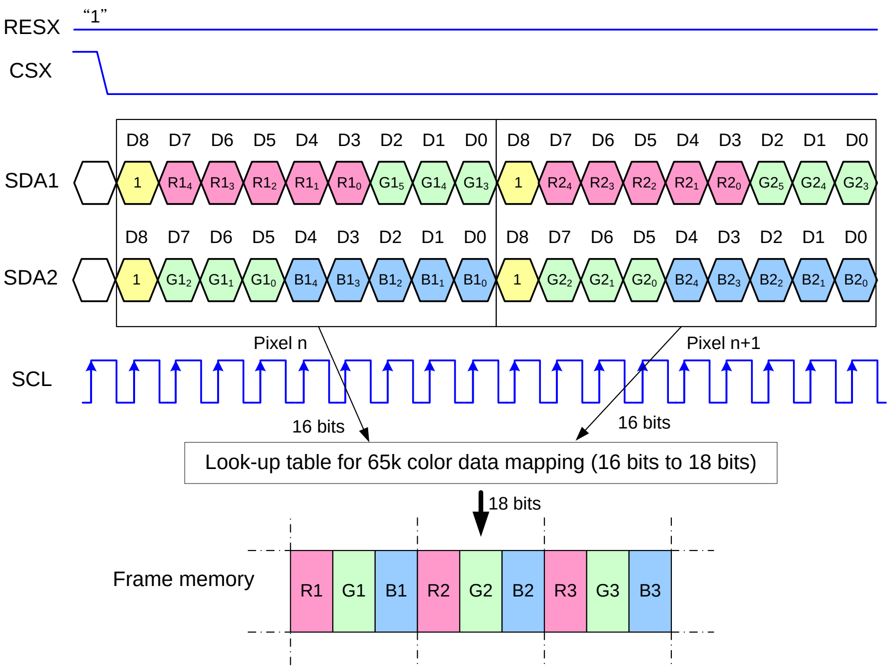
Note 1: Pixel data with the 24-bit color depth information
Note 2: The most significant bits are: Rx7, Gx7 and Bx7
Note 3: The least significant bits are: Rx0, Gx0 and Bx0

7.8.3 2 Data Lane Serial Interface

Different display data formats are available for two colors depth supported by the LCM listed below.

- 65k colors, RGB 5-6-5-bit input
- 262k colors, RGB 6-6-6-bit input
- 16M colors, RGB 8-8-8-bit input

7.8.3.1 Write data for 16-bit/pixel (RGB 5-6-5-bit input), 65K-Colors, 3Ah=”01h”

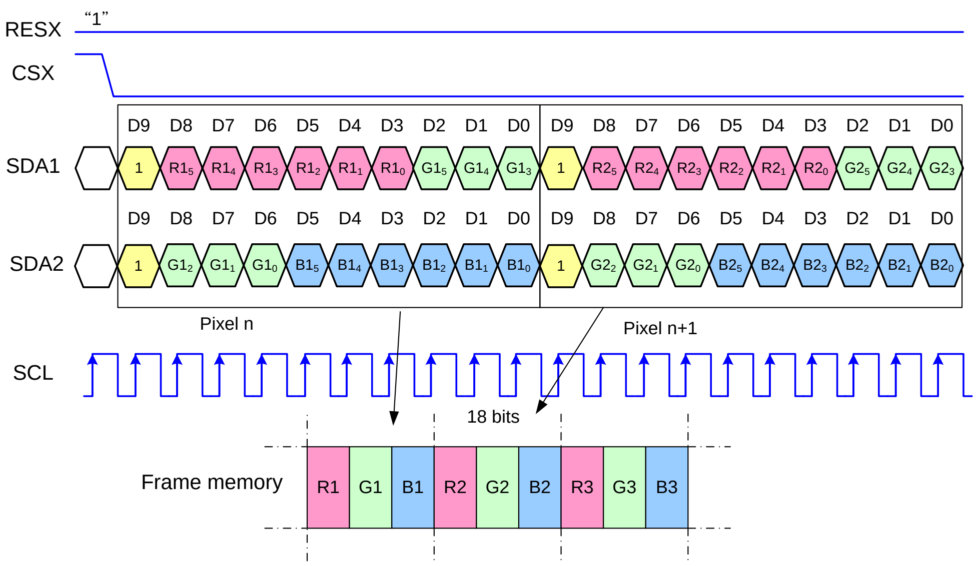


Note 1: Pixel data with the 16-bit color depth information

Note 2: The most significant bits are: Rx4, Gx5 and Bx4

Note 3: The least significant bits are: Rx0, Gx0 and Bx0

7.8.3.2 Write data for 18-bit/pixel (RGB 6-6-6-bit input), 262K-Colors, 3Ah=”02h”

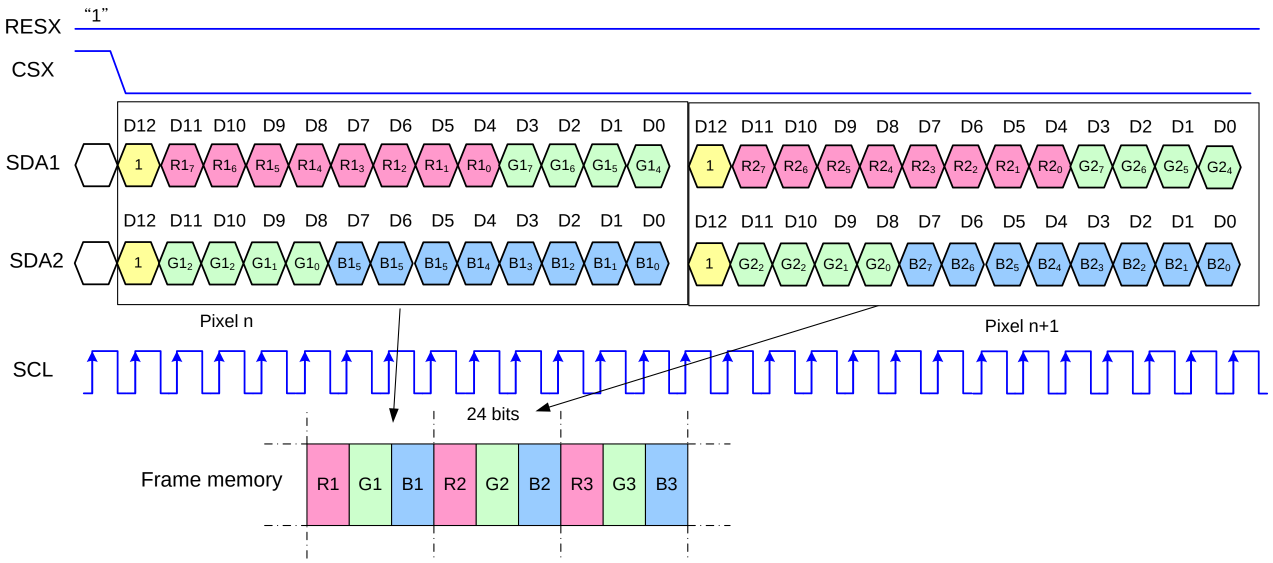


Note 1: Pixel data with the 18-bit color depth information

Note 2: The most significant bits are: Rx5, Gx5 and Bx5

Note 3: The least significant bits are: Rx0, Gx0 and Bx0

7.8.3.3 Write data for 24-bit/pixel (RGB 8-8-8-bit input), 16M-Colors, 3Ah=”03h”(TBD)



Note 1: Pixel data with the 24-bit color depth information

Note 2: The most significant bits are: Rx7, Gx7 and Bx7

Note 3: The least significant bits are: Rx0, Gx0 and Bx0

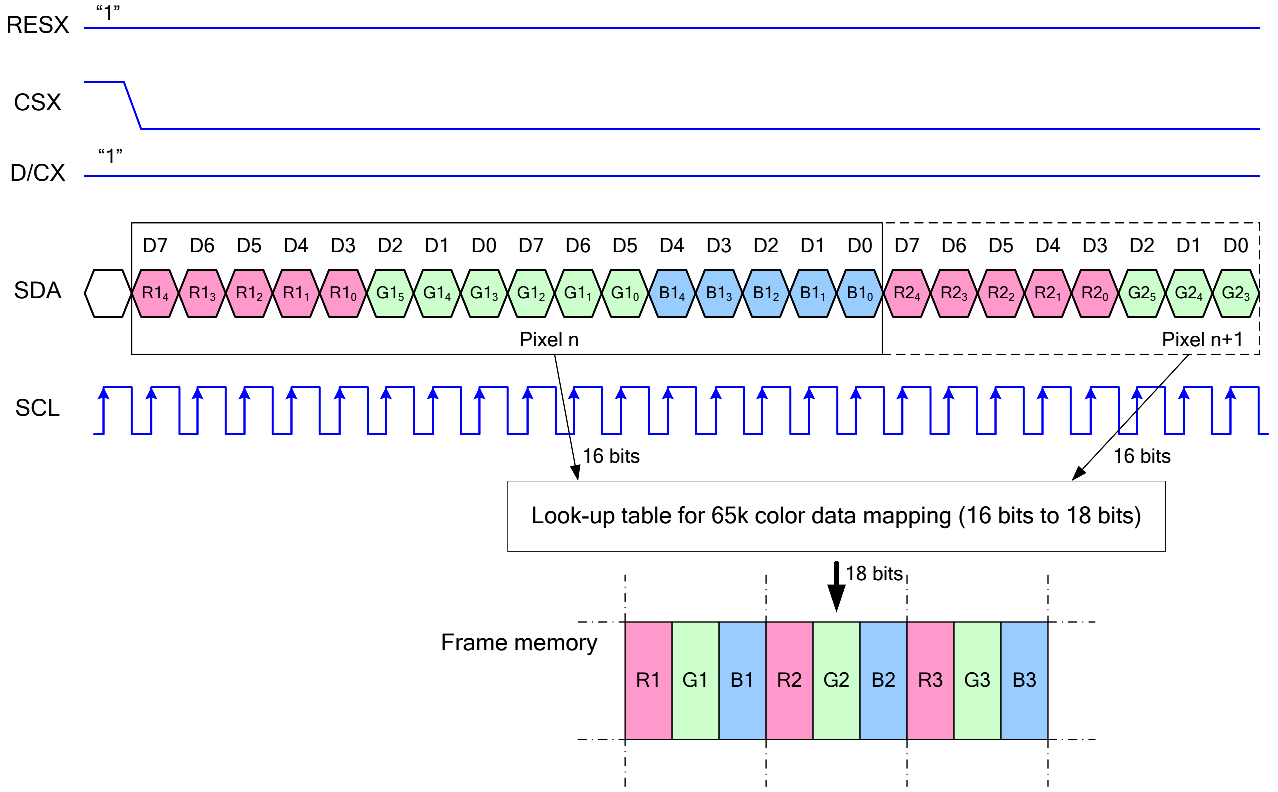
7.8.4 4-Line Serial Interface

Different display data formats are available for two colors depth supported by the LCM listed below.

65k colors, RGB 5-6-5-bit input

262k colors, RGB 6-6-6-bit input

7.8.4.1 Write data for 16-bit/pixel (RGB-5-6-5-bit input), 65K-Colors, 3Ah="01h"

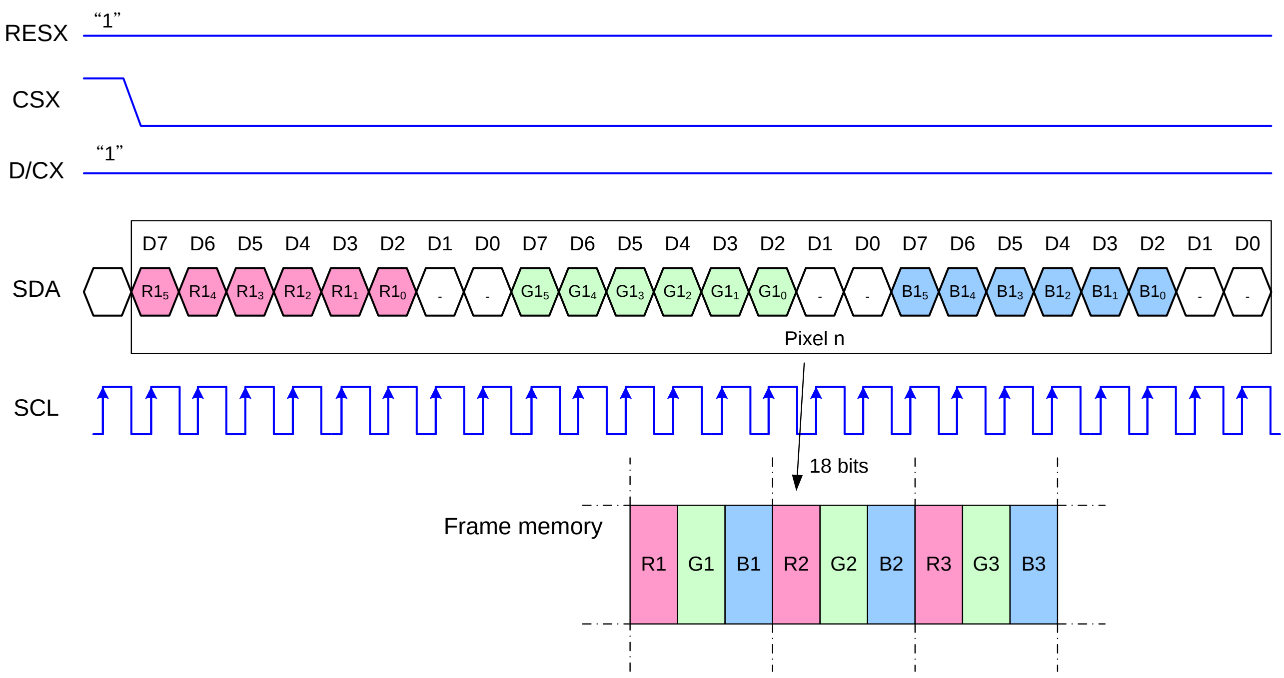


Note 1. Pixel data with the 16-bit color depth information

Note 2. The most significant bits are: Rx4, Gx5 and Bx4

Note 3. The least significant bits are: Rx0, Gx0 and Bx0

7.8.4.2 Write data for 18-bit/pixel (RGB-6-6-6-bit input), 262K-Colors, 3Ah=”02h”

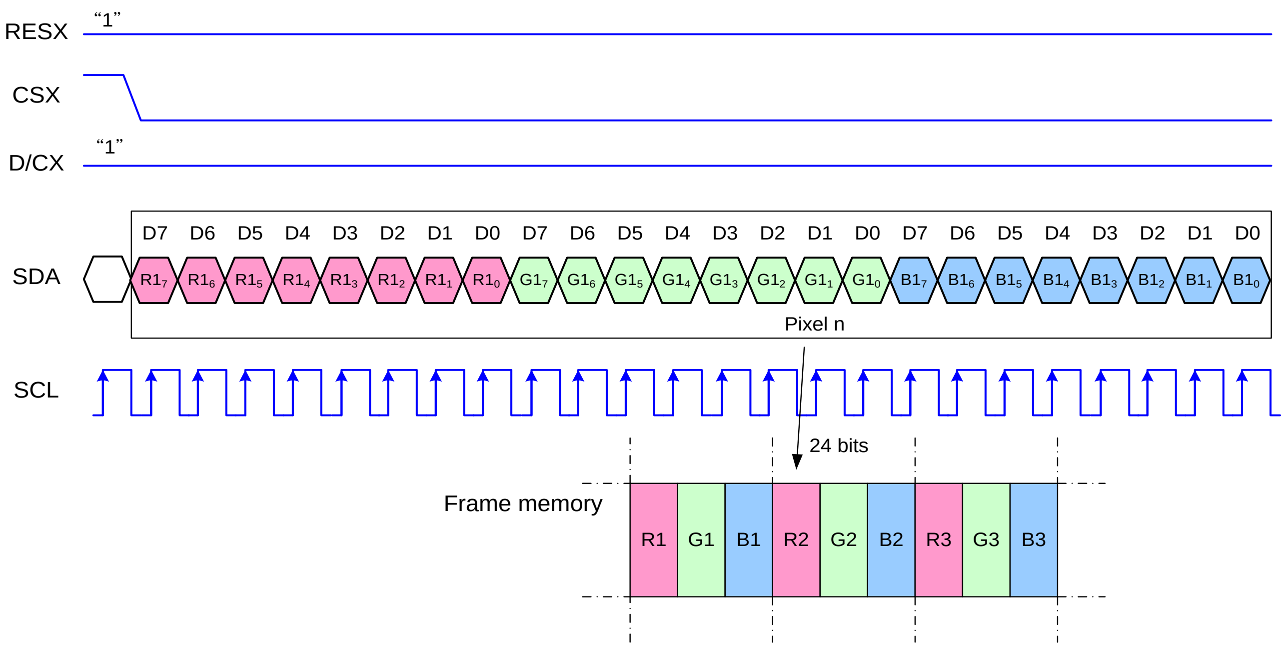


Note 1. Pixel data with the 18-bit color depth information

Note 2. The most significant bits are: Rx5, Gx5 and Bx5

Note 3. The least significant bits are: Rx0, Gx0 and Bx0

7.8.4.3 Write data for 24-bit/pixel (RGB-8-8-8-bit input), 16M-Colors, 3Ah=”03h”



Note 1. Pixel data with the 24-bit color depth information

Note 2. The most significant bits are: Rx7, Gx7 and Bx7

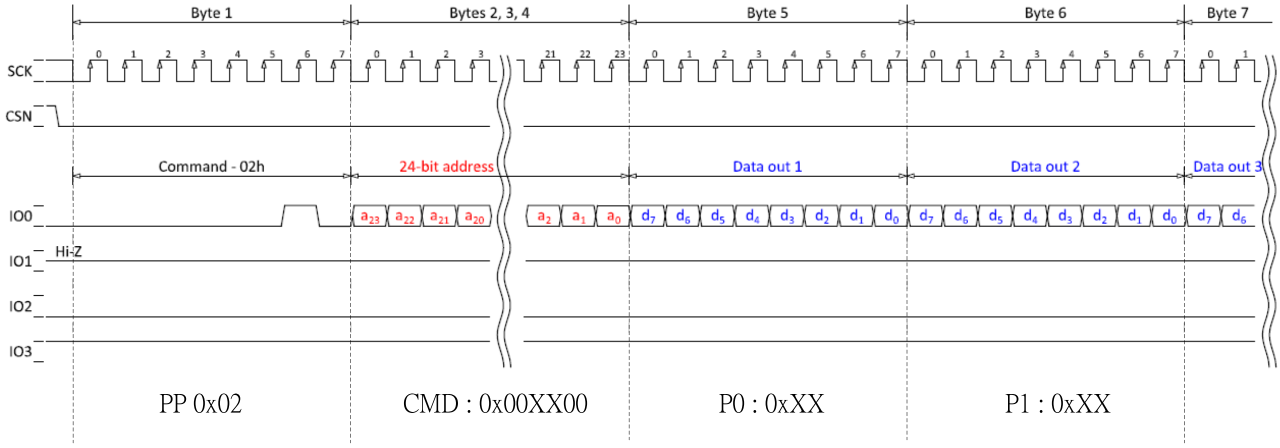
Note 3. The least significant bits are: Rx0, Gx0 and Bx0

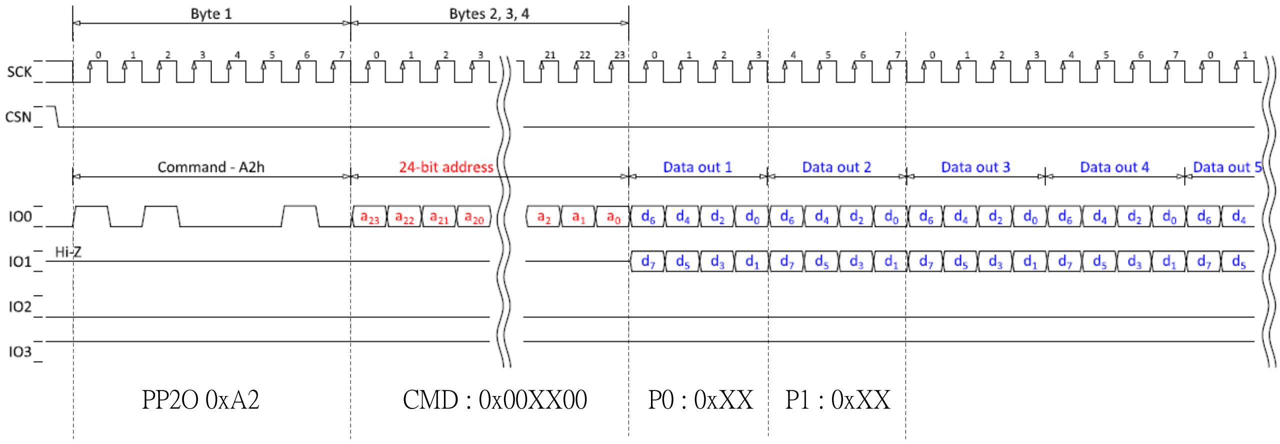
7.8.5 Quad-SPI Interface

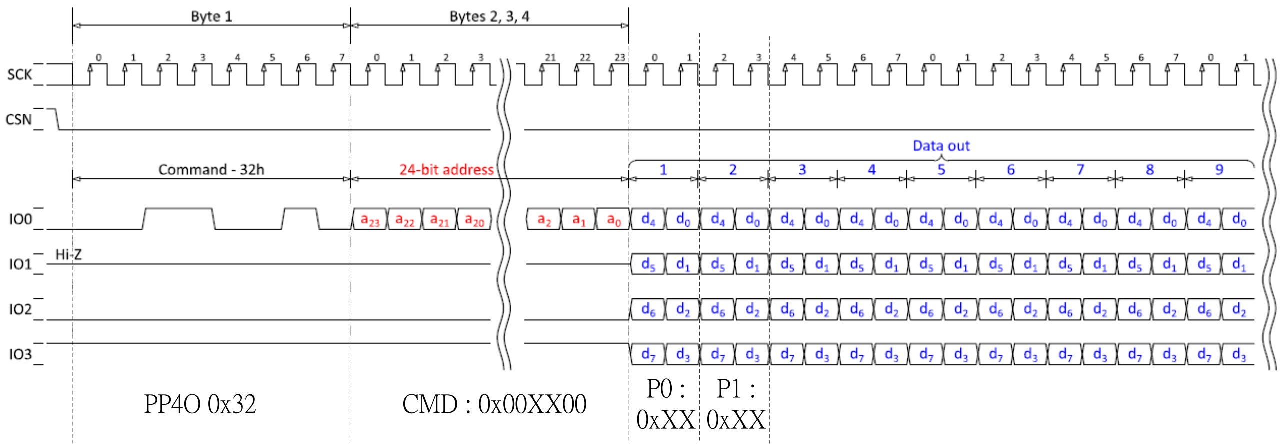
Pin Name	Description
CSX	Chip selection signal
RDX (SCL)	Clock signal
SDA	Serial input/output data lane 0
DCX	Serial input data lane 1
D0	Serial input data lane 2
D1	Serial input data lane 3

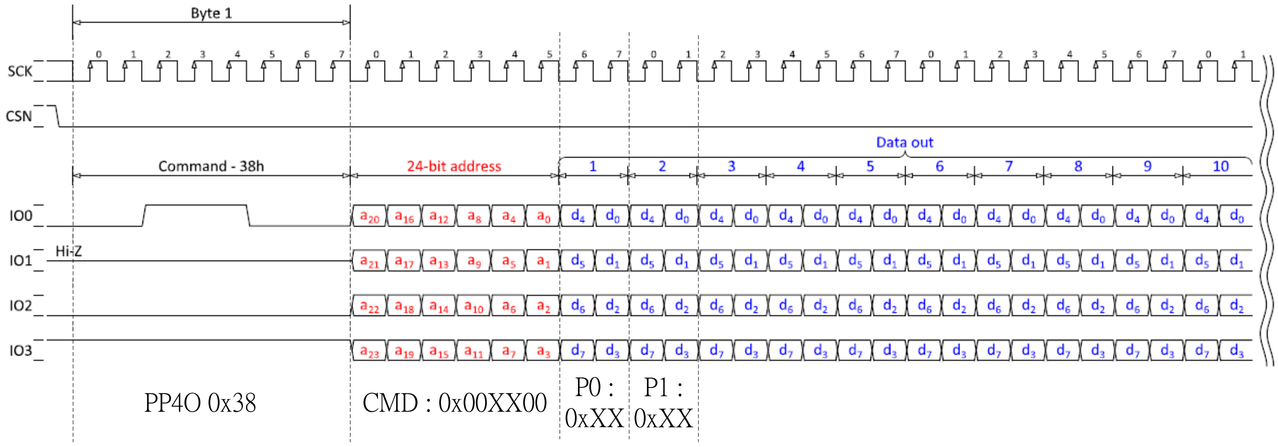
7.8.5.1 Command write mode:

When host writes commands or parameter to ST77922, host needs to send 1 byte of write command instruction (0x02、0xA2、0x32 or 0x38). Then host sends 3 bytes of AD[23:0] which is composed of 1 byte of 0x00, 1 byte of command address and 1 byte of 0x00. After host sending instruction and AD[23:0], the following data is parameter (are parameters). When the last bit of parameter has been sent, CSX pin should be returned “H” level.



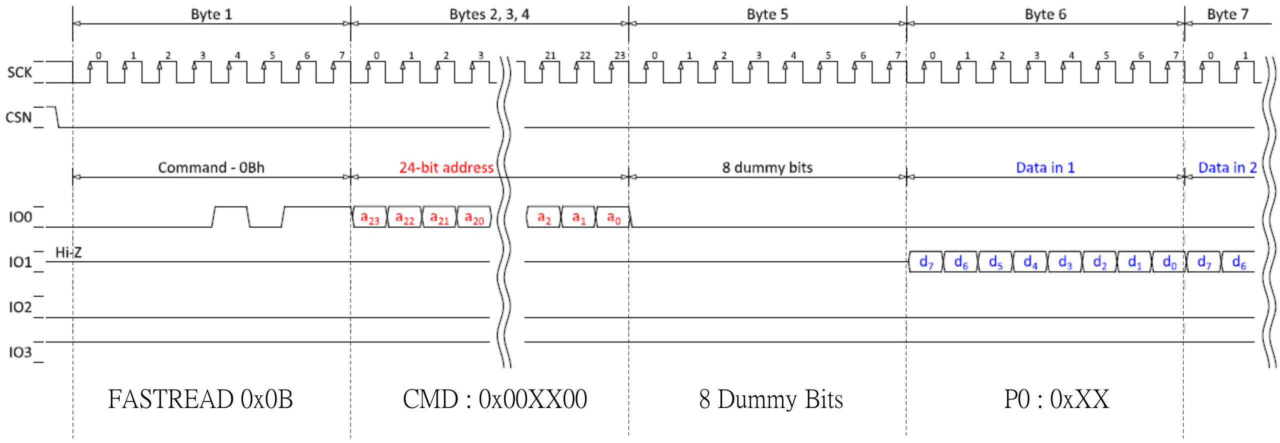






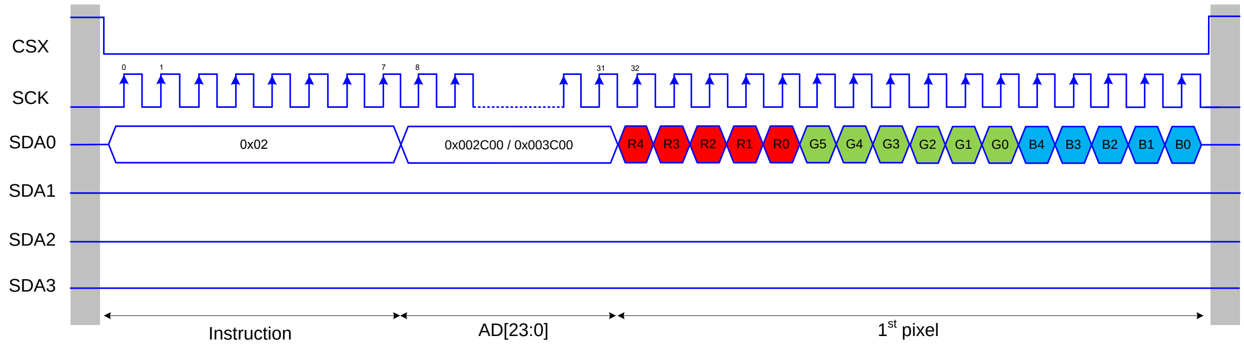
7.8.5.2 Read command mode

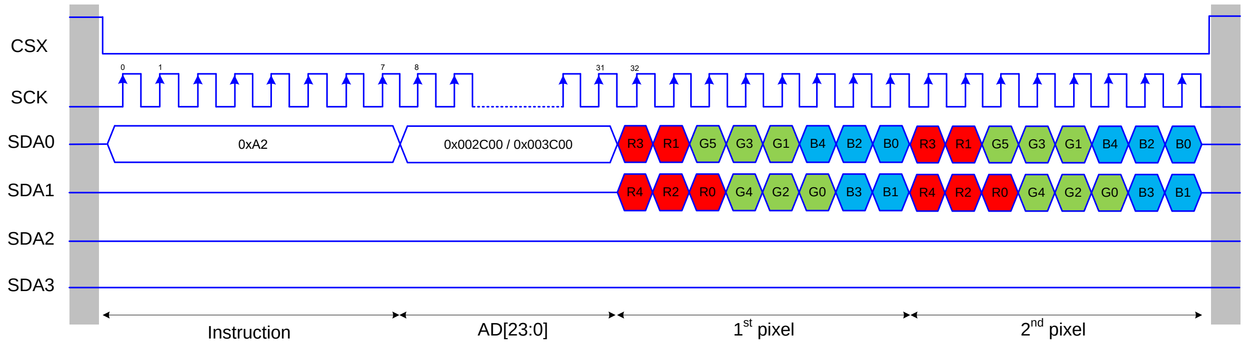
When host reads commands or parameter to ST77922, host needs to send 1 byte of write command instruction (0x0B). Then host sends 3 bytes of AD[23:0] which is composed of 1 byte of 0x00, 1 byte of command address and 1 byte of 0x00. After host sending read command and AD[23:0], the following output data is command address parameter (are parameters). When the last bit of parameter has been output, CSX pin should be returned "H" level.

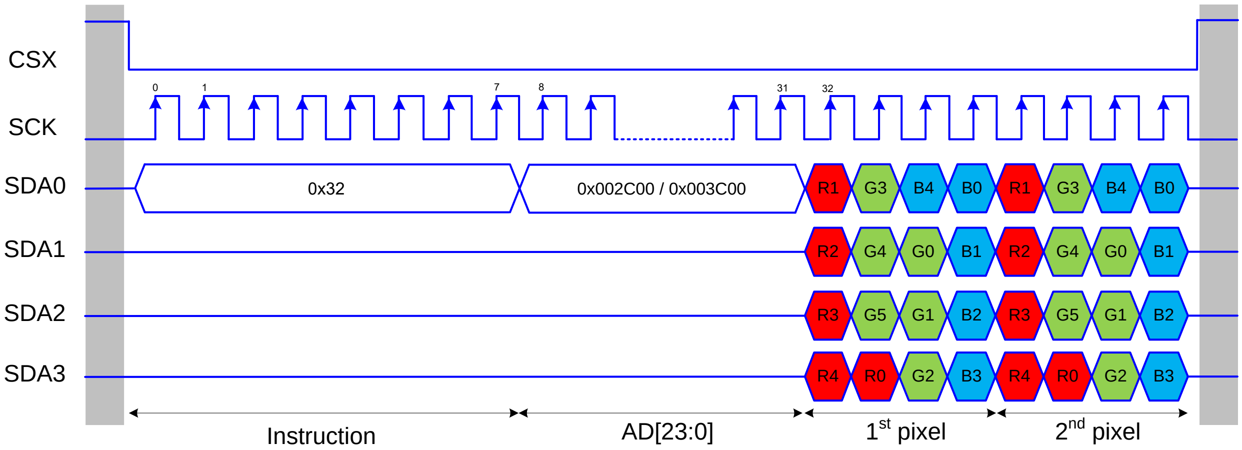


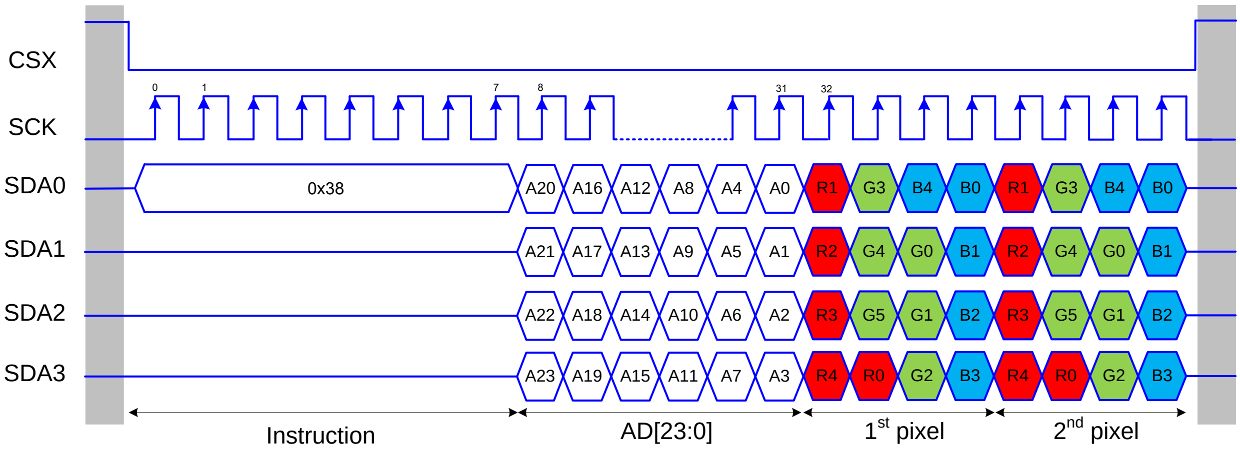
7.8.5.3 Color Format

QSPI RGB565

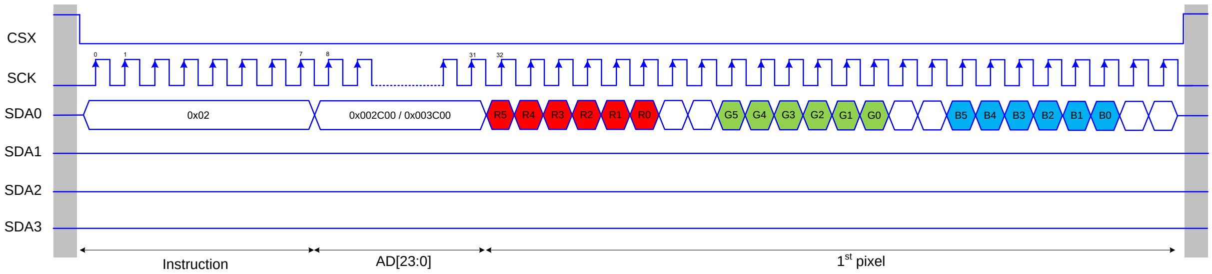


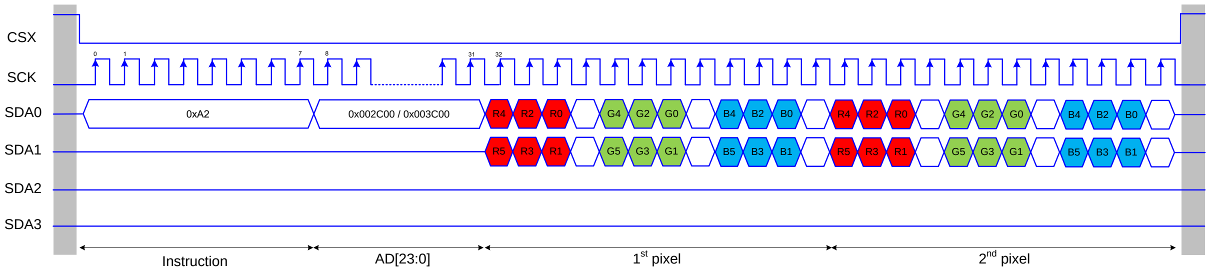


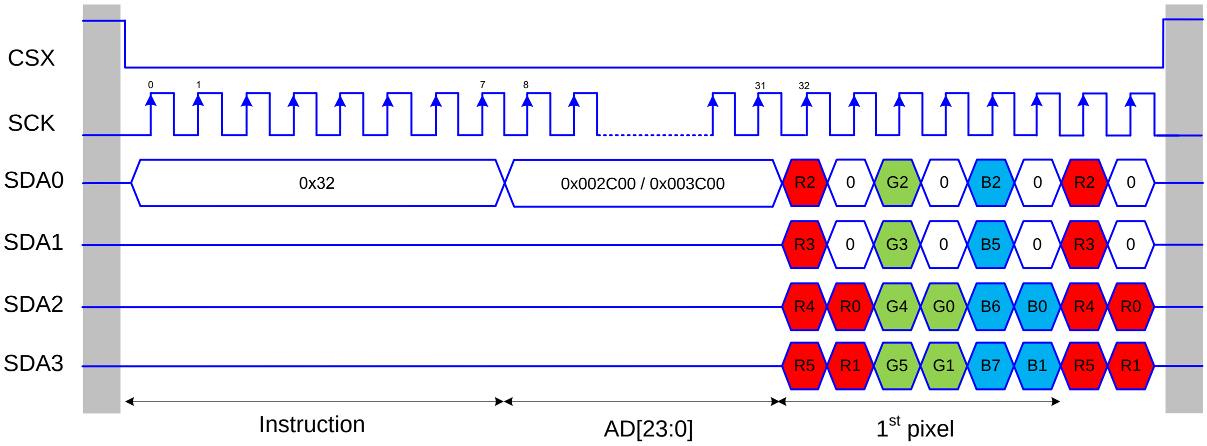


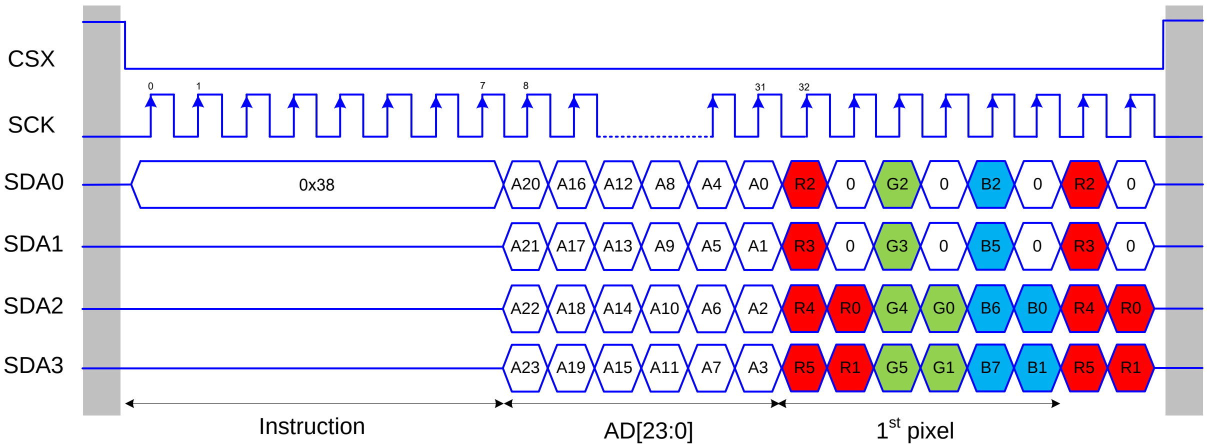


QSPI RGB666

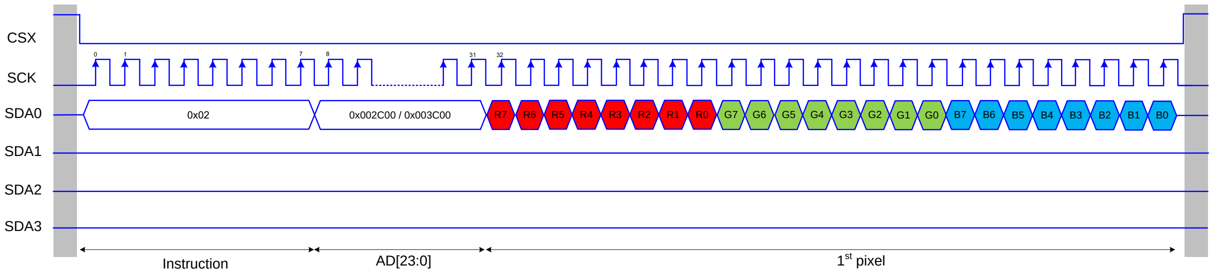

CSX
SCK
SDA0
SDA1
SDA2
SDA3
0x02
0x002C00 / 0x003C00
R5 R4 R3 R2 R1 R0
G5 G4 G3 G2 G1 G0
B5 B4 B3 B2 B1 B0
Instruction
AD[23:0]
1st pixel

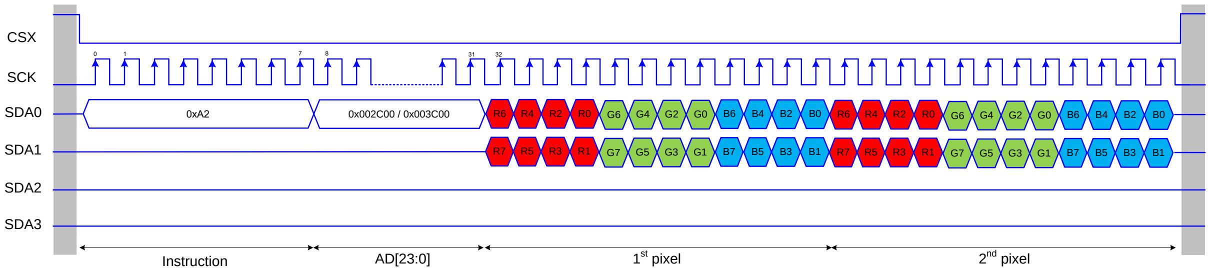

CSX
SCK
SDA0
SDA1
SDA2
SDA3
0xA2
0x002C00 / 0x003C00
R4 R2 R0 G4 G2 G0 B4 B2 B0
R5 R3 R1 G5 G3 G1 B5 B3 B1
Instruction
AD[23:0]
1st pixel
2nd pixel

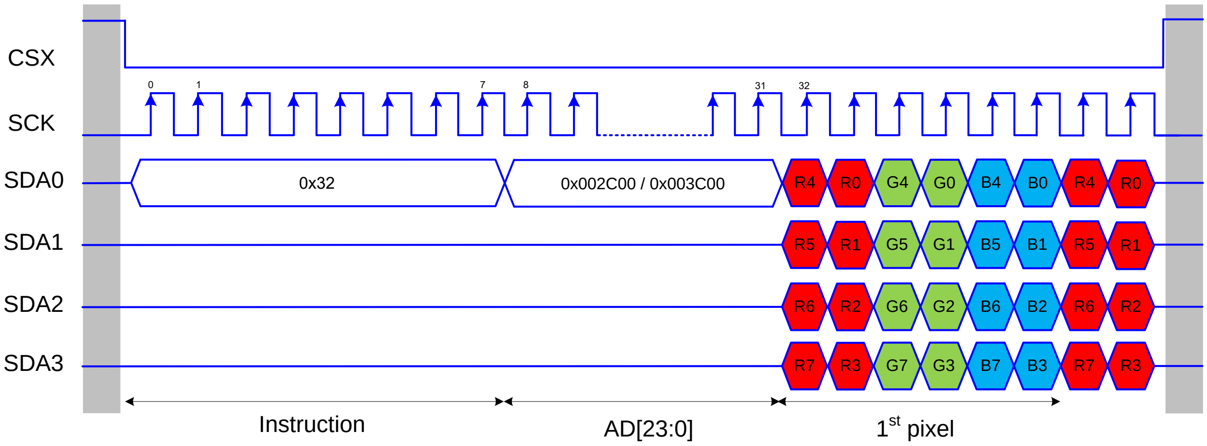

CSX
SCK
SDA0
SDA1
SDA2
SDA3
0x32
0x002C00 / 0x003C00
R2 0 G2 0 B2 0 R2 0
R3 0 G3 0 B5 0 R3 0
R4 R0 G4 G0 B6 B0 R4 R0
R5 R1 G5 G1 B7 B1 R5 R1
Instruction
AD[23:0]
1st pixel

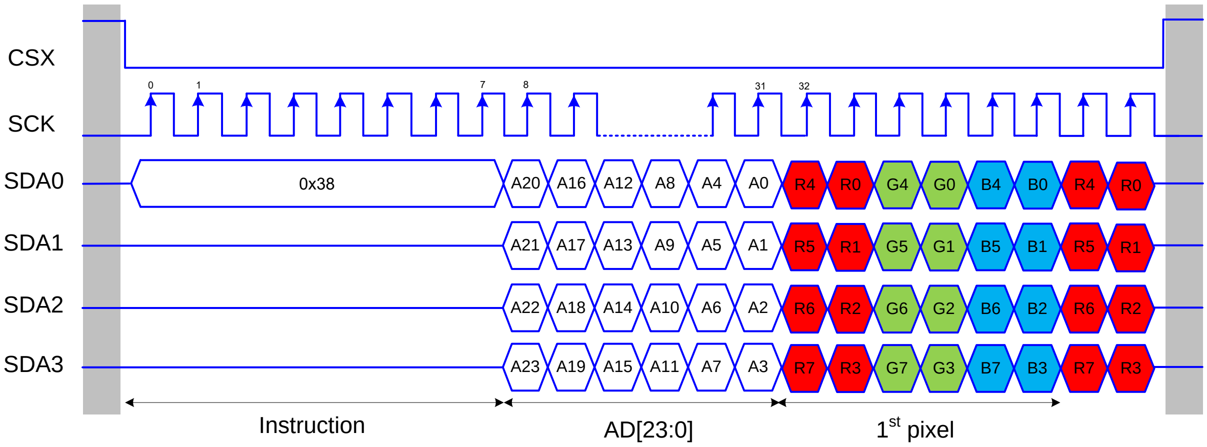

CSX
SCK
SDA0
SDA1
SDA2
SDA3
0x38
A20 A16 A12 A8 A4 A0
A21 A17 A13 A9 A5 A1
A22 A18 A14 A10 A6 A2
A23 A19 A15 A11 A7 A3
R2 0 G2 0 B2 0 R2 0
R3 0 G3 0 B5 0 R3 0
R4 R0 G4 G0 B6 B0 R4 R0
R5 R1 G5 G1 B7 B1 R5 R1
Instruction
AD[23:0]
1st pixel

QSPI RGB888


CSX
SCK
SDA0
SDA1
SDA2
SDA3
0x02
0x002C00 / 0x003C00
R7 R6 R5 R4 R3 R2 R1 R0 G7 G6 G5 G4 G3 G2 G1 G0 B7 B6 B5 B4 B3 B2 B1 B0
Instruction
AD[23:0]
1st pixel


CSX
SCK
SDA0
SDA1
SDA2
SDA3
0xA2
0x002C00 / 0x003C00
R6 R4 R2 R0 G6 G4 G2 G0 B6 B4 B2 B0 R6 R4 R2 R0 G6 G4 G2 G0 B6 B4 B2 B0
R7 R5 R3 R1 G7 G5 G3 G1 B7 B5 B3 B1 R7 R5 R3 R1 G7 G5 G3 G1 B7 B5 B3 B1
Instruction
AD[23:0]
1st pixel
2nd pixel


CSX
SCK
SDA0
SDA1
SDA2
SDA3
0x32
0x002C00 / 0x003C00
R4 R0 G4 G0 B4 B0 R4 R0
R5 R1 G5 G1 B5 B1 R5 R1
R6 R2 G6 G2 B6 B2 R6 R2
R7 R3 G7 G3 B7 B3 R7 R3
Instruction
AD[23:0]
1st pixel


CSX
SCK
SDA0
SDA1
SDA2
SDA3
0x38
A20 A16 A12 A8 A4 A0
A21 A17 A13 A9 A5 A1
A22 A18 A14 A10 A6 A2
A23 A19 A15 A11 A7 A3
R4 R0 G4 G0 B4 B0 R4 R0
R5 R1 G5 G1 B5 B1 R5 R1
R6 R2 G6 G2 B6 B2 R6 R2
R7 R3 G7 G3 B7 B3 R7 R3
Instruction
AD[23:0]
1st pixel

7.9 RGB Interface

7.9.1 RGB interface Selection

The color format selection of RGB Interface for ST77922 is selected by setting the command 3Ah, DB [1:0].

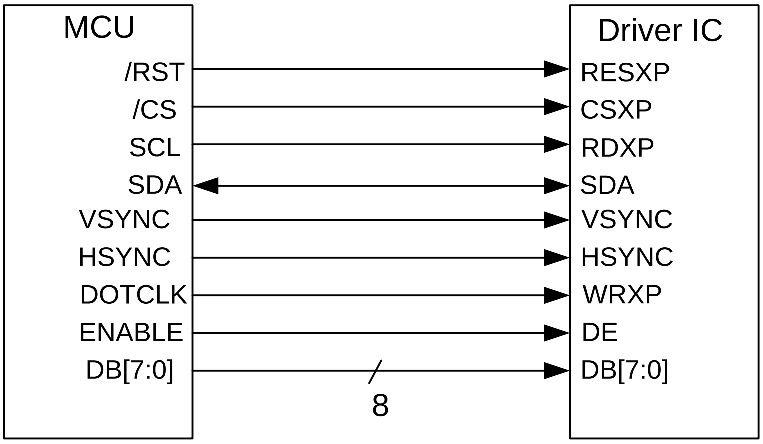
RGB Interface Mode	3Ah, DB[1:0]	Data pins
8-bit 16M RGB Interface	11	DB[7:0]
8-bit 262K RGB Interface	10	DB[7:0]
8-bit 65K RGB Interface	01	DB[7:0]

7.9.2 RGB Color Format

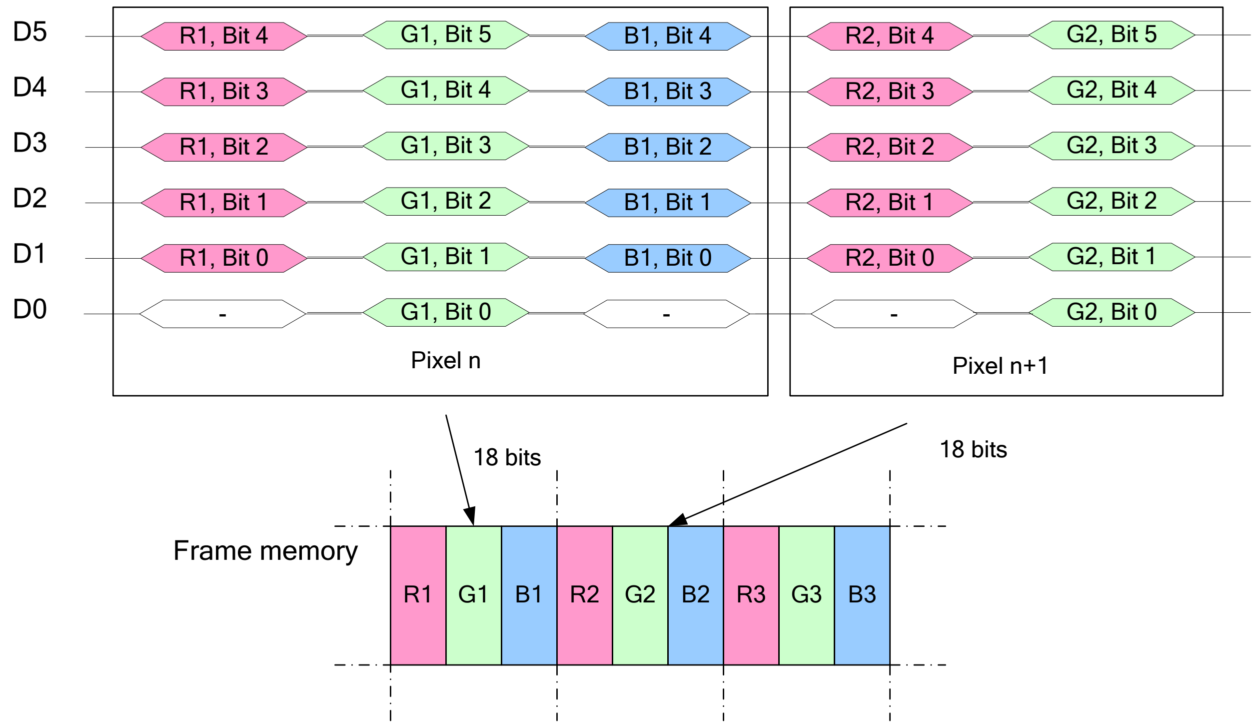
ST77922 supports two kinds of RGB interface, DE mode and HV mode, and 6bit data format. When DE mode is selected and the VSYNC, HSYNC, DOTCLK, DE, DB[7:0] pins can be used; when HV mode is selected and the VSYNC, HSYNC, DOTCLK, D[5:0] pins can be used. When using RGB interface, only serial interface can be selected.

8-bit RGB interface & 3-line serial interface hardware suggestion, IM[2:0]=100.
8-bit RGB interface & 4-line serial interface hardware suggestion, IM[2:0]=101.

8-bit RGB Interface



Write data for 6-bit/pixel (RGB 5-6-5-bit input), 65K-Colors



Write data for 6-bit/pixel (RGB 6-6-6-bit input), 262K-Colors

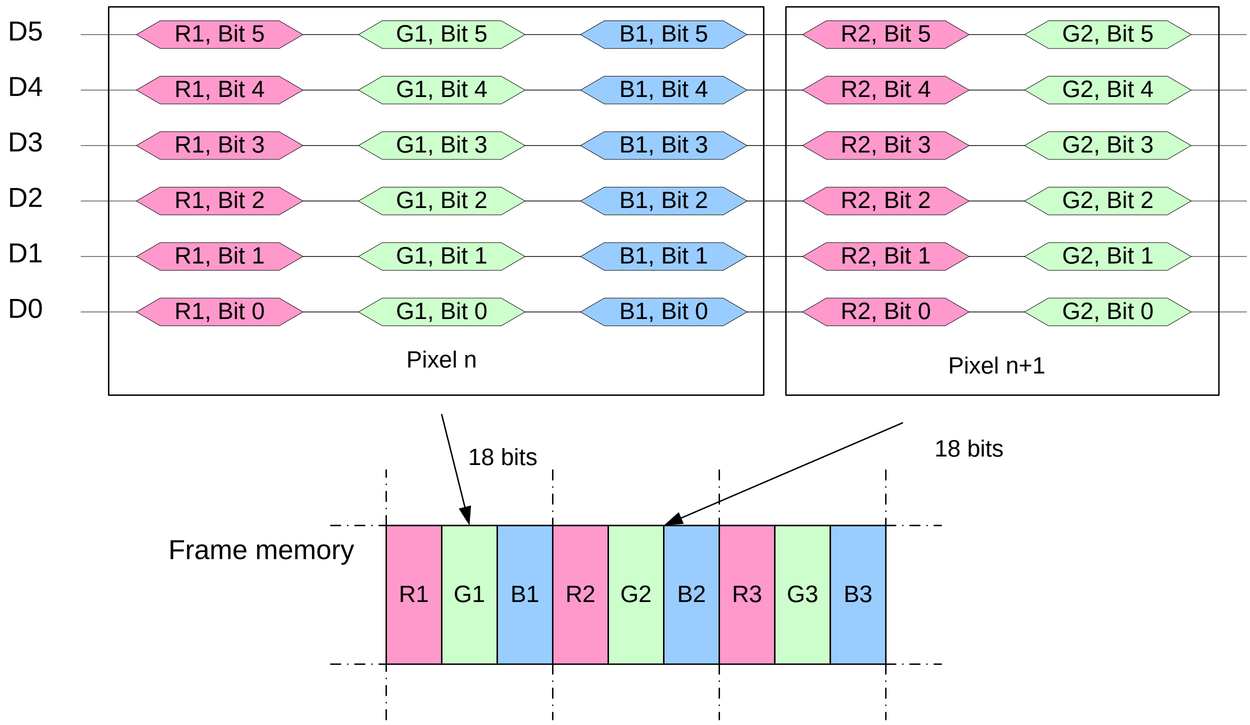


Figure 23 RGB Interface Data Format

7.9.3 RGB Interface Definition

The display operation via the RGB interface is synchronized with the VSYNC, HSYNC, and DOTCLK signals. The data can be written only within the specified area with low power consumption by using window address function. The back porch and front porch are used to set the RGB interface timing.

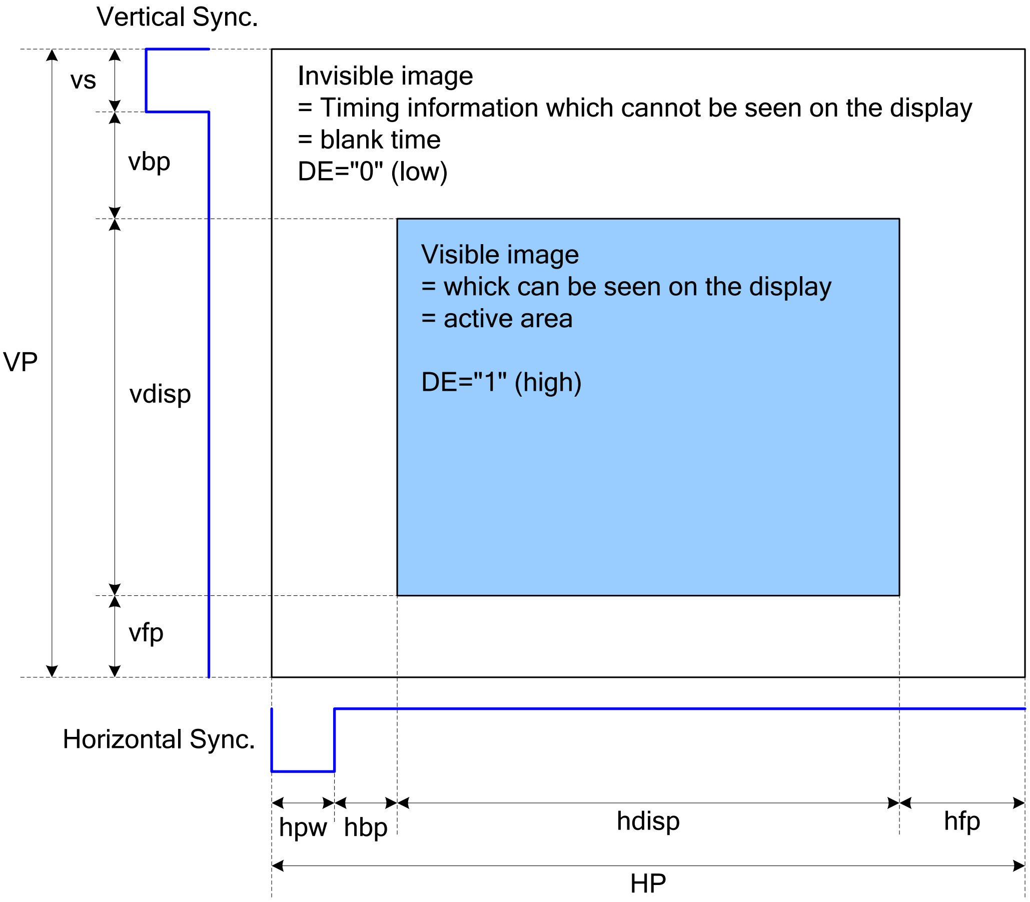


Figure 24 DRAM Access Area by RGB Interface

Please refer to the following table for the setting limitation of RGB interface signals.

8bit RGB interface:

Parameter	Symbol	Min.	Typ.	Max.	Unit
Horizontal Sync. Width	hpw	2	-	hpw+hbp=127	Clock
Horizontal Sync. Back Porch	hbp	2	-		Clock
Horizontal Sync. Front Porch	hfp	2	-	-	Clock
Vertical Sync. Width	vs	2	-	vs+vbp=127	Line
Vertical Sync. Back Porch	vbp	2	-		Line
Vertical Sync. Front Porch note2	vfp	2	-	1023	Line

Note:

1.Typical value are related to the setting of frame rate is 60Hz.

2.Touch Controller operation time was considered

3.The minimum values of this table mean the limitation of IC without considering the panel GIP

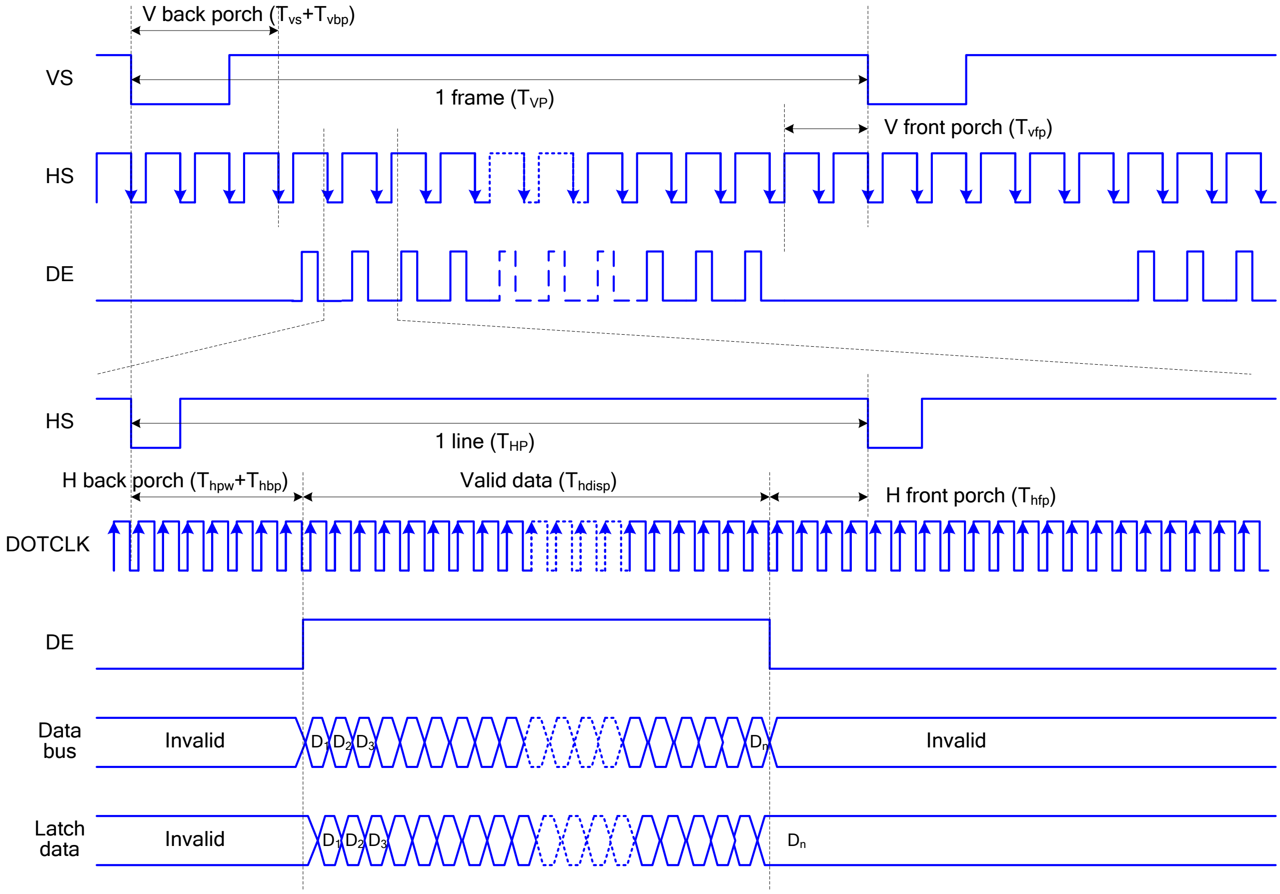
7.9.4 RGB Interface Mode Selection

ST77922 supports two kinds of RGB interface, DE mode and HV mode. Each mode also can select with ram and without ram. The table shown below uses command BDh, DB[7] to select with ram and without ram.

RCM	RGB Mode	WO	Data Path
0	DE mode	0	Ram
		1	Shift register (without Ram)
1	HV mode	0	Ram
		1	Shift register (without Ram)

7.9.5 RGB Interface Timing

The timing chart of RGB interface DE mode is shown as follows.



Note: The setting of front porch and back porch in host must match that in IC as this mode.

Figure 25 Timing Chart of Signals in RGB Interface DE Mode

The timing chart of RGB interface HV mode is shown as follows.

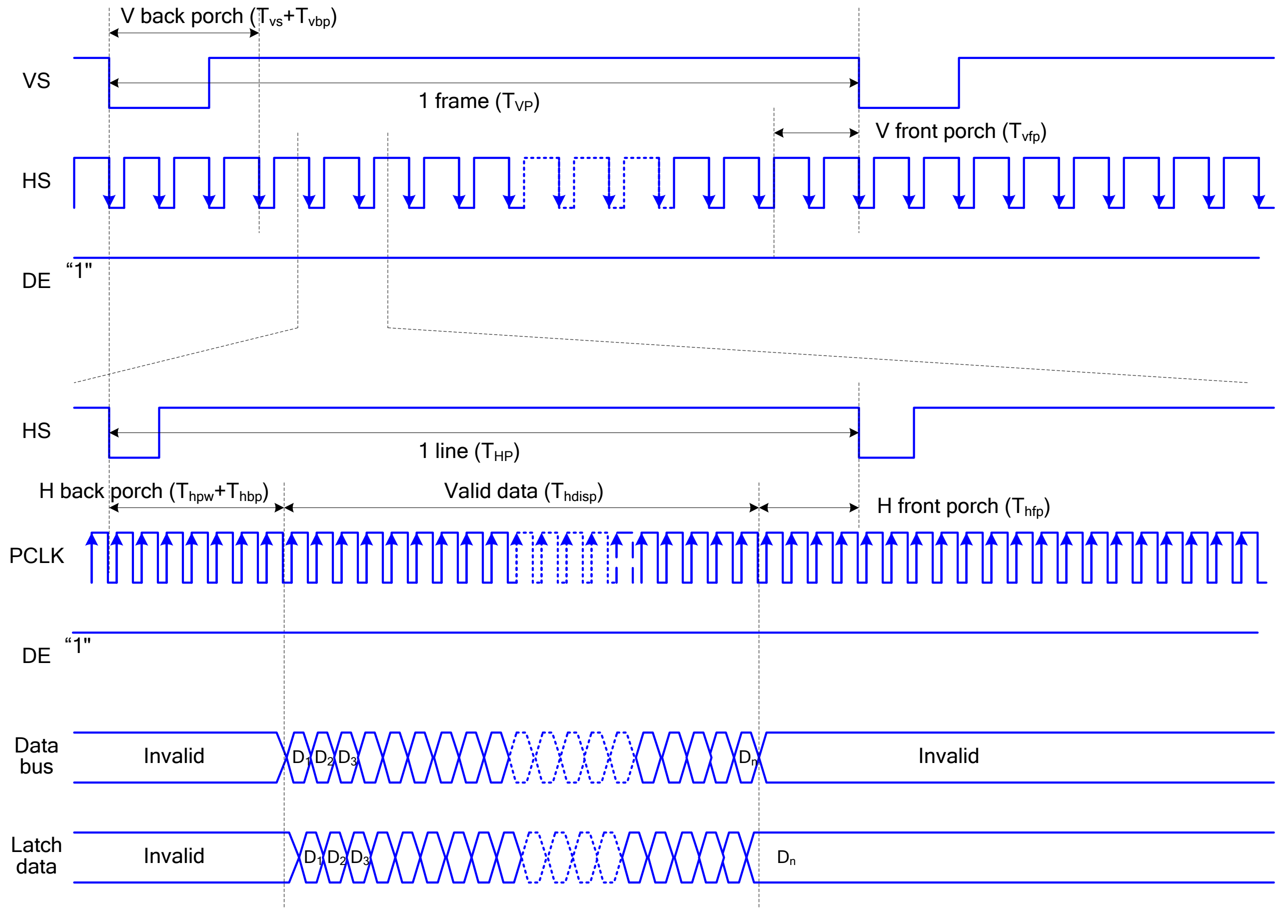


Figure 26 Timing chart of RGB interface HV mod

The following are the functions not available in RGB Input Interface mode.

Function	RGB Interface	I80 System Interface
Partial display	Not available	Available
Scroll function	Not available	Available
Interlaced scan	Not available	Available
Graphics operation function	Not available	Available

VSYNC, HSYNC, and DOTCLK signals must be supplied during a display operation period.

In RGB interface mode, the panel controlling signals are generated from DOTCLK, not the internal clock generated from the internal oscillator.

In 8-bit RGB interface mode, each of RGB dots are transferred in synchronization with DOTCLK signals. In other words, one pixel data needs to take three DOTCLKs to transfer.

In 8-bit RGB interface mode, the cycles of VSYNC, HSYNC, ENABLE, DOTCLK signals must be set correctly so that the data transfer is completed in units of pixels.

When switching between the internal operation mode and the external display interface operation mode, follow the sequences below in setting instruction.

In RGB interface mode, the front porch period continues until the next VSYNC input is detected after drawing one frame.

In RGB interface mode, a RAM address is set in the address counter every frame on the falling edge of VSYNC.

7.10 Mobile Industry Processor Interface (MIPI)

7.10.1 Display Serial Interface (DSI)

7.10.1.1 GENAL DESCRIPTION

The communication can be separated 2 different levels between the MCU and the display module:

1. Low level communication what is done on the interface level
2. High level communication what is done on the packet level

7.10.1.2 Interface Level Communication

The display module uses data and clock lane differential pairs for DSI (DSI-1M). Both differential lane pairs can be driven Low Power (LP) or High Speed (HS) mode.

Low Power mode means that each line of the differential pair is used in single end mode and a differential receiver is disable (A termination resistor of the receiver is disable) and it can be driven into a low power mode. High Speed mode means that differential pairs (The termination resistor of the receiver is enable) are not used in the single end mode. There are used different modes and protocols in each mode when there wanted to transfer information from the MCU to the display module and vice versa.

The State Codes of the High Speed (HS) and Low Power (LP) lane pair are defined below.

Lane Pair State	Line DC Voltage Levels		High Speed (HS)	Low Power	
	DATA_P	DATA_N	Burst Mode	CLOCK_P	CLOCK_N
HS-0	Low (HS)	High (HS)	Differential – 0	Note 1	Note1
HS-1	High (HS)	Low (HS)	Differential – 1	Note 1	Note 1
LP-00	Low (LP)	Low (LP)	Not Defined	Bridge	Space
LP-01	Low (LP)	High (LP)	Not Defined	HS – Request	Mark – 0
LP-10	High (LP)	Low (LP)	Not Defined	LP – Request	Mark – 1
LP-11	High (LP)	High (LP)	Not Defined	Stop	Note 2

Notes:

(1) Low-Power Receivers (LP-Rx) of the lane pair are checking the LP-00 state code, when the Lane Pair is in the High Speed (HS) mode.

(2 If Low-Power Receivers (LP-Rx) of the lane pair recognizes LP-11 state code, the lane pair returns to LP-11 of the Control Mode.

7.10.1.3 DSI-CLOCK Lanes

DSI-CLOCK_P/N lanes can be driven into three different power modes:

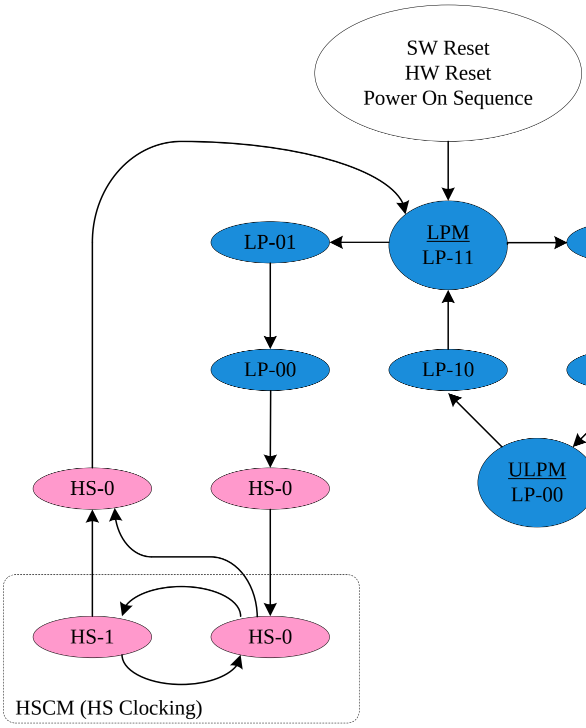
- Low Power Mode (LPM)
- Ultra Low Power Mode (ULPM)
- High Speed Clock Mode (HSCM)

Clock lanes are in a single end mode (LP = Low Power) when there is entering or leaving Low Power

Mode (LPM) or Ultra Low Power Mode (ULPM).

Clock lanes are in the single end mode (LP = Low Power) when there is entering in or leaving out High Speed Clock Mode (HSCM). These entering and leaving protocols are using clock lanes in the single end mode to generate an entering or leaving sequences.

The principal flow chart of the different clock lanes power modes is illustrated below.

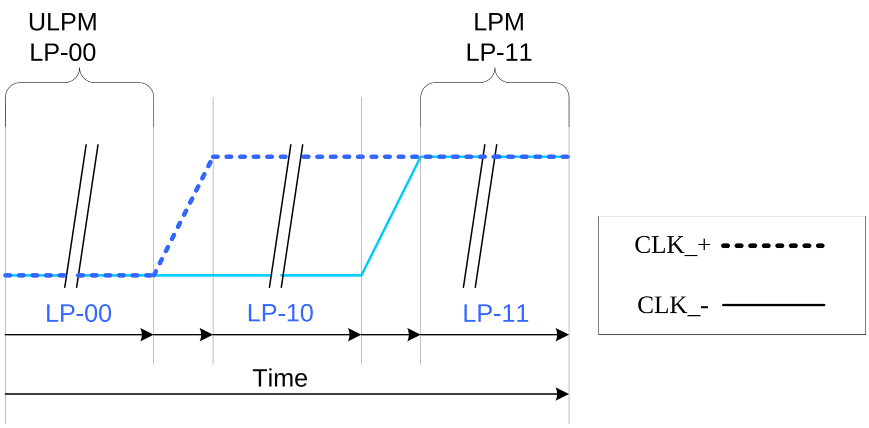


Flow chart of the different clock lanes

1. Low Power Mode (LPM)

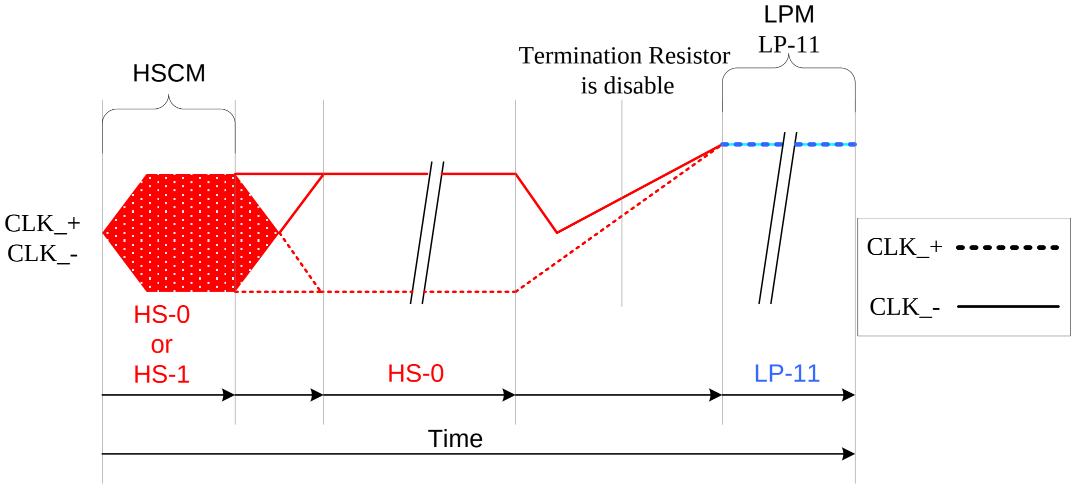
DSI-CLOCK_P/N lanes can be driven to the Low Power Mode (LPM), when DSI-CLOCK lanes are entering LP-11 State Code, in three different ways:

- After SW Reset, HW Reset or Power On Sequence =>LP-11 After DSI-CLOCK_P/N lanes are leaving Ultra Low Power Mode (ULPM, LP-00 State Code) =>LP-10
- LP-11 (LPM). This sequence is illustrated below.



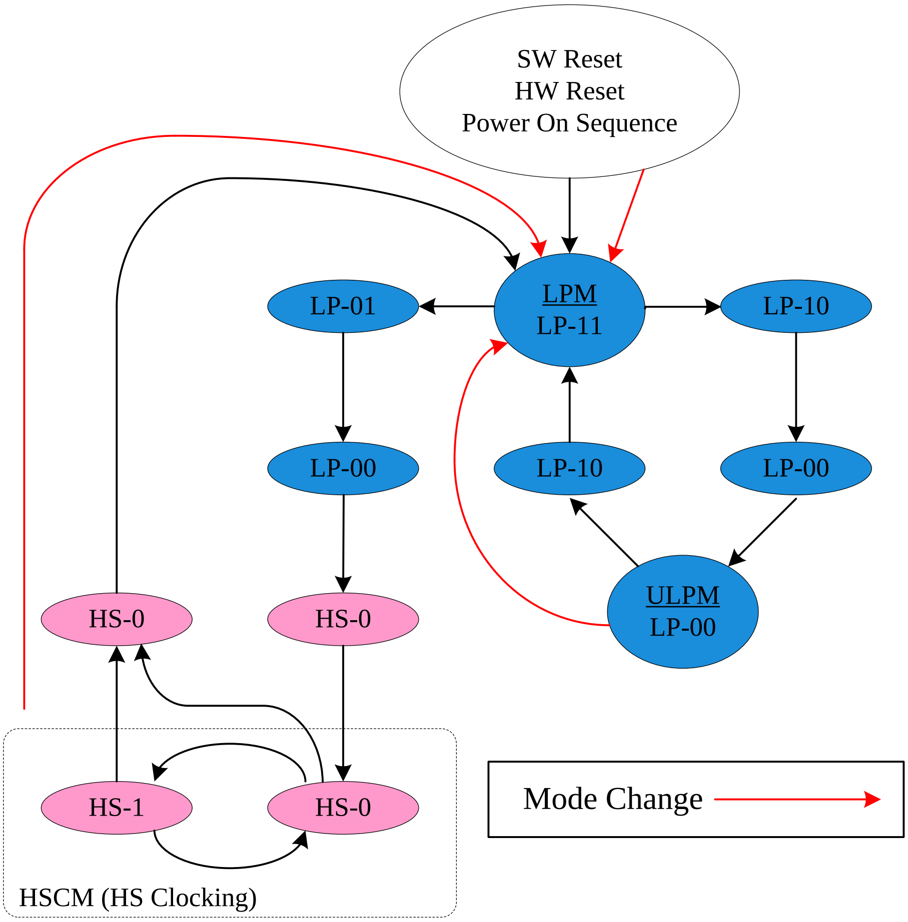
From ULPM to LPM

- After DSI-CLK+/- lanes are leaving High Speed Clock Mode (HSCM, HS-0 or HS-1 State Code) =>HS-0=>LP-11 (LPM). This sequence is illustrated below.



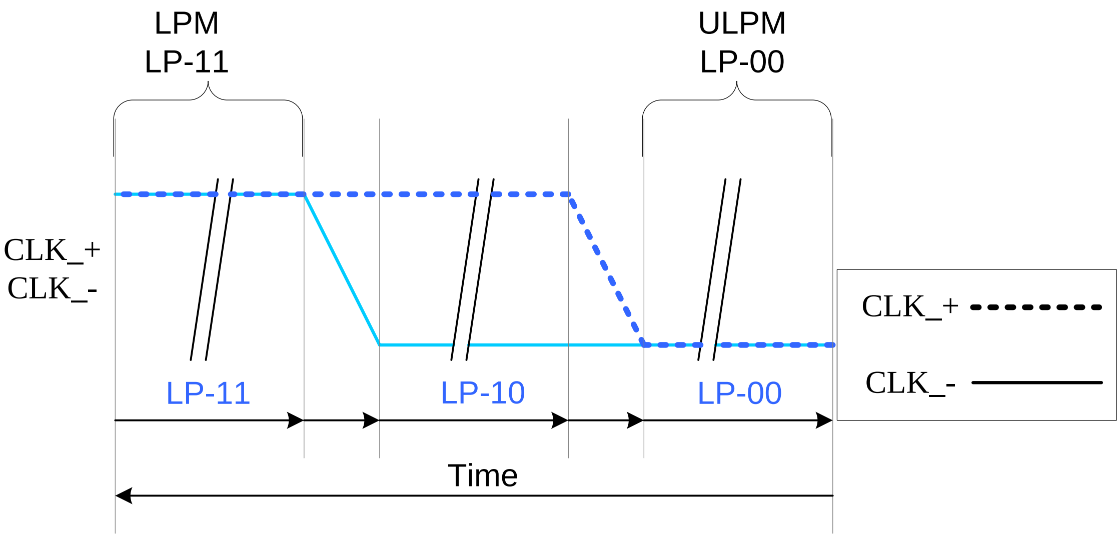
From High Speed Clock Mode (HSCM) to LPM

All three mode changes are illustrated a flow chart below.



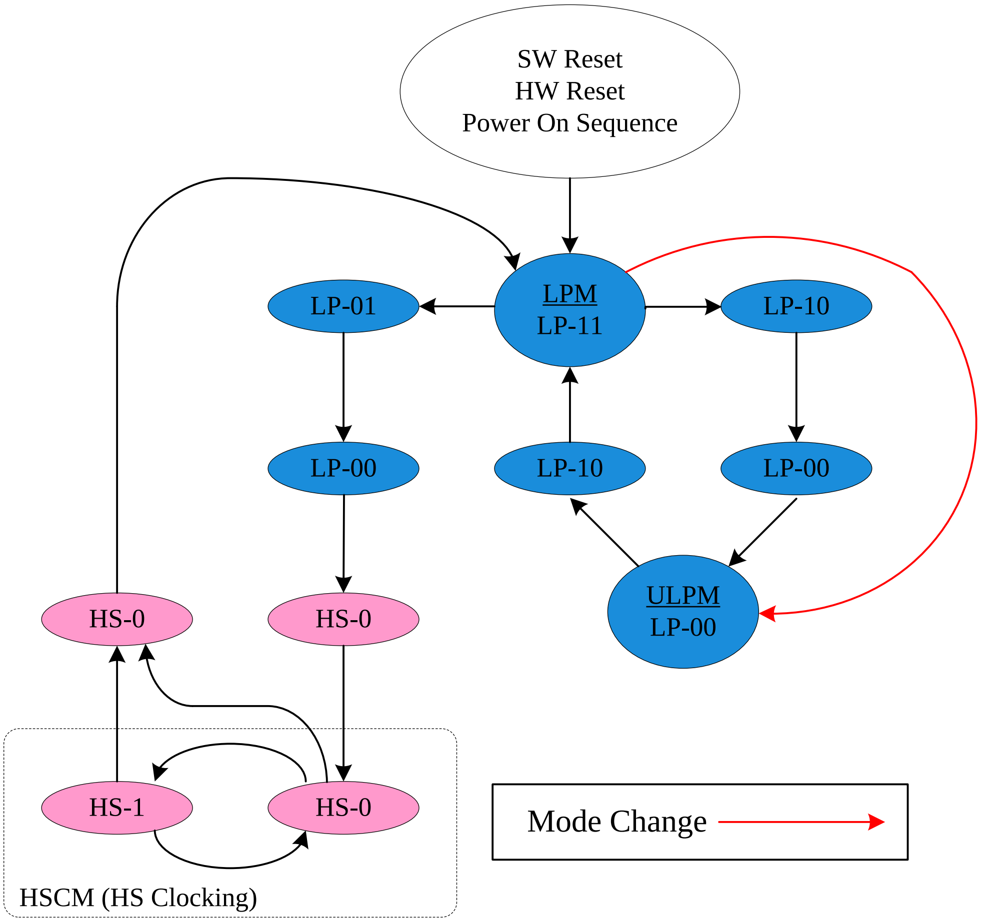
All Three Mode Changes to LPM on the Flow Chart

2. Ultra Low Power Mode (ULPM)



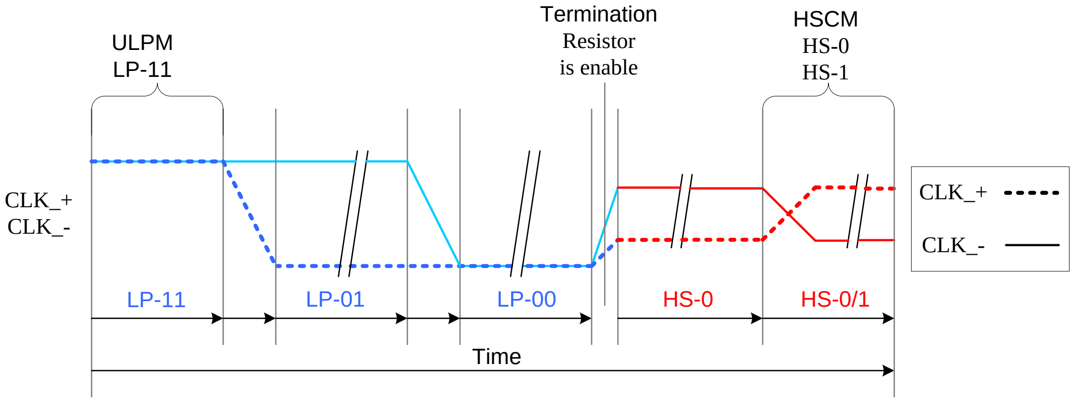
From LPM to ULPM

The mode change is also illustrated below.



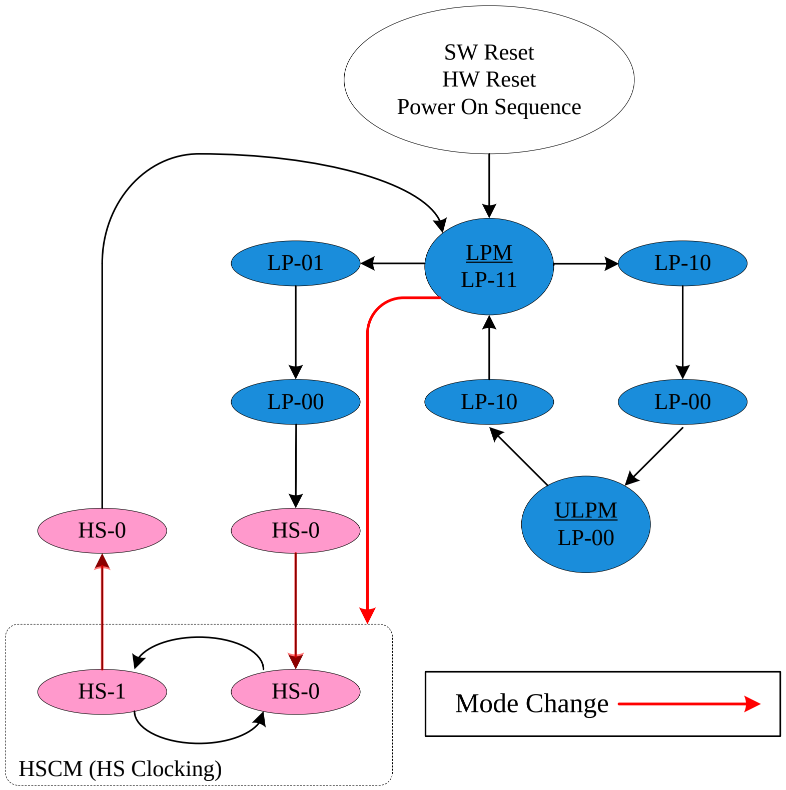
Mode Change from LPM to ULPM on the Flow Chart

3. High Speed Clock Mode (HSCM)



From LPM to HSCM

The mode change is also illustrated below.

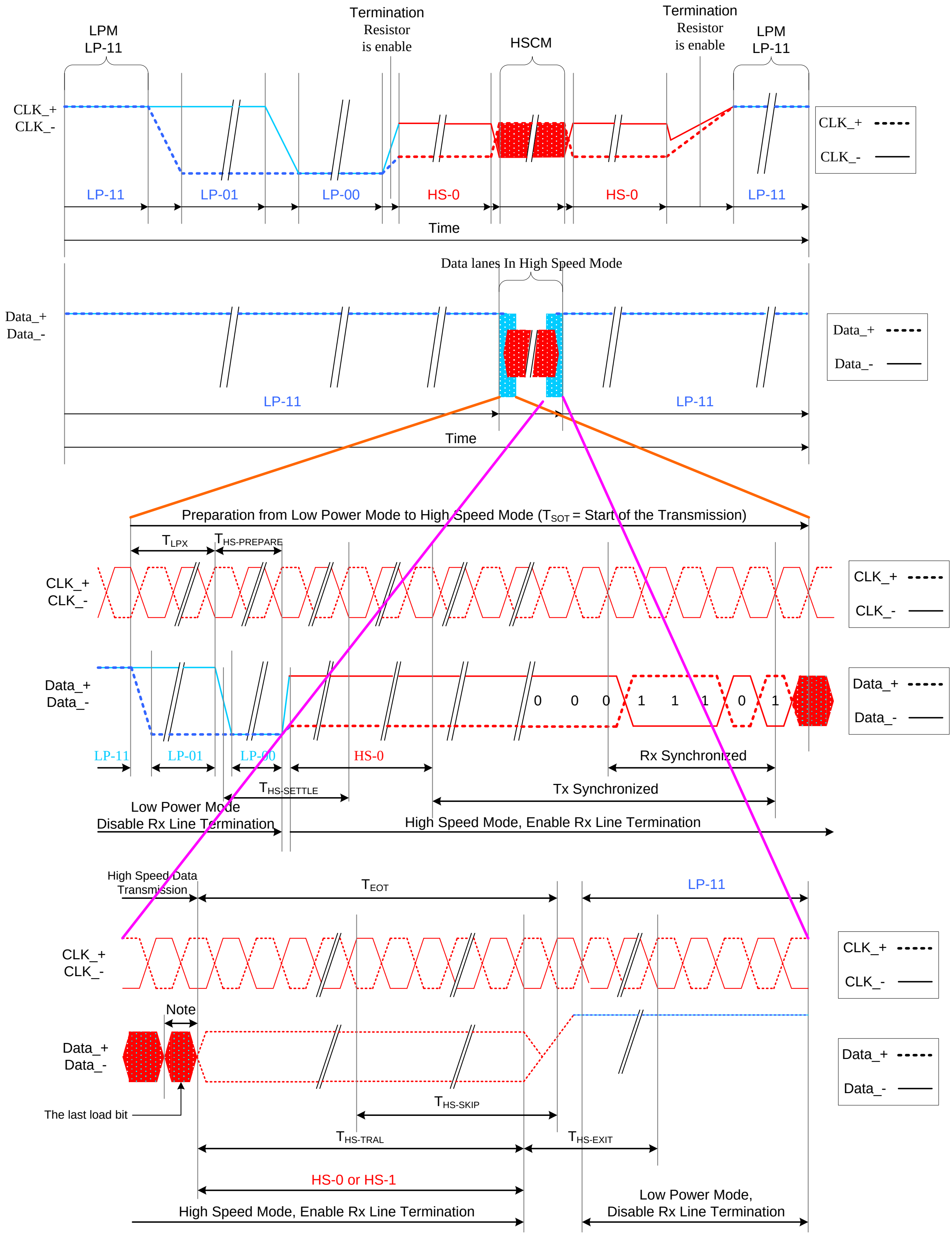


Mode Change from LPM to HSCM on the Flow Chart

The high speed clock (DSI-CLOCK_P/N) is started before high speed data is sent via DSI-DATA_P/N lanes. The high speed clock continues clocking after the high speed data sending has been stopped.

The burst of the high speed clock consists of:

- ●Even number of transitions
- ●Start state is HS-0
- ●End state is HS-0



High Speed Clock Burst

Note:

If the last load bit is HS-0, the transmitter changes from HS-0 to HS-1

If the last load bit is HS-1, the transmitter changes from HS-1 to HS-0

7.10.1.4 DSI-DATA Lanes

DSI-DATA_P/N Data Lanes can be driven in different modes which are:

- Escape Mode
- High-Speed Data Transmission
- Bus Turnaround Request

These modes and their entering codes are defined on the following table.

Mode	Entering Mode Sequence	Leaving Mode Sequence
Escape Mode	LP-11=>LP-10=>LP-00=>LP-01=>LP-00	LP-00=>LP-10=>LP-11(Mark-1)
High-Speed Data Transmission	LP-11=>LP-01=>LP-00=>HS-0	(HS-0 or HS-1) =>LP-11
Bus Turnaround Request	LP-11=>LP-10=>LP-00=>LP-10=>LP-00	High-Z, Note

1. **Escape Mode**

Data lanes (DSI-DATA_P/N) can be used in different Escape Modes when data lanes are in Low Power (LP) mode.

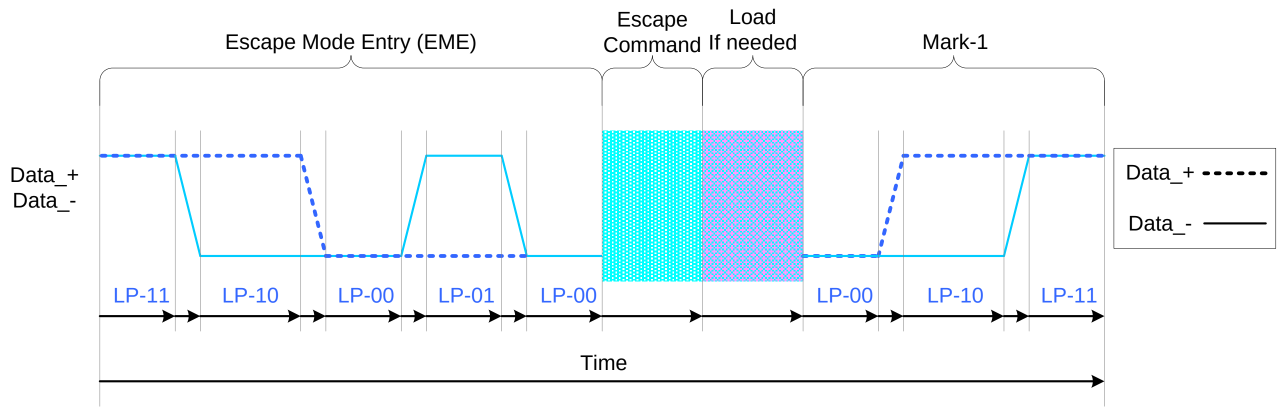
These Escape Modes are used to:

- Send “Low-Power Data Transmission” (LPDT) e.g. from the MCU to the display module
- Drive data lanes to “Ultra-Low Power State” (ULPS)
- Indicate “Remote Application Reset” (RAR), which is reset the display module
- Indicate “Tearing Effect”, which is used for a TE line event from the display module to the MCU
- Indicate (ACK), which is used for a non-error event from the display module to the MCU

The basic sequence of the Escape Mode is as follow

- Start: LP-11
- Escape Mode Entry (EME): LP-11 =>LP-10 =>LP-00 =>LP-01 =>LP-00
- Escape Command (EC), which is coded, when one of the data lanes is changing from low-to-high-to-low then this changed data lane is presenting a value of the current data bit (DSI-D0+ = 1, DSI-D0- = 0) e.g. when DSI-D0- is changing from low-to-high-to-low, the receiver is latching a data bit, which value is logical 0. The receiver is using this low-to-high-to-low transition for its internal clock.
- A load if it is needed
- Exit Escape (Mark-1) LP-00 =>LP-10 =>LP-11
- End: LP-11

This basic construction is illustrated below:



General Escape Mode Sequence

The number of the different Escape Commands (EC) is eight. These eight different Escape Commands (EC) can be divided 2 different groups: Mode or Trigger. The MCU is informing to the display module that it is controlling data lanes (RX_D0P/N) with the mode e.g. The MCU can inform to the display module that it can put data lanes in the low power mode. The MCU is waiting from the display module an event information, which has been set by the MCU, with the trigger e.g. when the display module reaches a new V-synch, the display module sent to the MCU a TE trigger (TEE), if the MCU has been requested it.

Escape commands are defined on the next table.

Escape command	Command Type Mode / Trigger	Entry command Pattern (First Last Bit Transmitted)
Low-Power Data	Mode	1110 0001 b
Ultra-Low Power Mode	Mode	0001 1110 b
Undefined-1, Note	Mode	1001 1111 b
Undefined-2, Note	Mode	1101 1110 b
Remote Application Reset	Trigger	0110 0010 b
Tearing Effect	Trigger	0101 1101 b
Acknowledge	Trigger	0010 0001 b
Uknown-5, Note	Trigger	1010 0000 b

Note: This Escape command support has not been implemented on the display module.

Low-Power Data Transmission (LPDT)

The MCU can send data to the display module in Low-Power Data Transmission (LPDT) mode when data lanes are entering in Escape Mode and Low-Power Data Transmission (LPDT) command has been sent to the display module. The display module is also using the same sequence when it is sending data to the MCU.

The Low Power Data Transmission (LPDT) is using a following sequence:

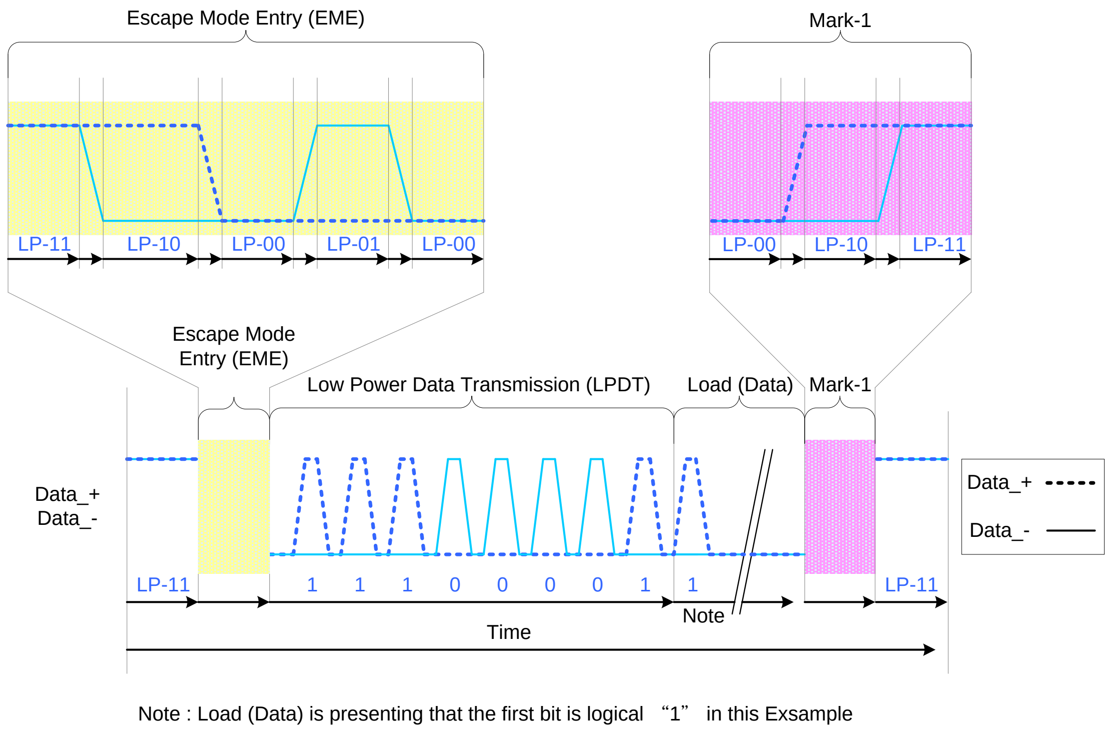
- Start: LP-11

- Escape Mode Entry (EME): LP-11 =>LP-10 =>LP-00 =>LP-01 =>LP-00
- Low-Power Data Transmission (LPDT) command in Escape Mode: 1110 0001 (First to Last bit)
- Load (Data):

  One or more bytes (8 bit)

  Data lanes are in pause mode when data lanes are stopped (Both lanes are low) between bytes
- Mark-1: LP-00 =>LP-10 =>LP-11
- End: LP-11

The Low-Power Data Transmission (LPDT) is as below,



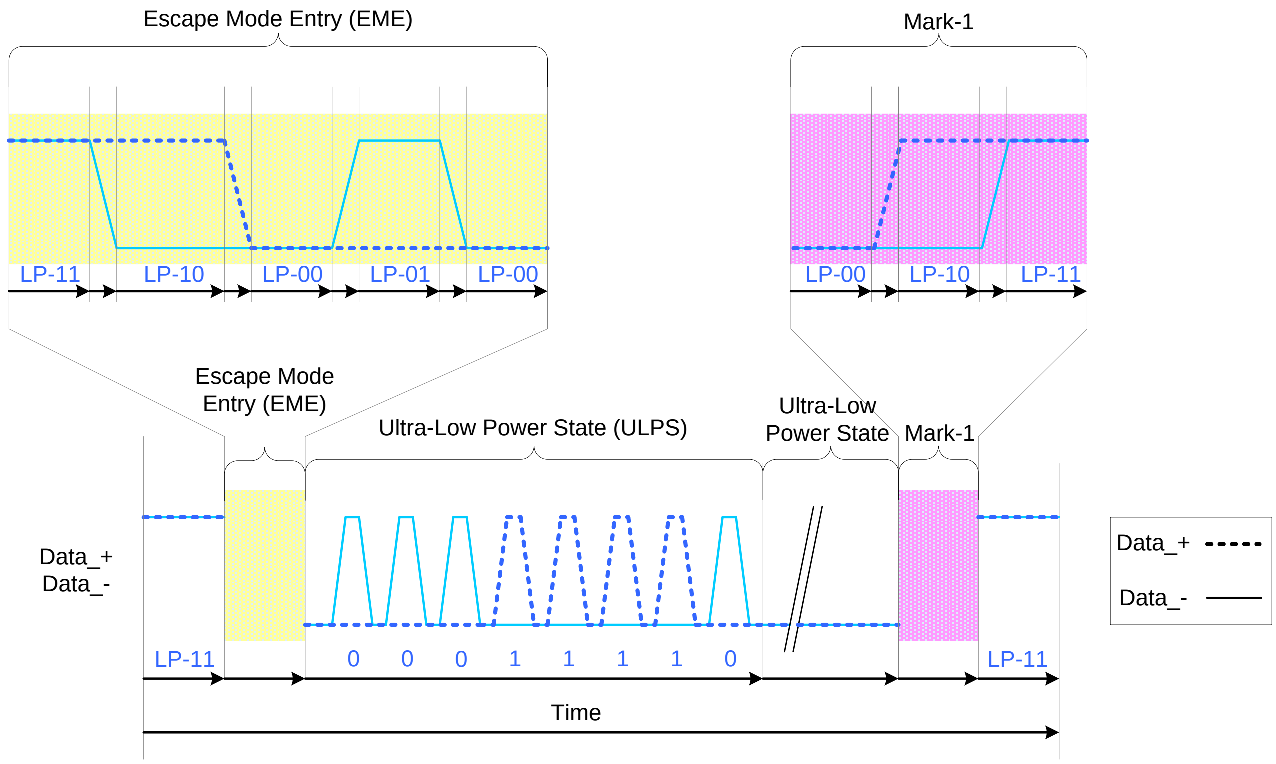
Low-Power Data Transmission (LPDT)

Ultra-Low Power State (ULPS)

The MCU can force data lanes in Ultra-Low Power State (ULPS) mode when data lanes are entering in Escape Mode. The Ultra-Low Power State (ULPS) is using a following sequence:

- Start: LP-11
- Escape Mode Entry (EME): LP-11 =>LP-10 =>LP-00 =>LP-01 =>LP-00
- Ultra-Low Power State (ULPS) command in Escape Mode: 0001 1110 (First to Last bit)
- Ultra-Low Power State (ULPS) when the MCU is keeping data lanes low
- Mark-1: LP-00 =>LP-10 =>LP-11
- End: LP-11

This sequence is illustrated for reference purposes below:



Ultra-Low Power State (ULPS)

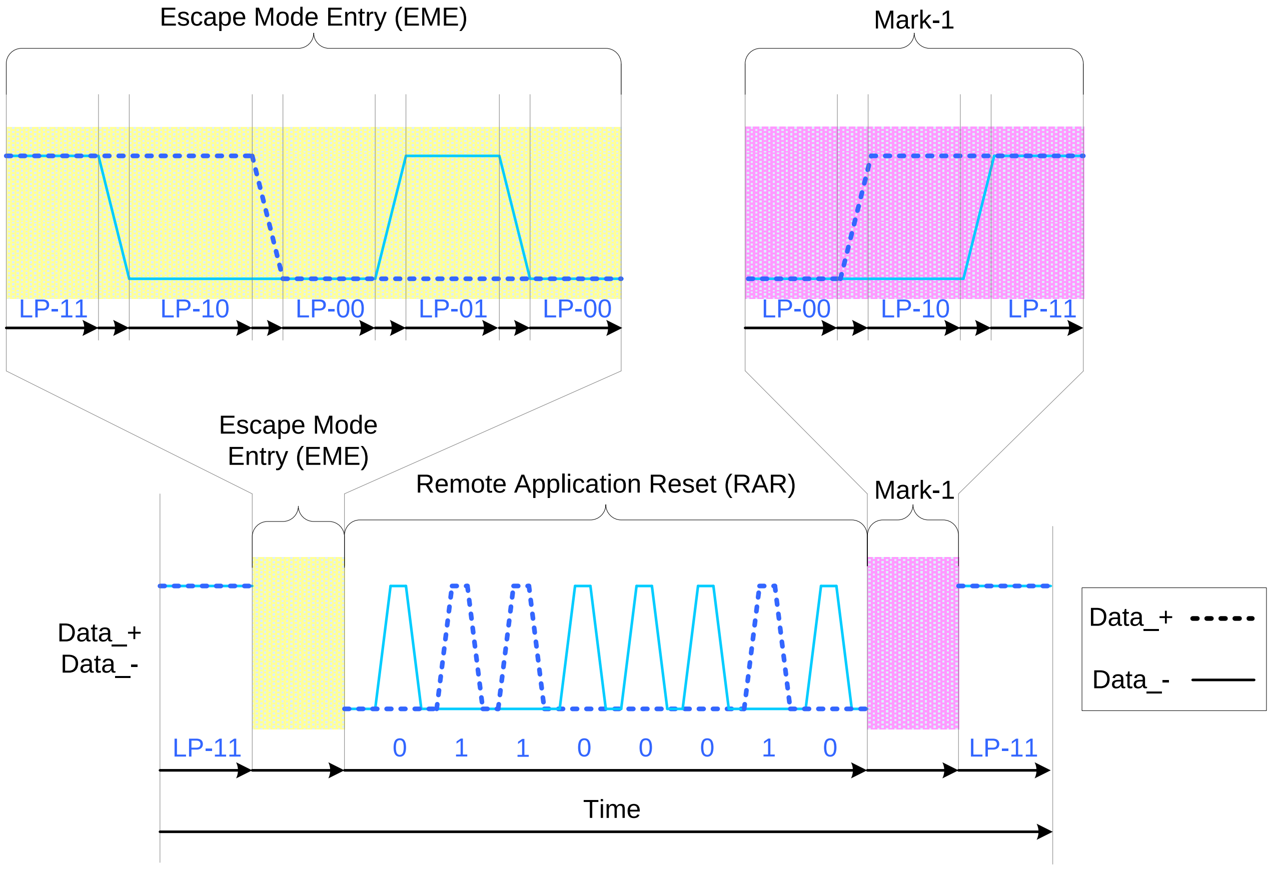
Remote Application Reset (RAR)

The MCU can inform to the display module that it should be reseted in Remote Application Reset (RAR) trigger when data lanes are entering in Escape Mode.

The Remote Application Reset (RAR) is using a following sequence:

- Start: LP-11
- Escape Mode Entry (EME): LP-11 =>LP-10 =>LP-00 =>LP-01 =>LP-00
- Remote Application Reset (RAR) command in Escape Mode: 0110 0010 (First to Last bit)
- Mark-1: LP-00 =>LP-10 =>LP-11
- End: LP-11

This sequence is illustrated for reference purposes below:



Remote Application Reset (RAR)

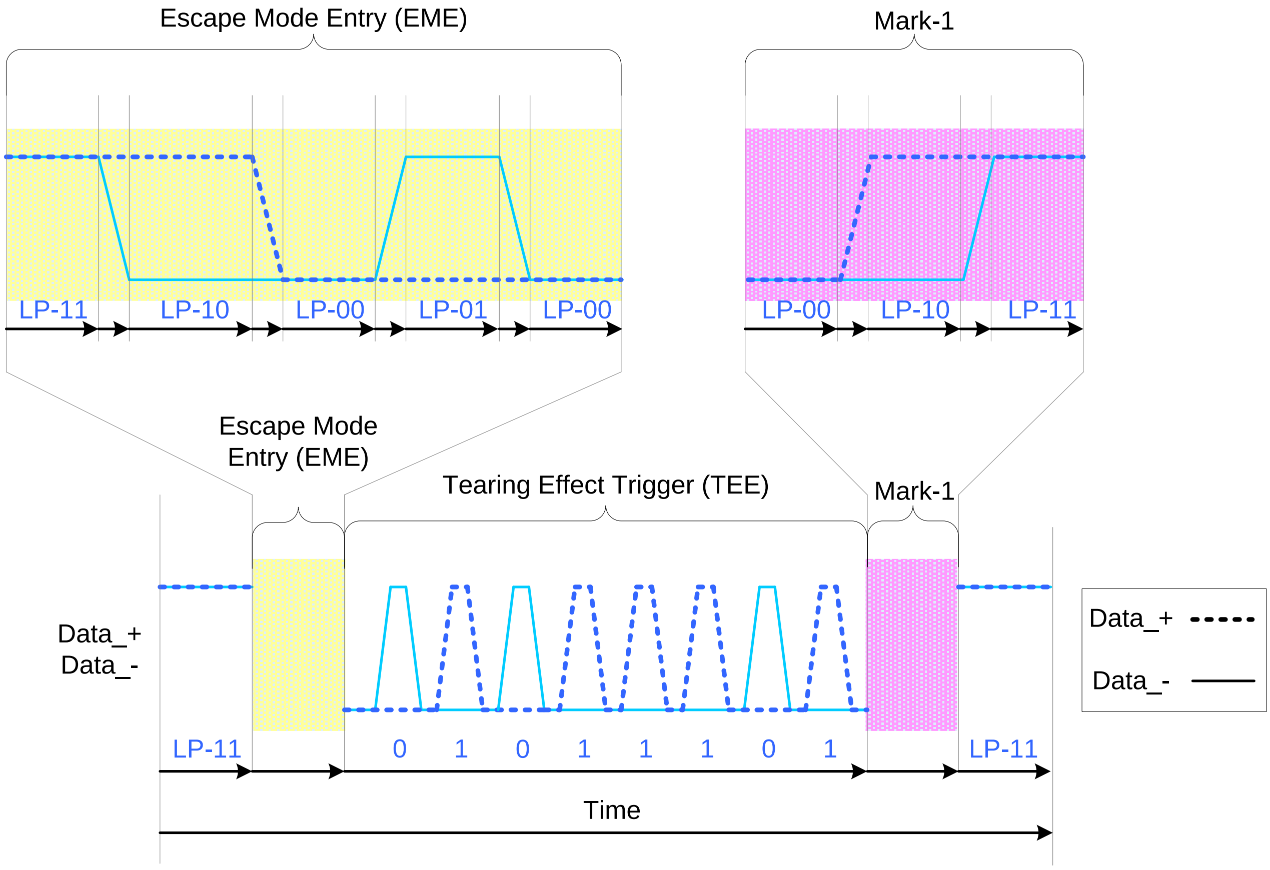
Tearing Effect (TEE)

The display module can inform to the MCU when a tearing effect event (New V-synch) has been appened on the display module by Tearing Effect (TEE).

The display module is sending the Tearing Effect (TEE) what is a following sequence:

- Start: LP-11
- Escape Mode Entry (EME): LP-11 =>LP-10 =>LP-00 =>LP-01 =>LP-00
- Tearing Effect (TEE) trigger in Escape Mode: 0101 1101 (First to Last bit)
- Mark-1: LP-00 =>LP-10 =>LP-11
- End: LP-11

This sequence is illustrated for reference purposes below:



Tearing Effect (TEE)

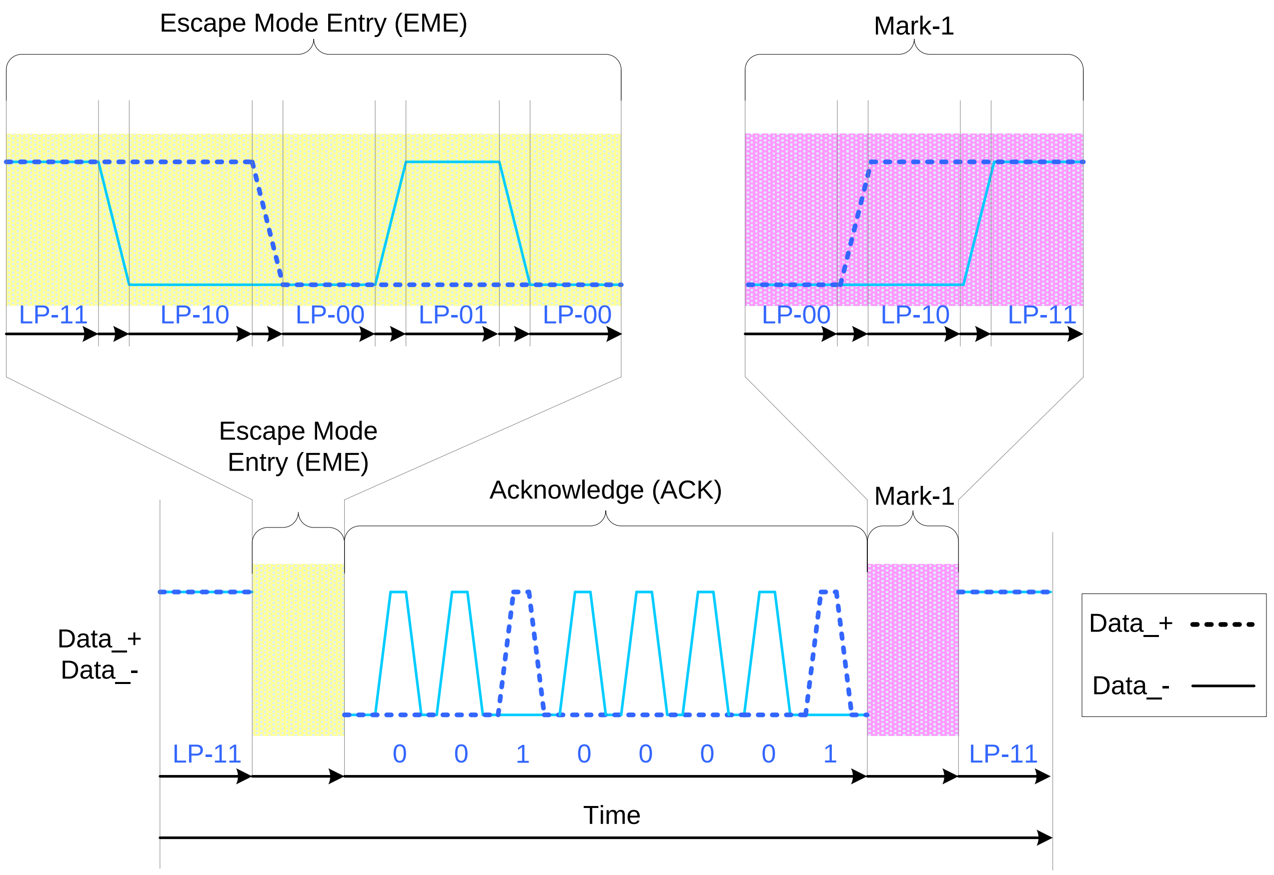
Acknowledge (ACK)

The display module can inform to the MCU when an error has not recognized on it by Acknowledge (ACK).

The display module is sending the Acknowledge (ACK) what is using a following sequence:

- Start: LP-11
- Escape Mode Entry (EME): LP-11 =>LP-10 =>LP-00 =>LP-01 =>LP-00
- Acknowledge (ACK) command in Escape Mode: 0010 0001 (First to Last bit)
- Mark-1: LP-00 =>LP-10 =>LP-11
- End: LP-11

This sequence is illustrated for reference purposes below:



Acknowledge (ACK)

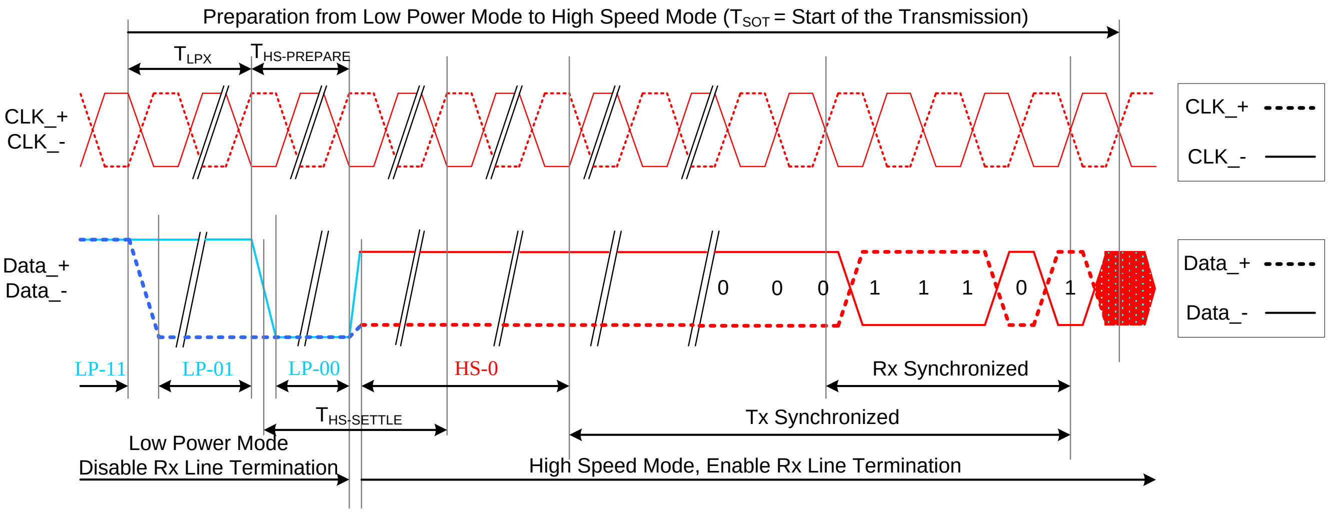
2. **High-Speed Data Transmission**

The display module is entering High-Speed Data Transmission (HSDT) when Clock lanes RX_CP/N have already been entered in the High-Speed Clock Mode (HSCM) by the MCU.

Data lanes of the display module are entering (T_{SOT}) in the High-Speed Data Transmission (HSDT) as follows:

- Start: LP-11
- HS-Request: LP-01
- HS-Settle: LP-00 => HS-0 (Rx: Lane Termination Enable)
- Rx Synchronization: 011101 (Tx (= MCU) Synchronization: 0001 1101)
- End: High-Speed Data Transmission (HSDT) – Ready to receive High-Speed Data Load

This same entering High-Speed Data Transmission (T_{SOT} of HSDT) sequence is illustrated below



Entering High-Speed Data Transmission (T_{SOT} of HSDT)

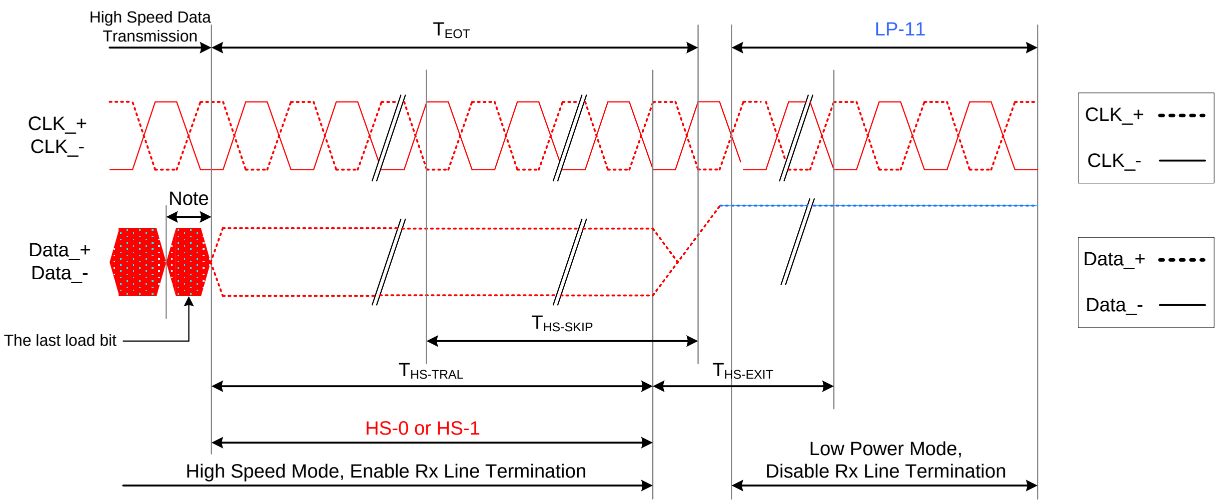
Leaving High-Speed Data Transmission

The display module is leaving the High-Speed Data Transmission (T_{EOT} of HSDT) when Clock lanes RX_CP/N are in the High-Speed Clock Mode (HSCM) by the MCU and this HSCM is kept until data lanes are in LP-11 mode.

Data lanes of the display module are leaving from the High-Speed Data Transmission (T_{EOT} of HSDT) as follows:

- Start: High-Speed Data Transmission (HSDT)
- Stops High-Speed Data Transmission
  - MCU changes to HS-1, if the last load bit is HS-0
  - MCU changes to HS-0, if the last load bit is HS-1
- End: LP-11 (Rx: Lane Termination Disable)

This same leaving High-Speed Data Transmission (T_{EOT} of HSDT) sequence is illustrated below

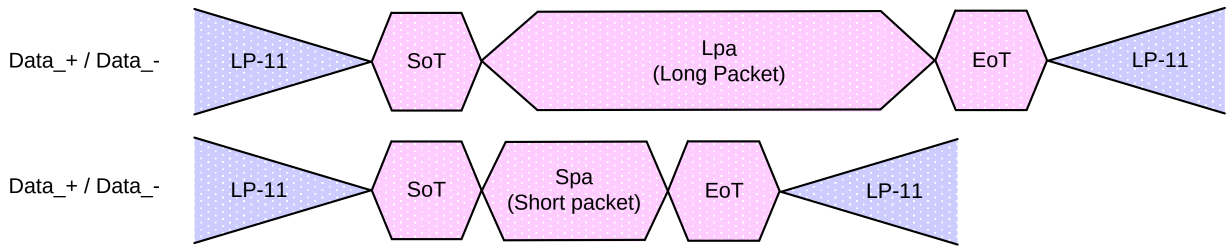


Leaving High-Speed Data Transmission (T_{EOT} of HSDT)

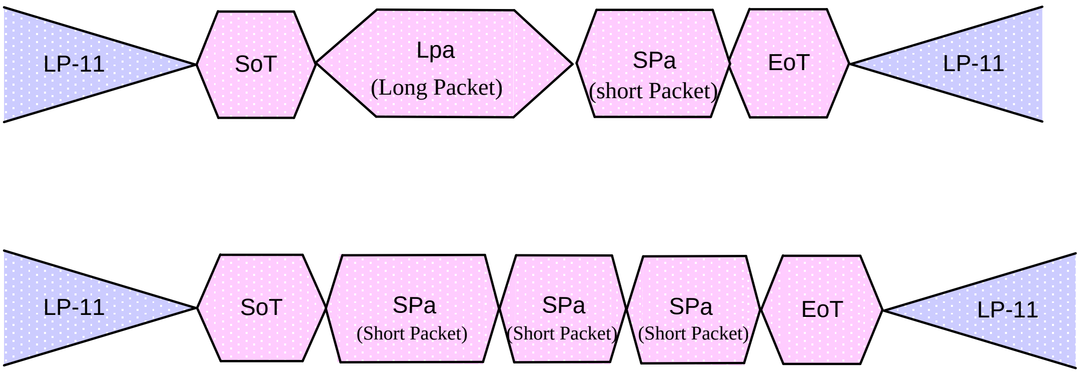
Burst of the High-Speed Data Transmission

The burst of the high-speed data transmission (HSDT) can consist of one data packet or several data packets. These data packets can be Long (Lpa) or Short (Spa) packets.

The single packet in High-Speed Data Transmission is illustrated for reference purposes below:

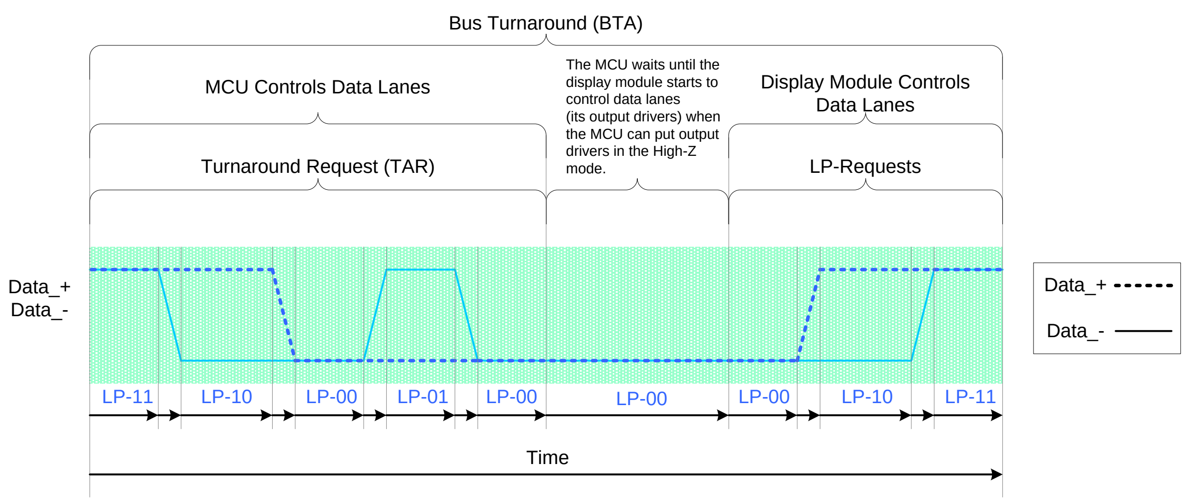


The multiple packets in High-Speed Data Transmission is illustrated for reference purposes below:



3. Bus Turnaround Request

The MCU which is controlling DSI-DATA_P/N Data Lanes, can start a bus turnaround procedure when it wants information from a receiver, which can be the MCU or Display Module. The MCU and Display Module are using the same sequence when this bus turnaround procedure is used. This sequence is described for reference purposes, when the MCU wants to do the bus turnaround procedure to Display Module, as follows.

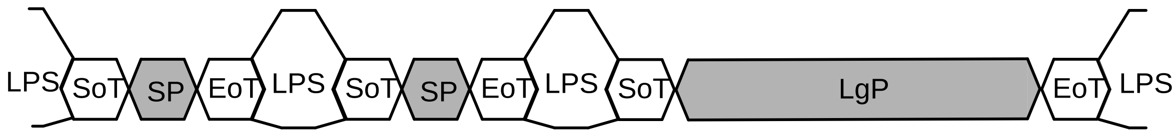


7.10.2 DSI protocol

The Protocol layer appends packet-protocol information and headers, and then sends complete bytes through the Lane Management layer to the PHY. Packets are serialized by the PHY and sent across the serial Link. The receiver side of a DSI Link performs the coverse of the transmitter side, decomposing the packet into parallel data, signal events and commands.

7.10.1.5 Multiple Packets per Transmission

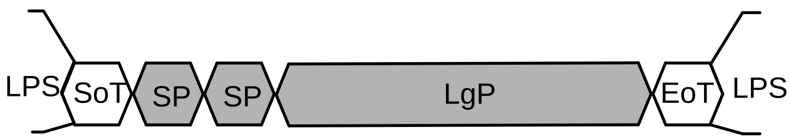
There are two modes of data transmission, HS and LP transmission modes, at the PHY layer. Before a HS transmission can be started, the transmitter PHY issues a SoT sequence to the receiver. After that, data or command packets can be transmitted in HS mode. Multiple packets may exist within a single HS transmission and the end of transmission is always signaled at the PHY layer using a dedicated EoT sequence. In order to enhance the overall robustness of the system, DSI defines a dedicated EoT packet (EoTp) at the protocol layer for signaling the end of HS transmission. For backwards compatibility with earlier DSI systems, the capability of generating and interpreting this EoTp can be enabled or disabled.



Separate Transimssions

Key:

LPS -- Low power state
SoT -- Start of Transmission
EoT -- End of Transmission
SP -- Short Packet
LgP -- Long Packet



Single Transimssions

7.10.1.6 Packet Composition

The first byte of the packet, the Data Identifier (DI), includes information specifying the type of the packet. For example, in Video Mode systems in a display application the logical unit for a packet may be one horizontal display line. Command Mode systems send commands and an associated set of parameters, with the number of parameters depending on the command type.

Packet sizes fall into two categories:

- **Short packets** are four bytes in length including the ECC. Short packets are used for most Command Mode commands and associated parameters. Other Short packets convey events like H Sync and V Sync edges. Because they are Short packets they can convey accurate timing information to logic at the peripheral.
- **Long packets** specify the payload length using a two-byte Word Count field. Payloads may be

from 0 to 2^{16}-1 bytes long. Therefore, a Long packet may be up to 65,541 bytes in length. Long packets permit transmission of large blocks of pixel or other data.

A special case of Command Mode operation is video-rate (update) streaming, which takes the form of an arbitrarily long stream of pixel or other data transmitted to the peripheral. As all DSI transactions use packets, the video stream shall be broken into separate packets. This “packetization” may be done by hardware or software. The peripheral may then reassemble the packets into a continuous video stream for display.

The Set Maximum Return Packet Size command allows the host processor to limit the size of response packets coming from a peripheral.

7.10.1.7 Endian Policy

All packet data traverses the interface as bytes. Sequentially, a transmitter shall send data LSB first, MSB last. For packets with multi-byte fields, the least significant byte shall be transmitted first unless otherwise specified.

Figure 12 shows a complete Long packet data transmission. Note, the figure shows the byte values in standard positional notation, i.e. MSB on the left and LSB on the right, while the bits are shown in chronological order with the LSB on the left, the MSB on the right and time increasing left to right.

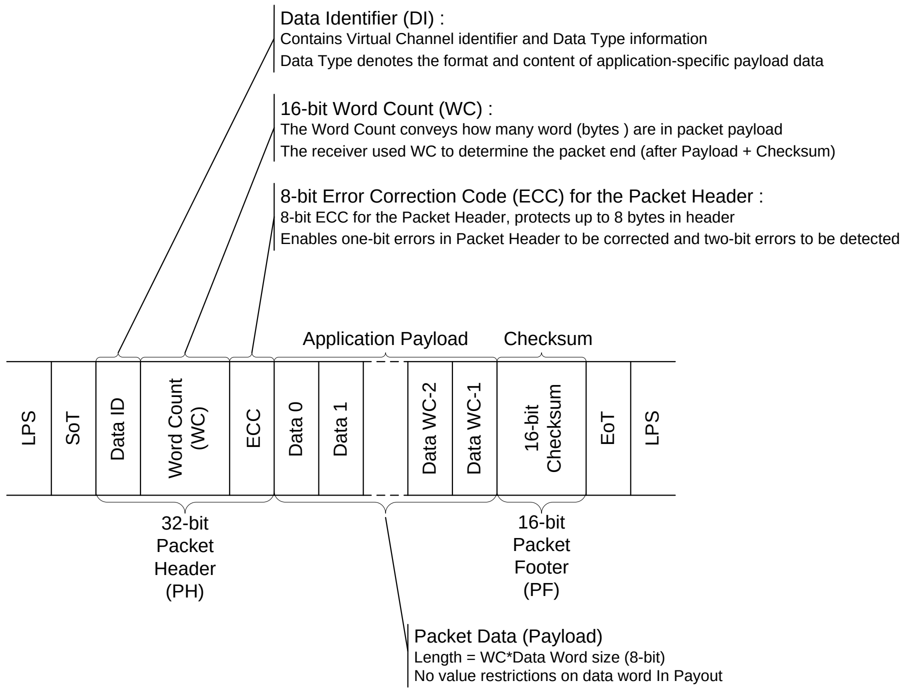
DI	WC (LS Byte)	WS (MS Byte)	ECC	Data	CRC (LS Byte)	CRC (MS Byte)
0x29	0x01	0x00	0x06	0x01	0x0E	0x1E
1 0 0 1 0 1 0 0	1 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0	0 1 1 0 0 0 0 0	1 0 0 0 0 0 0 0	0 1 1 1 0 0 0 0	0 1 1 1 1 0 0 0
LSB … MSB	LSB	MSB	LSB … MSB	LSB … MSB	LSB	MSB

Time →

Endian Example (Long Packet)

7.10.1.8 General Packet Structure(Long Packet Format)

A Long packet shall consist of three elements: a 32-bit Packet Header (PH), an application-specific Data Payload with a variable number of bytes, and a 16-bit Packet Footer (PF). The Packet Header is further composed of three elements: an 8-bit Data Identifier, a 16-bit Word Count, and 8-bit ECC. The Packet Footer has one element, a 16-bit checksum. Long packets can be from 6 to 65,541 bytes in length.



Long Packet Structure

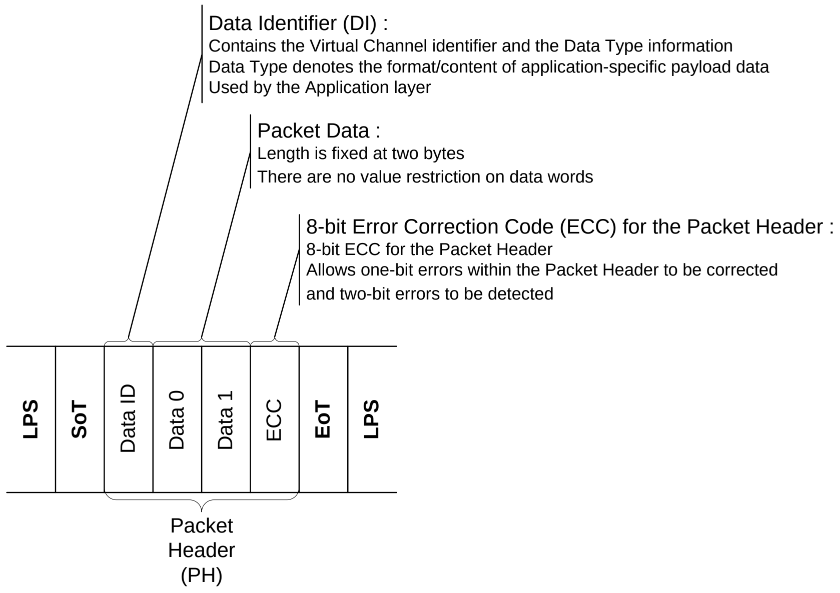
The Data Identifier defines the Virtual Channel for the data and the Data Type for the application specific payload data. See sections 7.8 through 7.10 for descriptions of Data Types. The Word Count defines the number of bytes in the Data Payload between the end of the Packet Header and the start of the Packet Footer. Neither the Packet Header nor the Packet Footer shall be included in the Word Count. The Error Correction Code (ECC) byte allows single-bit errors to be corrected and 2-bit errors to be detected in the Packet Header. This includes both the Data Identifier and Word Count fields. After the end of the Packet Header, the receiver reads the next Word Count * bytes of the Data Payload. Within the Data Payload block, there are no limitations on the value of a data word, i.e. no embedded codes are used. Once the receiver has read the Data Payload it reads the Checksum in the Packet Footer. The host processor shall always calculate and transmit a Checksum in the Packet Footer. Peripherals are not required to calculate a Checksum. Also note the special case of zero-byte Data Payload: if the payload has length 0, then the Checksum calculation results in (FFFFh). If the Checksum is not calculated, the Packet Footer shall consist of two bytes of all zeros (0000h). See section 9 for more information on calculating the Checksum. In the generic case, the length of the Data Payload shall be a multiple of bytes. In addition, each data format may impose additional restrictions on the length of the payload data, e.g. multiple of four bytes. Each byte shall be transmitted least significant bit first. Payload data may be transmitted in any byte order restricted only by data format requirements. Multi-byte elements such as Word Count and Checksum shall be transmitted least significant byte first

.

7.10.1.9 General Packet Structure(Short Packet Format)

A Short packet shall contain an 8-bit Data ID followed by two command or data bytes and an 8-bit ECC; a Packet Footer shall not be present. Short packets shall be four bytes in length. The Error Correction Code

(ECC) byte allows single-bit errors to be corrected and 2-bit errors to be detected in the Short packet.



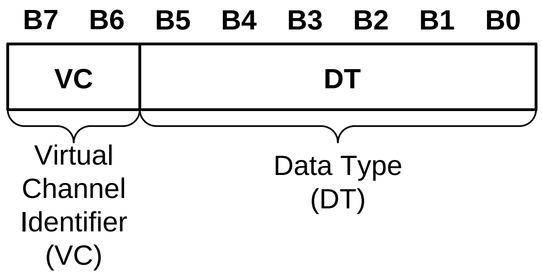
Short Packet Structure

7.10.1.10 Common Packet Elements

Long and Short packets have several common elements that are described in this section.

➢Data Identifier Byte

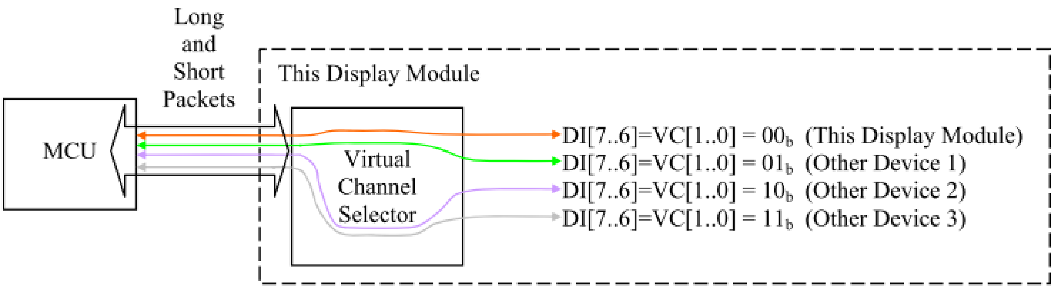
The first byte of any packet is the DI (Data Identifier) byte. Figure 15 shows the composition of the Data Identifier (DI) byte. DI[7:6]: These two bits identify the data as directed to one of four virtual channels.DI[5:0]: These six bits specify the Data Type.



Data Identifier Byte

Virtual Channel Identifier – VC field, DI[7:6]

A processor may service up to four peripherals with tagged commands or blocks of data, using the Virtual Channel ID field of the header for packets targeted at different peripherals. The Virtual Channel ID enables one serial stream to service two or more virtual peripherals by multiplexing packets onto a common transmission channel. Note that packets sent in a single transmission each have their own Virtual Channel assignment and can be directed to different peripherals. Although the DSI protocol permits communication with multiple peripherals, this specification only addresses the connection of a host processor to a single peripheral. Implementation details for connection to more than one physical peripheral are beyond the scope of this document.



Virtual Channel (VC) Configuration

Data Type Field DT[5:0]

The Data Type field specifies if the packet is a Long or Short packet type and the packet format. The Data Type field, along with the Word Count field for Long packets, informs the receiver of how many bytes to expect in the remainder of the packet. This is necessary because there are no special packet start / end sync codes to indicate the beginning and end of a packet. This permits packets to convey arbitrary data, but it also requires the packet header to explicitly specify the size of the packet. When the receiving logic has counted down to the end of a packet, it shall assume the next data is either the header of a new packet or the EoT (End of Transmission) sequence.

7.10.1.11 Error Correction Code

The Error Correction Code allows single-bit errors to be corrected and 2-bit errors to be detected in the Packet Header. The host processor shall always calculate and transmit an ECC byte. Peripherals shall support ECC in both forward- and reverse-direction communications.

Bits (P[7…0]) of the Error Correction Code (ECC) are defined, where the symbol '^' is presenting XOR function (Pn is '1' if there is odd number of '1's and Pn is '0' if there is even number of '1's), as follows.

P7 = 0

P6 = 0

P5 = D10^D11^D12^D13^D14^D15^D16^D17^D18^D19^D21^D22^D23

P4 = D4^D5^D6^D7^D8^D9^D16^D17^D18^D19^D20^D22^D23

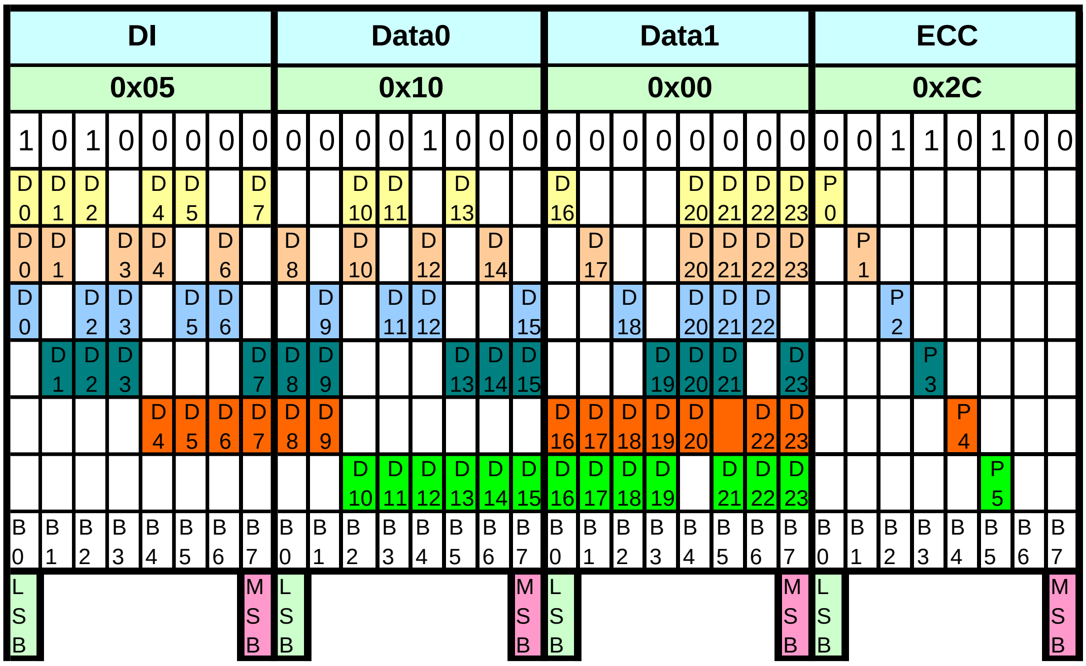
P3 = D1^D2^D3^D7^D8^D9^D13^D14^D15^D19^D20^D21^D23

P2 = D0^D2^D3^D5^D6^D9^D11^D12^D15^D18^D20^D21^D22

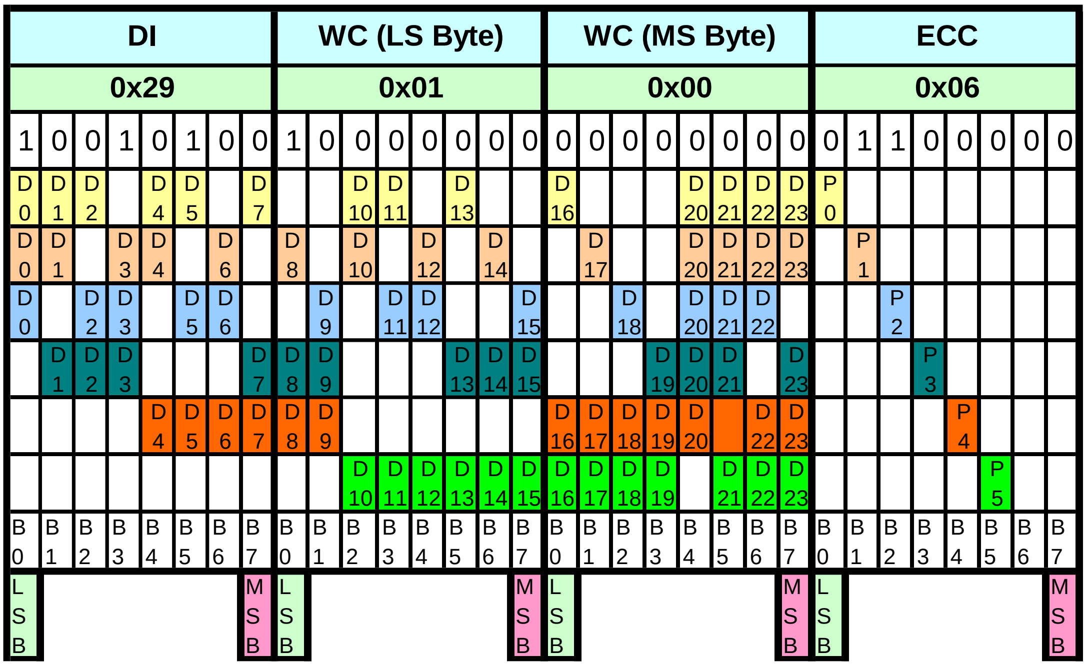
P1 = D0^D1^D3^D4^D6^D8^D10^D12^D14^D17^D20^D21^D22^D23

P0 = D0^D1^D2^D4^D5^D7^D10^D11^D13^D16^D20^D21^D22^D23

P7 and P6 are set to ‘0’ because Error Correction Code (ECC) is based on 64 bit value ([D63…0]), but this implementation is based on 24 bit value (D [23…0]). Therefore, there is only needed 6 bits (P [5…0]) for Error Correction Code (ECC).



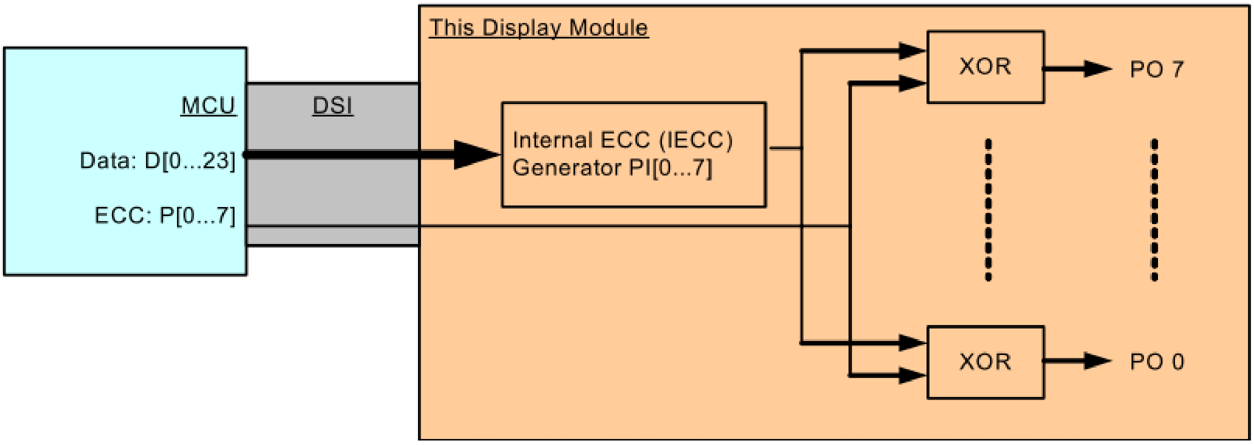
XOR Functionality on the Short Packet (Spa)



XOR Functionality on the Long Packet (Lpa)

The transmitter (The MCU or the Display Module) is sending data bits D[23:0] and Error Correction Code (ECC) P[7:0]. The receiver (The Display module or the MCU) is calculate an Internal Error Correction

Code (IECC) and compares the received Error Correction Code (ECC) and the Internal Error Correction Code (IECC). This comparison is done when each power bit of ECC and IECC have been done XOR function. The result of this function is PO[7:0].



Internal Error Correction Code (IECC) on the Display Module (The Receiver)

The sent data bits (D[23:0]) and ECC (P[7:0]) are received correctly, if a value of the PO[7:0]) is 00h. The sent data bits (D[23:0]) and ECC (P[7:0]) are not received correctly, if a value of the PO[7:0]) is not 00h.

ECC P[7:0]	1	1	0	0	0	0	0	0	03h
IECC PI[7:0]	1	1	0	0	0	0	0	0	03h
XOR(ECC,IECC	0	0	0	0	0	0	0	0	=00h => No Error
	LSB							MSB	

Internal XOR Calculation between ECC and IECC Values – No Error

ECC P[7:0]	1	1	0	0	0	0	0	0	03h
IECC PI[7:0]	1	1	1	1	0	0	0	0	0Fh
XOR(ECC,IECC	0	0	1	1	0	0	0	0	=0Ch => Error
	LSB							MSB	

Internal XOR Calculation between ECC and IECC Values – Error

Data Bit	PO7	PO6	PO5	PO4	PO3	PO2	PO1	PO0	Hex
D[0]	0	0	0	0	0	1	1	1	07h
D[1]	0	0	0	0	1	0	1	1	0Bh
D[2]	0	0	0	0	1	1	0	1	0Dh
D[3]	0	0	0	0	1	1	1	0	0Eh
D[4]	0	0	0	1	0	0	1	1	13h
D[5]	0	0	0	1	0	1	0	1	15h
D[6]	0	0	0	1	0	1	1	0	16h
D[7]	0	0	0	1	1	0	0	1	19h
D[8]	0	0	0	1	1	0	1	0	1Ah
D[9]	0	0	0	1	1	1	0	0	1Ch
D[10]	0	0	1	0	0	0	1	1	23h
D[11]	0	0	1	0	0	1	0	1	25h
D[12]	0	0	1	0	0	1	1	0	26h
D[13]	0	0	1	0	1	0	0	1	29h
D[14]	0	0	1	0	1	0	1	0	2Ah
D[15]	0	0	1	0	1	1	0	0	2Ch
D[16]	0	0	1	1	0	0	0	1	31h
D[17]	0	0	1	1	0	0	1	0	32h
D[18]	0	0	1	1	0	1	0	0	34h
D[19]	0	0	1	1	1	0	0	0	38h
D[20]	0	0	0	1	1	1	1	1	1Fh
D[21]	0	0	1	0	1	1	1	1	2Fh
D[22]	0	0	1	1	0	1	1	1	37h
D[23]	0	0	1	1	1	0	1	1	3Bh

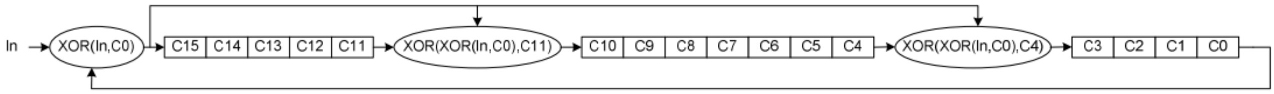
One error is detected if the value of the PO[7:0] is on the above table : One it Error Value of the Error Correction Code (ECC) and the receiver can correct this one bit error because this found value also defines what is a location of the corrupt bit e.g.

- PO [7…0] = 0Eh
- The bit of the data (D [23:0]), what is not correct, is D[3]

More than one error is detected if the value of the PO [7…0] is not on the above table: One Bit Error Value of the Error Correction Code (ECC) e.g. PO [7…0] = 0Ch.

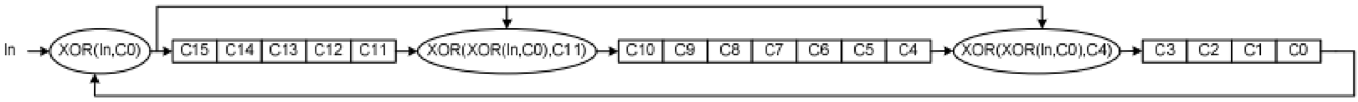
7.10.1.12 Packet Footer on the Long Packet

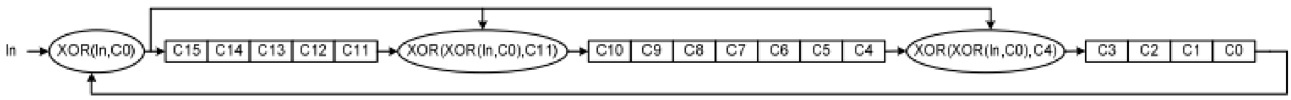
Packet Footer (PF) of the Long Packet (Lpa) is defined after the Packet Data (PD) of the Long Packet (Lpa). The Packet Footer (PF) is a checksum value what is calculated from the Packet Data of the Long Packet (Lpa). The checksum is using a 16-bit Cyclic Redundancy Check (CRC) value which is generated with a polynomial $X_{16} + X_{12} + X_{5} + X_{0}$ as it is illustrated below.



The 16-bit Cyclic Redundancy Check (CRC) generator is initialized to FFFFh before calculations. The Most Significant Bit (MSB) of the data byte of the Packet Data (PD) is the first bit what is inputted into the 16-bit Cyclic Redundancy Check (CRC).

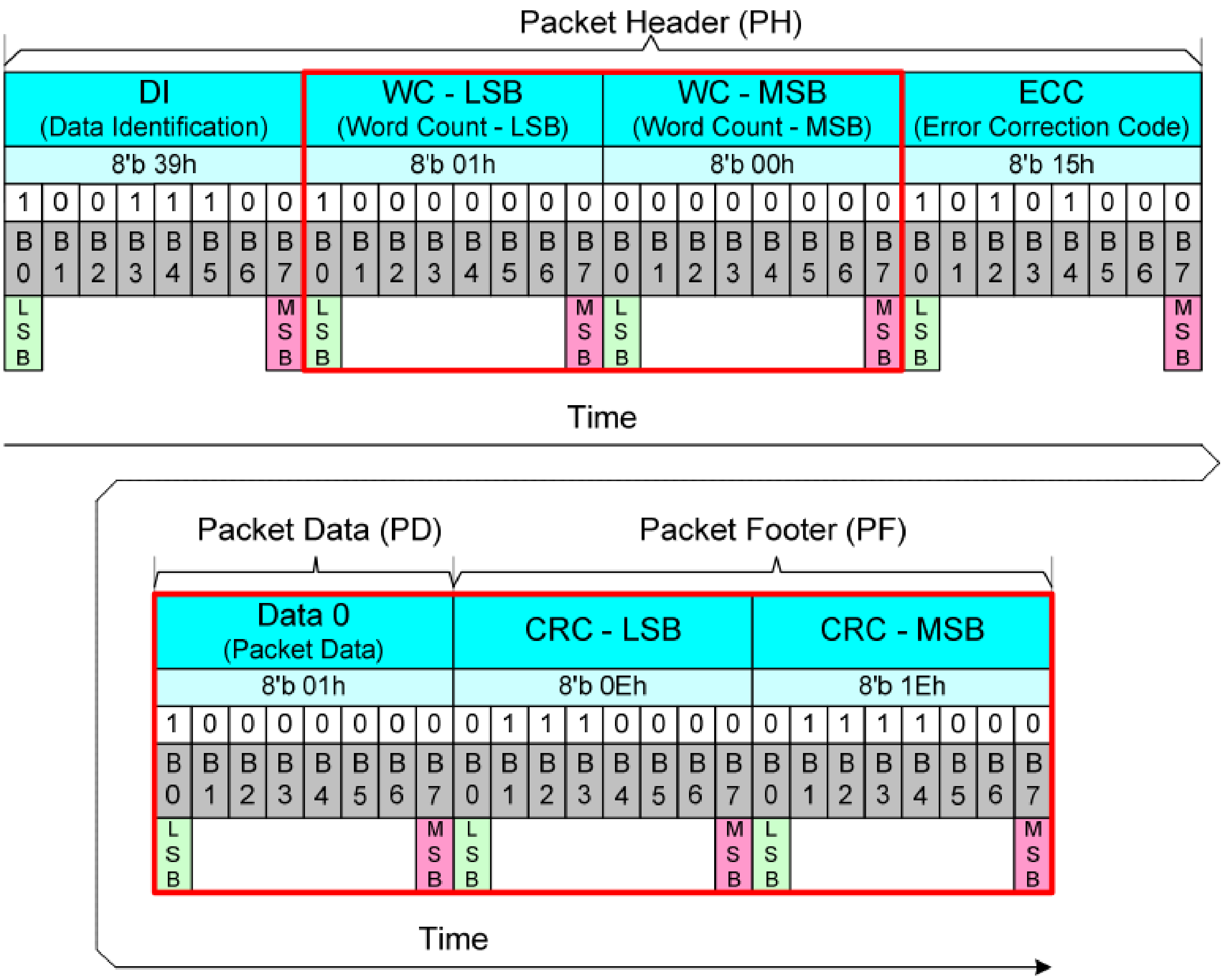
An example of the 16-bit Cyclic Redundancy Check (CRC), where the Packet Data (PD) of the Long Packet (Lpa) is 01h, is illustrated (step-by-step) below.





Step	In	XOR(In,C0)	C15	C14	C13	C12	C11	XOR(XOR(In,C0),C11(Step-1))	C10	C9	C8	C7	C6	C5	C4	XOR(XOR(In,C0),C4(Step-1))	C3	C2	C1	C0	C0
0	X	X	1	1	1	1	1	X	1	1	1	1	1	1	1	X	1	1	1	1	X
1	1(LSB)	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
2	0	1	1	0	1	1	1	0	0	1	1	1	1	1	1	0	0	1	1	1	1
3	0	1	1	1	0	1	1	0	0	0	1	1	1	1	1	0	0	0	1	1	1
4	0	1	1	1	1	0	1	0	0	0	0	1	1	1	1	0	0	0	0	1	1
5	0	1	1	1	1	1	0	0	0	0	0	0	1	1	1	0	0	0	0	0	0
6	0	0	0	1	1	1	1	0	0	0	0	0	0	1	1	1	1	0	0	0	0
7	0	0	0	0	1	1	1	1	1	0	0	0	0	0	1	1	1	1	0	0	0
8	0(MSB)	0	0	0	0	1	1	1	1	1	0	0	0	0	0	1	1	1	1	0	0
	1 byte	CRC result	0	0	0	1	1		1	1	0	0	0	0	0		1	1	1	0	
			MSB																	LSB	

A value of the Packet Footer (PF) is 1E0Eh in this example. This example (Command 01h has been sent) is illustrated below.



The receiver is calculated own checksum value from received Packet Data (PD). The receiver compares own checksum and the Packet Footer (PF) what the transmitter has sent. The received Packet Data (PD) and Packet Footer (PF) are correct if the own checksum of the receiver and Packet Footer (PF) is equal and vice versa the received Packet Data (PD) and Packet Footer(PF) are not correct if the own checksum of the receiver and Packet Footer (PF) are not equal.

7.10.1.13 Processor to Peripheral Direction Packet Data Types

The set of transaction types sent from the host processor to a peripheral, such as a display module, are show as below table.

Data type	Data type, binary	Description packet	Size
01h	00 0001	Sync Event, V Sync Start	Short
11h	01 0001	Sync Event, V Sync End	Short
21h	10 0001	Sync Event, H Sync Start	Short
31h	11 0001	Sync Event, H Sync End	Short
08h	00 1000	End of Transmission packet (EoTp)	Short
02h	00 0010	Color Mode (CM) Off Command	Short
12h	01 0010	Color Mode (CM) On Command	Short
22h	10 0010	Shut Down Peripheral Command	Short
32h	11 0010	Turn On Peripheral Command	Short
03h	00 0011	Generic Short WRITE, no parameters	Short
13h	01 0011	Generic Short WRITE, 1 parameter	Short
23h	10 0011	Generic Short WRITE, 2 parameters	Short
04h	00 0100	Generic READ, no parameters	Short
14h	01 0100	Generic READ, 1 parameter	Short
24h	10 0100	Generic READ, 2 parameters	Short
05h	00 0101	DCS Short WRITE, no parameters	Short
15h	01 0101	DCS Short WRITE, 1 parameter	Short
06h	00 0110	DCS READ, no parameters	Short
37h	11 0111	Set Maximum Return Packet Size	Short
09h	00 1001	Null Packet, no data	Long
19h	01 1001	Blanking Packet, no data	Long
39h	11 1001	DCS Long Write/write_LUT Command Packet	Long
0Eh	00 1110	Packed Pixel Stream, 16-bit RGB, 5-6-5 Format	Long
1Eh	01 1110	Packed Pixel Stream, 18-bit RGB, 6-6-6 Format	Long
2Eh	10 1110	Loosely Packed Pixel Stream, 18-bit RGB, 6-6-6 Format	Long
3Eh	11 1110	Packed Pixel Stream, 24-bit RGB, 8-8-8 Format	Long
X0h / XFh, unspecified	xx 0000 xx 1111	DO NOT USE All unspecified codes are reserved	

Data Types for Processor-sourced Packets

All detail function of data types is as below：

Sync event (H Start, H End, V Start, V End), Data Type = xx 0001 (x1h)

Sync event (H start, H end, V start, V end), data type=xx 0001 (x1h)		
Data type, hex	**Function description**	**Number of bytes**
01h	Sync Event, V Sync Start	Short
11h	Sync Event, V Sync End	Short
21h	Sync Event, H Sync Start	Short
31h	Sync Event, H Sync End	Short
Note: In order to represent timing information as accurately as possible a V Sync Start event represents the start of the VSA and also implies an H Sync Start event for the first line of the VSA. Similarly, a V Sync End event implies an H Sync Start event for the last line of the VSA.		

Color mode status (Color Mode On, Color Mode Off)		
Data type, hex	**Function description**	**Number of bytes**
02h	Color Mode On that switches a Video Mode display module to a low-color mode for power saving.	Short
12h	Color Mode Off that switches a Video Mode display module from low-color display to normal display.	Short

Display status (shutdown command, turn-on command)		
Data type, hex	**Function description**	**Number of bytes**
22h	Shutdown Peripheral command that turns off the display in a Video Mode display for power saving.	Short
32h	Turn On Peripheral command that turns on the display in Video Mode display for normal display.	Short
Note: When use shutdown command, interface shall remain powered in order to receive the turn-on, or wake-up, command.		

DCS command setting		
Data type, hex	**Function description**	**Number of bytes**
05/15h	DCS Short Write command is used to write a single data byte to a peripheral such as a display module. If a parameter is not required, the parameter byte shall be 00h.	Short
06h	DCS Read command, the returned data may be of Short or Long packet format.	Short

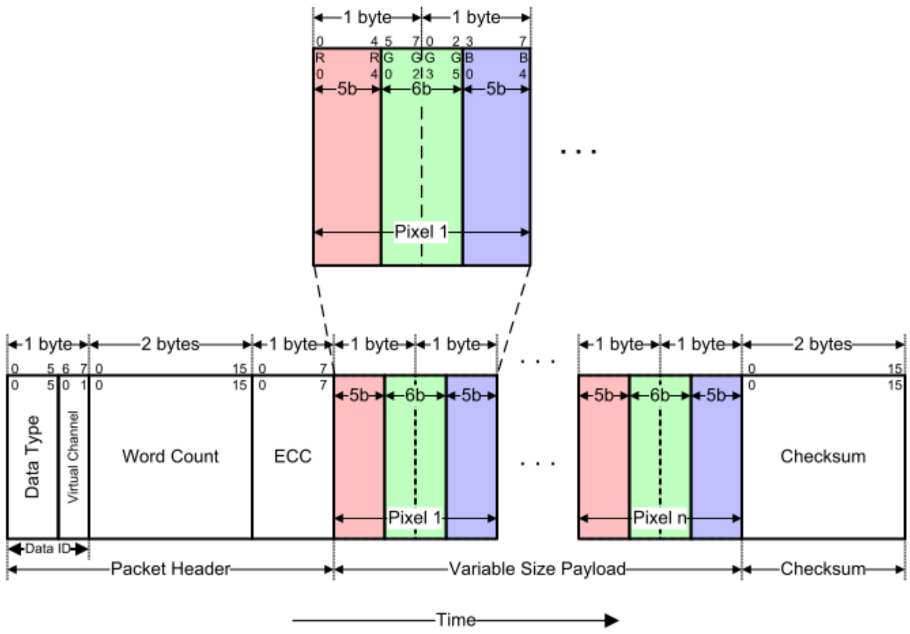
39h	DCS Long Write/ Write _ LUT Command is used to send larger blocks of data to a display module that implements the Display Command Set.	Long

Return packet size setting		
Data type, hex	**Function description**	**Number of bytes**
37h	Set Maximum Return Packet Size that specifies the maximum size of the payload in a Long packet transmitted from peripheral back to the host processor.	Short
Note: The two-byte value is transmitted with LS byte first. And during a power-on or Reset sequence, the Maximum Return Packet Size shall be set by the peripheral to a default value of one.		

Variable data packet		
Data type, hex	**Function description**	**Number of bytes**
09h	Null Packet is a mechanism for keeping the serial Data Lane(s) in High-Speed mode while sending dummy data.	Short
19h	Blanking packet is used to convey blanking timing information in a Long packet.	Short
Note: (1) When Null Packet, the Payload Data belong “null” Data, actual data values sent are irrelevant because the peripheral does not capture or store the data. (2) When Blanking packet, the packet represents a period between active scan lines of a Video Mode display,		

Data stream format – 16bit Format

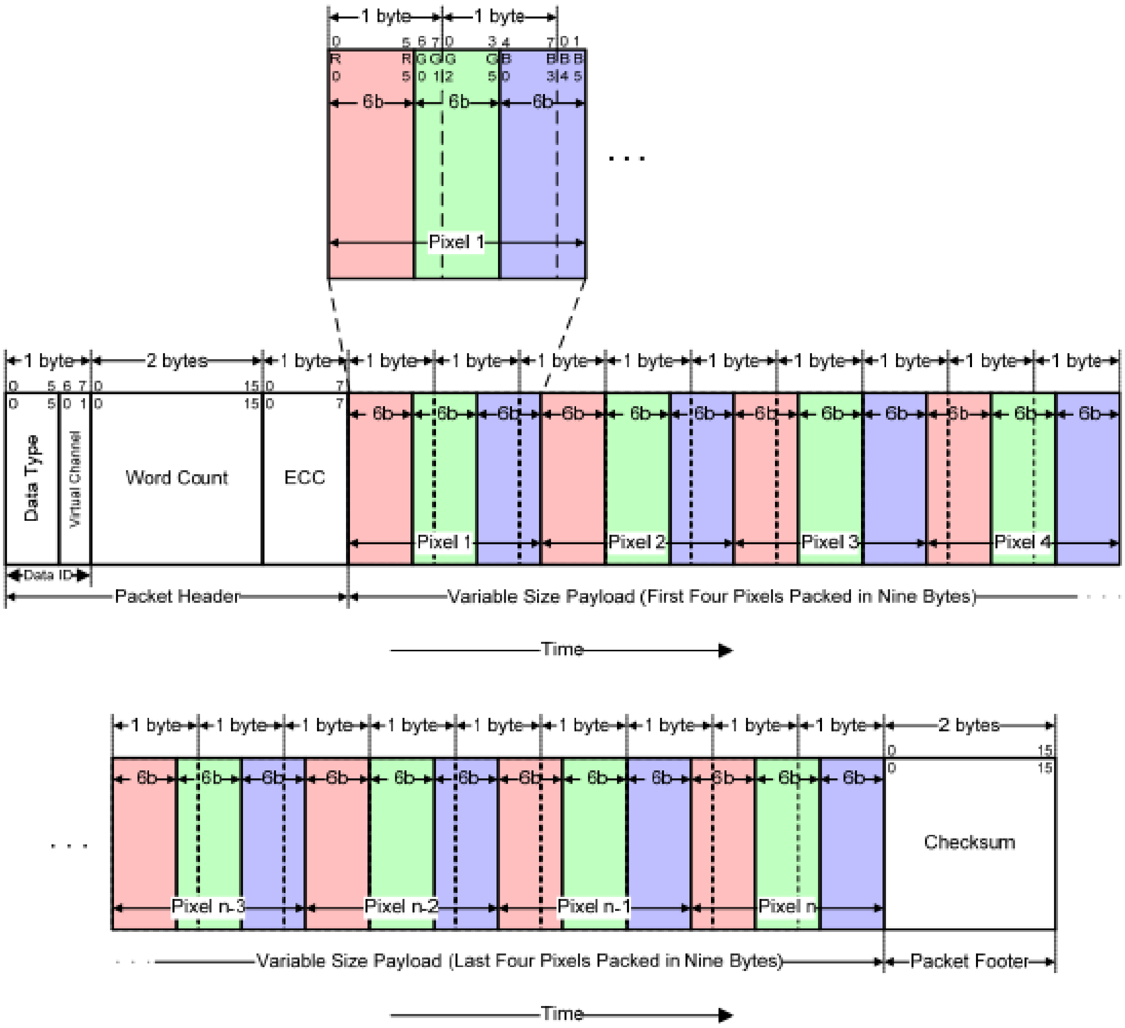
Data stream format – 16bit Format		
Data type, hex	**Function description**	**Number of bytes**
0Eh	Packed Pixel Stream 16-Bit Format is a Long packet used to transmit image data formatted as 16-bit pixels to a Video Mode display module. Pixel format is five bits red, six bits green, five bits blue, in that order.	Long



Note: That the "Green" component is split across two bytes. Within a color component, the LSB is sent first, the MSB last.

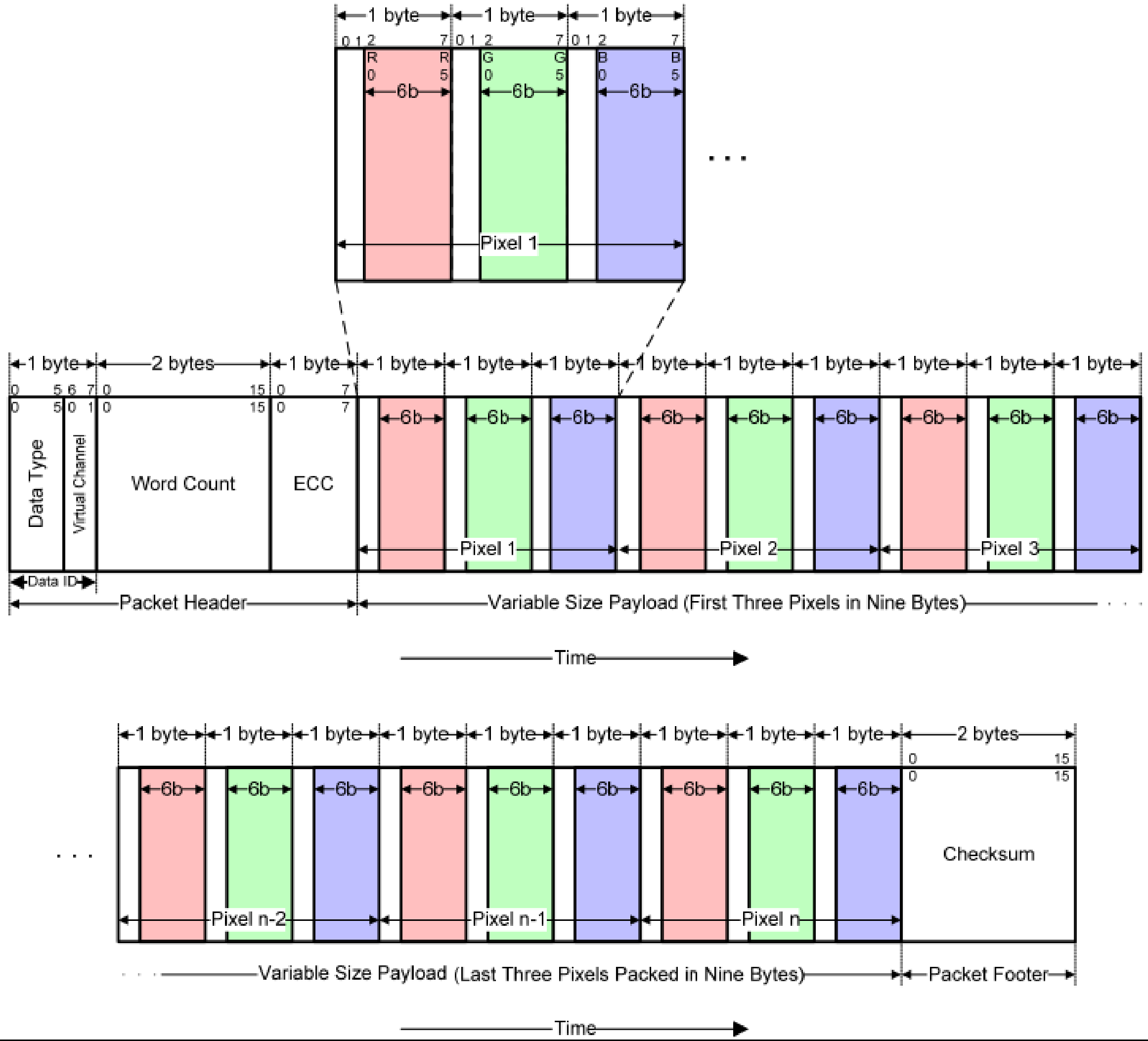
Data stream format – 18bit Format (mode1)

Data stream format – 18bit Format(Mode1)		
Data type, hex	**Function description**	**Number of bytes**
1Eh	Packed Pixel Stream 18-Bit Format is a Long packet used to transmit image data formatted as 18-bit pixels to a Video Mode display module. Pixel format is six bits red, six bits green, six bits blue, in that order.	Long



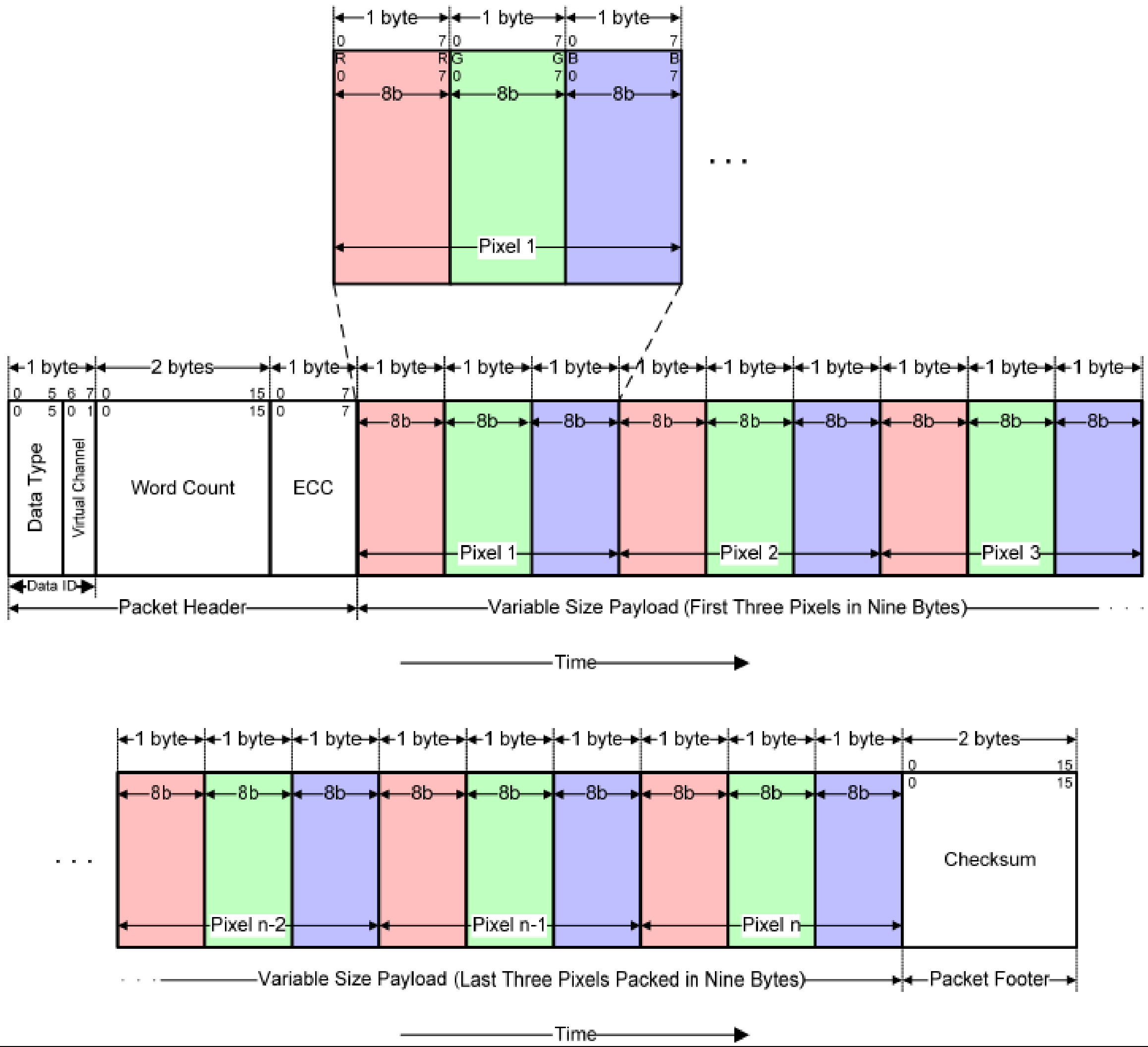
Note: Within a color component, the LSB is sent first and the MSB last and pixel boundaries only line up with byte boundaries every four pixels (nine bytes). Preferably, display modules employing this format have a horizontal extent (width in pixels) evenly divisible by four, so no partial bytes remain at the end of the display line data. It is possible to send pixel data that represent a line width that is not a multiple of four pixels, but display logic on the receiver end shall dispose of the extra bits of the partial byte at the end of active display and ensure a “clean start” for the next line.

Data stream format – 18bit Format(mode2)

<table>
<tr><th colspan="3">Data stream format – 18bit Format(Mode2)</th></tr>
<tr><th>Data type, hex</th><th>Function description</th><th>Number of bytes</th></tr>
<tr><td>2Eh</td><td>In the 18-bit Pixel Loosely Packed format, each R, G, or B color component is six bits but is shifted to the upper bits of the byte, such that the valid pixel bits occupy bits [7:2] of each byte. Bits[1:0] of each payload byte representing active pixels are ignored.</td><td>Long</td></tr>
<tr><td colspan="3">
</td></tr>
<tr><td colspan="3">Note: Within a color component, the LSB is sent first, the MSB last and With this format, pixel boundaries line up with byte boundaries every three bytes.</td></tr>
</table>

Data stream format – 24bit Format

Data stream format –24bit Format		
Data type, hex	**Function description**	**Number of bytes**
3Eh	Packed Pixel Stream 24-Bit Format is used to transmit image data formatted as 24-bit pixels to a Video Mode display module. Pixel format is (8 bits) red, (8 bits) green and (8 bits) blue.	Long



Note: Within a color component, the LSB is sent first, the MSB last and With this format, pixel boundaries line up with byte boundaries every three bytes.

7.10.1.14 Peripheral-to-Processor (Reverse Direction) LP Transmissions

All Command Mode systems require bidirectional capability for returning READ data, acknowledge, or error information to the host processor. Multi-Lane systems shall use Lane 0 for all peripheral-to-processor transmissions; other Lanes shall be unidirectional. Reverse-direction signaling shall only use LP (Low Power) mode of transmission.

Peripheral-to-processor transactions are of four basic types:

- Tearing Effect is a Trigger message sent to convey display timing information to the host processor. Trigger messages ate signal byte packets sent by a peripheral's PHY layer in response to a signal form the DSI protocol layer.
- Acknowledge is a Trigger Message sent when the current transmission, as well as all preceding transmissions since the last peripheral to host communication.
- Acknowledge and Error Report is a Short packet sent if any errors were detected in preceding transmission from the host processor. Once reported, accumulated errors in the error register are cleared.
- Response to Read Request may be Short or Long packet that returns data requested by the preceding READ command from the processor.

In general, if the host processor completes a transmission to the peripheral with BTA asserted, the peripheral shall respond with one or more appropriate packet(s), and then return bus ownership to the host processor. If BTA is not asserted following a transmission from the host processor, the peripheral shall not communicate an Acknowledge or error information back to the host processor.

Interpretation of processor-to-peripheral transactions with BTA asserted, and the expected responses, are as follows:

- Following a non-Read command in which no error was detected, the peripheral shall respond with Acknowledge.
- Following a Read request in which no error was detected, the peripheral shall send the requested READ data.
- Following a Read request in which the ECC error was detected and corrected, the Peripheral shall send the requested READ data in a Long or Short packet, followed by a 4-byte (Acknowledge with Error Report) packet in the same LP transmission. The Error Report shall have the ECC Error flag set.
- Following a non-Read command in which the ECC error was detected and corrected, the peripheral shall proceed to execute the command, and shall respond to BTA by sending a 4-byte (Acknowledge with Error Report) packet, the Error Report shall have the ECC Error flag set.
- Following any command in which SoT Error, SoT Sync Error, EoT Sync Error, LP Transmit Sync Error, checksum error or DSI VC ID Invalid was detected, or the DSI command was not

recognized, the peripheral shall send a 4-byte Acknowledge with Error Report response, with the appropriate error flags set in the two-byte error field. Only the ACK/Error Report packet shall be transmitted; no read or write accesses shall take place on the peripheral in response.

An error report is a Short packet comprised of two bytes following the DI byte, with an ECC byte following the Error Report bytes. By convention, detection and reporting of each error type is signified by setting the corresponding bit to “1”. Table 18 shows the bit assignment for all error reporting.

Bit	Error Report Bit Description
0	SoT Error
1	SoT Sync Error
2	EoT Sync Error
3	Escape Mode Entry Command Error
4	Low-Power Transmit Sync Error
5	HS Receive Timeout Error
6	False Control Error
7	Reserved
8	ECC Error, single-bit (detected and corrected)
9	ECC Error, multi-bit (detected, not corrected)
10	Checksum Error (Long packet only)
11	DSI Data Type Not Recognized
12	DSI VC ID Invalid
13	Invalid Transmission Length
14	Reserved
15	DSI Protocol Violation

The table as below presents the complete set of peripheral-to-processor Data Types

Data type, hex	Data type, binary	Description packet	Size
02h	00 0010	Acknowledge and Error Report	Short
08h	00 1000	End of Transmission packet (EoTp)	Short
11h	01 0001	Generic Short READ Response, 1 byte returned	Short
12h	01 0010	Generic Short READ Response, 2 bytes returned	Short
1Ah	01 1010	Generic Long READ Response	Short
1Ch	01 1100	DCS Long READ Response	Short
21h	10 0001	DCS Short READ Response, 1 byte returned	Short
22h	10 0010	DCS Short READ Response, 2 bytes returned	Short

Data Types for Peripheral-sourced Packets

Acknowledge types		
Data type, hex	**Function description**	**Number of bytes**
02h	Get Acknowledge with Error report when Error occurs from processor transmission.	4 bytes
Note: When processor transmits complete Payload, following signal by BTA, peripheral must respond to processor. With error Acknowledge with error report, Without error Acknowledge.		

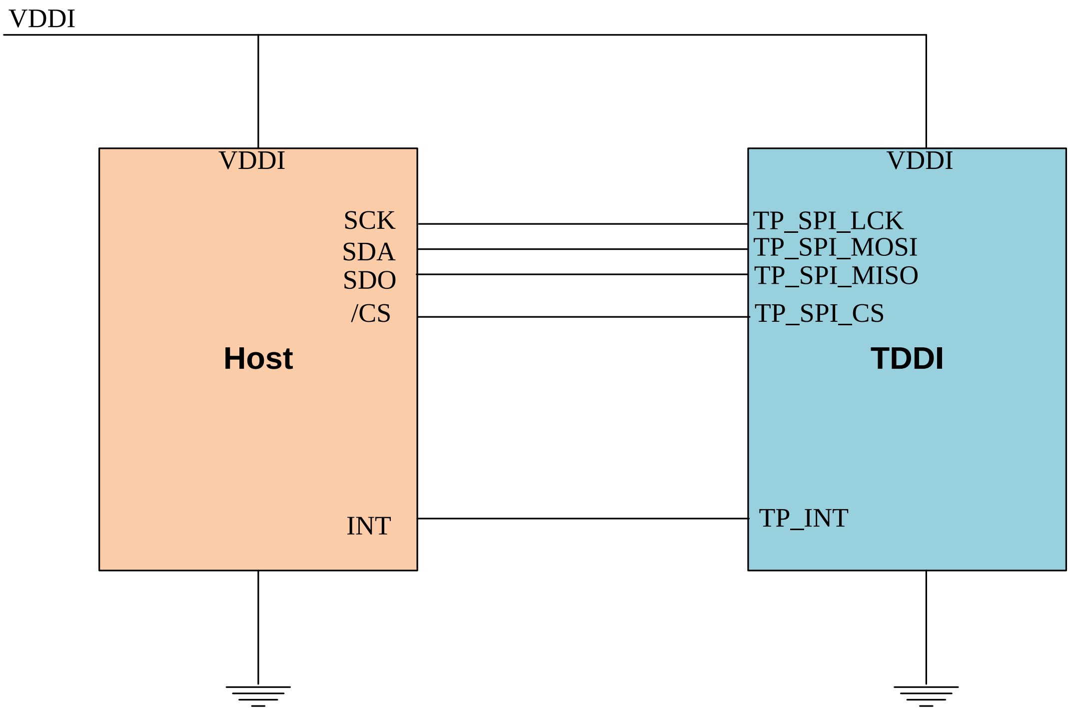
Generic Read types		
Data type, hex	**Function description**	**Number of bytes**
11h, 12h	This is the Generic Short Read Response, 1 or 2bytes, respectively.	4 bytes
1Ah	This is the long-packet response to Generic Long Read Request.	Up to 65541 bytes (DI + WC + ECC + DCS CMD + Payload DATA + PF)
Note: If the peripheral is Checksum capable, is shall return a calculated two-byte Checksum appended to the N-byte payload data. If the peripheral does not support Checksum, it shall return 0000h.If the DCS command itself is possibly corrupt, due to an uncorrectable ECC error, SoT or SoT Sync error, the requested READ data packet shall not be sent after the Acknowledge with Error Report packet be sent.		

DCS Read types		
Data type, hex	**Function description**	**Number of bytes**
21h, 22h	This is the DCS Short Read Response, 1 or 2bytes, respectively..	4 bytes
1Ch	This is the long-packet response to DCS Long Read Request.	Up to 65541 bytes (DI + WC + ECC + DCS CMD + Payload DATA + PF)
Note: If the peripheral is Checksum capable, is shall return a calculated two-byte Checksum appended to the N-byte payload data. If the peripheral does not support Checksum, it shall return 0000h.If the DCS command itself is possibly corrupt, due to an uncorrectable ECC error, SoT or SoT Sync error, the requested READ data packet shall not be sent after the Acknowledge with Error Report packet be sent.		

7.11 Touch Interface

7.11.1 SPI

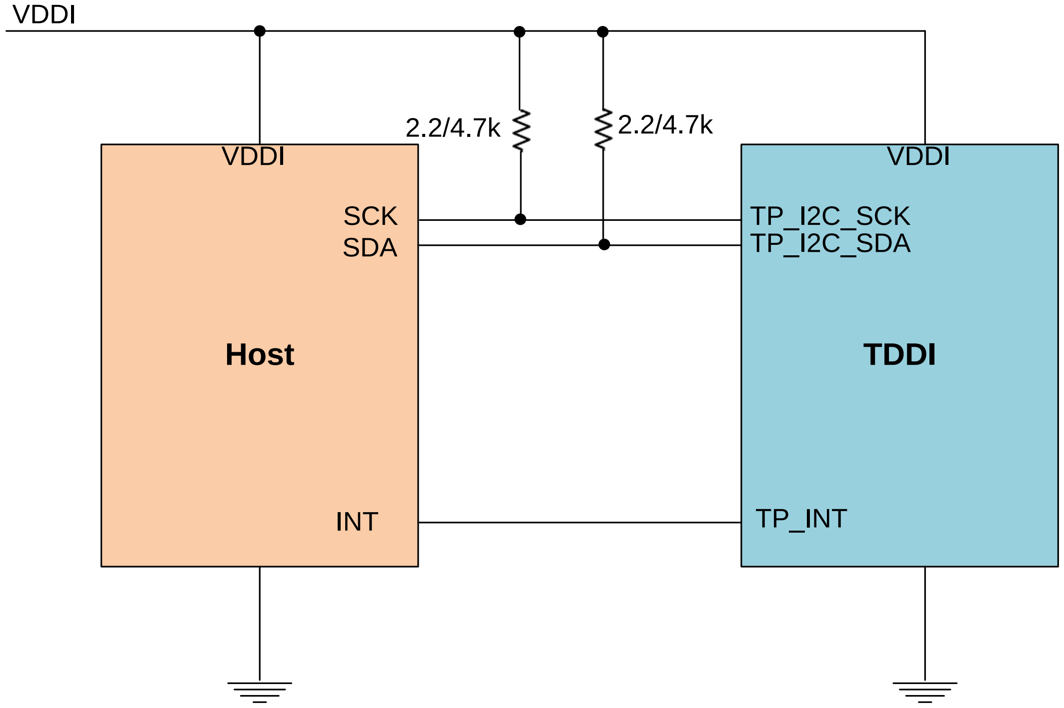
ST77922 operates as a SPI slave device and data length is 16-bit. SS, SCK, and MOSI are Schmitt trigger inputs. The data on MOSI is latched on rising edge of SCK and the data on MISO is outputted on falling edge of SCK.



7.11.2 I2C

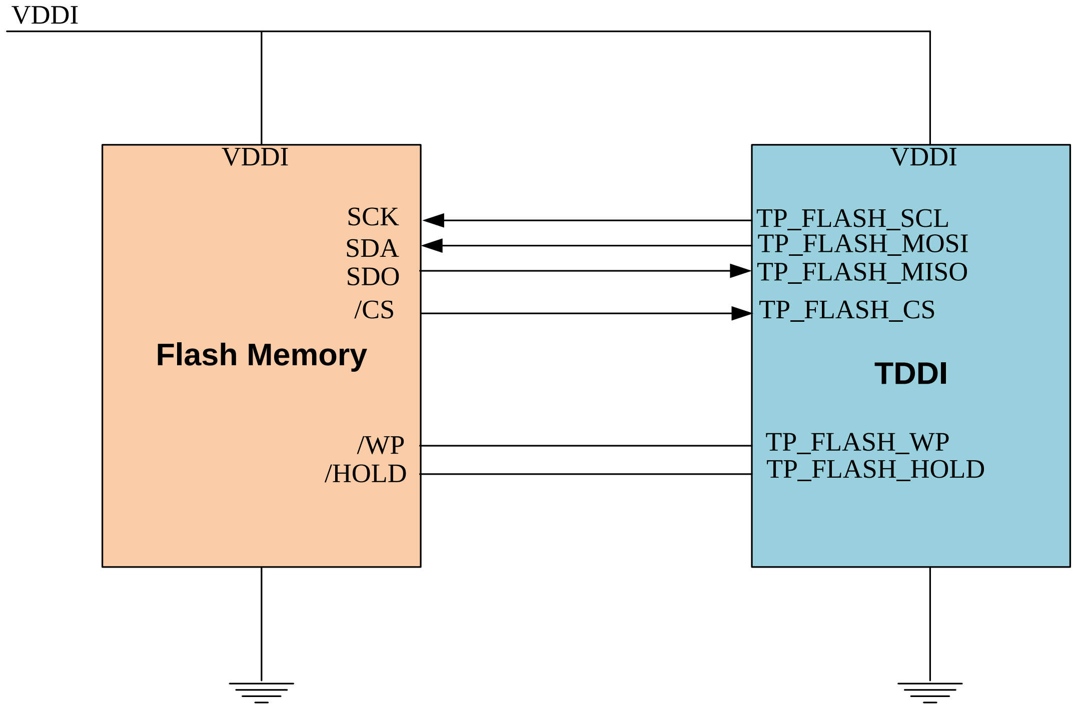
ST77922 protocol supports operating speeds of up to 400 kb/s with 7-bit addressing and 8-bit data bytes. The SCK, MOSI(SDA), and IRQ pins are typically used in an I2C interface.

The values of the pull-up resistors should be chosen to ensure that the rise times of the MOSI(SDA) and SCK signals are within the limits set by the I2C specification. These values depend on what other devices, if any, are on the I2Cbus. External pull-up resistors are of 2.2 kΩ or 4.7 kΩ.



7.11.3 Flash Memory

The master mode Flash serial peripheral Interface (SPI) communicate with Flash Memory and fetch the FW for Embedded MCU to achieve the touch function.



8 FUNCTION DESCRIPTION

8.1 Display Data RAM

8.1.1 Configuration

The display module has an integrated 200x400x24-bit graphic type static 1/2RAM. This bit memory allows storing on-chip a 400xRGBx400 image with an 24-bpp resolution (16M-color). There will be no abnormal visible effect on the display when there is a simultaneous Panel Read and Interface Read or Write to the same location of the Frame Memory.

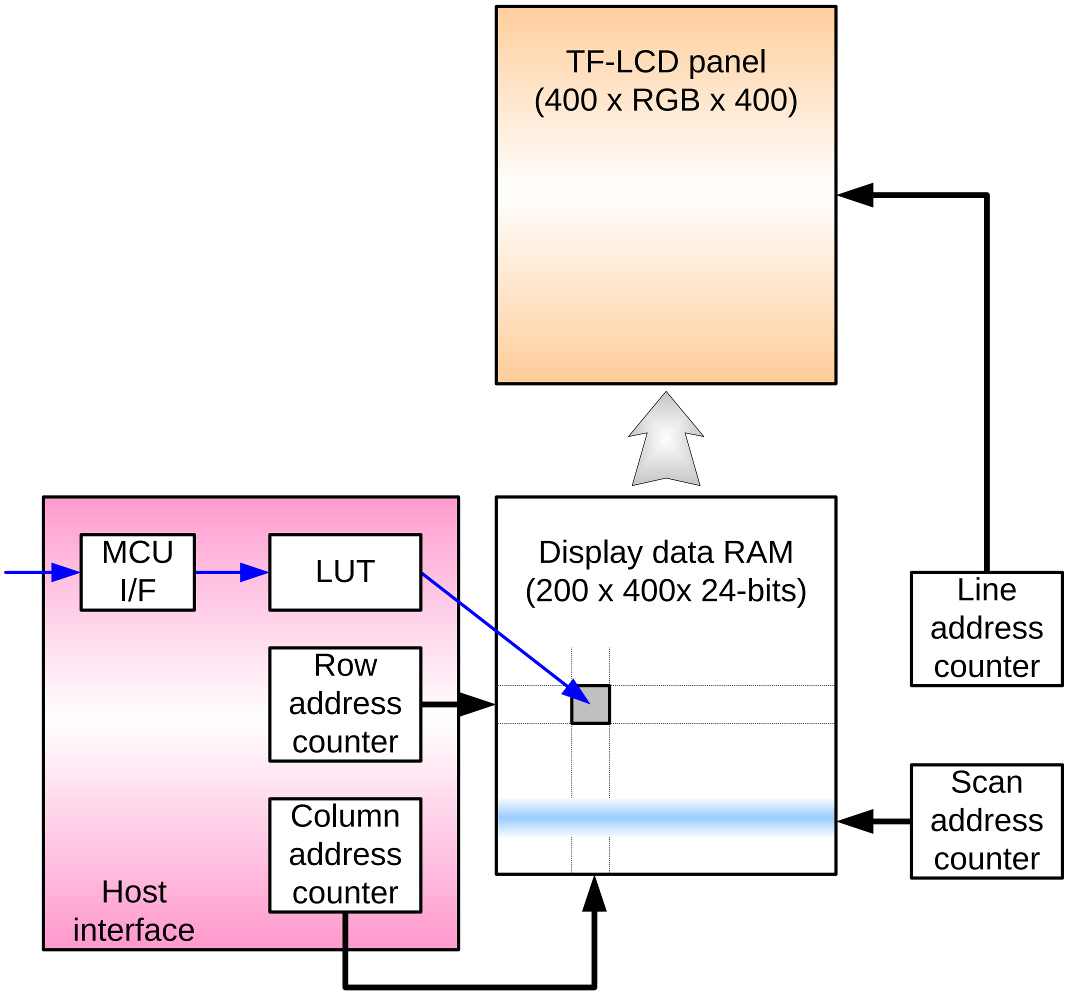


Figure 27 Display data RAM organization

8.1.2 Memory to display address mapping

RGB alignment											
Data control command		Column									
(MADCTR) MX=0		0			1				399		
(MADCTR) MX=1		399			398				0		
Color		R	G	B	R	G	B		R	G	B
Page / Data											
(MADCTR) MY=0	(MADCTR) MY=1										
0	399										
1	398										
2	397										
3	396										
4	395										
5	394										
6	393										
7	:										
:	:										
:	7										
393	6										
394	5										
395	4										
396	3										
397	2										
398	1										
399	0										
Source output		0	1	2	3	4	5		597	598	599

8.2 Address Control

The address counter sets the addresses of the display data RAM for writing and reading.

Data is written pixel-wise into the RAM matrix of DRIVER. The data for one pixel or two pixels is collected (RGB 6-6-6-bit), according to the data formats. As soon as this pixel-data information is complete the "Write access" is activated on the RAM. The locations of RAM are addressed by the address pointers. The address ranges are X=0 to X=399 (18Fh) and Y=0 to Y=399 (18Fh). Addresses outside these ranges are not allowed. Before writing to the RAM, a window must be defined that will be written. The window is programmable via the command registers XS, YS designating the start address and XE, YE designating the end address.

For example the whole display contents will be written, the window is defined by the following values: XS=0 (0h) YS=0 (0h) and XE=399 (18Fh), YE=399 (18Fh).

For flexibility in handling a wide variety of display architectures, the commands "CASET, RASET and MADCTL", define flags MX and MY, which allows mirroring of the X-address and Y-address. All combinations of flags are allowed. Show the available combinations of writing to the display RAM. When MX, MY will be changed the data bust be rewritten to the display RAM.

For each image condition, the controls for the column and row counters apply as below

Condition	Column Counter	Row Counter
When RAMWR/RAMRD command is accepted	Return to "Start Column (XS)"	Return to "Start Row (YS)"
Complete Pixel Read / Write action	Increment by 1	No change
The Column counter value is larger than "End Column (XE)"	Return to "Start Column (XS)"	Increment by 1
The Column counter value is larger than "End Column (XE)" and the Row counter value is larger than "End Row (YE)"	Return to "Start Column (XS)"	Return to "Start Row (YS)"

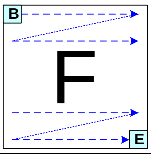
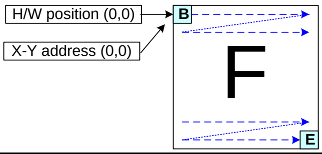
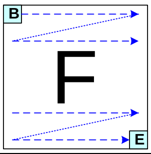
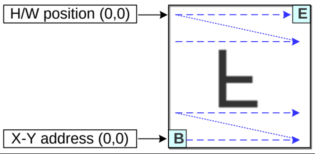
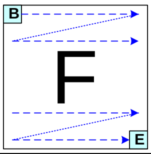
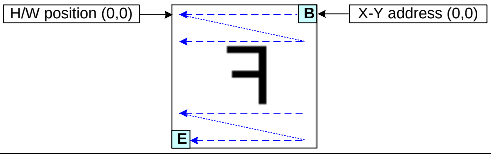
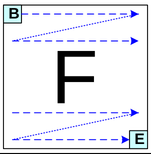
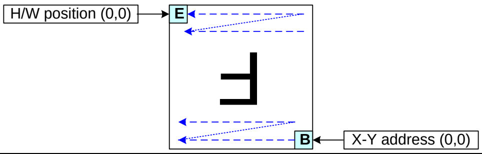
Display Data Direction	MADCTR Parameter		Image in the Host (MPU)	Image in the Driver (DDRAM)
	MX	MY		
Normal	**0**	**0**		
Y-Mirror	**0**	**1**		
X-Mirror	**1**	**0**		
X-Mirror Y-Mirror	**1**	**1**		

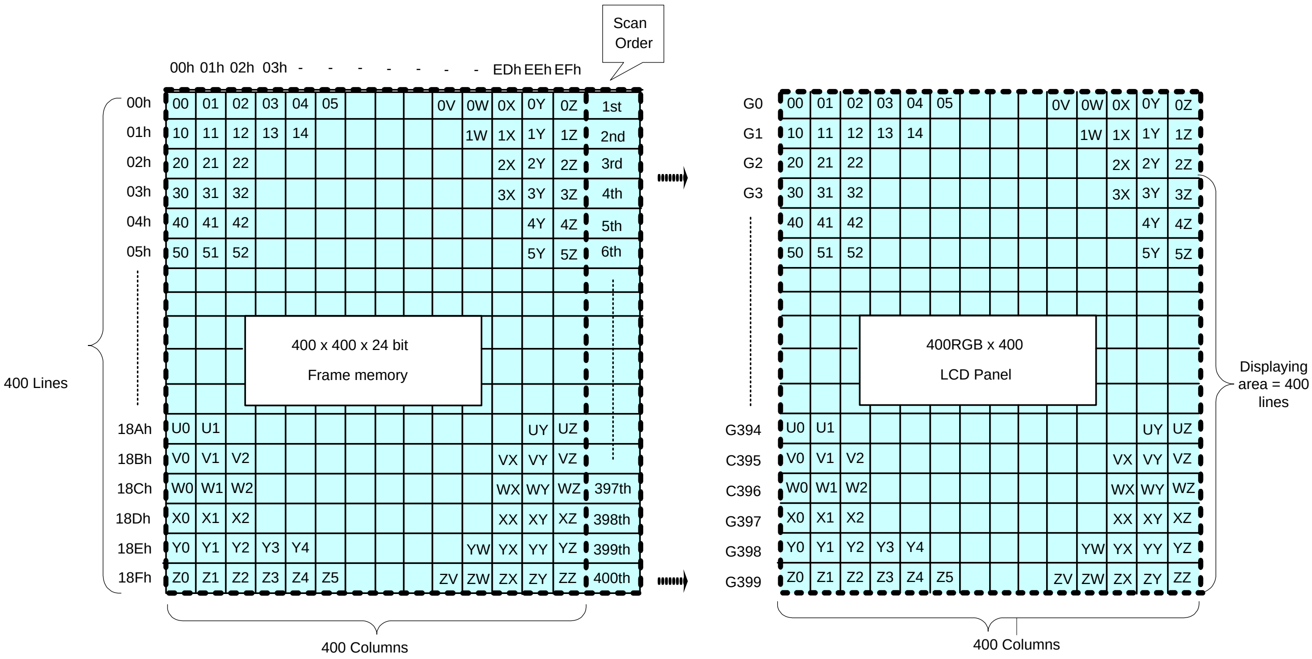
Figure 28 Display data RAM organization

8.3 Normal Display On or Partial Mode On, Vertical Scroll Off

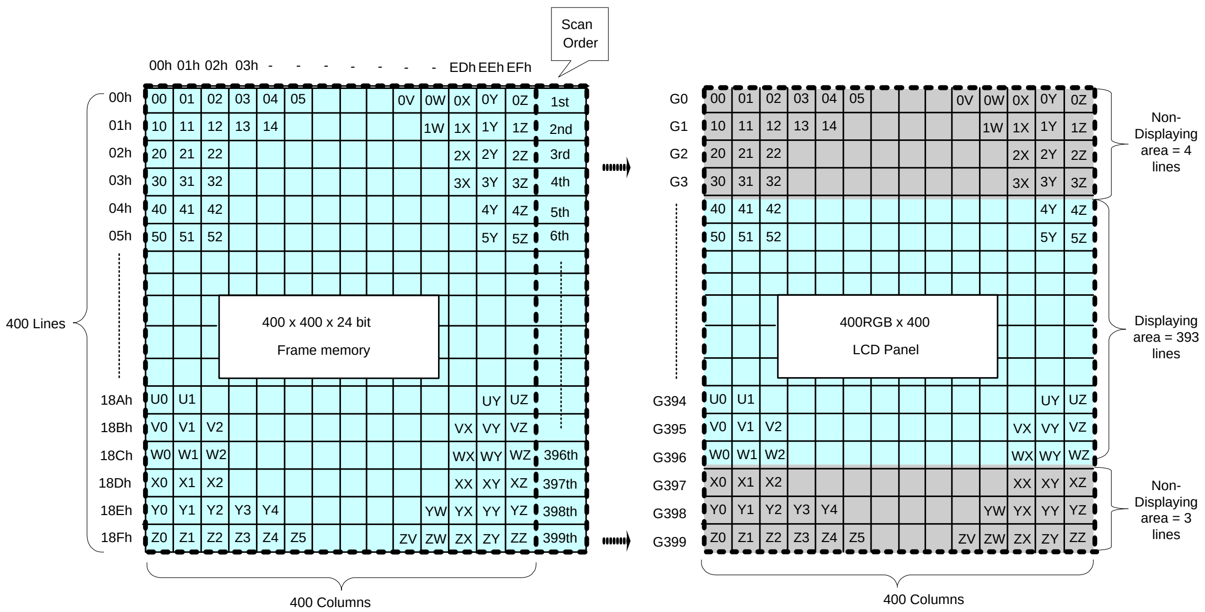
In this mode, contents of the frame memory within an area where column address is 00h to 83h and row address is 00h to 83h is displayed.

To display a dot on leftmost top corner, store the dot data at (column address, row address) = (0,0).

Example1) Normal Display On



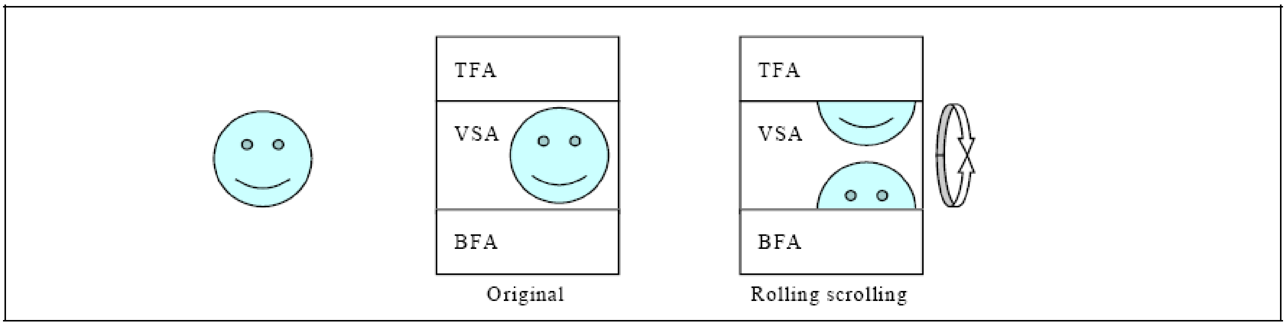
Example2) Partial Display On: PSL[15:0] = 0004h, PEL[15:0] = 013Ch, MADCTR (ML)=0



8.4 Vertical Scroll Mode

8.4.1 Rolling scroll

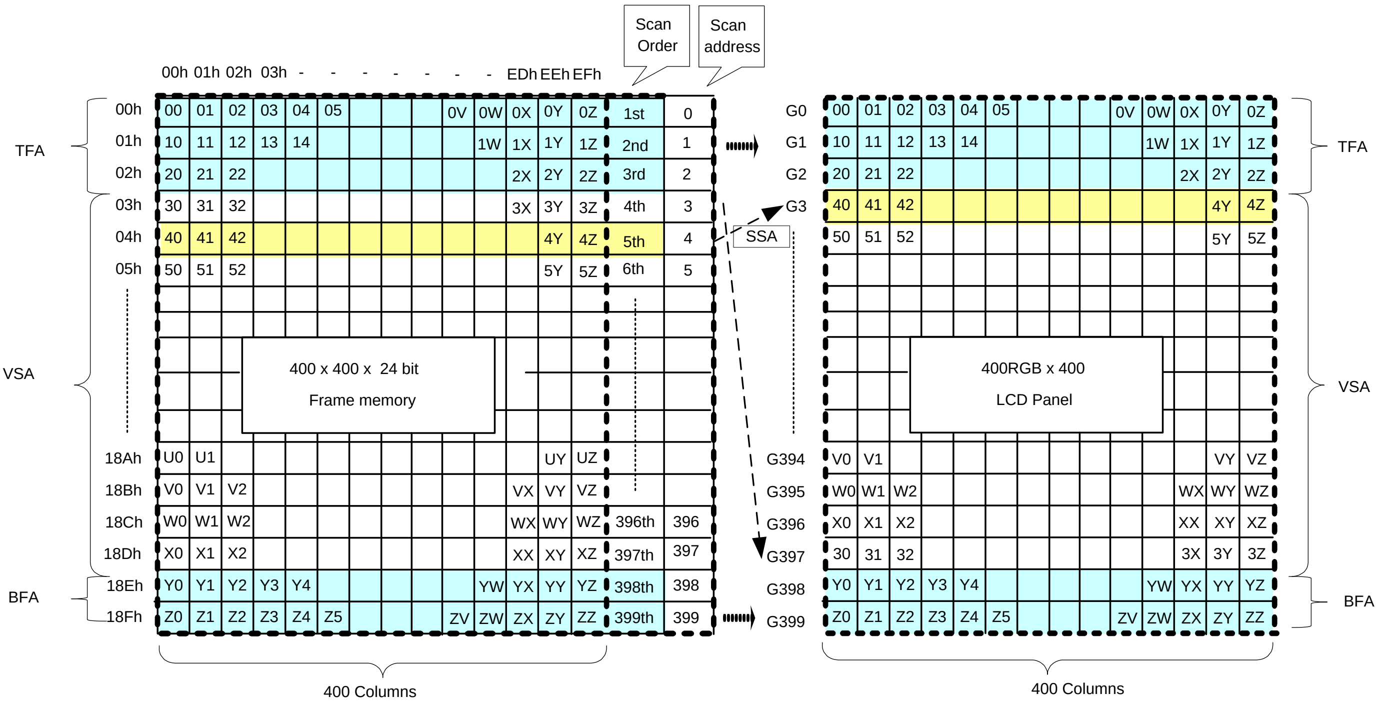
There is just one types of vertical scrolling, which are determined by the commands “Vertical Scrolling Definition” (33h) and “Vertical Scrolling Start Address” (37h).



Rolling Scroll Definition

When Vertical Scrolling Definition Parameters (TFA+VSA+BFA) =400. In this case, ‘rolling’ scrolling is applied as shown below. All the memory contents will be used.

Example: Panel size=400 x 400, TFA =3, VSA=395, BFA=2, SSA=4, MADCTR ML=0: Rolling Scroll



8.4.2 Vertical Scroll Example

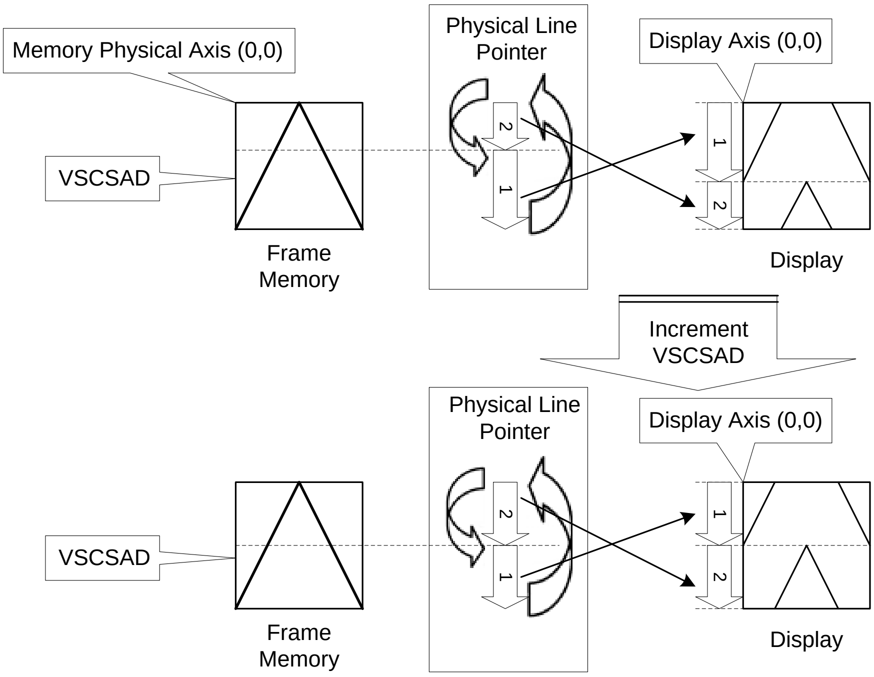
There are 2 types of vertical scrolling, which are determined by the commands “Vertical Scrolling Definition” (33h) and “Vertical Scrolling Start Address” (37h).

Case 1: TFA + VSA + BFA≠Panel total scan lines. In this case, scrolling is applied as shown below.

N/A. Do not set TFA + VSA + BFA≠Panel total scan lines. In that case, unexpected picture will be shown.

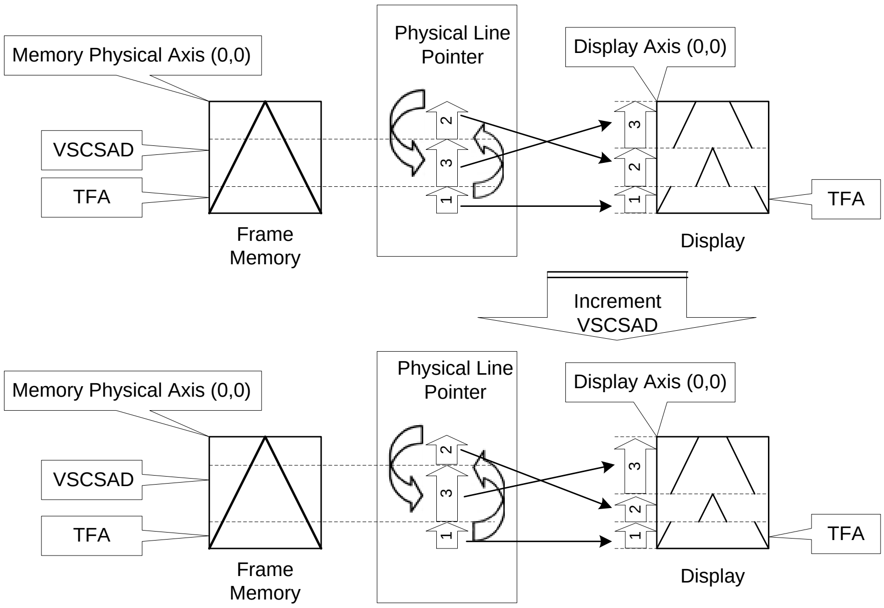
Case 2: TFA + VSA + BFA= Panel total scan lines

Example1) When MADCTR parameter ML=”0”, TFA=0, VSA=390, BFA=0 and VSCSAD=40.



Display of Vertical Scroll Example 1

Example2) When MADCTR parameter ML=”1”, TFA=60, VSA=340, BFA=0 and VSCSAD=160.



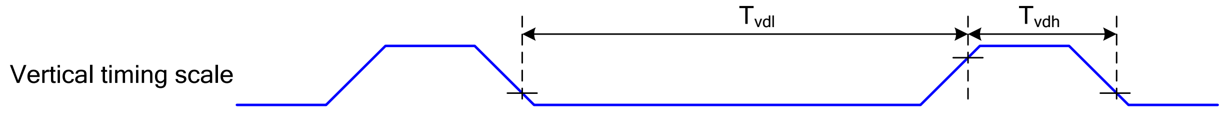
Display of Vertical Scroll Example 2

8.5 Tearing Effect

The Tearing Effect output line supplies to the MPU a Panel synchronization signal. This signal can be enabled or disabled by the Tearing Effect Line Off & On commands. The mode of the Tearing Effect signal is defined by the parameter of the Tearing Effect Line On command. The signal can be used by the MPU to synchronize Frame Memory Writing when displaying video images.

8.5.1 Tearing effect line modes

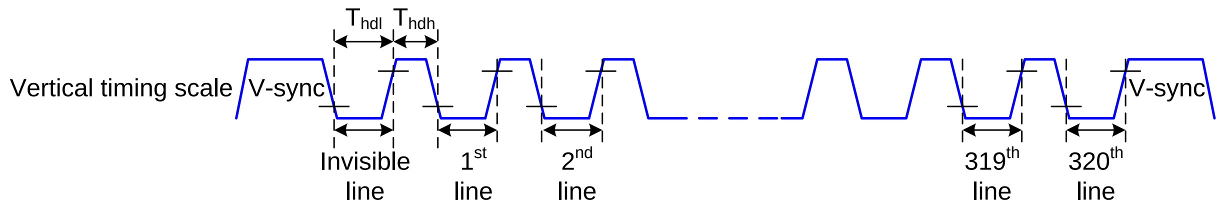
Mode 1, the Tearing Effect Output signal consists of V-Blanking Information only:



tvdh= The LCD display is not updated from the Frame Memory

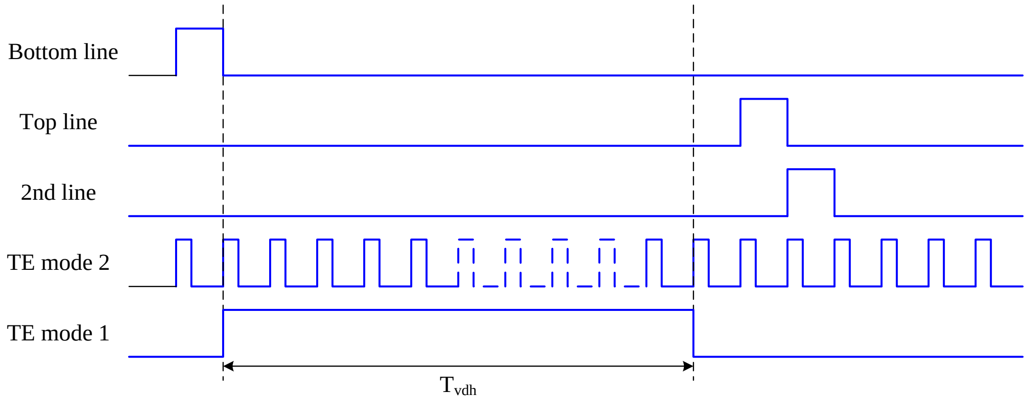
tvdl= The LCD display is updated from the Frame Memory (except Invisible Line – see above)

Mode 2, the Tearing Effect Output signal consists of V-Blanking and H-Blanking Information, there is one V-sync and 390 H-sync pulses per field.



thdh= The LCD display is not updated from the Frame Memory

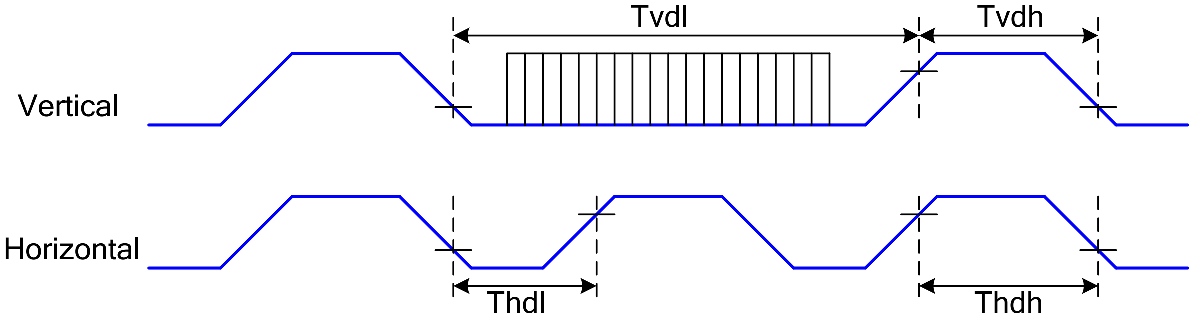
thdl= The LCD display is updated from the Frame Memory (except Invisible Line – see above)



Note: During Sleep In Mode, the Tearing Output Pin is active Low.

8.5.2 Tearing effect line timings

The Tearing Effect signal is described below:

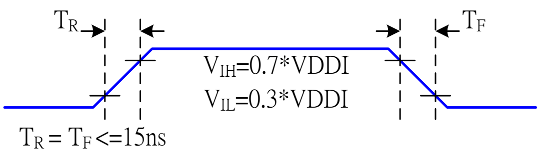


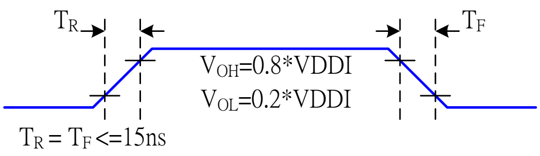
Symbol	Parameter	min	max	unit	description
tvdl	Vertical Timing Low Duration	13	-	ms	
tvdh	Vertical Timing High Duration	1000	-	µs	
thdl	Horizontal Timing Low Duration	16	-	µs	
thdh	Horizontal Timing Low Duration	-	500	µs	

Table AC characteristics of Tearing Effect Signal Idle Mode Off (Frame Rate = 60 Hz, Ta=25°C)

Note: The timings in Table 15 apply when MADCTL ML=0 and ML=1

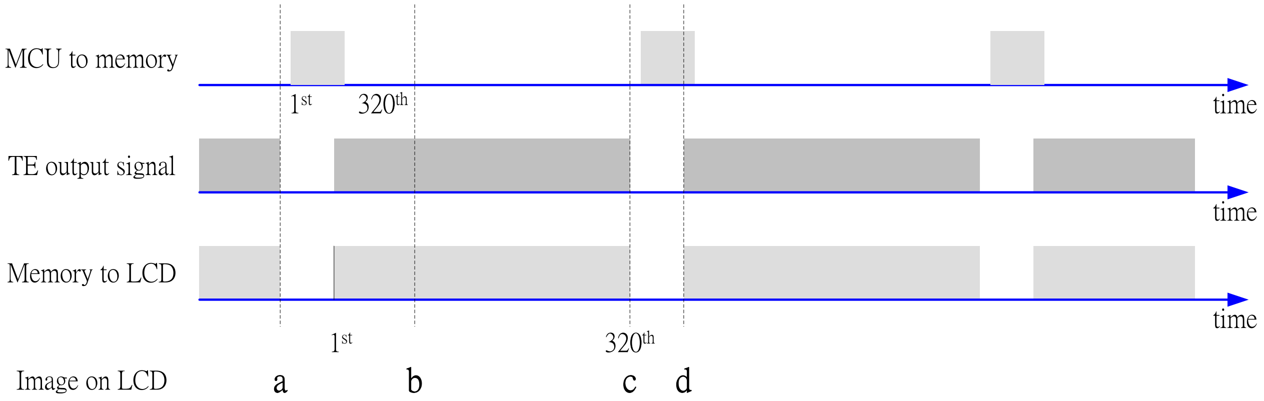
The signal's rise and fall times (tf, tr) are stipulated to be equal to or less than 15ns.



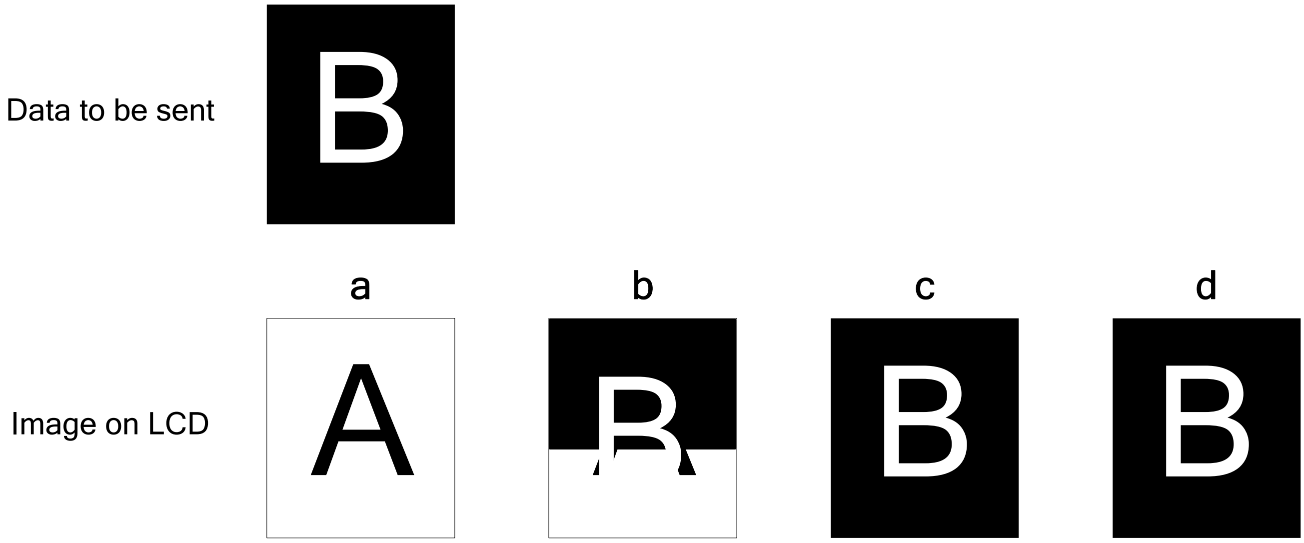


The Tearing Effect Output Line is fed back to the MPU and should be used as shown below to avoid Tearing Effect:

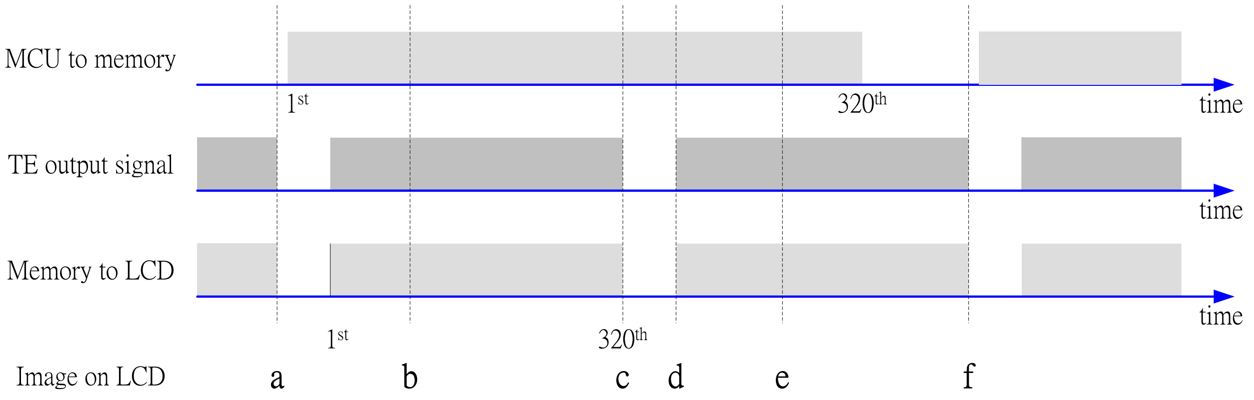
8.5.3 Example 1: MPU Write is faster than panel read



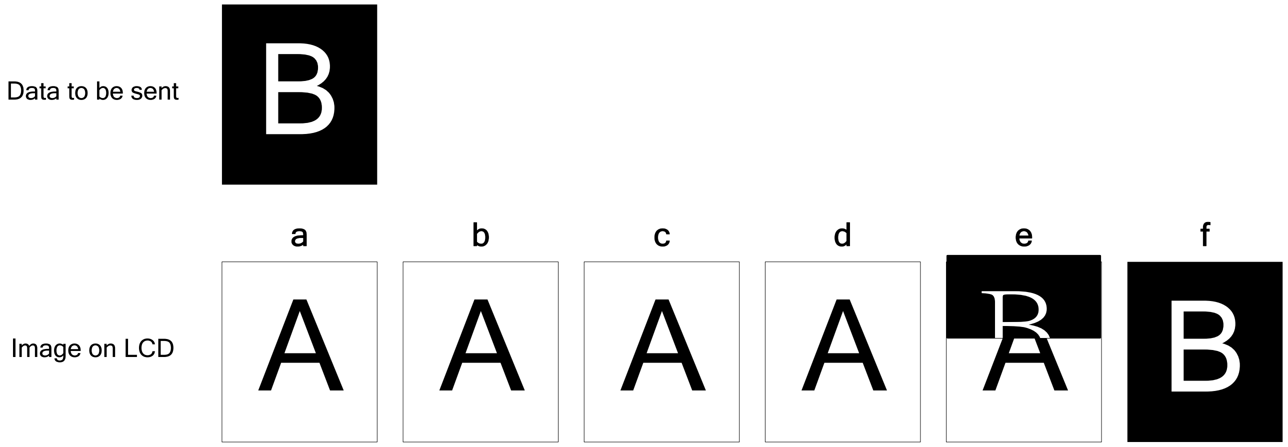
Data write to Frame Memory is now synchronized to the Panel Scan. It should be written during the vertical sync pulse of the Tearing Effect Output Line. This ensures that data is always written ahead of the panel scan and each Panel Frame refresh has a complete new image:



8.5.4 Example 2: MPU write is slower than panel read

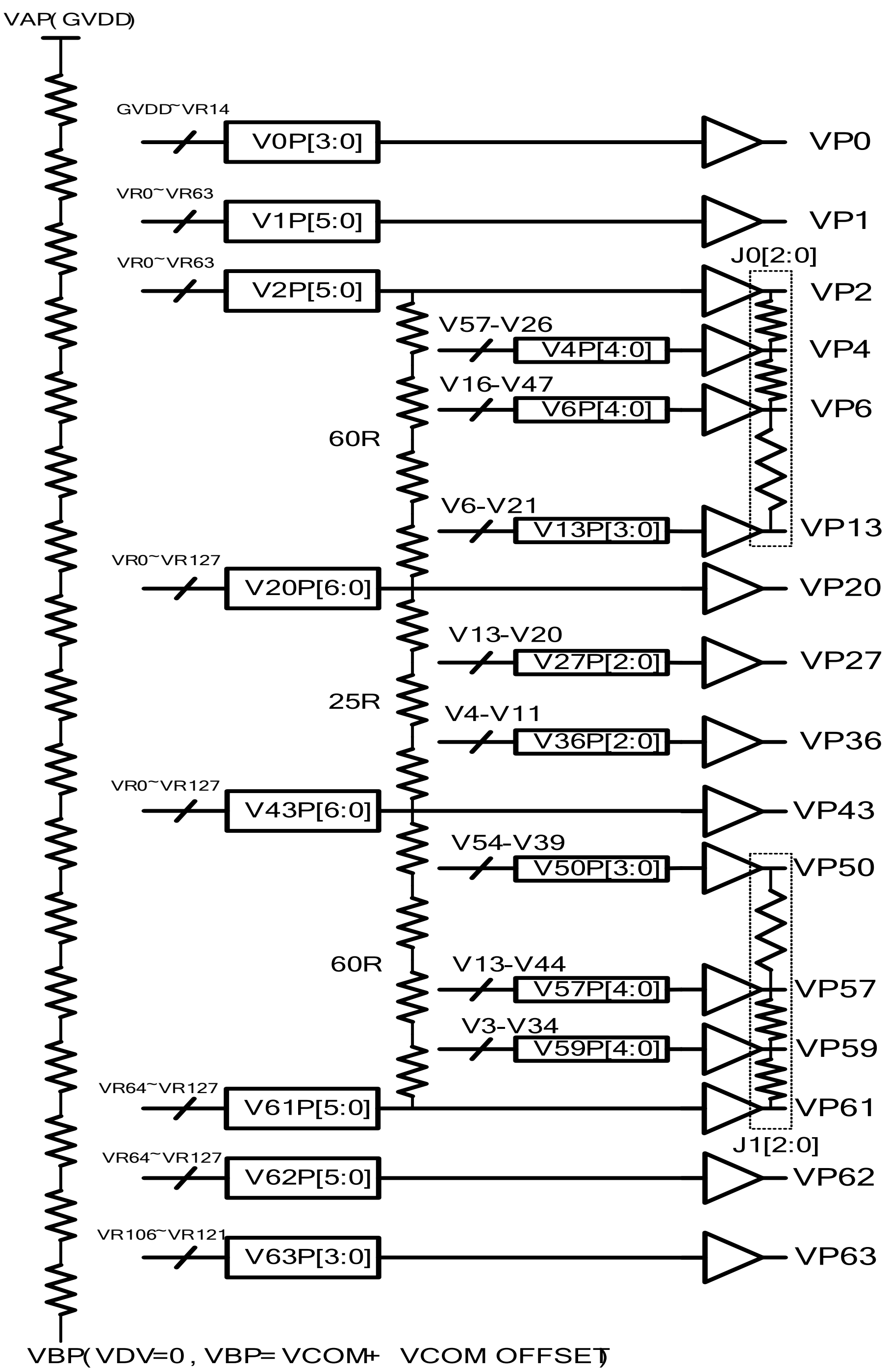


The MPU to Frame Memory write begins just after Panel Read has commenced i.e. after one horizontal sync pulse of the Tearing Effect Output Line. This allows time for the image to download behind the Panel Read pointer and finishing download during the subsequent Frame before the Read Pointer "catches" the MPU to Frame memory write position.

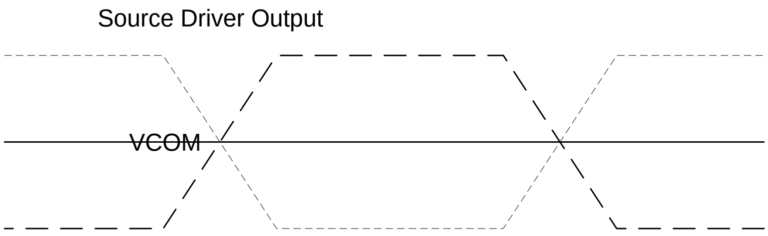


8.6 Gamma Correction

ST77922 incorporate the gamma correction function to display 16M colors for the LCD panel. The gamma correction is performed with 3 groups of registers, which are gradient adjustment, contrast adjustment and fine- adjustment registers for positive and negative polarities.



Gray scale Voltage Generation (Positive)



Relationship between Source Output and VCOM

Percentage adjustment:

VJ0P[2:0], VJ1P[2:0], VJ0N[2:0], VJ1N[2:0] these register are used to adjust the voltage level of interpolation point. The following table is the detail description.

VJ0P[2:0]/VJ0N[2:0]:

	00h	01h	02h	03h	04h	05h	06h	07h
VP3/VN3	50%,18	56%,20	50%,18	60%,22	42%,15	65%,23	45%,16	70%,25
VP5/VN5	50%,18	44%,16	50%,18	42%,15	65%,23	52%,19	40%,14	33%,12
VP7/VN7	86%,30	71%,25	80%,28	66%,23	88%,31	70%,25	76%,27	60%,21
VP8/VN8	71%,25	57%,20	63%,22	49%,17	61%,21	52%,18	58%,20	46%,16
VP9/VN9	57%,20	40%,14	49%,17	34%,12	60%,21	41%,15	47%,16	30%,11
VP10/VN10	43%,15	29%,10	34%,12	23%,8	46%,16	25%,9	36%,13	20%,7
VP11/VN11	29%,10	17%,6	20%,7	14%,5	32%,11	26%,9	23%,8	12%,4
VP12/VN12	14%,5	6%,2	9%,3	6%,2	20%,7	11%,4	17%,6	3%,1

VJ1P[2:0]/VJ1N[2:0]:

	00h	01h	02h	03h	04h	05h	06h	07h
VP51/VN51	86%,30	86%,30	86%,30	89%,31	77%,27	92%,32	83%,29	95%,33
VP52/VN52	71%,25	71%,25	77%,27	80%,28	63%,22	69%,24	75%,26	83%,29
VP53/VN53	57%,20	60%,21	63%,22	69%,24	48%,17	54%,19	66%,23	72%,25
VP54/VN54	43%,15	46%,16	46%,16	51%,18	35%,12	41%,14	55%,19	60%,21
VP55/VN55	29%,10	34%,12	31%,11	37%,13	23%,8	40%,14	26%,9	43%,15
VP56/VN56	14%,5	17%,6	14%,5	20%,7	9%,3	23%,8	11%,4	26%,9
VP58/VN58	50%,18	56%,20	47%,17	47%,17	53%,19	59%,21	45%,16	42%,15
VP60/VN60	50%,18	50%,18	50%,18	53%,19	42%,15	45%,16	55%,20	60%,21

voltage level percentage adjustment description

Source voltage of positive gamma level

Gamma level	Related Register	Formula
VP0	V0P[3:0]	(VAP-VBP)*(129R-V0P[3:0]R)/129R+VBP
VP1	V1P[5:0]	(VAP-VBP)*(128R-V1P[5:0]R)/129R+VBP
VP2	V2P[5:0]	(VAP-VBP)*(128R-V2P[5:0]R)/129R+VBP
VP3	VJ0P[2:0]	(VP2-VP4)* VJ0P[2:0]+VP4
VP4	V4P[4:0]	(VP2-VP20)*(57R-V4P[4:0])/60R+VP20
VP5	VJ0P[2:0]	(VP4-VP6)* VJ0P[2:0]+VP6
VP6	V6P[4:0]	(VP2-VP20)*(47R-V6P[4:0])/60R+VP20
VP7	VJ0P[2:0]	(VP6-VP13)* VJ0P[2:0]+VP13
VP8	VJ0P[2:0]	(VP6-VP13)* VJ0P[2:0]+VP13
VP9	VJ0P[2:0]	(VP6-VP13)* VJ0P[2:0]+VP13
VP10	VJ0P[2:0]	(VP6-VP13)* VJ0P[2:0]+VP13
VP11	VJ0P[2:0]	(VP6-VP13)* VJ0P[2:0]+VP13
VP12	VJ0P[2:0]	(VP6-VP13)* VJ0P[2:0]+VP13
VP13	V13P[3:0]	(VP2-VP20)*(21R-V13P[3:0])/60R+VP20
VP14	--	(VP13-VP20)/(20-13)*(20-14)+VP20
VP15	--	(VP13-VP20)/(20-13)*(20-15)+VP20
VP16	--	(VP13-VP20)/(20-13)*(20-16)+VP20
VP17	--	(VP13-VP20)/(20-13)*(20-17)+VP20
VP18	--	(VP13-VP20)/(20-13)*(20-18)+VP20
VP19	--	(VP13-VP20)/(20-13)*(20-19)+VP20
VP20	V20P[6:0]	(VAP-VBP)*(128R-V20P[6:0]R)/129R+VBP
VP21	--	(VP20-VP27)/(27-20)*(27-21)+VP27
VP22	--	(VP20-VP27)/(27-20)*(27-22)+VP27
VP23	--	(VP20-VP27)/(27-20)*(27-23)+VP27
VP24	--	(VP20-VP27)/(27-20)*(27-24)+VP27
VP25	--	(VP20-VP27)/(27-20)*(27-25)+VP27
VP26	--	(VP20-VP27)/(27-20)*(27-26)+VP27
VP27	V27P[2:0]	(VP20-VP43)*(20R-V27P[2:0])/25R+VP43
VP28	--	(VP27-VP36)/(36-27)*(36-28)+VP36
VP29	--	(VP27-VP36)/(36-27)*(36-29)+VP36
VP30	--	(VP27-VP36)/(36-27)*(36-30)+VP36
VP31	--	(VP27-VP36)/(36-27)*(36-31)+VP36
VP32	--	(VP27-VP36)/(36-27)*(36-32)+VP36
VP33	--	(VP27-VP36)/(36-27)*(36-33)+VP36
VP34	--	(VP27-VP36)/(36-27)*(36-34)+VP36
VP35	--	(VP27-VP36)/(36-27)*(36-35)+VP36
VP36	V36P[2:0]	(VP20-VP43)*(11R-V36P[2:0])/25R+VP43
VP37	--	(VP36-VP43)/(43-36)*(43-37)+VP43
VP38	--	(VP36-VP43)/(43-36)*(43-38)+VP43
VP39	--	(VP36-VP43)/(43-36)*(43-39)+VP43
VP40	--	(VP36-VP43)/(43-36)*(43-40)+VP43
VP41	--	(VP36-VP43)/(43-36)*(43-41)+VP43
VP42	--	(VP36-VP43)/(43-36)*(43-42)+VP43
VP43	V43P[6:0]	(VAP-VBP)*(128R-V43P[6:0]R)/129R+VBP
VP44	--	(VP43-VP50)/(50-43)*(50-44)+VP50
VP45	--	(VP43-VP50)/(50-43)*(50-45)+VP50
VP46	--	(VP43-VP50)/(50-43)*(50-46)+VP50
VP47	--	(VP43-VP50)/(50-43)*(50-47)+VP50
VP48	--	(VP43-VP50)/(50-43)*(50-48)+VP50
VP49	--	(VP43-VP50)/(50-43)*(50-49)+VP50
VP50	V50P[3:0]	(VP43-VP61)*(54R-V50P[3:0])/60R+VP61
VP51	VJ1P[2:0]	(V5P0-VP57)*VJ1P[2:0]+VP57
VP52	VJ1P[2:0]	(VP50-VP57)* VJ1P[2:0]+VP57

VP53	VJ1P[2:0]	(VP50-VP57)* VJ1P[2:0]+VP57
VP54	VJ1P[2:0]	(VP50-VP57)* VJ1P[2:0]+VP57
VP55	VJ1P[2:0]	(VP50-VP57)* VJ1P[2:0]+VP57
VP56	VJ1P[2:0]	(VP50-VP57)* VJ1P[2:0]+VP57
VP57	V57P[4:0]	(VP43-VP61)*(44R-V57P[4:0])/60R+VP61
VP58	VJ1P[2:0]	(VP57-VP59)* VJ1P[2:0]+VP59
VP59	V59P[4:0]	(VP43-VP61)*(34R-V59P[4:0])/60R+VP61
VP60	VJ1P[2:0]	(VP59-VP61)* VJ1P[2:0]+VP61
VP61	V61P[5:0]	(VAP-VBP)*(64R-V61P[5:0]R)/129R+VBP
VP62	V62P[5:0]	(VAP-VBP)*(64R-V62P[5:0]R)/129R+VBP
VP63	V63P[3:0]	(VAP-VBP)*(23R-V63P[3:0]R)/129R+VBP

Source voltage of negative gamma level

Gamma level	Related Register	Formula
VN0	V0N[3:0]	VBN-(VBN-VAN)*(129R-V0N[3:0]R)/129R
VN1	V1N[5:0]	VBN-(VBN-VAN)*(128R-V1N[5:0]R)/129R
VN2	V2N[5:0]	VBN-(VBN-VAN)*(128R-V2N[5:0]R)/129R
VN3	VJ0N[2:0]	(VN2-VN4)*VJ0N[2:0]+VN4
VN4	V4N[4:0]	(VN2-VN20)*(57R-V4N[4:0])/60R+VN20
VN5	VJ0N[2:0]	(VN4-VN6)* VJ0N[2:0]+VN6
VN6	V6N[4:0]	(VN2-VN20)*(47R-V6N[4:0])/60R+VN20
VN7	VJ0N[2:0]	(VN6-VN13)* VJ0N[2:0]+VN13
VN8	VJ0N[2:0]	(VN6-VN13)* VJ0N[2:0]+VN13
VN9	VJ0N[2:0]	(VN6-VN13)* VJ0N[2:0]+VN13
VN10	VJ0N[2:0]	(VN6-VN13)* VJ0N[2:0]+VN13
VN11	VJ0N[2:0]	(VN6-VN13)* VJ0N[2:0]+VN13
VN12	VJ0N[2:0]	(VN6-VN13)* VJ0N[2:0]+VN13
VN13	V13N[3:0]	(VN2-VN20)*(21R-V13N[3:0])/60R+VN20
VN14	--	(VN13-VN20)/(20-13)*(20-14)+VN20
VN15	--	(VN13-VN20)/(20-13)*(20-15)+VN20
VN16	--	(VN13-VN20)/(20-13)*(20-16)+VN20
VN17	--	(VN13-VN20)/(20-13)*(20-17)+VN20
VN18	--	(VN13-VN20)/(20-13)*(20-18)+VN20
VN19	--	(VN13-VN20)/(20-13)*(20-19)+VN20
VN20	V20N[6:0]	VBN-(VBN-VAN)*(128R-V20N[6:0]R)/129R
VN21	--	(VN20-VN27)/(27-20)*(27-21)+VN27
VN22	--	(VN20-VN27)/(27-20)*(27-22)+VN27
VN23	--	(VN20-VN27)/(27-20)*(27-23)+VN27
VN24	--	(VN20-VN27)/(27-20)*(27-24)+VN27
VN25	--	(VN20-VN27)/(27-20)*(27-25)+VN27
VN26	--	(VN20-VN27)/(27-20)*(27-26)+VN27
VN27	V27N[2:0]	(VN20-VN43)*(20R-V27N[2:0])/25R+VN43
VN28	--	(VN27-VN36)/(36-27)*(36-28)+VN36
VN29	--	(VN27-VN36)/(36-27)*(36-29)+VN36
VN30	--	(VN27-VN36)/(36-27)*(36-30)+VN36
VN31	--	(VN27-VN36)/(36-27)*(36-31)+VN36
VN32	--	(VN27-VN36)/(36-27)*(36-32)+VN36
VN33	--	(VN27-VN36)/(36-27)*(36-33)+VN36
VN34	--	(VN27-VN36)/(36-27)*(36-34)+VN36
VN35	--	(VN27-VN36)/(36-27)*(36-35)+VN36
VN36	V36N[2:0]	(VN20-VN43)*(11R-V36N[2:0])/25R+VN43
VN37	--	(VN36-VN43)/(43-36)*(43-37)+VN43
VN38	--	(VN36-VN43)/(43-36)*(43-38)+VN43
VN39	--	(VN36-VN43)/(43-36)*(43-39)+VN43

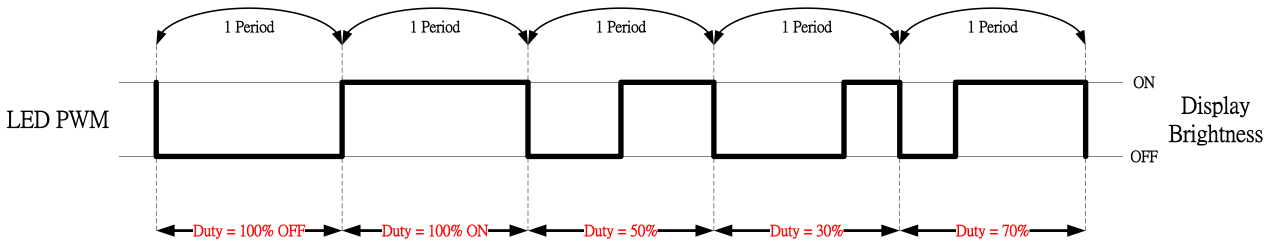
VN40	--	(VN36-VN43)/(43-36)*(43-40)+VN43
VN41	--	(VN36-VN43)/(43-36)*(43-41)+VN43
VN42	--	(VN36-VN43)/(43-36)*(43-42)+VN43
VN43	V43N[6:0]	VBN-(VBN-VAN)*(128R-V43N[6:0]R)/129R
VN44	--	(VN43-VN50)/(50-43)*(50-44)+VN50
VN45	--	(VN43-VN50)/(50-43)*(50-45)+VN50
VN46	--	(VN43-VN50)/(50-43)*(50-46)+VN50
VN47	--	(VN43-VN50)/(50-43)*(50-47)+VN50
VN48	--	(VN43-VN50)/(50-43)*(50-48)+VN50
VN49	--	(VN43-VN50)/(50-43)*(50-49)+VN50
VN50	V50N[3:0]	(VN43-VN61)*(54R-V50N[3:0])/60R+VN61
VN51	VJ1N[2:0]	(V5N0-VN57)*VJ1N[2:0]+VN57
VN52	VJ1N[2:0]	(VN50-VN57)* VJ1N[2:0]+VN57
VN53	VJ1N[2:0]	(VN50-VN57)* VJ1N[2:0]+VN57
VN54	VJ1N[2:0]	(VN50-VN57)* VJ1N[2:0]+VN57
VN55	VJ1N[2:0]	(VN50-VN57)* VJ1N[2:0]+VN57
VN56	VJ1N[2:0]	(VN50-VN57)* VJ1N[2:0]+VN57
VN57	V57N[4:0]	(VN43-VN61)*(44R-V57N[4:0])/60R+VN61
VN58	VJ1N[2:0]	(VN57-VN59)* VJ1N[2:0]+VN59
VN59	V59N[4:0]	(VN43-VN61)*(34R-V59N[4:0])/60R+VN61
VN60	VJ1N[2:0]	(VN59-VN61)* VJ1N[2:0]+VN61
VN61	V61N[5:0]	VBN-(VBN-VAN)*(64R-V61N[5:0]R)/129R
VN62	V62N[5:0]	VBN-(VBN-VAN)*(64R-V62N[5:0]R)/129R
VN63	V63N[3:0]	VBN-(VBN-VAN)*(23R-V63N[3:0]R)/129R

8.7 Brightness Control Block

There is an external output signal from brightness block, LEDPWM to control the LED driver IC in order to control display brightness.

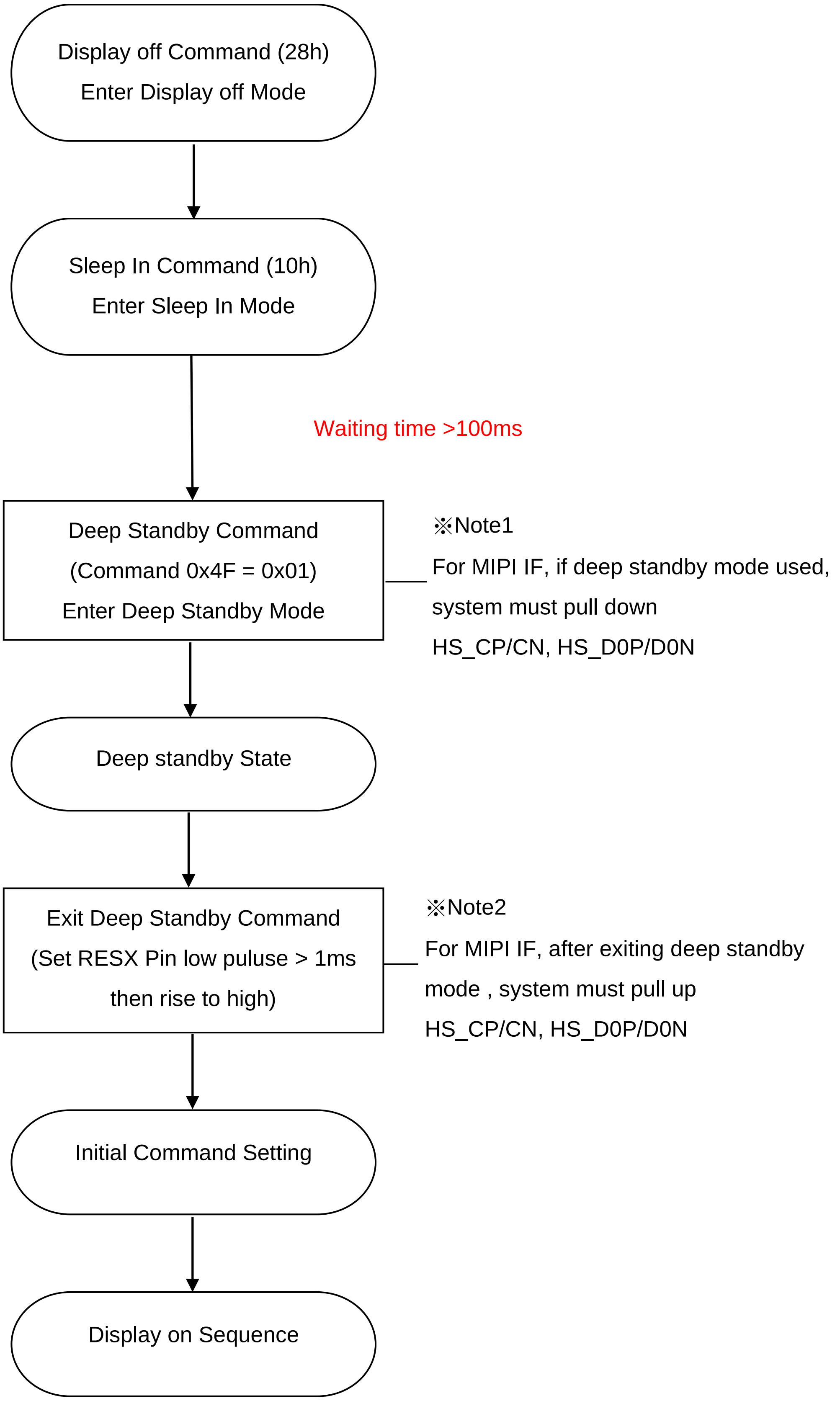
There are register bits, R51h, DBV[7:0] for display brightness of manual brightness setting. The LEDPWM duty is calculated as DBV[7:0] / 255 x Period (affected by OSC frequency).

For example: LEDPWM period = 3 ms, and DBV[7:0] = ‘200’. Then LEDPWM duty = 200 / 255 = 78.1%. Correspond to the LEDPWM period = 3 ms, the high-level of LEDPWM (high effective) = 2.344 ms, and the low-level of LEDPWM = 0.656 m



LEDPWM Output Duty

8.8 Deep Standby mode



9 POWER DEFINITION

9.1 Power Level

6 level modes are defined they are in order of Maximum Power consumption to Minimum Power Consumption

1. Normal Mode On (full display), Idle Mode Off, Sleep Out.

In this mode, the display is able to show maximum 262,144 colors.

2. Normal Mode Off (full display), Idle Mode Off, Sleep Out.

In this mode part of the display is used with maximum 262,144 colors.

3. Normal Mode On (full display), Idle Mode On, Sleep Out.

In this mode, the full display area is used but with 8 colors.

4. Normal Mode Off (full display), Idle Mode On, Sleep Out.

In this mode, part of the display is used but with 8 colors.

5. Sleep In Mode

In this mode, the DC:DC converter, internal oscillator and panel driver circuit are stopped. Only the MCU interface and memory works with VDDI power supply. Contents of the memory are safe.

6. Power Off Mode

In this mode, both VDD and VDDI are removed.

Note: Transition between modes 1-5 is controllable by MCU commands. Mode 6 is entered only when both Power supplies are removed.

9.2 Power ON/OFF Sequence

VDDI and VDD can be applied in any order.

VDD and VDDI can be power down in any order.

During power off, if LCD is in the Sleep Out mode, VDD and VDDI must be powered down minimum 120msec after RESX has been released.

During power off, if LCD is in the Sleep In mode, VDDI or VCI can be powered down minimum 0msec after RESX has been released.

CSX can be applied at any timing or can be permanently grounded. RESX has priority over CSX.

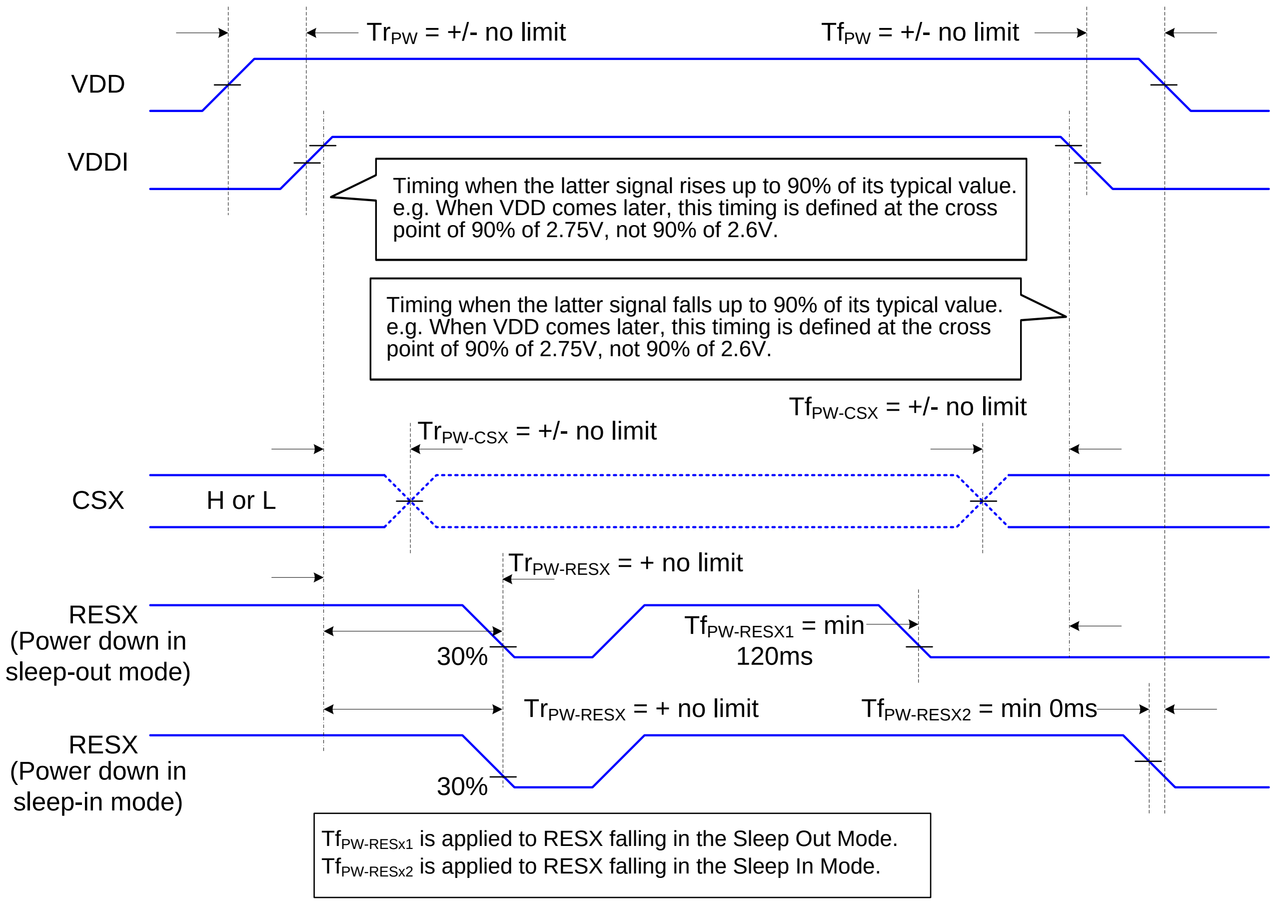
Note 1: There will be no damage to the display module if the power sequences are not met.

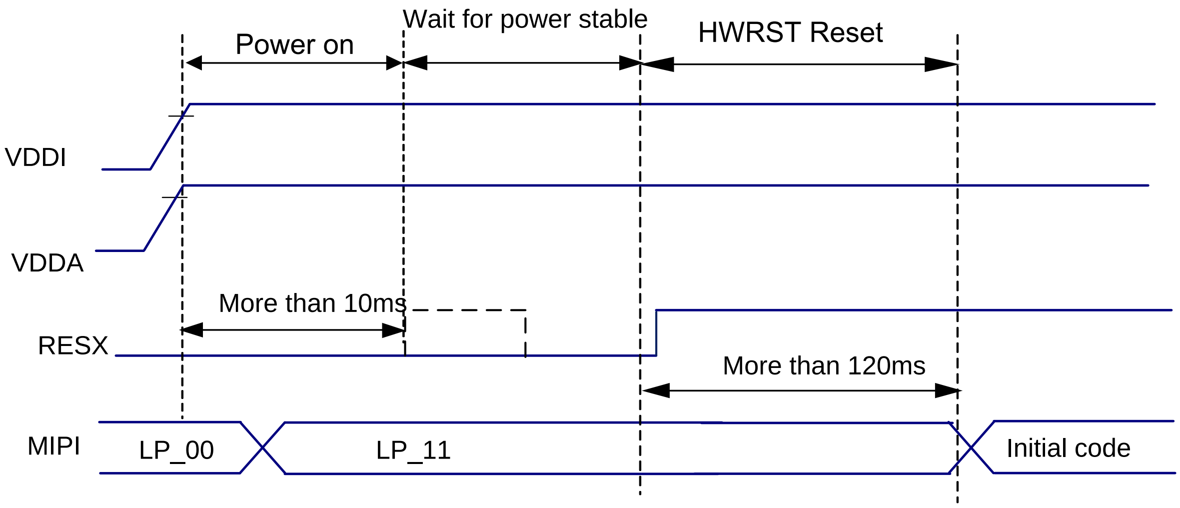
Note 2: There will be no abnormal visible effects on the display panel during the Power On/Off Sequences.

Note 3: There will be no abnormal visible effects on the display between end of Power On Sequence and before receiving Sleep Out command. Also between receiving Sleep In command and Power Off Sequence.

Note 4: If RESX line is not held stable by host during Power On Sequence as defined in the sequence below, then it will be necessary to apply a Hardware Reset (RESX) after Host Power On Sequence is complete to ensure correct operation. Otherwise function is not guaranteed.

The power on/off sequence is illustrated below



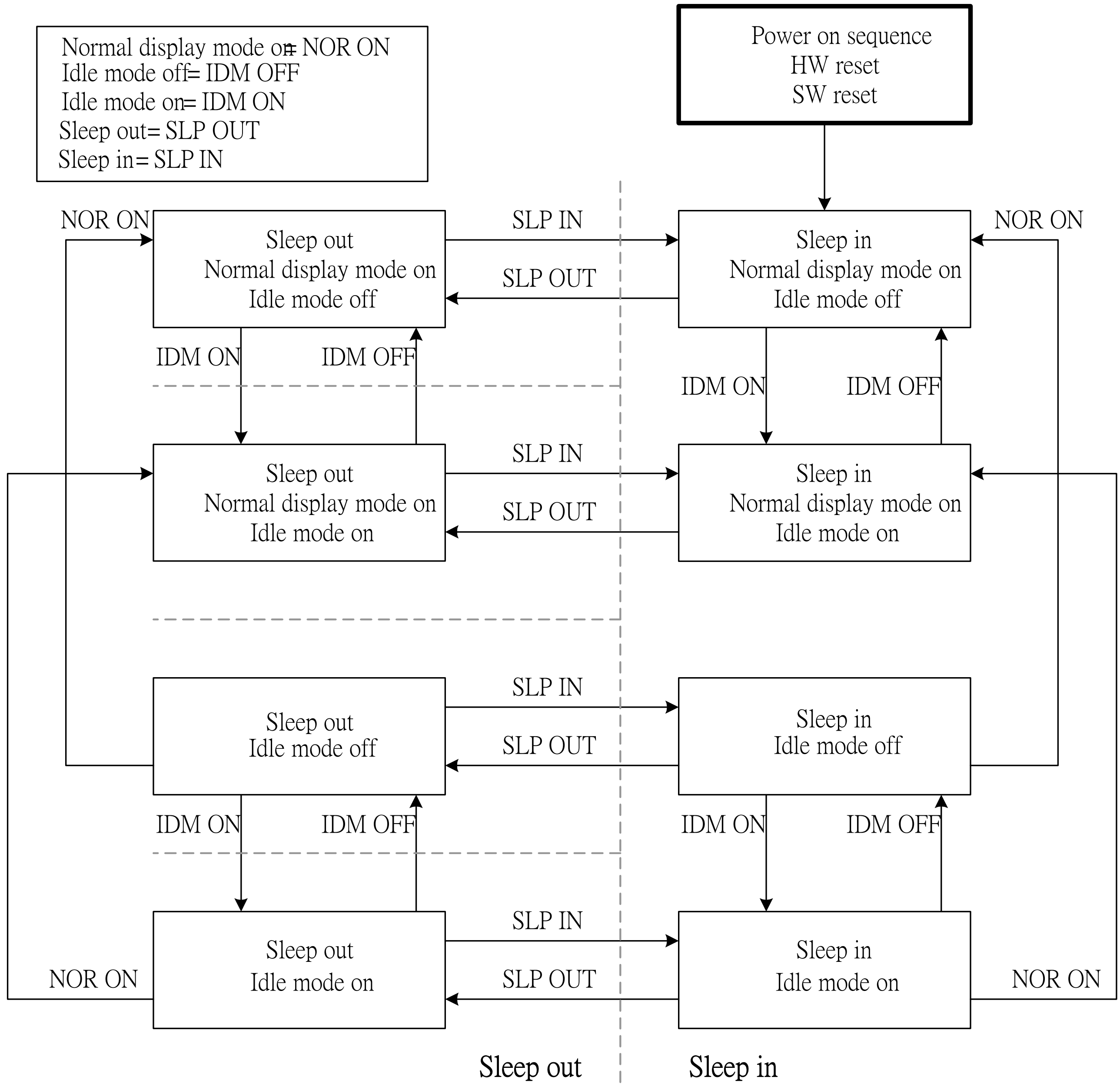


9.3 Uncontrolled Power OFF

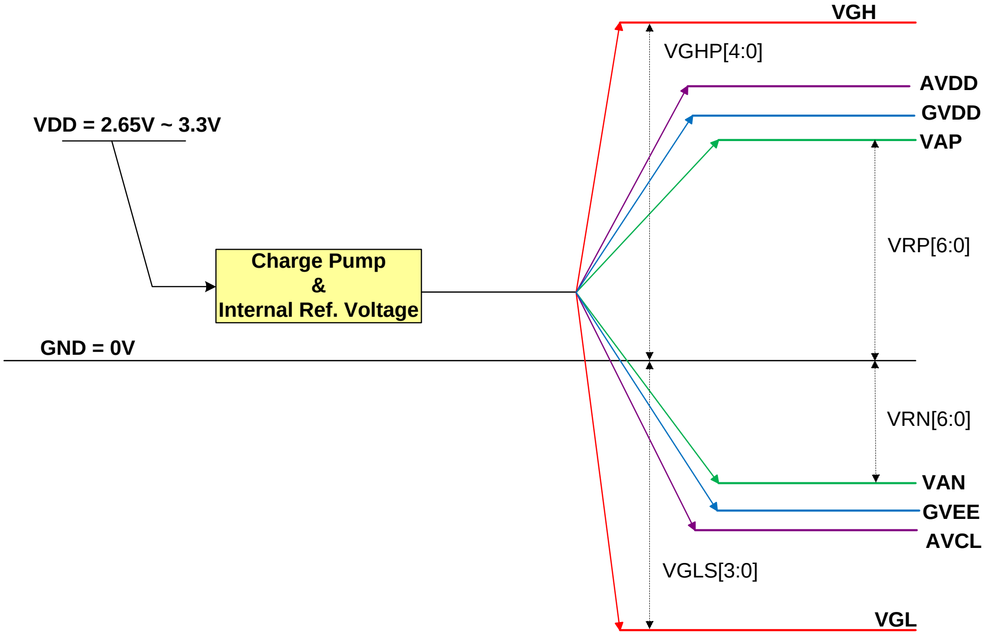
The uncontrolled power-off means a situation which removed a battery without the controlled power off sequence. It will neither damage the module or the host interface.

If uncontrolled power-off happened, the display will go blank and there will not any visible effect on the display (blank display) and remains blank until "Power On Sequence" powers it up.

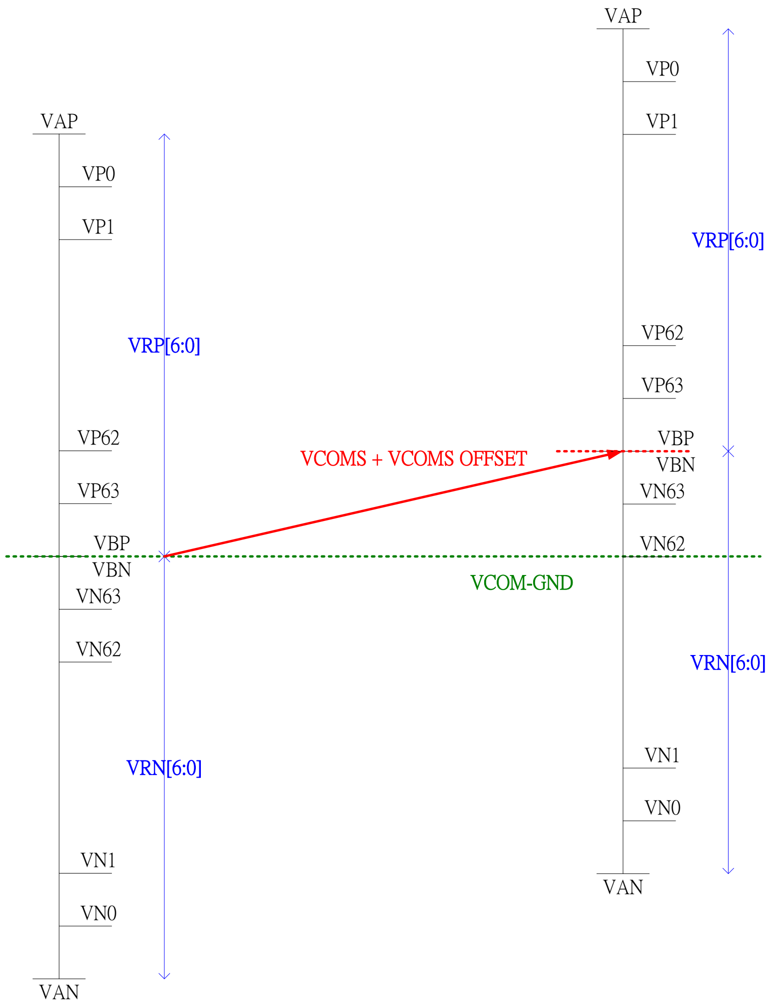
9.4 Power Flow Chart



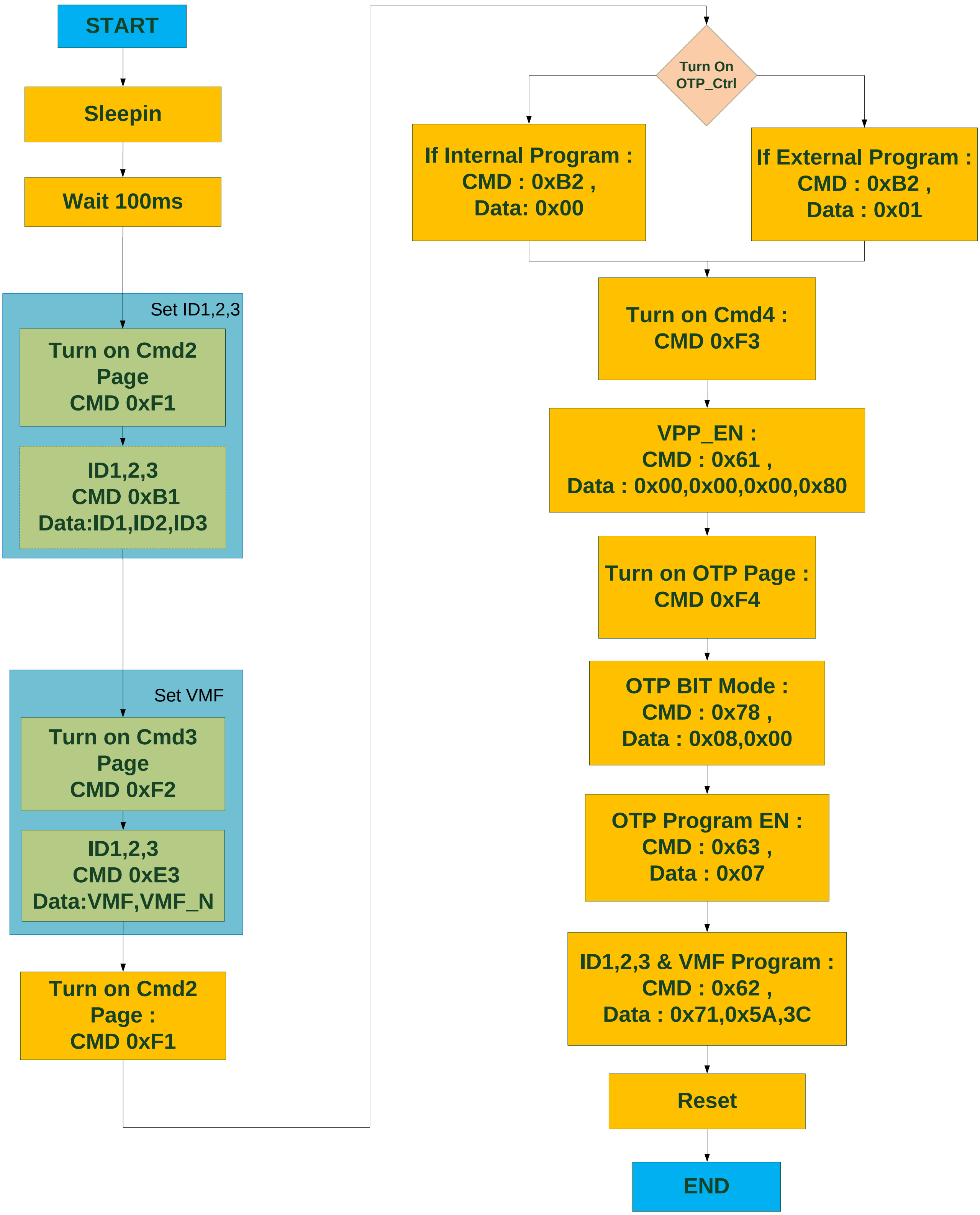
9.5 Voltage Generation



9.6 Relationship about source voltage

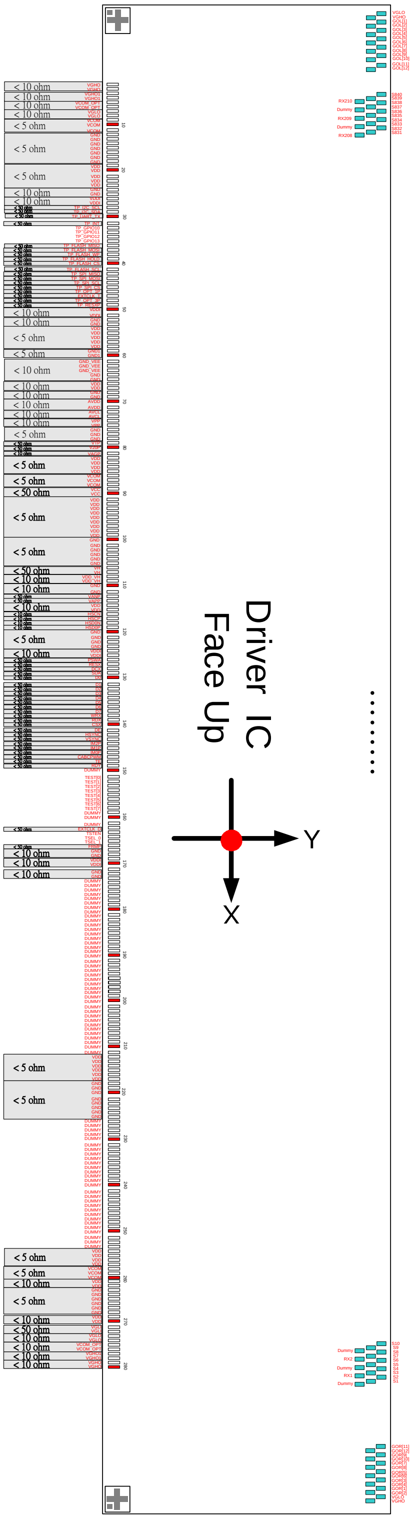


10 NVM PROGRAMMING FLOW



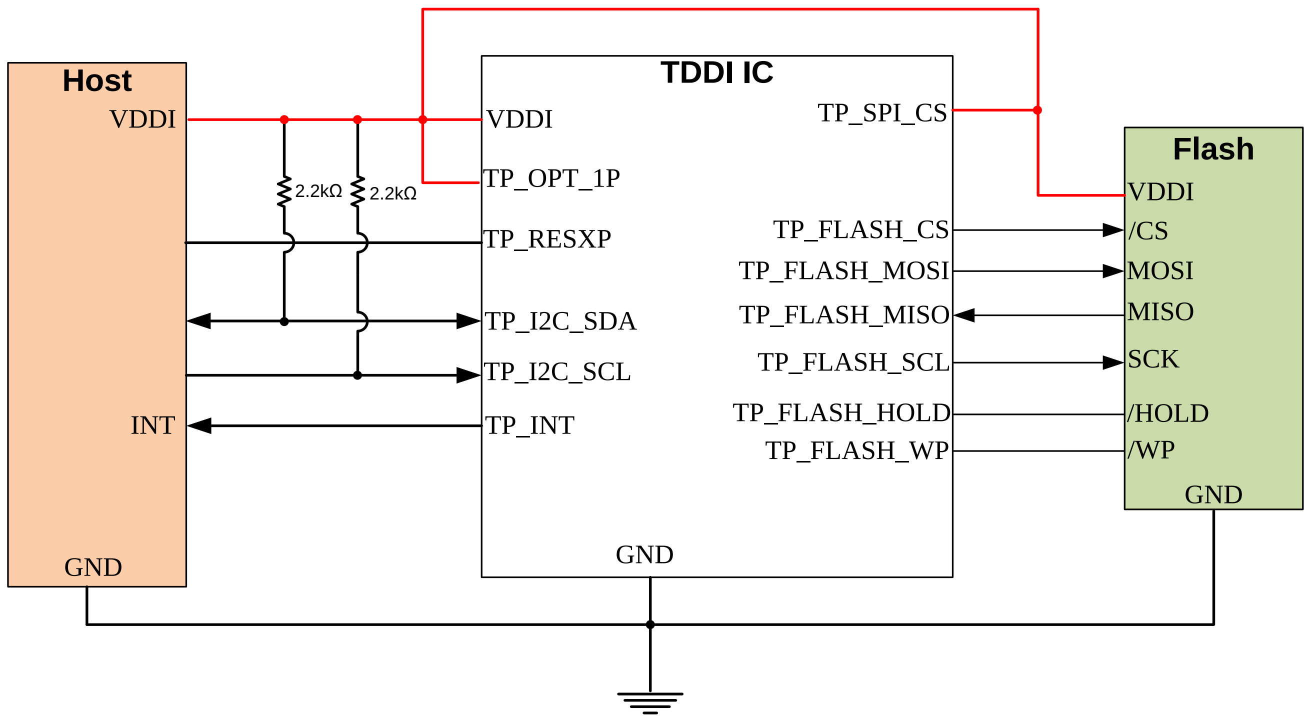
11 APPLICATION NOTE

11.1 Layout Resistance Suggestion

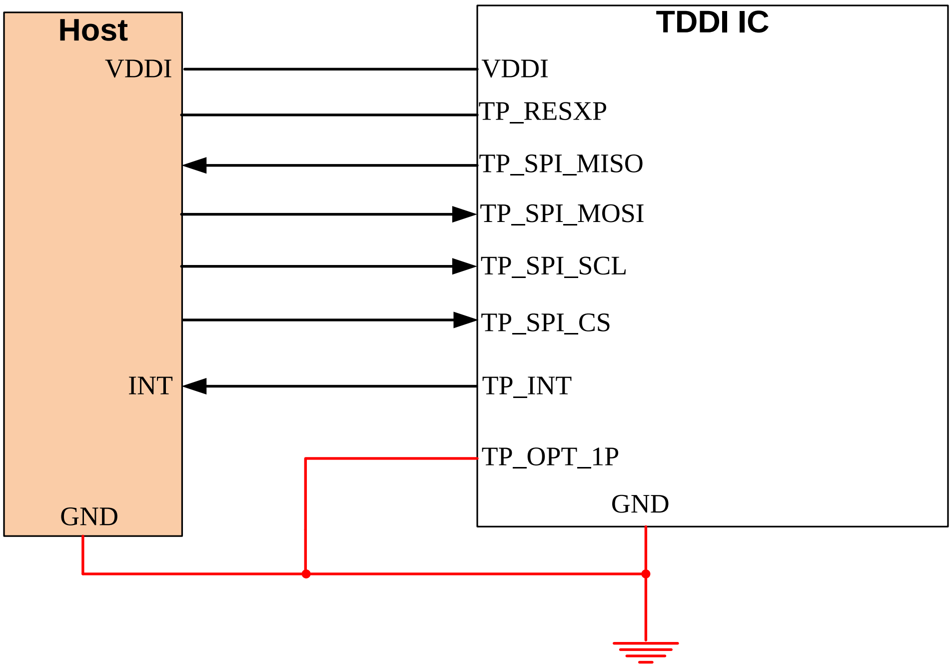


11.2 Touch Application circuit

11.2.1 With Flash Mode Application Circuit

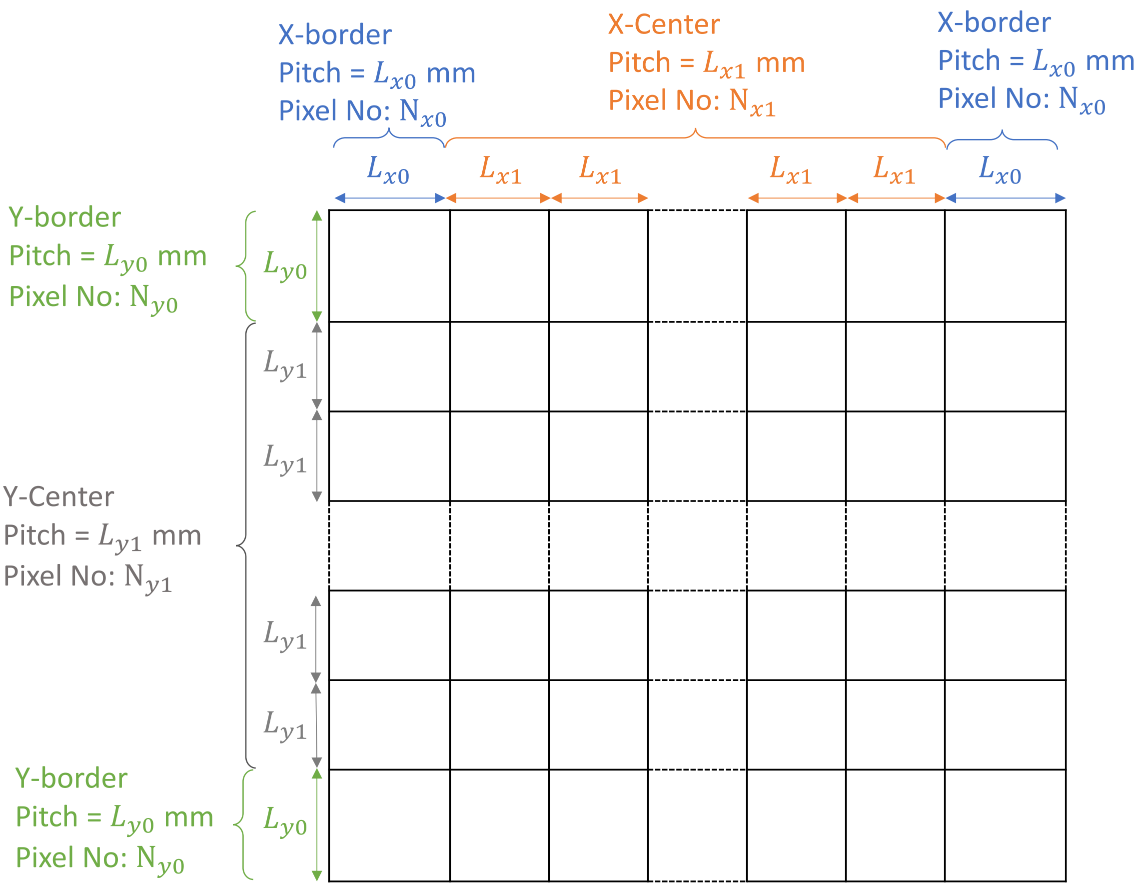


11.2.2 No-Flash Mode Application Circuit



11.3 Touch Electrode Application

The pitch of X-axis & Y-axis will be defined by two sizes, the leftmost channel and rightmost channel are the border channels and pitch should be the same. Others are the center channels where the pitch of center channels should be all the same which can be different pitches with border channels.



12 COMMAND

12.1 Page Set Table

PAGE SET Table														
Instruction	**D/CX**	**WRX**	**RDX**	**PNUM**	**D7**	**D6**	**D5**	**D4**	**D3**	**D2**	**D1**	**D0**	**Hex**	**Function**
SWCMD1	0	↑	1	-	1	1	1	1	0	0	0	0	(F0h)	Switch to CMD1
SWCMD2	0	↑	1	-	1	1	1	1	0	0	0	1	(F1h)	Switch to CMD2
SWCMD3	0	↑	1	-	1	1	1	1	0	0	1	1	(F2h)	Switch to CMD3

SWCMD1 (F0h): Switch to CMD1

F0H	Switch to CMD1											
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
SWCMD1	0	↑	1	-	1	1	1	1	0	0	0	0	(F0h)
Parameter	No Parameter												
Description	This command is used to select the function of Command 1												
Register availability	(see table below)												

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

SWCMD2 (F1h): Switch to CMD2

F1H	Switch to CMD2												
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
CK	0	↑	1	-	1	1	1	1	0	0	0	1	(F1h)
Parameter	No Parameter												
Description	This command is used to select the function of Command 2												
Register availability	(see table below)												

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

SWCMD3 (F2h): Switch to CMD3

F2H	Switch to CMD3											
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
CK	0	↑	1	-	1	1	1	1	0	0	1	0	(F2h)
Parameter	No Parameter												
Description	This command is used to select the function of Command 3												

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

12.2 Command Table 1

COMMAND Table 1														
Instruction	**D/CX**	**WRX**	**RDX**	**PNUM**	**D7**	**D6**	**D5**	**D4**	**D3**	**D2**	**D1**	**D0**	**Hex**	**Function**
NOP	0	↑	1	-	0	0	0	0	0	0	0	0	(00h)	No Operation
SWRESET	0	↑	1	-	0	0	0	0	0	0	0	1	(01h)	Software Reset
RDDID	0	↑	1	-	0	0	0	0	0	1	0	0	(04h)	Read Display ID
	1	1	↑		-	-	-	-	-	-	-	-		Dummy read
	1	1	↑		ID1[7:0]									ID1 read
	1	1	↑		ID2[7:0]									ID2 read
	1	1	↑		ID3[7:0]									ID3 read
RDNUMED	0	↑	1	-	0	0	0	0	0	1	0	1	(05h)	Read Number of Errors on DSI
	1	1	↑	-	-	-	-	-	-	-	-	-		Dummy read
	1	1	↑	-	DSI_ERR_NUM[7:0]									
RDDST	0	↑	1	-	0	0	0	0	1	0	0	1	(09h)	Read Display Status
	1	1	↑		-	-	-	-	-	-	-	-		Dummy read
	1	1	↑		MY	MX	-	ML	RGB	MH	-	-		
	1	1	↑		BSTON	-	IFPF[1:0]		IDMON	PTLON	SLOUT	NORON		
	1	1	↑		VSSON	-	INVON	-	-	DISON	TEON	-		
	1	1	↑		-	-	TELOM	-	-	-	-	-		
RDDPM	0	↑	1	-	0	0	0	0	1	0	1	0	(0Ah)	Read Display Power Mode
	1	1	↑		-	-	-	-	-	-	-	-		Dummy read
	1	1	↑		BSTON	IDMON	PLTON	SLPOUT	NORON	DISON	-	-		
RDD MADCTL	0	↑	1	-	0	0	0	0	1	0	1	1	(0Bh)	Read Display MADCTL
	1	1	↑		-	-	-	-	-	-	-	-		Dummy read
	1	1	↑		MY	MX	-	ML	BGR	MH	-	-		
RDD COLMOD	0	↑	1	-	0	0	0	0	1	1	0	0	(0Ch)	Read Display Pixel Format
	1	1	↑		-	-	-	-	-	-	-	-		Dummy read
	1	1	↑		-	-	-	-	-	-	IFPF.1-0			

COMMAND Table 1														
Instruction	**D/CX**	**WRX**	**RDX**	**PNUM**	**D7**	**D6**	**D5**	**D4**	**D3**	**D2**	**D1**	**D0**	**Hex**	**Function**
RDDIM	0	↑	1	-	0	0	0	0	1	1	0	1	(0Dh)	Read Display Image Mode
	1	1	↑		-	-	-	-	-	-	-	-		Dummy read
	1	1	↑		VSSON	-	INVON	-	-	-	-	-		
RDDSM	0	↑	1	-	0	0	0	0	1	1	1	0	(0Eh)	Read Display Signal Mode
	1	1	↑		-	-	-	-	-	-	-	-	-	Dummy read
	1	1	↑		TEON	TELOM	-	-	-	-	-	EDSI		
RDDSDR	0	↑	1	-	0	0	0	0	1	1	1	1	(0Fh)	Read Display Sef-Diagnostic
	1	1	↑		-	-	-	-	-	-	-	-		Dummy read
	1	1	↑		RLD	-	-	-	-	-	-	CCR		
SLPIN	0	↑	1	-	0	0	0	1	0	0	0	0	(10h)	Sleep in
SLPOUT	0	↑	1	-	0	0	0	1	0	0	0	1	(11h)	Sleep out
PTLON	0	↑	1	-	0	0	0	1	0	0	1	0	(12h)	Partial On
NORON	0	↑	1	-	0	0	0	1	0	0	1	1	(13h)	Normal On
INVOFF	0	↑	1	-	0	0	1	0	0	0	0	0	(20h)	Display inversion off
INVON	0	↑	1	-	0	0	1	0	0	0	0	1	(21h)	Display inversion on
DISPOFF	0	↑	1	-	0	0	1	0	1	0	0	0	(28h)	Display off
DISPON	0	↑	1	-	0	0	1	0	1	0	0	1	(29h)	Display on
CASET	0	↑	1	4	0	0	1	0	1	0	1	0	(2Ah)	Column Address Set
	1	↑	1		-	-	-	-	-	-	XS9	XS8		X address start: 0≦XS≦X
	1	↑	1		XS7	XS6	XS5	XS4	XS3	XS2	XS1	XS0		
	1	↑	1		-	-	-	-	-	-	XE9	XE8		X address start: S≦XE≦X
	1	↑	1		XE7	XE6	XE5	XE4	XE3	XE2	XE1	XE0		
RASET	0	↑	1	4	0	0	1	0	1	0	1	1	(2Bh)	Row Address Set
	1	↑	1		-	-	-	-	-	YS10	YS9	YS8		Y address start: 0≦YS≦Y
	1	↑	1		YS7	YS6	YS5	YS4	YS3	YS2	YS1	YS0		

COMMAND Table 1													
Instruction	**D/CX**	**WRX**	**RDX**	**PNUM**	**D7**	**D6**	**D5**	**D4**	**D3**	**D2**	**D1**	**D0**	**Hex**	**Function**
	1	↑	1		-	-	-	-	-	-	YE9	YE8		Y address start: S≦YE≦Y
	1	↑	1		YE7	YE6	YE5	YE4	YE3	YE2	YE1	YE0		
RAMWR	0	↑	1	-	0	0	1	0	1	1	0	0	(2Ch)	Memory Write
	1	↑	1		D1[7:0]									Write data
	1	↑	1		Dx[7:0]									
	1	↑	1		Dn[7:0]									
RAMRD	0	↑	1	-	0	0	1	0	1	1	1	0	(2Eh)	Memory Read
	1	1	↑		-	-	-	-	-	-	-	-		Dummy read
	1	1	↑		D1[7:0]									Read data
	1	1	↑		Dx[7:0]									
	1	1	↑		Dn[7:0]									
RPLTAR	0	↑	1	4	0	0	1	1	0	0	0	0	(30h)	Partial Area
	1	1	↑		-	-	-	-	-	SR[10:8]				
	1	1	↑		SR[7:0]									
	1	1	↑		-	-	-	-	-	ER[10:8]				
	1	1	↑		ER[7:0]									
VSCRDEF	0	↑	1	6	0	0	1	1	0	0	1	1	(33h)	Vertical Scrolling Definition
	1	↑	1		-	-	-	-	-	TFA[10:8]				
	1	↑	1		TFA[7:0]									
	1	↑	1		-	-	-	-	-	VSA[10:8]				
	1	↑	1		VSA[7:0]									
	1	↑	1		-	-	-	-	-	BFA[10:8]				
	1	↑	1		BFA[7:0]									
TEOFF	0	↑	1	-	0	0	1	1	0	1	0	0	(34h)	Tearing Effect Line off
TEON	0	↑	1	1	0	0	1	1	0	1	0	1	(35h)	Tearing Effect Line on
	1	↑	1		-	-	-	-	-	-	-	TELO		
MADCTL	0	↑	1	1	0	0	1	1	0	1	1	0	(36h)	Memory Data Access Control
	1	↑	1		MY	MX	-	ML	RGB	MH	-	-		
VSCRSADD	0	↑	1	2	0	0	1	1	0	1	1	1	(37h)	Vertical Scrolling Start Address
	1	↑	1		-	-	-	-	-	VSP[10:8]				
	1	↑	1		VSP[7:0]									
IDMOFF	0	↑	1	-	0	0	1	1	1	0	0	0	(38h)	Idle Mode off

COMMAND Table 1														
Instruction	**D/CX**	**WRX**	**RDX**	**PNUM**	**D7**	**D6**	**D5**	**D4**	**D3**	**D2**	**D1**	**D0**	**Hex**	**Function**
IDMON	0	↑	1	-	0	0	1	1	1	0	0	1	(39h)	Idle Mode on
COLMOD	0	↑	1	1	0	0	1	1	1	0	1	0	(3Ah)	Interface Pixel Format
	1	↑	1		-	-	-	-	-	-	IFPF[1:0]			
RAMWRC	0	↑	1	-	0	0	1	1	1	1	0	0	(3Ch)	Memory Write Continue
	1	↑	1					D1[7:0]						Write data
	1	↑	1					Dx[7:0]						
	1	↑	1		Dn[7:0]									
RAMRDC	0	↑	1	-	0	0	1	1	1	1	1	0	(3Eh)	Memory Write Continue
	1	1	↑		-	-	-	-	-	-	-	-		Dummy read
	1	1	↑		D1[7:0]									Read data
	1	1	↑		Dx[7:0]									
	1	1	↑		Dn[7:0]									
SPIRDMD	0	↑	1	-	0	0	1	1	0	0	1	1	(42h)	SPI Read Mode
	1	↑	1		SPI_RD_CAP[7:0]									
TESLWR	0	↑	1	2	0	1	0	0	0	1	0	0	(44h)	Write Tear Scan Line
	1	↑	1		-	-	-	-	TE_SL[11:8]					
	1	↑	1		TE_SL[7:0]									
TESLRD	0	↑	1	2	0	1	0	0	0	1	0	1	(45h)	Read Tear Scan Line
	1	1	↑		-	-	-	-				-		Dummy Read
	1	1	↑		-	-	-	-	TE_SL[11:8]					
	1	1	↑		TE_SL[7:0]									
RAMCLACT	0	↑	1	-	0	1	0	0	1	1	0	0	(4Ch)	Memory Clear Act
RAMCLRSET	0	↑	1	3	0	1	0	0	1	1	0	1	(4Dh)	Memory Clear Setting
	1	↑	1		CLNRAM_R[7:0]									
	1	↑	1		CLNRAM_G[7:0]									
	1	↑	1		CLNRAM_B[7:0]									
RAMCLRBUSY	0	↑	1	-	0	1	0	0	1	1	1	0	(4Eh)	Memory Clear Busy
	1	1	↑		-	-	-	-	-	-	-	-		Dummy Read

COMMAND Table 1														
Instruction	D/CX	WRX	RDX	PNUM	D7	D6	D5	D4	D3	D2	D1	D0	Hex	Function
	1	1	↑		-	-	-	-	-	-	-	CLR_BUSY		
DSTB	0	↑	1	-	0	1	0	0	1	1	1	1	(4Fh)	Deep Standby Mode
	1	↑	1		-	-	-	-	-	-	-	DSTB		
WRDISBV	0	↑	1	-	0	1	0	1	0	0	0	1	(51h)	Write Display Brightness
	1	↑	1		DBV[7:0]									
RDDISBV	0	↑	1	-	0	1	0	1	0	0	1	0	(52h)	Read Display Brightness
	1	1	↑		-	-	-	-	-	-	-	-		Dummy read
	1	1	↑		DBV[7:0]									
WRCTRLD	0	↑	1	1	0	1	0	1	0	0	1	1	(53h)	Write CTRL Display
	1	↑	1		-	-	-	-	-	BL	-	-		
RDCTRLD	0	↑	1	-	0	1	0	1	0	1	0	0	(54h)	Read CTRL Display
	1	1	↑		-	-	-	-	-	-	-	-		Dummy read
	1	1	↑		-	-	-	-	-	BL	-	-		
WRSRECTRL	0	↑	1	1	0	1	0	1	0	1	0	1	(55h)	Write Content Adaptive Brightness Control
	0	↑	1		-	SRE_EN	-	-	-	-	-	-		
RDSRECTRL	0	↑	1	-	0	1	0	1	0	1	1	0	(56h)	Read Content Adaptive Brightness Control
	1	1	↑		-	-	-	-	-	-	-	-		Dummy read
	1	1	↑		-	SRE_EN	-	-	-	-	-	-		
RDFCS	0	↑	1	-	0	1	1	0	1	1	0	0	(AAh)	Read First Checksum
	1	1	↑											Dummy read
	1	1	↑		FCS[7:0]									

COMMAND Table 1														
Instruction	**D/CX**	**WRX**	**RDX**	**PNUM**	**D7**	**D6**	**D5**	**D4**	**D3**	**D2**	**D1**	**D0**	**Hex**	**Function**
RDCCS	0	↑	1	-	0	1	1	0	1	1	0	1	(AFh)	Read Continuous Checksum
	1	1	↑											Dummy read
	1	1	↑		CCS[7:0]									
RAMMD	0	↑	1	1	0	1	1	0	1	1	1	1	(D0h)	RAM Mode
	1	↑	1		Scan_ram_spwr	-	-	-	-	-	Video_mode[1:0]			
RDID1	0	↑	1	-	1	1	0	1	1	0	1	0	(DAh)	Read ID1
	1	1	↑		-	-	-	-	-	-	-	-		Dummy read
	1	1	↑		ID1[7:0]									
RDID2	0	↑	1	-	1	1	0	1	1	0	1	1	(DBh)	Read ID2
	1	1	↑		-	-	-	-	-	-	-	-		Dummy read
	1	1	↑		ID2[7:0]									
RDID3	0	↑	1	-	1	1	0	1	1	1	0	0	(DCh)	Read ID3
	1	1	↑		-	-	-	-	-	-	-	-		Dummy read
	1	1	↑		ID3[7:0]									

NOP (00h)

00H	NOP (No Operation)											
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
NOP	0	↑	1	-	0	0	0	0	0	0	0	0	(00h)
Parameter	No Parameter												-
Description	This command is empty command. "-" Don't care												
Restriction													
Register Availability	see table below												
Default	see table below												
Flow Chart													

Register Availability:

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default:

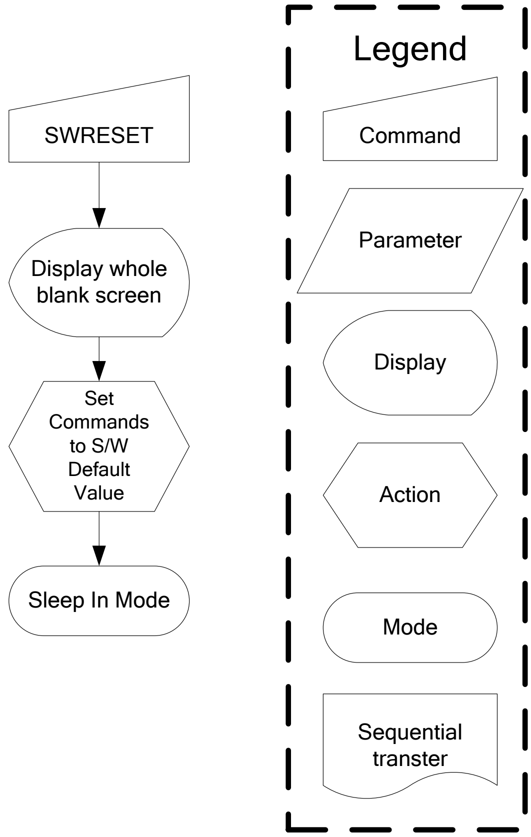
Status	Default Value
Power On Sequence	N/A
S/W Reset	N/A
H/W Reset	N/A

SWRESET (01h): Software Reset

01H	SWRESET (Software Reset)												
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
SWRESET	0	↑	1	-	0	0	0	0	0	0	0	1	(01h)
Parameter	No Parameter												-
Description	"-" Don't care - When the Software Reset command is written, it causes software reset. It resets the commands and parameters to their S/W Reset default values. - Frame memory contents are unaffected by this command.												
Restriction	It will be necessary to wait 5msec before sending new command following software reset. The display module loads all display suppliers' factory default values to the registers during this 5msec. If software reset is sent during sleep in mode, it will be necessary to wait 120msec before sending sleep out command. Software reset command cannot be sent during sleep out sequence.												
Register Availability	see table below												

Register Availability:

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes

<table>
<tr><td></td><td><table><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td><table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>N/A</td></tr><tr><td>S/W Reset</td><td>N/A</td></tr><tr><td>H/W Reset</td><td>N/A</td></tr></table></td></tr>
<tr><td>Flow Chart</td><td>
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RDDID (04h): Read Display ID

04H	RDDID (Read Display ID)											
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDDID	0	↑	1	-	0	0	0	0	0	1	0	0	(04h)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	-
2nd parameter	1	1	↑	-		ID1[6:0]							
3rd parameter	1	1	↑	-		ID2[6:0]							
4th parameter	1	1	↑	-		ID3[6:0]							

Description

-This read byte returns 24-bit display identification information.

-The 1st parameter is dummy data

-The 2nd parameter (ID1[6:0]): LCD module’s manufacturer ID.

-The 3rd parameter (ID2[6:0]): LCD module/driver version ID

-The 4th parameter (ID3[6:0]): LCD module/driver ID.

-Commands RDID1/2/3(DAh, DBh, DCh) read data correspond to the parameters 2,3,4 of the command 04h, respectively.

“-“ Don’t care

Restriction

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value		
	ID1	ID2	ID3
Power On Sequence	See description	See description	See description
S/W Reset	See description	See description	See description
H/W Reset	See description	See description	See description

RDDST (09h): Read Display Status

09H	RDDST (Read Display Status)												
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDDST	0	↑	1	-	0	0	0	0	1	0	0	1	(09h)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	
2nd parameter	1	1	↑	-	MY	MX	-	ML	RGB	MH	-	-	
3rd parameter	1	1	↑	-	BOTON	-	IFPF[1:0]		IDMON	PTLON	SLOUT	NORON	

4th parameter	1	1	↑	-	VSSON	-	INVON	-	-	DISON	TEON	-	
5th parameter	1	1	↑	-	-	-	TELOM	-	-	-	-	-	

Description

This command indicates the current status of the display as described in the table below:

Bit	Description	Value
BSTON	Booster Voltage Status	'1' =Booster on, '0' =Booster off
MY	Row Address Order (MY)	'1' =Decrement, (Bottom to Top, when MADCTL (36h) D7='1') '0' =Increment, (Top to Bottom, when MADCTL (36h) D7='0')
MX	Column Address Order (MX)	'1' =Decrement, (Right to Left, when MADCTL (36h) D6='1') '0' =Increment, (Left to Right, when MADCTL (36h) D6='0')
ML	Scan Address Order (ML)	'0' =Decrement, (LCD refresh Top to Bottom, when MADCTL (36h) D4='0') '1'=Increment, (LCD refresh Bottom to Top, when MADCTL (36h) D4='1')
RGB	RGB/ BGR Order (RGB)	'1' =BGR, (When MADCTL (36h) D3='1') '0' =RGB, (When MADCTL (36h) D3='0')
MH	Horizontal Order	'0' =Decrement, (LCD refresh Left to Right, when MADCTL (36h) D2='0') '1' =Increment, (LCD refresh Right to Left, when MADCTL (36h) D2='1')
IFPF[1:0]	Interface Color Pixel Format Definition	"01" = 16-bit / pixel, "10" = 18-bit / pixel, "11" = 24-bit / pixel, others are not defined.
IDMON	Idle Mode On/Off	'1' = On, "0" = Off
SLPOUT	Sleep In/Out	'1' = Out, "0" = In
NORON	Display Normal Mode On/Off	'1' = Normal Display,
INVON	Inversion Status	'1' = On, "0" = Off
DISON	Display On/Off	'1' = On, "0" = Off
TEON	Tearing effect line on/off	'1' = On, "0" = Off
TELOM	Tearing effect line mode	'0' = mode1, '1' = mode2

"-" Don't care

RDDPM (0Ah): Read Display Power Mode

0AH	RDDPM (Read Display Power Mode)												
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDDPM	0	↑	1	-	0	0	0	0	1	0	1	-	(0Ah)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	-
2nd parameter	1	1	↑	-	BSTON	IDMON	PLTON	SLPOUT	NORON	DISON	-	-	

Description	This command indicates the current status of the display as described in the table below:

Bit	Description	Value
BSTON	Booster Voltage Status	'1' =Booster on, '0' =Booster off
IDMON	Idle mode on/off	'1' = Idle Mode On, '0' = Idle Mode Off
PTLON	Partial mode on/off	'1' =Partial mode on, '0' =Partial mode off,
SLPOUT	Sleep in/out	'1' =Sleep out, '0' =Sleep in,
NORON	Display normal mode on/off	'1' = Normal display, '0' = Partial display,
DISON	Display on/off	'1' =Display on, '0' =Display off,

"-" Don't care

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value (D7 to D0)
Power On Sequence	0000-1000(08h)
S/W Reset	0000-1000(08h)
H/W Reset	0000-1000(08h)

RDDMADCTL (0Bh): Read Display MADCTL

0BH	RDDMADCTL (Read Display MADCTL)												
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDDMADCTL	0	↑	1	-	0	0	0	0	1	0	1	1	(0Bh)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	-
2nd parameter	1	1	↑	-	MY	MX	-	ML	RGB	MH	-	-	

Description

This command indicates the current status of the display as described in the table below:

Bit	Description	Value
MY	Row Address Order (MY)	'1' =Decrement, (Bottom to Top, when MADCTL (36h) D7='1') '0' =Increment, (Top to Bottom, when MADCTL (36h) D7='0')

MX	Column Address Order (MX)	‘1’ =Decrement, (Right to Left, when MADCTL (36h) D6=’1’) ‘0’ =Increment, (Left to Right, when MADCTL (36h) D6=’0’)
ML	Scan Address Order (ML)	‘0’ =Decrement, (LCD refresh Top to Bottom, when MADCTL (36h) D4=’0’) ‘1’=Increment, (LCD refresh Bottom to Top, when MADCTL (36h) D4=’1’)
RGB	RGB/ BGR Order (RGB)	‘1’ =BGR, (When MADCTL (36h) D3=’1’) ‘0’ =RGB, (When MADCTL (36h) D3=’0’)
MH	Horizontal Order	‘0’ =Decrement, (LCD refresh Left to Right, when MADCTL (36h) D2=’0’) ‘1’ =Increment, (LCD refresh Right to Left, when MADCTL (36h) D2=’1’)

“-“ Don’t care

Restriction: There is one dummy parameter when using Parallel interface.

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value (D7 to D0)
Power On Sequence	0000-0000 (00h)
S/W Reset	No change
H/W Reset	0000-0000 (00h)

RDDCOLMOD (0Ch): Read Display Pixel Format

0CH	RDDCOLMOD (Read Display Pixel Format)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDDCOLMOD	0	↑	1	-	0	0	0	0	1	1	0	0	(0Ch)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	-
2nd parameter	1	1	↑	-	-	-	-	-	-	IFPF[1:0]			

Description

This command indicates the current status of the display as described in the table below:

Bit	Description	Value
IFPF[1:0]	Control interface color format	‘01’ = 16 bit/pixel ‘10’ = 18 bit/pixel ‘11’ = 24 bit/pixel

“-“ Don’t care

Restriction

There is one dummy parameter when using Parallel interface.

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	0000-0011 (24 bit/pixel)
S/W Reset	No change
H/W Reset	0000-0011 (24 bit/pixel)

RDDIM (0Dh): Read Display Image Mode

0DH	RDDIM (Read Display Image Mode)											
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDDIM	0	↑	1	-	0	0	0	0	1	1	0	1	(0Dh)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	-
2nd parameter	1	1	↑	-	VSSON	-	INVON	-	-	-	-	-	

Description

This command indicates the current status of the display as described in the table below:

-VSSON: Vertical scrolling on/off

-INVON: Inversion on/off

Others are no define and invalid

"-" Don't care

Restriction

There is one dummy parameter when using Parallel interface.

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	0000-0000
S/W Reset	0000-0000
H/W Reset	0000-0000

RDDSM (0Eh): Read Display Signal Mode

0EH	RDDSM (Read Display Signal Status)											
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDDSM	0	↑	1	-	0	0	0	0	1	1	1	0	(0Eh)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	-
2nd parameter	1	1	↑	-	TEON	TELOM	-	-	-	-	-	EDSI	

Description

This command indicates the current status of the display as described in the table below:

Bit	Description	Value
TEON	Tearing effect line on/off	'1' = ON, '0' = OFF,
TELOM	Tearing effect line mode	'1' = mode2, '0' = mode1,
EDSI	Error on DSI	'1' = Error, '0' = No Error,

	"-" Don't care
Restriction	There is one dummy parameter when using Parallel interface.
Register availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>0000-0000</td></tr><tr><td>S/W Reset</td><td>0000-0000</td></tr><tr><td>H/W Reset</td><td>0000-0000</td></tr></table>

RDDSDR (0Fh): Read Display Self-Diagnostic Result

0FH	RDBST (Read Busy Status)												
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDBST	0	↑	1	-	0	0	0	0	1	1	1	1	(0Fh)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	-
2nd parameter	1	1	↑	-	RLD	-	-	-	-	-	-	CCR	

Description	This command indicates the current status of the display as described in the table below: -RLD: This bit is the Register Loading Detection -CCR: The cmd1 checksum compare result. 0 = Checksum is the same. 1 = Checksum is not the same "-" Don't care
Restriction	There is one dummy parameter when using Parallel interface.
Register availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>0000-0000</td></tr><tr><td>S/W Reset</td><td>0000-0000</td></tr><tr><td>H/W Reset</td><td>0000-0000</td></tr></table>

SLPIN (10h): Sleep in

10H	SLPIN (Sleep In)											
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
SLPIN	0	↑	1	-	0	0	0	1	0	0	0	0	(10h)
parameter	No Parameter												
Description	-This command causes the LCD module to enter the minimum power consumption mode. -In this mode the DC/DC converter is stopped, internal oscillator is stopped, and panel scanning is stopped. -MCU interface and memory are still working and the memory keeps its contents. -Dimming function does not work when there is changing mode from Sleep OUT to Sleep IN. "-" Don't care												
Restriction	-This command has no effect when module is already in sleep in mode. Sleep in mode can only be left by the sleep out command (11h). -It will be necessary to wait 5msec before sending any new commands to a display module following this command to allow time for the supply voltages and clock circuits to stabilize. -It will be necessary to wait 120msec after sending sleep out command (when in sleep in mode) before sending an sleep in command.												

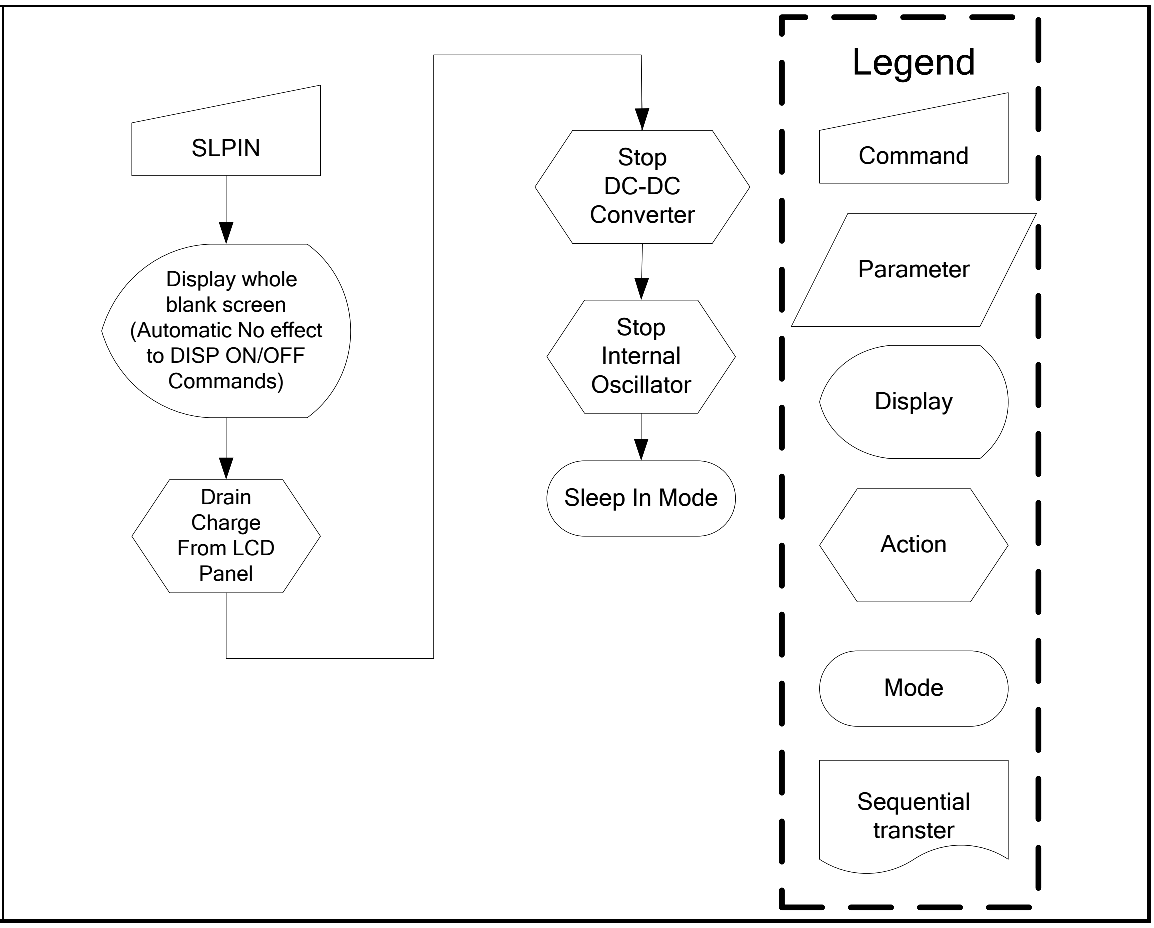
Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	Sleep in mode
S/W Reset	Sleep in mode
H/W Reset	Sleep in mode

Flow Chart



SLPOUT (11h): Sleep Out

11H	SLPOUT (Sleep Out)											
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
SLPOUT	0	↑	1	-	0	0	0	1	0	0	0	1	(11h)

parameter	No Parameter
Description	-This command turn off sleep mode. -In this mode the DC/DC converter is enabled, internal display oscillator is started, and panel scanning is started.
Restriction	-This command has no effect when module is already in sleep out mode. Sleep out mode can only be left by the sleep in command (10h). -It will be necessary to wait 5msec before sending any new commands to a display module following this command to allow time for the supply voltages and clock circuits to stabilize. -It will be necessary to wait 120msec after sending sleep out command (when in sleep in mode) before sending an sleep in command. -The display module runs the self-diagnostic functions after this command is received.

Register availability

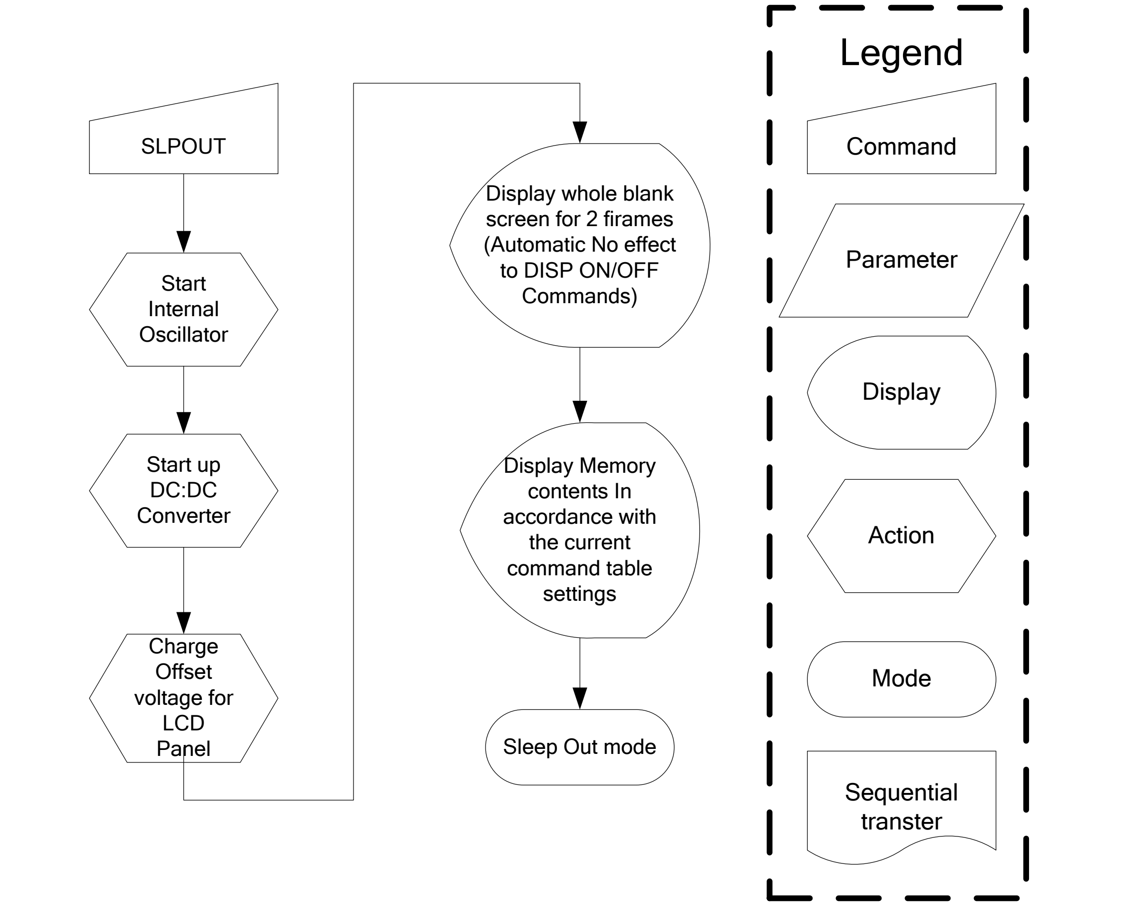
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes

Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	Sleep in mode
S/W Reset	Sleep in mode
H/W Reset	Sleep in mode

Flow Chart



PTLON (12h): Partial Display Mode On

12H	PTLON (Partial Display Mode on)												
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
PTLON	0	↑	1	-	0	0	0	1	0	0	1	0	(12h)
parameter	No Parameter												
Description	-This command turns on Partial mode. The Partial mode window us described by the Partial Area (30h,31h). -To leave Partial mode, the Normal Display Mode On command (13h) should be written. “-“ Don’t care												
Restriction	This command has no effect when partial mode is active.												
Register													

availability		<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>
Default		<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>Normal display mode on</td></tr><tr><td>S/W Reset</td><td>Normal display mode on</td></tr><tr><td>H/W Reset</td><td>Normal display mode on</td></tr></table>
Flow Chart		See Partial Area (30h/31h)

NORON (13h): Normal On

12H	NORON (Normal On)												
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
NORON	0	↑	1	-	0	0	0	1	0	0	1	1	(13h)
parameter	No Parameter												
Description	-This command turns the display to Normal On mode. -Normal display mode on means Normal off mode off. -Exit from NORON by the NOROFF command. “-“ Don’t care												
Restriction	This command has no effect when Normal On mode is active.												
Register availability	(see table below)												
Default	(see table below)												
Flow Chart													

Register availability:

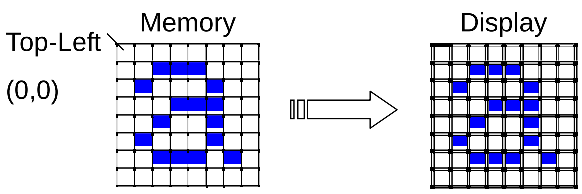
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

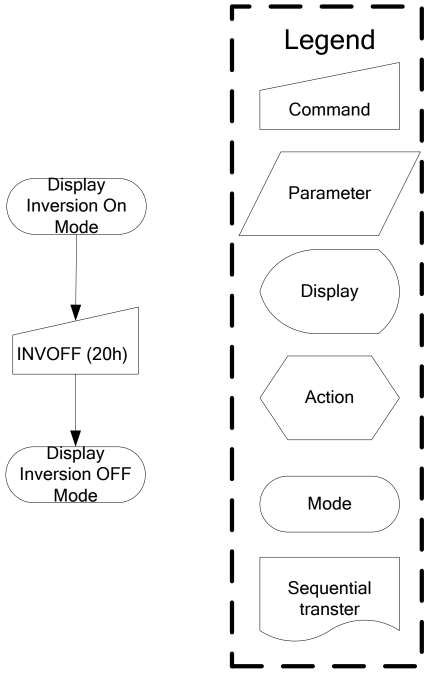
Default:

Status	Default Value
Power On Sequence	Normal display mode on
S/W Reset	Normal display mode on
H/W Reset	Normal display mode on

INVOFF (20h): Display Inversion Off

20H	INVOFF (Display Inversion Off)												
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
INVOFF	0	↑	1	-	0	0	1	0	0	0	0	0	(20h)
parameter	No Parameter												
Description	-This command is used to recover from display inversion mode. “-“ Don’t care (Example)												

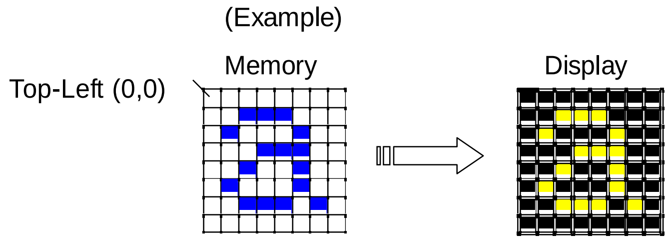


<table>
<tr><td>Restriction</td><td>This command has no effect when module is already in inversion off mode.</td></tr>
<tr><td>Register availability</td><td>
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table>
</td></tr>
<tr><td>Default</td><td>
<table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power On Sequence</td><td>Display inversion off</td></tr>
<tr><td>S/W Reset</td><td>Display inversion off</td></tr>
<tr><td>H/W Reset</td><td>Display inversion off</td></tr>
</table>
</td></tr>
<tr><td>Flow Chart</td><td>


</td></tr>
</table>

INVON (21h): Display Inversion On

21H	INVON (Display Inversion On)												
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
INVON	0	↑	1	-	0	0	1	0	0	0	0	1	(21h)
parameter	No Parameter												
Description	-This command is used to recover from display inversion mode. "-" Don't care												
Restriction	This command has no effect when module is already in inversion on mode.												

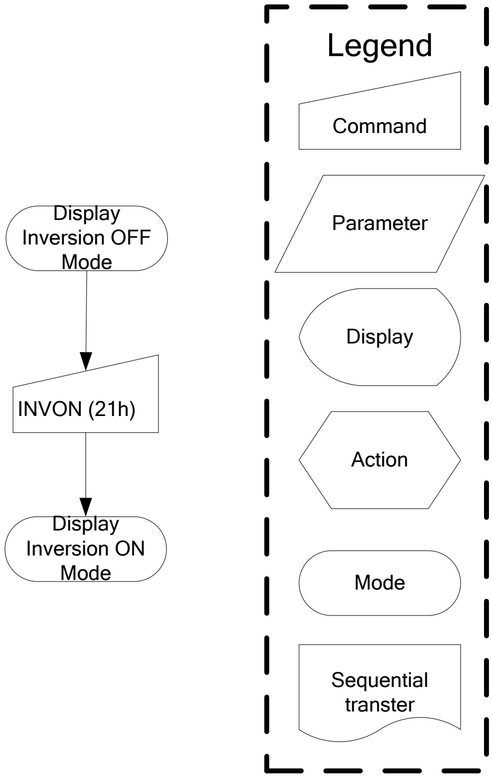


Register availability

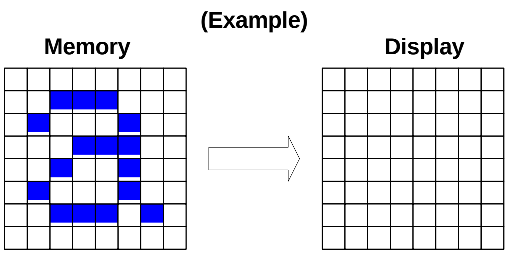
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

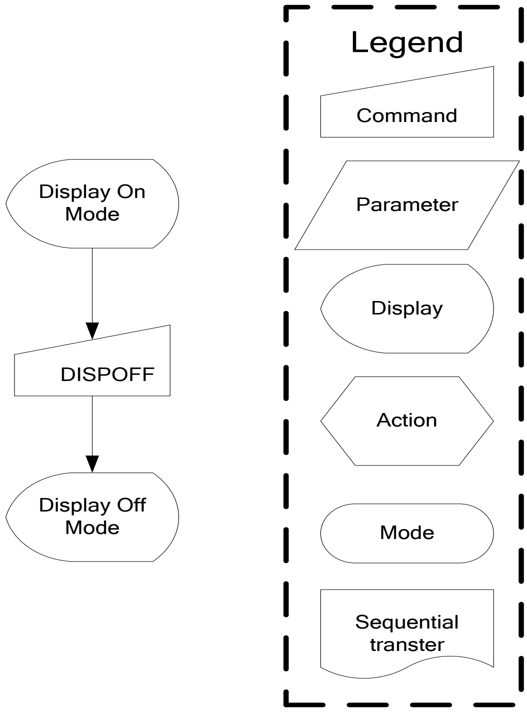
Default

Status	Default Value
Power On Sequence	Display inversion off
S/W Reset	Display inversion off
H/W Reset	Display inversion off

Flow Chart	

DISPOFF (28h): Display Off

28H	DISPOFF (Display Off)												
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
DISPOFF	0	↑	1	-	0	0	1	0	1	0	0	0	(28h)
parameter	No Parameter												
Description	- This command is used to enter into DISPLAY OFF mode. In this mode, the output from Frame Memory is disabled and blank page inserted. - This command makes no change of contents of frame memory. - This command does not change any other status. - There will be no abnormal visible effect on the display. - Exit from this command by Display On (29h) 												
Restriction	This command has no effect when module is already in display off mode.												

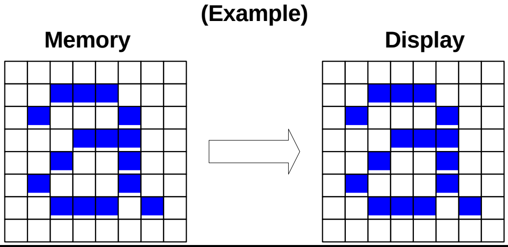
<table>
<tr><td>Register availability</td><td>
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table>
</td></tr>
<tr><td>Default</td><td>
<table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power On Sequence</td><td>Display off</td></tr>
<tr><td>S/W Reset</td><td>Display off</td></tr>
<tr><td>H/W Reset</td><td>Display off</td></tr>
</table>
</td></tr>
<tr><td>Flow Chart</td><td>


</td></tr>
</table>

DISPON (29h): Display On

29H	DISPON (Display On)											
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
DISPO N	0	↑	1	-	0	0	1	0	1	0	0	1	(29h)
parameter	No Parameter												
Description	- This command is used to recover from DISPLAY OFF mode. - Output from the Frame Memory is enabled. - This command makes no change of contents of frame memory. - This command does not change any other status.												
Restriction	This command has no effect when module is already in display on mode.												

(Example)

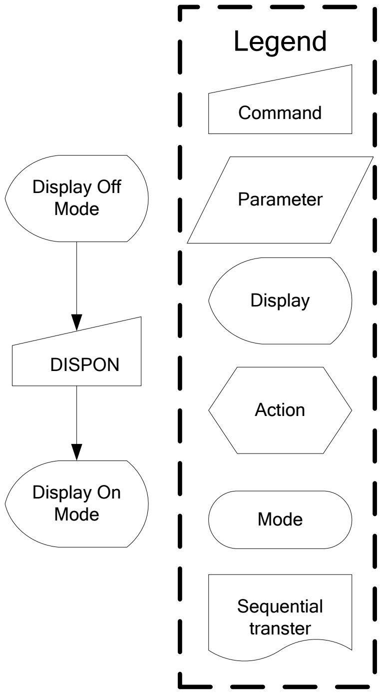


Register availability

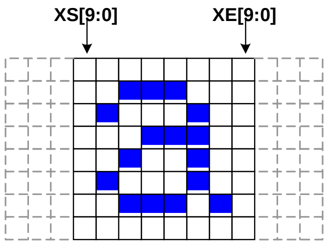
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	Display off
S/W Reset	Display off
H/W Reset	Display off

Flow Chart	

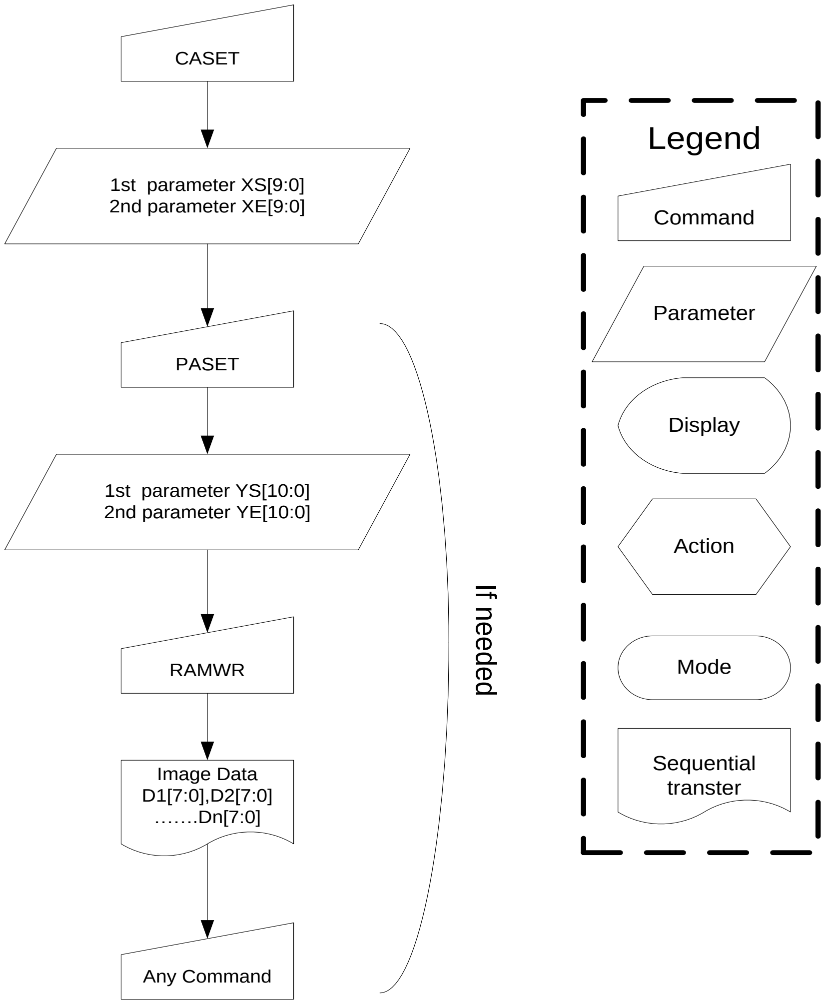
CASET (2Ah): Column Address Set

2AH	CASET (Column Address Set)												
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
CASET	0	↑	1	-	0	0	1	0	1	0	1	0	(2Ah)
1st parameter	1	↑	1	-	-	-	-	-	-	-	XS[9:8]		
2nd parameter	1	↑	1	-	XS[7:0]								
3rd parameter	1	↑	1	-	-	-	-	-	-	-	XE[9:8]		
4th parameter	1	↑	1	-	XE[7:0]								
Description	-The value of XS [9:0] and XE [9:0] are referred when RAMWR command comes. -Each value represents one column line in the Frame Memory. 												
Restriction	XS [9:0] always must be equal to or less than XE [9:0] XS [9:0] & "XE [9:0] – XS [9:0] +1" always must be set as a multiple of 4. When XS [9:0] or XE [9:0] is greater than maximum address like below, data of out of range will be ignored. (Parameter range: 0 < XS [9:0] < XE [9:0] < 559 (022Fh))												
Register availability	Status | Availability												

	Normal Mode On, Idle Mode Off, Sleep Out	Yes
	Normal Mode On, Idle Mode On, Sleep Out	Yes
	Partial Mode On, Idle Mode Off, Sleep Out	Yes
	Partial Mode On, Idle Mode On, Sleep Out	Yes
	Sleep In	Yes

Default	Status	Default Value	
	Power On Sequence	XS[9:0]=0000h	XE[9:0]=0243h
	S/W Reset	XS[9:0]=0000h	XE[9:0]=0243h
	H/W Reset	XS[9:0]=0000h	XE[9:0]=0243h

Flow Chart



RASET (2Bh): Row Address Set

2BH	RASET (Row Address Set)											
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RASET	0	↑	1	-	0	0	1	0	1	0	1	1	(2Bh)
1st parameter	1	↑	1	-	-	-	-	-	-	YS[10:8]			
2nd parameter	1	↑	1	-	YS[7:0]								

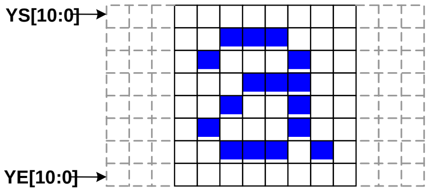
3rd parameter	1	↑	1	-	-	-	-	-	-	YE[10:8]	
4th parameter	1	↑	1	-	YE[7:0]						

Description

-This command is used to define area of frame memory where MCU can access.

-The value of YS [10:0] and YE [10:0] are referred when RAMWR command comes.

-Each value represents one page line in the Frame Memory.



Restriction

YS [10:0] always must be equal to or less than YE [9:0]

When YS [10:0] or YE [10:0] is greater than maximum address like below, data of out of range will be ignored.

(Parameter range: 0 < YS [10:0] < YE [10:0] < 1023 (03FFh))

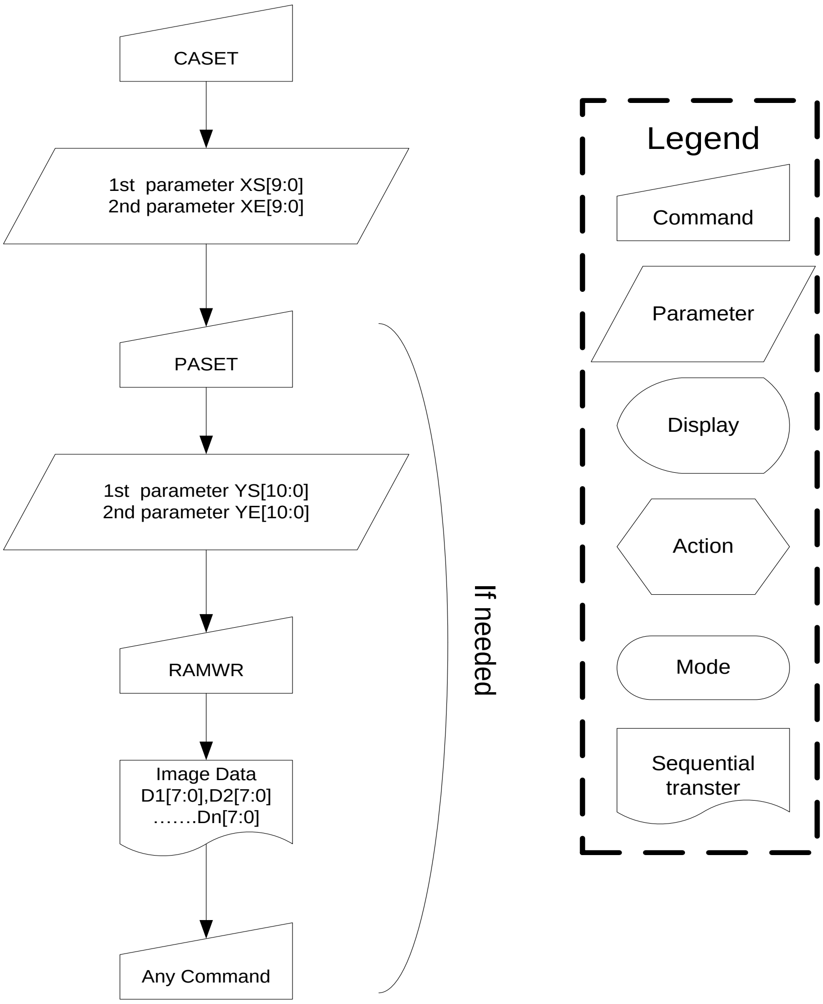
Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value	
Power On Sequence	YS[10:0]=0000h	YE[10:0]=0243h
S/W Reset	YS[10:0]=0000h	YE[10:0]=0243h,
H/W Reset	YS[10:0]=0000h	YE[10:0]=0243h

Flow Chart	



RAMWR (2Ch): Memory Write

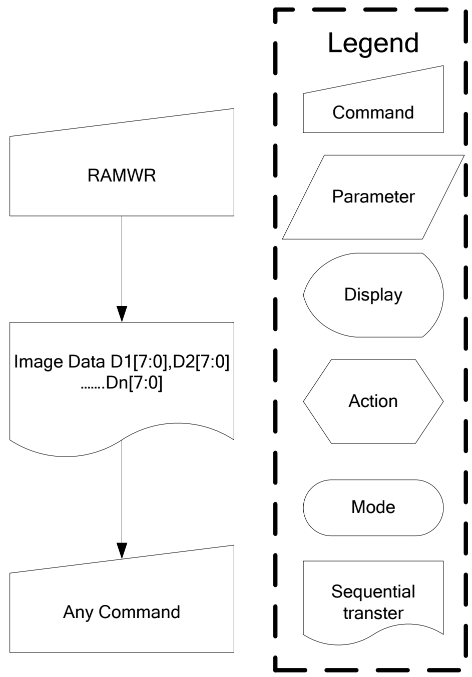
2CH	RAMWR (Memory Write)												
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RAMWR	0	↑	1	-	0	0	1	0	1	1	0	0	(2Ch)
1st parameter	1	↑	1	-	D1[7:0]								-
…	1	↑	1	-	.								-
N parameter	1	↑	1	-	Dn[7:0]								-
Description	-This command is used to transfer data from MCU to frame memory. -When this command is accepted, the column register and the page register are reset to the start column/start page positions. -The start column/start page positions are different in accordance with MADCTL setting. -Sending any other command can stop frame write.												
Restriction	In all color modes, there is no restriction on length of parameters.												
Register availability	Status: Normal Mode On, Idle Mode Off, Sleep Out — Availability: Yes												

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes

	Normal Mode On, Idle Mode On, Sleep Out	Yes
	Partial Mode On, Idle Mode Off, Sleep Out	Yes
	Partial Mode On, Idle Mode On, Sleep Out	Yes
	Sleep In	Yes

	Status	Default Value
Default	Power On Sequence	Contents of memory is set randomly
	S/W Reset	Contents of memory is not cleared
	H/W Reset	Contents of memory is not cleared

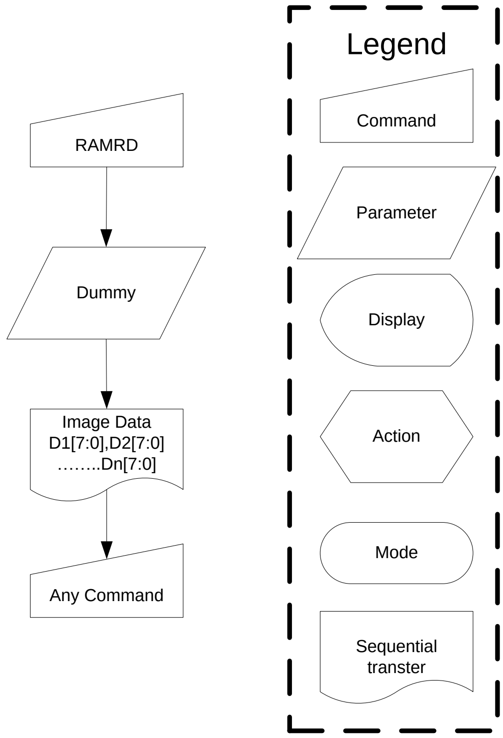
Flow Chart



RAMRD (2Eh): Memory Read

2CH	RAMRD (Memory Read)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RAMRD	0	↑	1	-	0	0	1	0	1	1	1	0	(2Eh)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	-
2nd parameter	1	1	↑	-	D1[7:0]								-
:	1	1	↑	:	:								:
(N+1)th parameter	1	1	↑	-	Dn[7:0]								-
Description	-This command is used to transfer data from frame memory to MCU. -When this command is accepted, the column register and the row register are reset to the Start Column/Start Row positions. -The Start Column/Start Row positions are different in accordance with MADCTL setting. -Then D[7:0] is read back from the frame memory and the column register and the row register incremented												

<table>
<tr><td></td><td>-Frame Read can be cancelled by sending any other command.
-The data color coding is fixed to 18-bit in reading function. Please see section 9.8 “Data color coding” for color coding (18-bit cases), when there is used 8, 9 data lines for image data.

Note1: The Command 3Ah should be set to 66h when reading pixel data from frame memory.</td></tr>
<tr><td>Restriction</td><td></td></tr>
<tr><td>Register availability</td><td>
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table>
</td></tr>
<tr><td>Default</td><td>
<table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power On Sequence</td><td>Contents of memory is set randomly</td></tr>
<tr><td>S/W Reset</td><td>Contents of memory is not cleared</td></tr>
<tr><td>H/W Reset</td><td>Contents of memory is not cleared</td></tr>
</table>
</td></tr>
<tr><td>Flow Chart</td><td>


</td></tr>
</table>

RPTLAR (30h): Partial Area

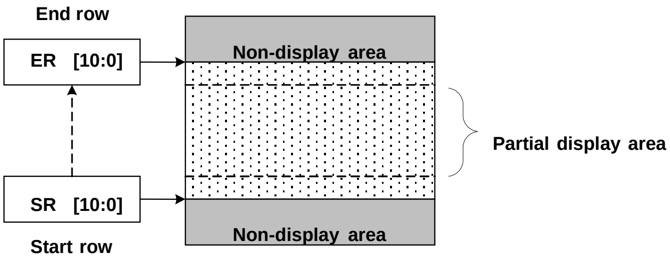
30H	PTLAR (Partial Area)												
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
PTLAR	0	↑	1	-	0	0	1	1	0	0	0	0	(30h)
1st parameter	1	↑	1	-	-	-	-	-	-	SR[10:8]			
2nd parameter	1	↑	1	-	SR[7:0]								
3rd parameter	1	↑	1	-	-	-	-	-	-	ER[10:8]			
4th parameter	1	↑	1	-	ER[7:0]								

Description

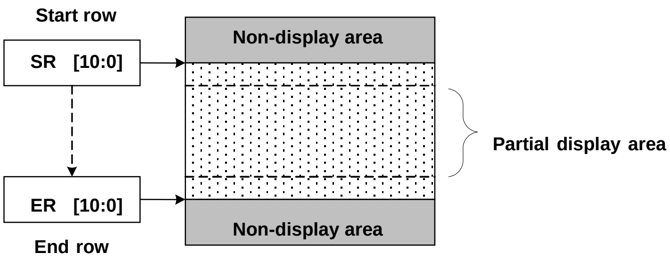
-This command defines the partial mode's display area.

-There are 4 parameters associated with this command, the first defines the Start Row (SR) and the second the End Row (ER), as illustrated in the figures below. PSL and PEL refer to the Frame Memory row address counter.

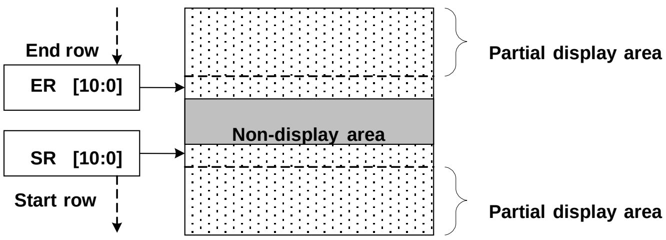
-If End Row > Start Row, when MADCTL ML='1'



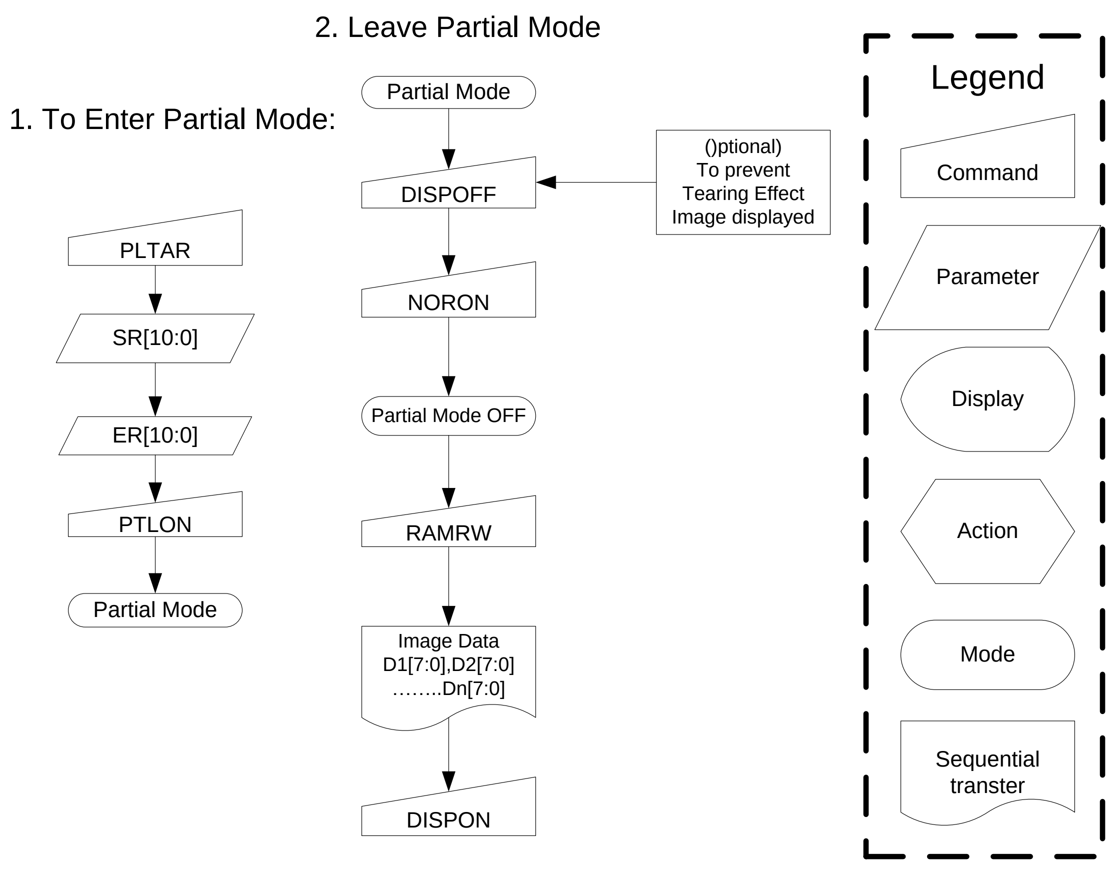
-If End Row > Start Row, when MADCTL ML='0'



-If End Row < Start Row, when MADCTL ML='0'



-If End Row = Start Row then the Partial Area will be one row deep.

<table>
<tr><td>Restriction</td><td>Each detail initial value by the display resolution will be updated.</td></tr>
<tr><td>Register availability</td><td>
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table>
</td></tr>
<tr><td>Default</td><td>
<table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power On Sequence</td><td>SR[10:0]=0000h, ER=018Fh</td></tr>
<tr><td>S/W Reset</td><td>SR[10:0]=0000h, ER=018Fh</td></tr>
<tr><td>H/W Reset</td><td>SR[10:0]=0000h, ER=018Fh</td></tr>
</table>
</td></tr>
<tr><td>Flow Chart</td><td>


</td></tr>
</table>

VSCRDEF (33h): Vertical Scrolling Definition

33H	VSCRDEF (Vertical Scrolling Definition)												
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
VSCRDEF	0	↑	1	-	0	0	1	1	0	0	1	1	(33h)

1st parameter	1	↑	1	-	-	-	-	-	-	TFA[10:8]			
2nd parameter	1	↑	1	-	TFA[7:0]								
3rd parameter	1	↑	1	-	-	-	-	-	-	VSA[10:8]			
4th parameter	1	↑	1	-	VSA[7:0]								
5th parameter	1	↑	1		-	-	-	-	-	BFA[10:8]			
6th parameter	1	↑	1		BFA[7:0]								

Description

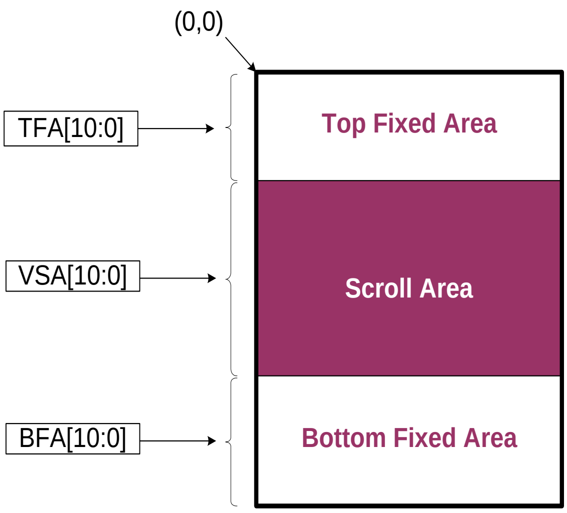
-This command just defines the Vertical Scrolling Area of the display and not performs vertical scroll

-The 1st & 2nd parameter TFA [10:0] describes the Top Fixed Area (in No. of lines from Top of the Frame Memory and Display).

-The 3rd & 4th parameter VSA [10:0] describes the height of the Vertical Scrolling Area (in No. of lines of the Frame Memory [not the display] from the Vertical Scrolling Start Address) The first line appears immediately after the bottom most line of the Top Fixed Area.

-The 4th & 5th parameter BFA [10:0] describes the Bottom Fixed Area (in No. of lines from Bottom of the Frame Memory and Display).

TFA, VSA and BFA refer to the Frame Memory Line Pointer



Restriction

The condition is TFA+VSA+BFA = YS[10:0]+YE[10:0], otherwise Scrolling mode is undefined.

Register availability

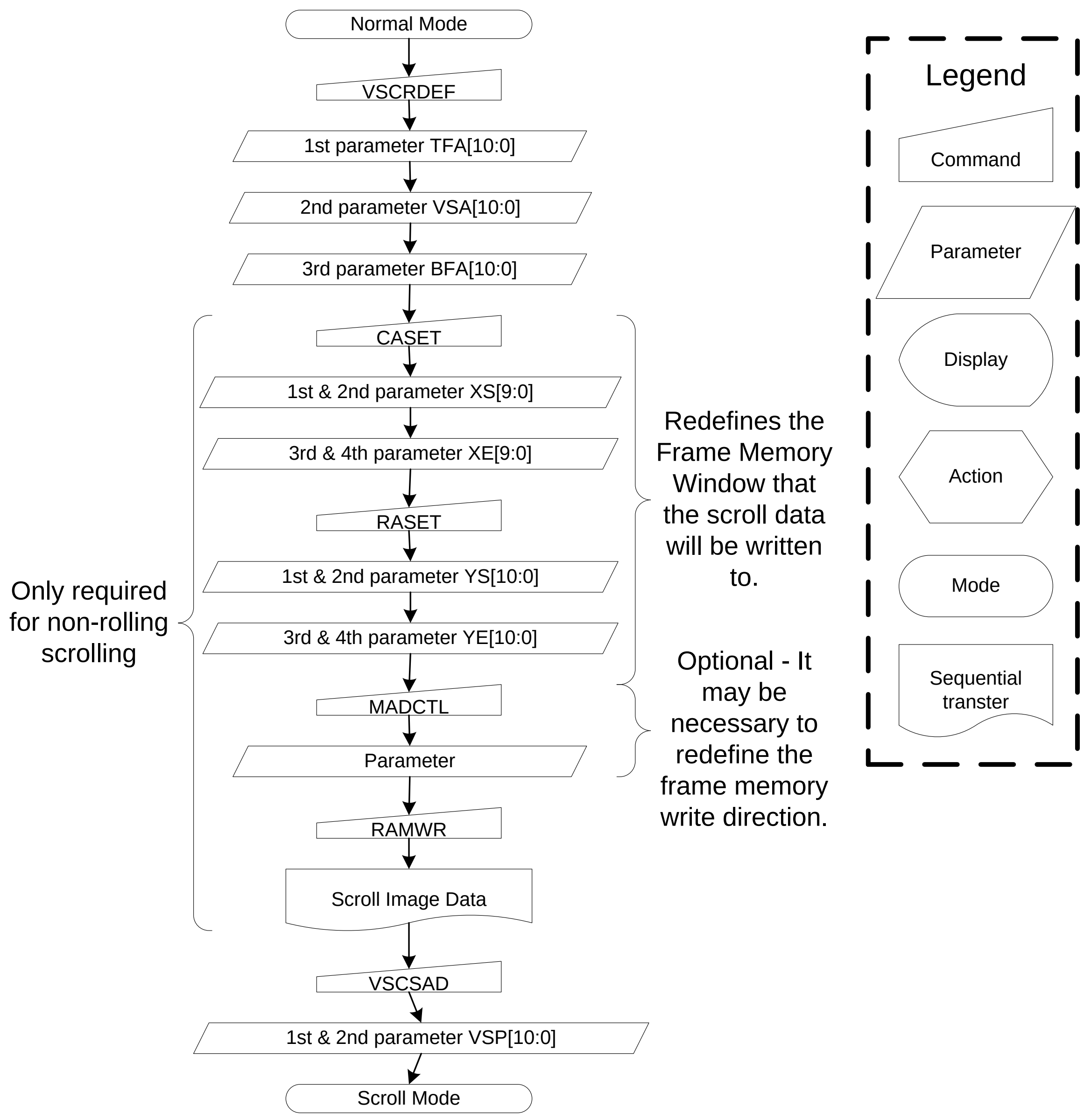
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value		
Power On Sequence	TFA[10:0] = 0000h	VSA[10:0] = 0000h	BFA[10:0] = 0000h
S/W Reset	TFA[10:0] = 0000h	VSA[10:0] = 0000h	BFA[10:0] = 0000h
H/W Reset	TFA[10:0] = 0000h	VSA[10:0] = 0000h	BFA[10:0] = 0000h

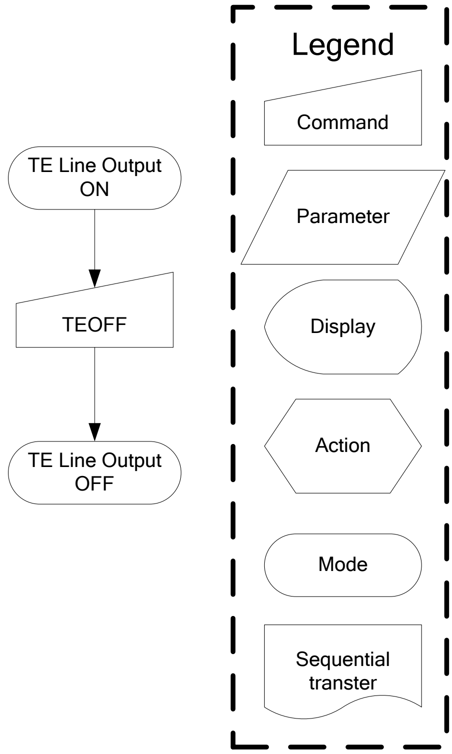
Flow Chart

1. TO Enter Vertical Scroll Mode:



TEOFF (34h): Tearing Effect Line OFF

34H	TEOFF (Tearing Effect Line OFF)											
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
TEOFF	0	↑	1	-	0	0	1	1	0	1	0	0	(34h)

<table>
<tr><td>parameter</td><td>No Parameter</td></tr>
<tr><td>Description</td><td>-This command is used to turn OFF (Active Low) the Tearing Effect output signal from the TE signal line.</td></tr>
<tr><td>Restriction</td><td>This command has no effect when tearing effect output is already off..</td></tr>
<tr><td>Register availability</td><td>
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table>
</td></tr>
<tr><td>Default</td><td>
<table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power On Sequence</td><td>Off</td></tr>
<tr><td>S/W Reset</td><td>Off</td></tr>
<tr><td>H/W Reset</td><td>Off</td></tr>
</table>
</td></tr>
<tr><td>Flow Chart</td><td>


</td></tr>
</table>

TEON (35h): Tearing Effect Line On

35H	TEON (Tearing Effect Line On)											
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
TEON	0	↑	1	-	0	0	1	1	0	1	0	1	(35h)
parameter	1	↑	1	-	0	0	0	0	0	0	0	TE_MD	-

Description

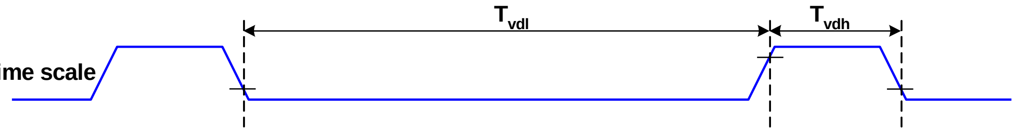
-This command is used to turn ON the Tearing Effect output signal from the TE signal line.

-This output is not affected by changing MADCTL bit ML.

-The Tearing Effect Line On has one parameter, which describes the mode of the Tearing Effect Output Line:

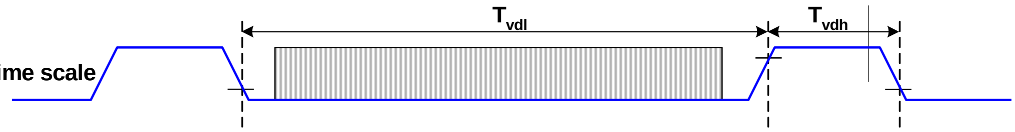
-When TEM ='0': The Tearing Effect output line consists of V-Blanking information only

Vertical time scale



-When TEM ='1': The Tearing Effect output Line consists of both V-Blanking and H-Blanking information

Vertical time scale



Note: During Sleep In Mode with Tearing Effect Line On, Tearing Effect Output pin will be active Low.

Restriction

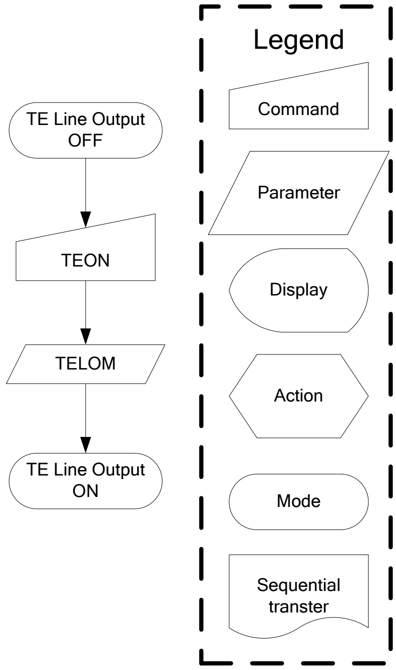
This command has no effect when tearing effect output is already on.

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	Off
S/W Reset	Off
H/W Reset	Off

Flow Chart	

MADCTL (36h): Memory Data Access Control

36H	MADCTL (Memory Data Access Control)												
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
MADCTL	0	↑	1	-	0	0	1	1	0	1	1	0	(36h)
parameter	1	↑	1	-	MY	MX	-	ML	RGB	MH	-	-	

Description

-This command defines read/ write scanning direction of frame memory.

Bit	NAME	DESCRIPTION
D7	MY	Page Address Order
D6	MX	Column Address Order
D4	ML	Line Address Order
D3	RGB	RGB/BGR Order
D2	MH	Display Data Latch Order

-Bit Assignment

Bit D7- Page Address Order

“0” = Top to Bottom (When MADCTL D7=”0”).

“1” = Bottom to Top (When MADCTL D7=”1”).

Bit D6- Column Address Order

“0” = Left to Right (When MADCTL D6=”0”).

“1” = Right to Left (When MADCTL D6=”1”).

Bit D4- Line Address Order

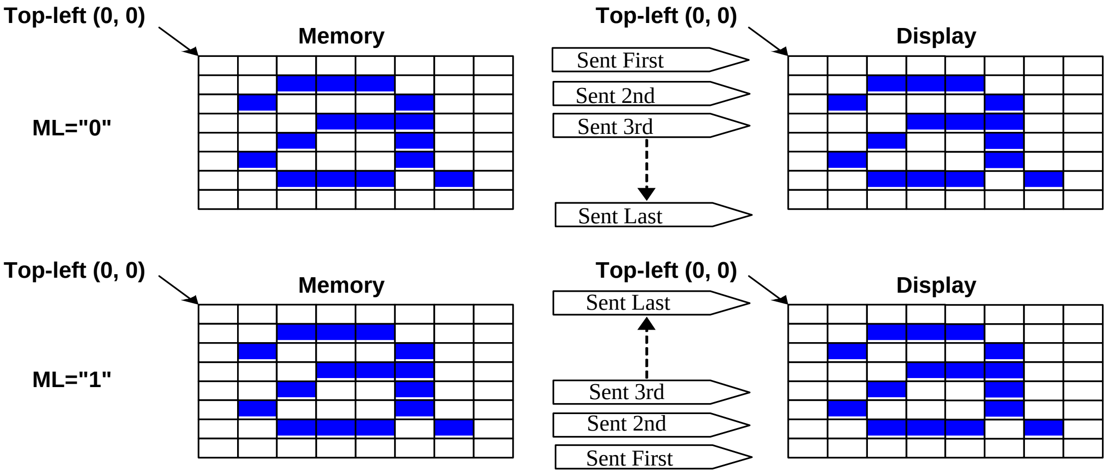
“0” = LCD Refresh Top to Bottom (When MADCTL D4=”0”)

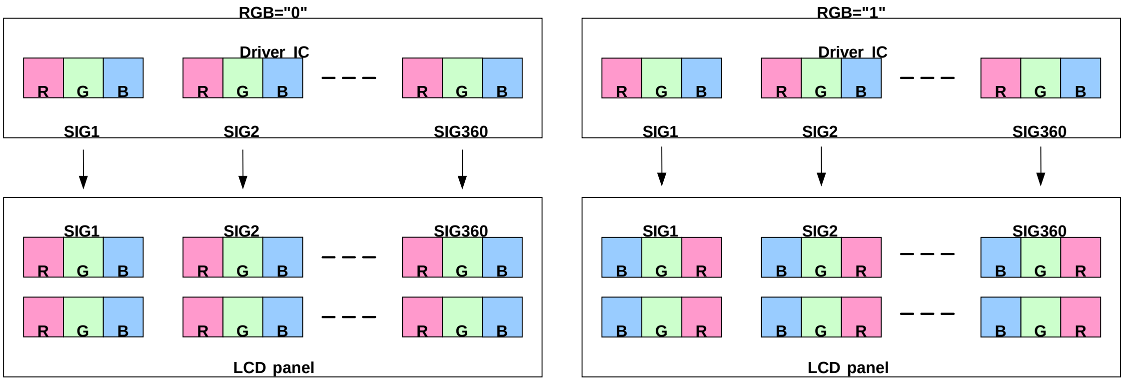
“1” = LCD Refresh Bottom to Top (When MADCTL D4=”1”)

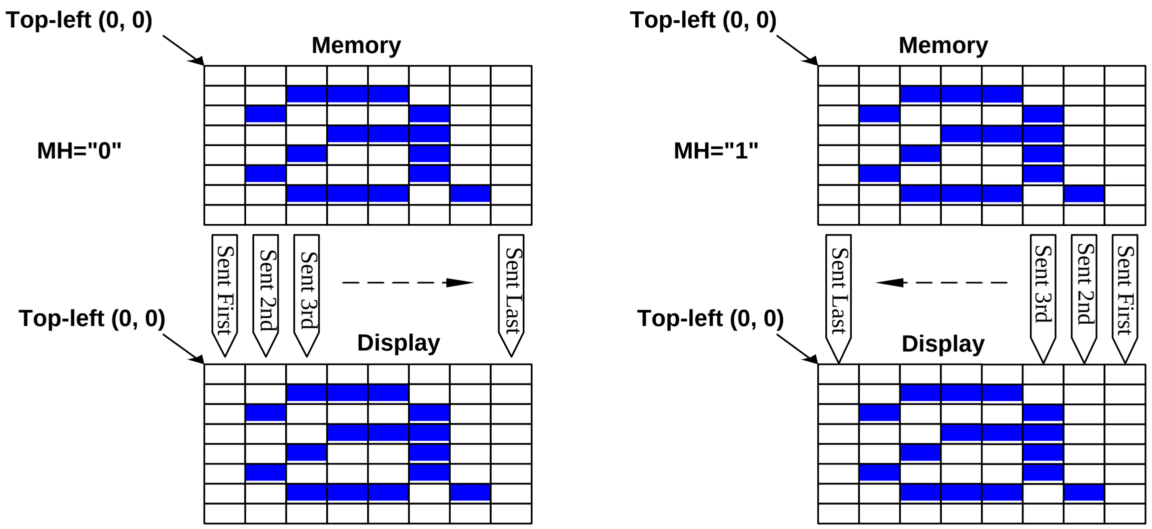
Bit D3- RGB/BGR Order

“0” = RGB (When MADCTL D3=”0”)

“1” = BGR (When MADCTL D3=”1”)





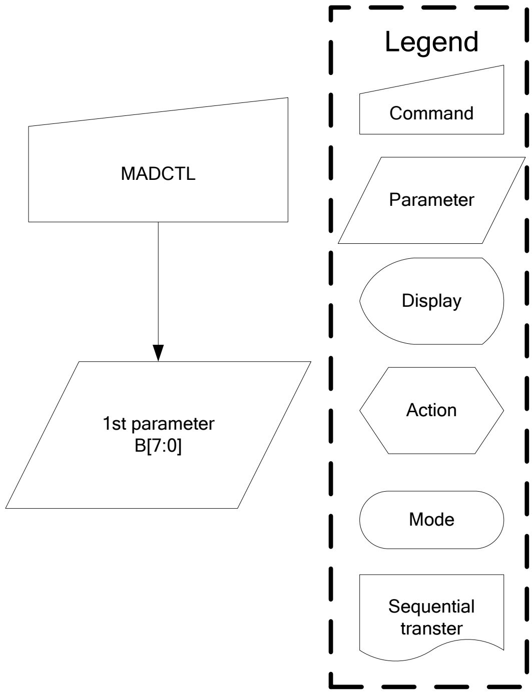


Register availability

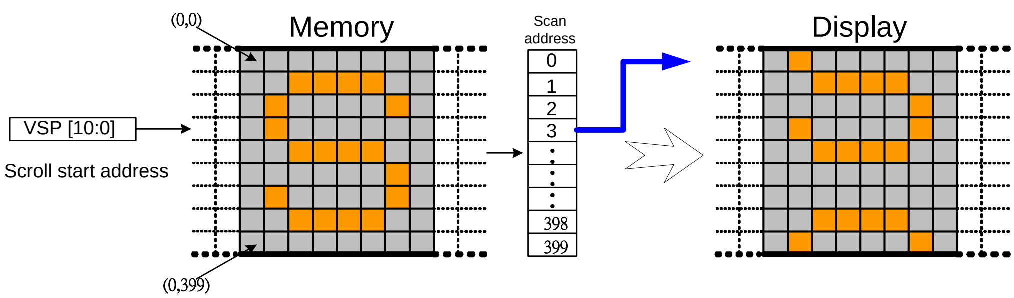
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

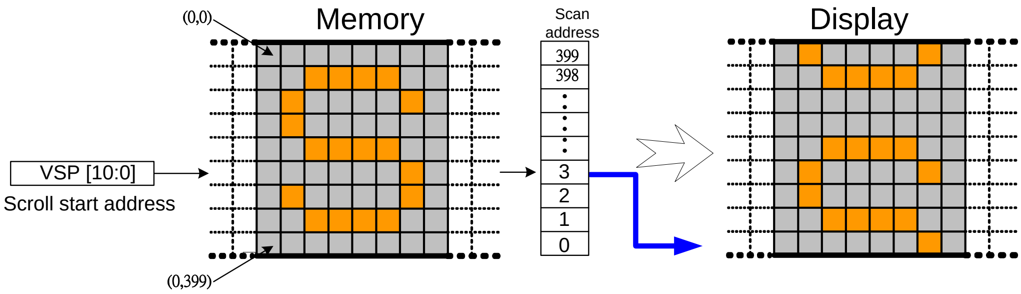
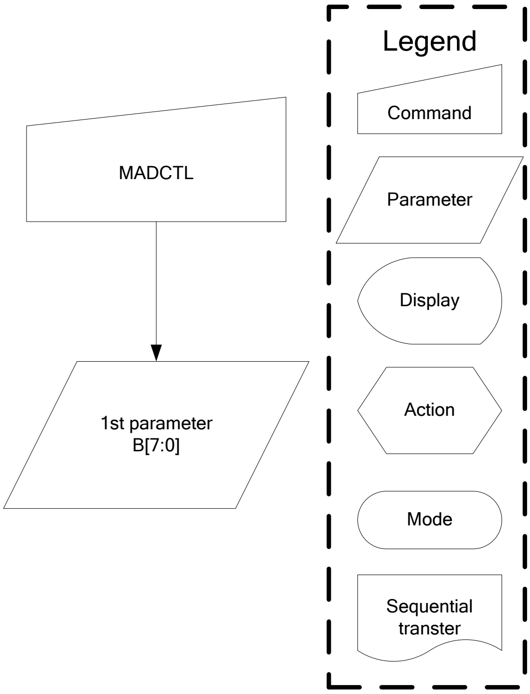
Default

Status	Default Value
Power On Sequence	0000h
S/W Reset	No change
H/W Reset	0000h

Flow Chart	

VSCSAD (37h): Vertical Scroll Start Address of RAM

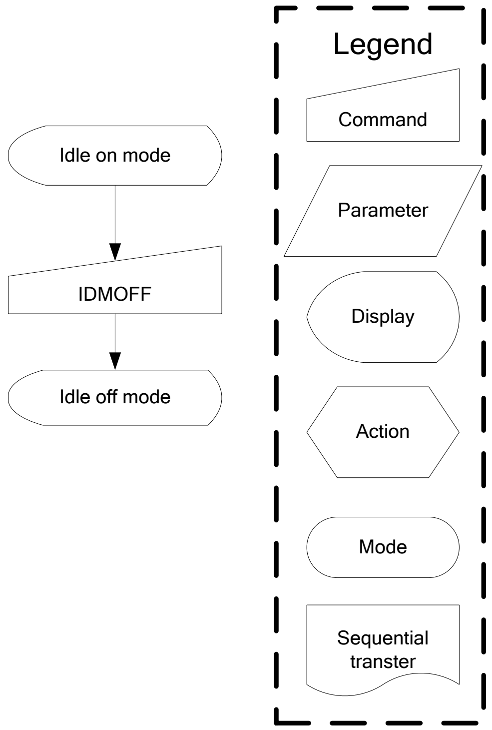
37H	VSCSAD (Vertical Scroll Start Address of RAM)											
Inst / Para	D/CX	WRX	RDX	D8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
VSCSAD	0	↑	1	-	0	0	1	1	0	1	1	1	(37h)
1ST parameter	1	↑	1	-	-	-	-	-	-	VSP[10:8]			
2ND parameter	1	↑	1	-	VSP[7:0]								
Description	-This command is used together with Vertical Scrolling Definition (33h). -These two commands describe the scrolling area and the scrolling mode. -The Vertical Scrolling Start Address command has one parameter which describes which line in the Frame Memory will be written as the first line after the last line of the Top Fixed Area on the display as illustrated below: When ML=0 Example: When Top Fixed Area = Bottom Fixed Area = 00, vertical Scrolling Area = 390 and VSP = '3'  When ML=1 Example:												

	When Top Fixed Area = Bottom Fixed Area = 00, vertical Scrolling Area = 400 and VSP = '3'  NOTE: When new Pointer position and Picture Data are sent, the result on the display will happen at the next Panel Scan to avoid tearing effect. VSP refers to the Frame Memory line Pointer
Register availability	Since the value of the vertical scrolling start address is absolute (with reference to the frame memory), it must not enter the fixed area (defined by Vertical Scrolling Definition (33h)- otherwise undesirable image will be displayed on the panel)
Default	(see table below)
Flow Chart	

Status	Default Value
Power On Sequence	0000h
S/W Reset	No change
H/W Reset	0000h

IDMOFF (38h): Idle Mode Off

38H	IDMOFF (Idle Mode Off)												
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
IDMOFF	0	↑	1	-	0	0	1	1	1	0	0	0	(38h)

<table>
<tr><td>parameter</td><td>No Parameter</td></tr>
<tr><td>Description</td><td>-This command is used to recover from Idle mode on.
-In the idle off mode,
1. LCD can display 65k ,262k or 16m colors.
2. Normal frame frequency is applied.</td></tr>
<tr><td>Restriction</td><td>This command has no effect when module is already in idle off mode</td></tr>
<tr><td>Register availability</td><td>
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table>
</td></tr>
<tr><td>Default</td><td>
<table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power On Sequence</td><td>Idle mode off</td></tr>
<tr><td>S/W Reset</td><td>Idle mode off</td></tr>
<tr><td>H/W Reset</td><td>Idle mode off</td></tr>
</table>
</td></tr>
<tr><td>Flow Chart</td><td>


</td></tr>
</table>

IDMON (39h): Idle Mode On

39H	IDMON (Idle Mode On)											
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
IDMON	0	↑	1	-	0	0	1	1	1	0	0	1	(39h)
parameter	No Parameter												

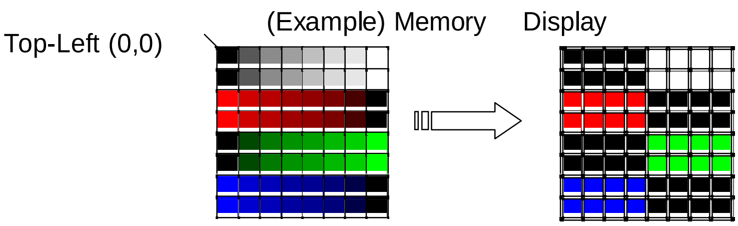
Description

-This command is used to enter into Idle mode on.

-There will be no abnormal visible effect on the display mode change transition.

-In the idle on mode,

1. Color expression is reduced. The primary and the secondary colors using MSB of each R,G and B in the Frame Memory, 8 color depth data is displayed.
2. 8-Color mode frame frequency is applied.
3. Exit from IDMON by Idle Mode Off (38h) command



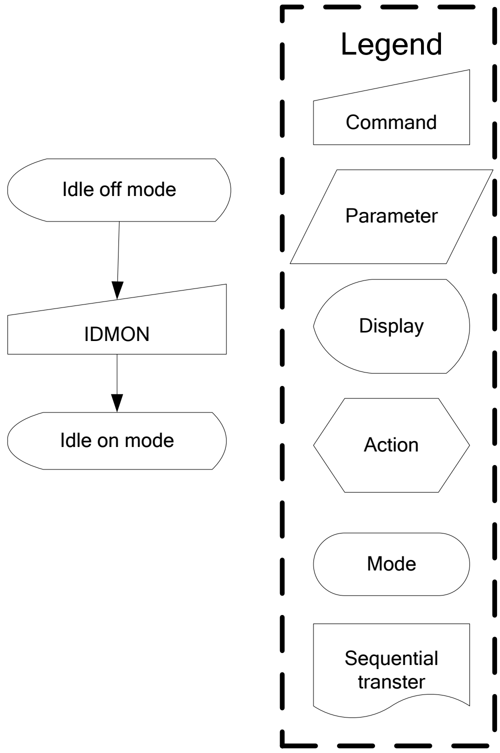
Color	R5 R4 R3 R2 R1 R0	G5 G4 G3 G2 G1 G0	B5 B4 B3 B4 B1 B0
Black	0xxxxx	0xxxxx	0xxxxx
Blue	0xxxxx	0xxxxx	1xxxxx
Red	1xxxxx	0xxxxx	0xxxxx
Magenta	1xxxxx	0xxxxx	1xxxxx
Green	0xxxxx	1xxxxx	0xxxxx
Cyan	0xxxxx	1xxxxx	1xxxxx
Yellow	1xxxxx	1xxxxx	0xxxxx
White	1xxxxx	1xxxxx	1xxxxx

Restriction

This command has no effect when module is already in idle off mode

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>Idle mode off</td></tr><tr><td>S/W Reset</td><td>Idle mode off</td></tr><tr><td>H/W Reset</td><td>Idle mode off</td></tr></table>
Flow Chart	

COLMOD (3Ah): Interface Pixel Format

3AH	COLMOD (Interface Pixel Format)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
COLMOD	0	↑	1	-	0	0	1	1	1	0	1	0	(3Ah)
Parameter	1	↑	1	-	-	-			-	-	IFPF[1:0]		

Description

This command is used to define the format of RGB picture data, which is to be transferred via the MCU interface. The formats are shown in the table:

1st parameter:

Bit	Name	Description
IFPF[1:0]	Control interface color format	'01' = 16bit/pixel '10' = 18bit/pixel '11' = 24bit/pixel

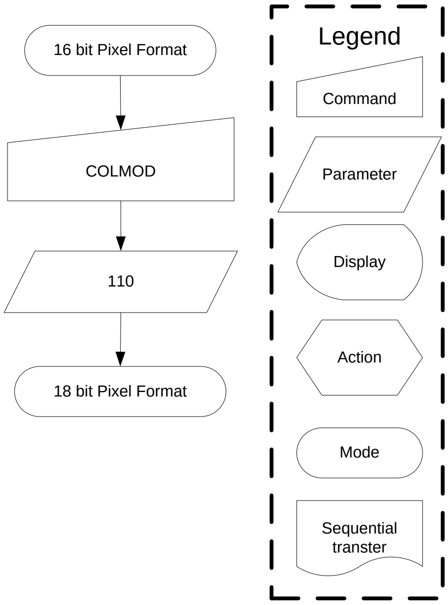
Note1: In 16-bit/Pixel,18-bit/Pixel,24-bit/Pixel mode, the LUT is applied to transfer data into the Frame Memory.

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	24bit/pixel
S/W Reset	No change
H/W Reset	24bit/pixel

Flow Chart	

WRMEMC (3Ch): Write Memory Continue

3CH	WRMEMC (Write Memory Continue)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
WRMEMC	0	↑	1	-	0	0	1	0	1	1	0	0	(3Ch)
1ST parameter	1	↑	1	-	D1[7:0]								-
⋮	1	↑	1	-	Dx[7:0]								-
Nth parameter	1	↑	1	-	Dn[7:0]								-
Description	-This command transfers image data from the host processor to the display module's frame memory continuing from the pixel location following the previous write memory continue or memory write command. - Data is written continuing from the pixel location after the write range of the previous memory write or write memory continue. The column register is then incremented and pixels are written to the frame memory until the column register equals the end column (XE) value. The column register is then reset to XS and the page register is incremented. Pixels are written to the frame memory until the page register equals the end page (YE) value and the column register equals the XE value, or the host processor sends another command. If the number of pixels exceeds (XE-XS+1)*(YE-YS+1) the extra pixels are ignored.												

Condition	Column	Page
Command 2C is accepted	Return to “Start Column”	Return to “Start Page”
Read/Write RAM action	Increment by 1	No change
Column value is large than “End Column”	Return to “Start Column”	Increment by 1
Page value is large than “End Page”	Return to “Start Column”	Return to “Start Page”

Restriction

A memory write should follow a column address set or page address set to define the write address. Otherwise, data written with write memory continue is written to undefined addresses.

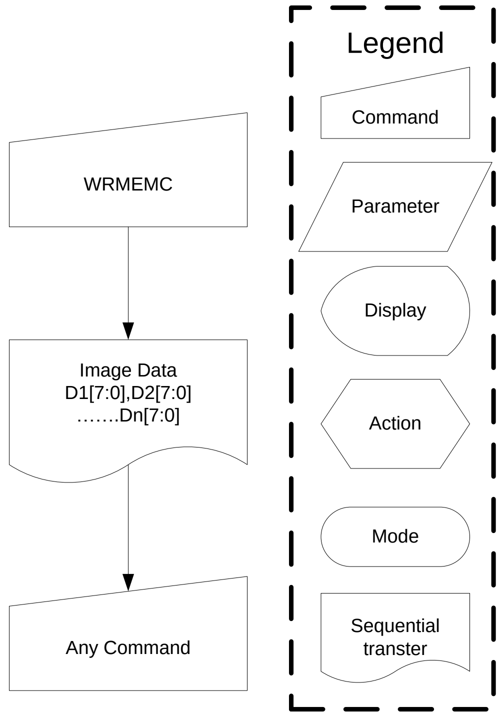
Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	Contents of memory is set randomly
S/W Reset	Contents of memory is not cleared
H/W Reset	Contents of memory is not cleared

Flow Chart



RDMEMC (3Eh): Read Memory Continue

3EH	RDMEMC (Read Memory Continue)											
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDMEMC	0	↑	1	-	0	0	1	1	1	1	1	0	(3Eh)
1ST parameter	1	1	↑	-	-	-	-	-	-	-	-	-	-
2nd parameter	1	1	↑	-	D1[7:0]								-
⋮	1	1	↑		Dx[7:0]								
(N+1)th parameter	1	1	↑	-	Dn[7:0]								-
Description	-This command transfers image data from the host processor to the display module's frame memory continuing from the pixel location following the previous read memory continue or memory read command. - Pixels are read continuing from the pixel location after the read range of the previous memory read or read memory continue. The column register is then incremented and pixels are read from the frame memory until the column register equals the end column (XE) value. The column register is then reset to XS and the page register is incremented. Pixels are read from the frame memory until the page register equals the end page (YE) value and the column register equals the XE value, or the host processor sends another command.												
Restriction	Regardless of the color mode set in interface pixel format, the pixel format returned by read memory continue is always 18-bit so there is no restriction on the length of data												
Register availability	(see table below)												
Default	(see table below)												

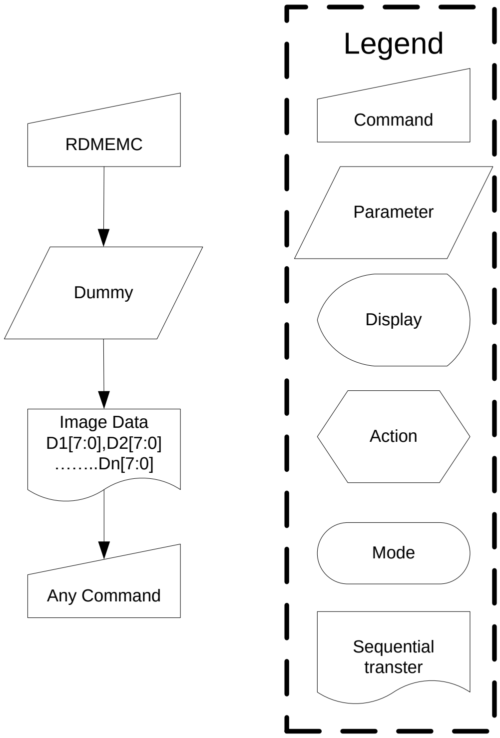
Register availability:

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default:

Status	Default Value
Power On Sequence	Contents of memory is set randomly
S/W Reset	Contents of memory is not cleared

	H/W Reset	Contents of memory is not cleared

Flow Chart	

SPIRDMD (42h): SPI Read Mode

42H	SPIRDMD (SPI Read Mode)												
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
COLMOD	0	↑	1	-	0	1	0	0	0	0	1	0	(42h)
Parameter	1	↑	1	-	SPI_RD_CAP[7:0]								
Description	- This command for SPI Read Mode. IF SPI Read ,SPI_RD_CAP[7:0]=0x5A												
Register availability													

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes

Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

TESLWR (44h): Write Tear Scanline

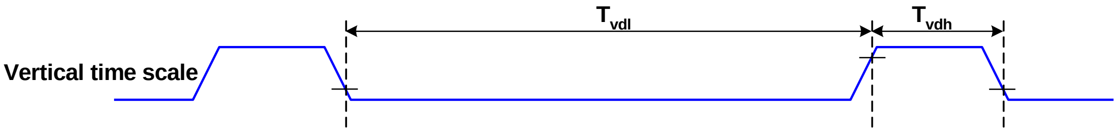
44H	STE (Write Tear Scanline)											
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
TESLWR	0	↑	1	-	0	1	0	0	0	1	0	0	(44h)
1st parameter	1	↑	1	-	-	-	-	-	TE_SL[11:8]				
2nd parameter	1	↑	1	-	TE_SL[7:0]								

Description

-This command turns on the display module's Tearing Effect output signal on the TE signal line when the display module reaches line N.

-The tearing effect line on has one parameter that describes the tearing effect output line mode.

-The tearing effect output line consist of V-blanking information only.



Note that set tear scanline with N=0 is equivalent to tearing effect line on with TEM=0.

The tearing effect output line shall be active low when the display module is in sleep mode

Restriction

This command takes effect on the frame following the current frame. Therefore, if the tear effect (TE) output is already on, the TE output shall continue to operate as programmed by the previous tearing effect line on or set tear scanline command until the end of the frame

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes

<table>
<tr><td></td><td><table><tr><td>Sleep In</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td><table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power On Sequence</td><td>0000h</td></tr>
<tr><td>S/W Reset</td><td>0000h</td></tr>
<tr><td>H/W Reset</td><td>0000h</td></tr>
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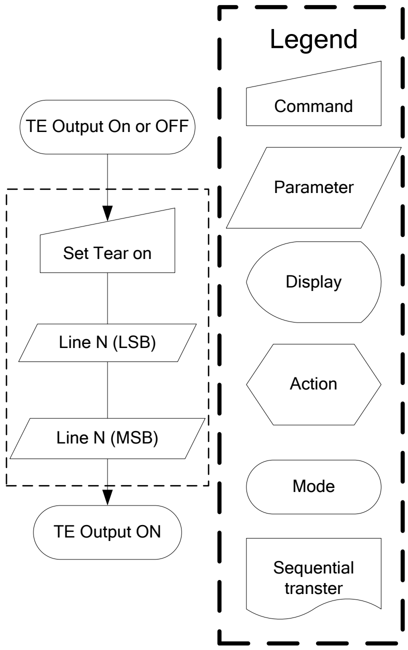
<tr><td>Flow Chart</td><td>
</td></tr>
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TESLRD (45h): Read Tear Scanline

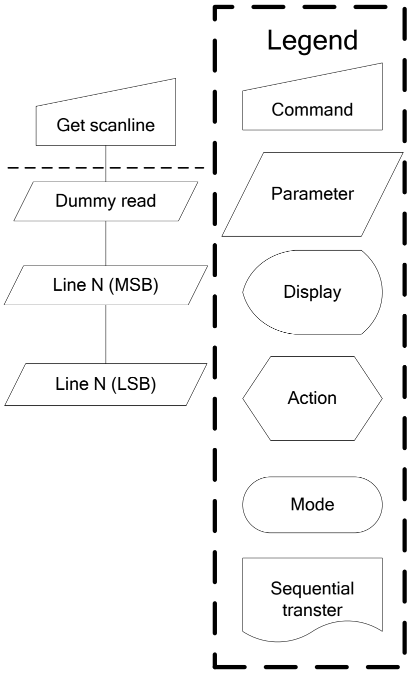
45H	TESLRD (Read Tear Scanline)											
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
TESLRD	0	↑	1	-	0	1	0	0	0	1	0	1	(45h)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	
2nd parameter	1	1	↑	-	-	-	-	-	TE_SL[11:8]				
3rd parameter	1	1	↑	-	TE_SL[7:0]								
Description	-The display module returns the current scanline ,N, used to update the display device. The total number of scanlines on a display device is defined as VSYNC+VBP+VACT+VFP. The first scanline is defined as the first line of V Sync and is denoted as Line 0. -When in sleep in mode, the value returned by get scanline is undefined. Note: that Set Tear Scan Line with N = 0 is equivalent to Tearing Effect Line ON with M = 0.												
Restriction	-												

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	0000h
S/W Reset	0000h
H/W Reset	0000h

Flow Chart	

RAMCLACT (4Ch): Memory Clear Act

4CH	RAMCLACT (Memory Clear Act)											
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RAMCLACT	0	↑	1	-	0	1	0	0	1	1	0	0	(4Ch)
parameter	No Parameter												
Description	-This command is fill all pixels data in RAM.												
Restriction													

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	
S/W Reset	
H/W Reset	

Flow Chart

RAMCLSET (4Dh): Memory Clear Setting

4DH	RAMCLSET (Memory Clear Setting)											
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RAMCLSETR	0	↑	1	-	0	1	0	0	1	1	0	1	(4Dh)
1st parameter	1	↑	1	-	CLNRAM_R[7:0]								
2nd parameter	1	↑	1	-	CLNRAM_G[7:0]								
3rd parameter	1	↑	1	-	CLNRAM_B[7:0]								
Description	CLNRAM_R/G/B[7:0]: Red/Green/Blue subpixel data setting when using RAMCLRACT(0x4Ch) function												

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	000000h
S/W Reset	000000h
H/W Reset	000000h

RAMCLRBUSY (4Eh): Memory Clear Busy

45H	RAMCLRBUSY (Memory Clear Busy)												
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
TESLRD	0	↑	1	-	0	1	0	0	1	1	1	0	(4Eh)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	CLR_BUSY	
Description	- CLR_BUSY : Display memory is inoperable due to memory clear action												
Restriction	-												

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes

		Partial Mode On, Idle Mode On, Sleep Out	Yes
		Sleep In	Yes

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

(Default)

DTSB (4Fh): Deep Standby Mode on

4FH	DTSB (Deep Standby Mode on)												
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
CDCCTR	0	↑	1	-	0	1	0	0	1	1	1	1	(4Fh)
1st parameter	1	↑	1	-	-	-	-	-	-	-	-	DSTB	
Description	-When DSTB =1 : Deep standby Mode ON Exit deep standby mode by pulling low “RESX” pin at least 1ms												
Restriction	-Only effect at Sleep in mpde												

Register availability

Status	Availability
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

WRDISBV (51h): Write Display Brightness

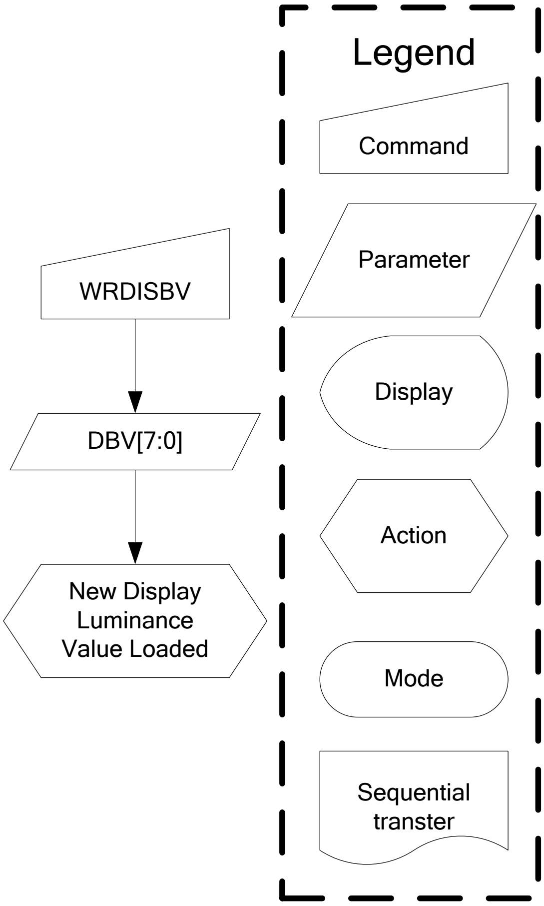
51H	WRDISBV (Write Display Brightness)											
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
WRDISBV	0	↑	1	-	0	1	0	1	0	0	0	1	(51h)
Parameter	1	↑	1	-	DBV[7:0]								
Description	-This command is used to adjust the brightness value of the display. -It should be checked what the relationship between this written value and output brightness of the display is. This relationship is defined on the display module specification. -In principle relationship is that 00h value means the lowest brightness and FFh value means the highest brightness.												
Restriction													
Register availability	(see table below)												
Default	(see table below)												

Register availability:

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default:

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart	

RDDISBV (52h): Read Display Brightness

52H	RDDISBV (Read Display Brightness Value)												
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDDISBV	0	↑	1	-	0	1	0	1	0	0	1	0	(52h)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	
2nd parameter	1	1	↑	-	DBV[7:0]								(00h)
Description	-This command returns the brightness value of the display. -It should be checked what the relationship between this returned value and output brightness of the display. This relationship is defined on the display module specification is. -In principle the relationship is that 00h value means the lowest brightness and FFh value means the highest brightness. -DBV[7:0] is reset when display is in sleep in mode. -DBV[7:0] is '0' when bit BCTRL of write CTRL display command (53h) is '0' -DBV[7:0] IS manual set brightness specified with write CTRL display command (53h) when bit BCTRL is '1'												
Restriction	-												
Register availability													

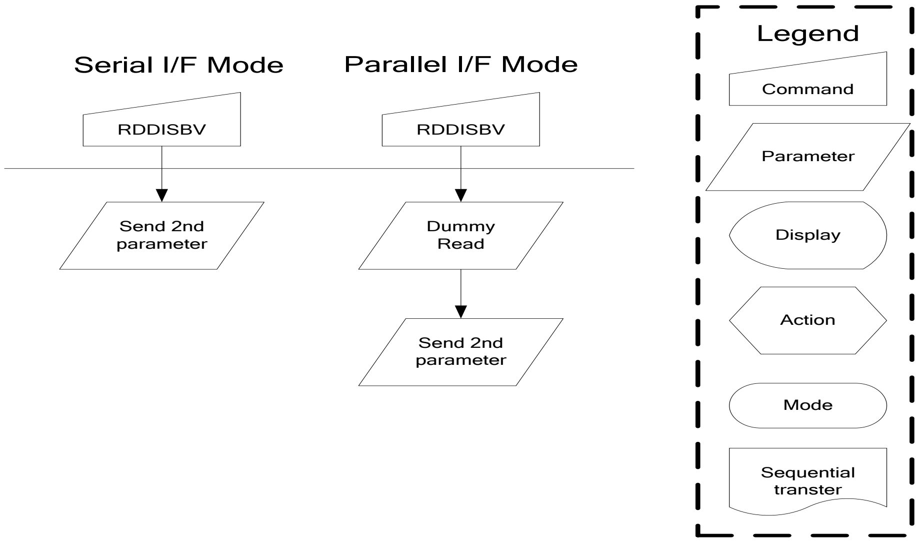
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes

Status	Availability
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	0000h
S/W Reset	0000h
H/W Reset	0000h

Flow Chart



WRCTRLD (53h): Write CTRL Display

53H	WRCTRLD (Write CTRL Display)												
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
WRCTRLD	0	↑	1	-	0	1	0	1	0	0	1	1	(53h)
Parameter	1	↑	1	-	-	-	-	-	-	BL	-	-	-
Description	-This command is used to control display brightness. -BL: Backlight Control On/Off 0 = Off (Completely turn off backlight circuit. Control lines must be low.) 1 = On												
Restriction													

Register availability

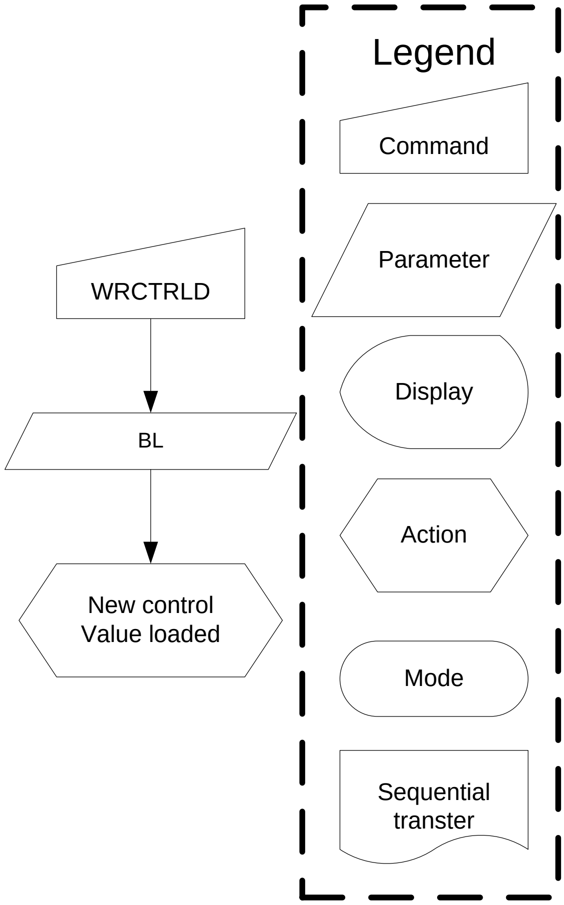
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes

	Partial Mode On, Idle Mode On, Sleep Out	Yes
	Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart



RDCTRLD (54h): Read CTRL Display

54H	RDCTRLD (Read CTRL value Display)												
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDCTRLD	0	↑	1	-	0	1	0	1	0	1	0	0	(54h)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	
2nd parameter	1	1	↑	-	-	-	-	-	-	BL	-	-	
Description	-This command returns ambient light and brightness control values. -BL: Backlight Control On/Off 0 = Off 1 = On												

<table>
<tr><td>Restriction</td><td>-</td></tr>
<tr><td>Register availability</td><td>
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table>
</td></tr>
<tr><td>Default</td><td>
<table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power On Sequence</td><td>00h</td></tr>
<tr><td>S/W Reset</td><td>00h</td></tr>
<tr><td>H/W Reset</td><td>00h</td></tr>
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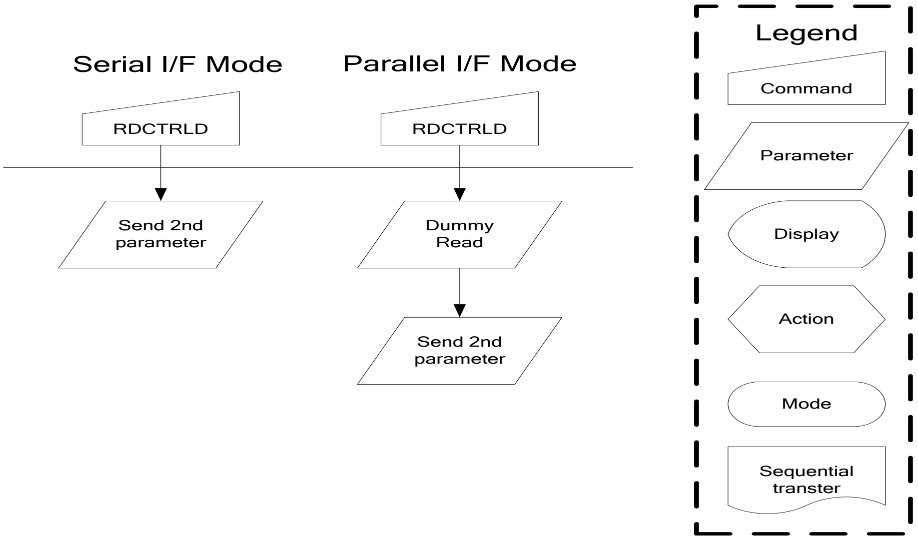
<tr><td>Flow Chart</td><td>


</td></tr>
</table>

WRSRECTRL (55h): Write Content Adaptive Brightness Control

55H	WRSRECTRL (Write Content Adaptive Brightness Control)												
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
WRCTRLD	0	↑	1	-	0	1	0	1	0	1	0	1	(55h)
Parameter	1	↑	1	-	-	SRE	-	-	-	-	-	-	
Description	-This command is used to control SRE. -SRE: CABC Sunlight Readable Enhance function enable 0 = Off 1 = On												
Restriction													

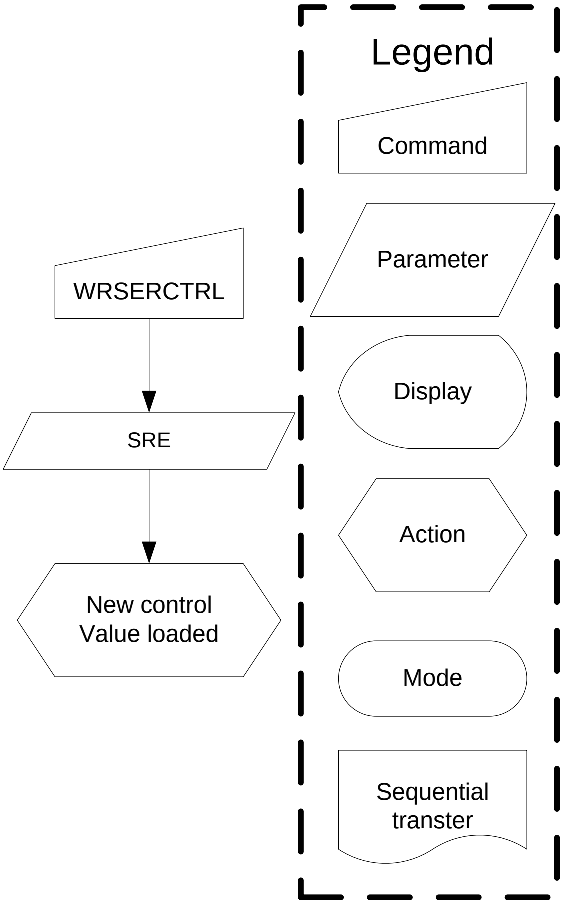
Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart



RDSRECTRL (56h): Read Content Adaptive Brightness Control

56H	RDSRECTRL (Read Content Adaptive Brightness Control)											
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDCTRLD	0	↑	1	-	0	1	0	1	0	1	1	0	(56h)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	

2nd parameter	1	1	↑	-	-	SRE	-	-	-	-	-	-	

Description

-This command returns SRE Status

-SRE: CABC Sunlight Readable Enhance function enable

0 = Off

1 = On

Restriction

-

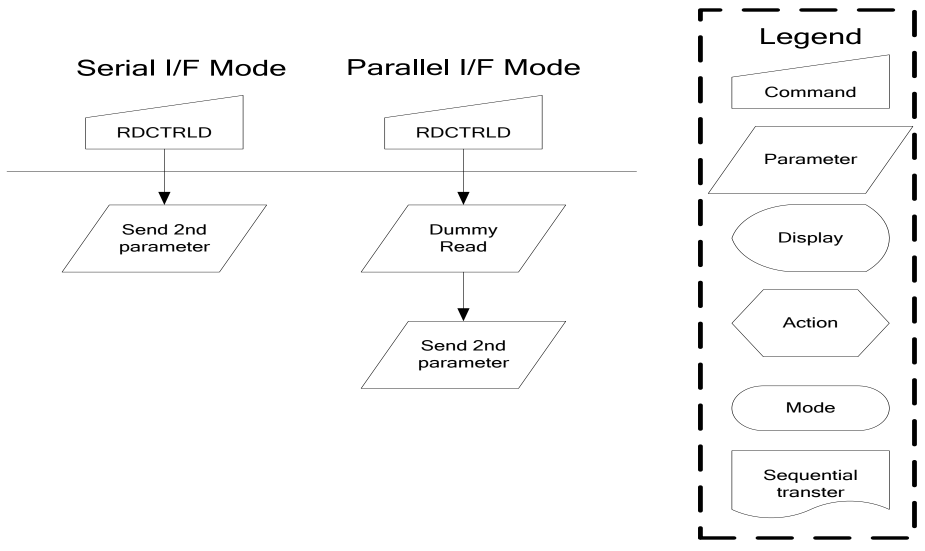
Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart



RDFCS (AAh): Read First Checksum

AAH	RDFCS (Read First Checksum)												
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDCTRLD	0	↑	1	-	1	0	1	0	1	0	1	0	(AAh)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	
2nd parameter	1	1	↑	-	FCS[7:0]								

Description	-Read the first checksum that has been calculated from user commands after write access to these commands has been done.
Restriction	-only in sleep out mode

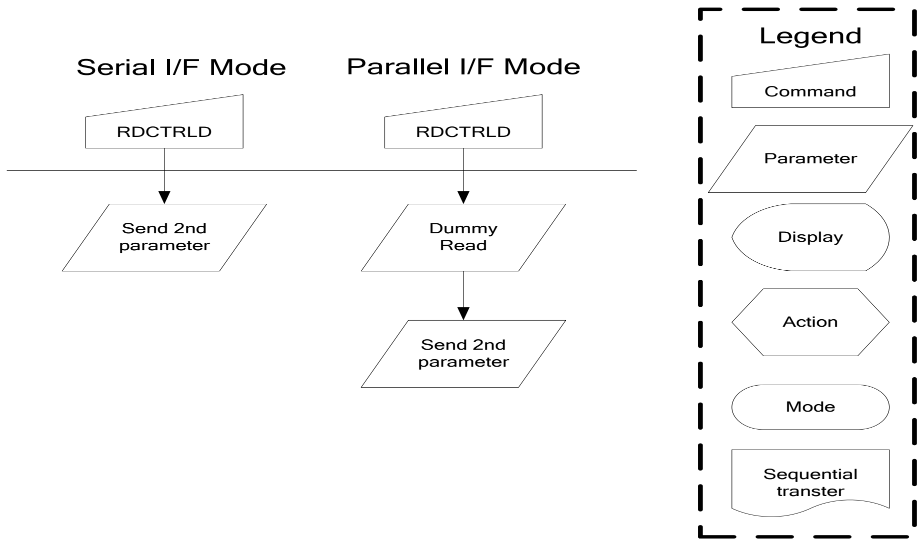
Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart



RDCCS (AFh): Read Continue Checksum

AFH	RDCCS (Read Continue Checksum)											
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDCTRLD	0	↑	1	-	1	0	1	0	1	1	1	1	(AFh)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	
2nd parameter	1	1	↑	-	FCS[7:0]								
Description	- Read the continue checksum that has been calculated continuously after the first checksum has calculated from user commands after write access to these commands has been done.												
Restriction	-only in sleep out mode												
Register													

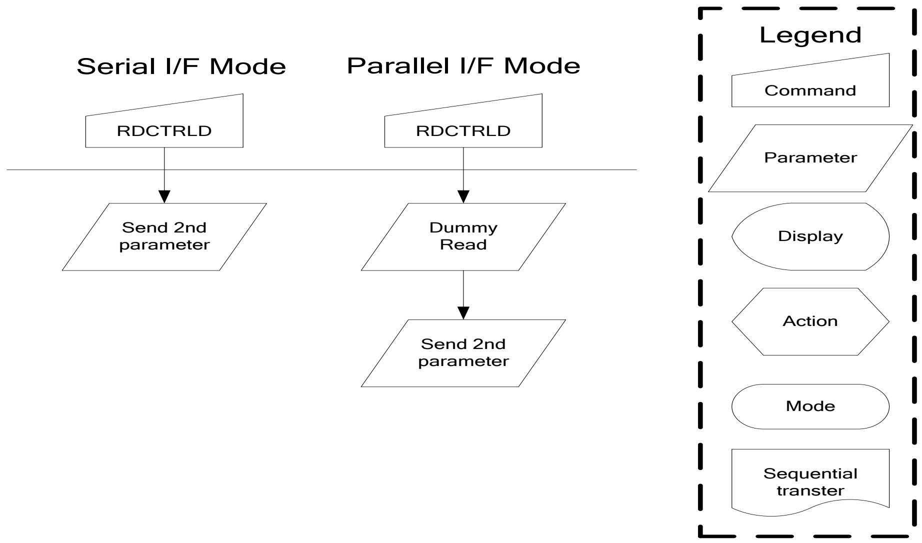
availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart



RAMMD (D0h): RAM Mode

D0H	RAMMD (RAM Mode)												
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
WRCTRLD	0	↑	1	-	1	1	0	1	0	0	0	0	(D0h)
Parameter	1	↑	1	-	scan_ ram_spwr	-	-	-	-	-	Video_mode [1:0]		
Description	-scan_ram_spwr: When Video_mode[1:0] = 00 or 01 , this bit need set 1 -Video_mode[1:0]: 0 = with RAM Compression On Ex. Resolution 280x571 1 = with RAM Compression Off 2 = without RAM Ex. Resolution 280x1280												
Restriction													
Register availability													
Default													

Register availability:

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default:

Status	Default Value
Power On Sequence	81h
S/W Reset	81h
H/W Reset	81h

RDID1 (DAh): Read ID1

DAH	RDID1 (Read ID1)												
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDID1	0	↑	1	-	1	1	0	1	1	0	1	0	(DAh)
1st parameter	1	1	↑	-	-								
2nd parameter	1	1	↑	-	ID1[7:0]								

Description	-This read byte identifies the LCD module's manufacturer. '-': Don't care.
Register availability	(see table below)
Default	(see table below)

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

RDID2 (DBh): Read ID2

DBH	RDID2 (Read ID2)											
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDID2	0	↑	1	-	1	1	0	1	1	0	1	1	(DBh)
1st parameter	1	1	↑	-	-								
2nd parameter	1	1	↑	-	ID2[7:0]								
Description	This read byte is used to track the LCD module/driver IC version. '-': Don't care.												

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

RDID3 (DCh): Read ID3

DCH	RDID3 (Read ID3)											
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDID3	0	↑	1	-	1	1	0	1	1	1	0	0	(DCh)
1st parameter	1	1	↑	-	-								
2nd parameter	1	1	↑	-	ID3[7:0]								
Description	This read byte identifies the LCD module/driver. '-': Don't care.												
Register availability	(see table below)												
Default	(see table below)												

Register availability:

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default:

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

12.3 Command Table 2

COMMAND Table 2														
Instruction	**D/CX**	**WRX**	**RDX**	**PNUM**	**D7**	**D6**	**D5**	**D4**	**D3**	**D2**	**D1**	**D0**	**Hex**	**Function**
PWMCTRL	0	↑	1	3	0	1	1	0	0	0	0	0	(60h)	PWM Control
	1	↑	1		-	0	0	0	-	0	0	0		
	1	↑	1		-	-	PWM_CLK_SEL[5:0]							
	1	↑	1		-	-	-	-	0	0	0	0		
LBCTRL	0	↑	1	1	0	1	1	0	0	1	1	0	(66h)	Line Buffer Control
	1	↑	1		-	-	-	-	-	-	SSI	-		
TIMCTRL	0	↑	1	12	0	1	1	1	0	0	0	0	(70h)	TIMING CTRL
	1	↑	1		-	-	-	-	-	-	RTN[9:8]			
	1	↑	1		RTN[7:0]									
	1	↑	1		-	Y_Res[10:8]			-	-	X_Res[9:8]			
	1	↑	1		X_Res[7:0]									
	1	↑	1		Y_Res[7:0]									
	1	↑	1		-	-	-	VBP[8]	-	-	VFP[9:8]			
	1	↑	1		VBP[7:0]									
	1	↑	1		VFP[7:0]									
	1	↑	1		-	-	0	0	-	-	0	0		
	1	↑	1		0	0	0	0	0	0	0	0		
	1	↑	1		-	-	-	-	-	-	0	0		
	1	↑	1		0	0	0	1	1	0	1	0		
DISPMODE	0	↑	1	1	0	1	1	1	0	0	0	1	(71h)	Display Mode
	1	↑	1		1	1	Nline[1:0]		0	-	Hsync_sel	Disp_ mode		
NVMID	0	↑	1	3	1	0	1	1	0	0	0	1	(B1h)	NVM ID Setting
	1	↑	1		ID1_PROG_DATA[7:0]									
	1	↑	1		ID2_PROG_DATA[7:0]									
	1	↑	1		ID3_PROG_DATA[7:0]									
GAMCTRP	0	↑	1	16	1	0	1	1	0	1	1	1	(B7h)	GAMCTRP
	1	↑	1		-				VC0P[3:0]					
	1	↑	1		-	-	VC1P[5:0]							
	1	↑	1		-	-	VC2P[5:0]							
	1	↑	1		-	-	-	VC4P[4:0]						

COMMAND Table 2														
Instruction	**D/CX**	**WRX**	**RDX**	**PNUM**	**D7**	**D6**	**D5**	**D4**	**D3**	**D2**	**D1**	**D0**	**Hex**	**Function**
	1	↑	1		-	-	-	VC6P[4:0]						
	1	↑	1			-			VC13P[3:0]					
	1	↑	1		-	VC20P[6:0]								
	1	↑	1		-	-			-	VC27P[2:0]				
	1	↑	1			-				VC36P[2:0]				
	1	↑	1		-	VC43P[6:0]								
	1	↑	1		-	-			VC50P[3:0]					
	1	↑	1		-	-	-	VC57P[4:0]						
	1	↑	1		-	-	-	VC59P[4:0]						
	1	↑	1		-	-	VC61P[5:0]							
	1	↑	1		-	-	VC62P[5:0]							
	1	↑	1		-	-	-	-	VC63P[3:0]					
GAMCTRN	0	↑	1	16	1	0	1	1	1	0	0	0	(B8h)	GAMCTRN
	1	↑	1		-				VC0N[3:0]					
	1	↑	1		-	-	VC1N[5:0]							
	1	↑	1		-	-	VC2N[5:0]							
	1	↑	1		-	-	-	VC4N[4:0]						
	1	↑	1		-	-	-	VC6N[4:0]						
	1	↑	1		-	-	-	-	VC13N[3:0]					
	1	↑	1		-	VC20N[6:0]								
	1	↑	1		-	-	-	-	-	VC27N[2:0]				
	1	↑	1			-	-	-	-	VC36N[2:0]				
	1	↑	1		-	VC43N[6:0]								
	1	↑	1		-	-	-	-	VC50N[3:0]					
	1	↑	1		-	-	-	VC57N[4:0]						
	1	↑	1		-	-	-	VC59N[4:0]						
	1	↑	1		-	-	VC61N[5:0]							
	1	↑	1		-	-	VC62N[5:0]							
	1	↑	1		-	-	-	-	VC63N[3:0]					
GAMCTR	0	↑	1	2	1	0	1	1	1	0	0	0	(B9h)	GAMCTR
	1	↑	1		-	AJ0P[2:0]				AJ1P[2:0]				
	1	↑	1		-	AJ0N[2:0]				AJ1N[2:0]				
VCOM_SSI	0	↑	1	3	1	0	1	1	1	1	1	0	(BEh)	VCOM_SSI Control
	1	↑	1		-	VCMP[6:0]								

COMMAND Table 2														
Instruction	**D/CX**	**WRX**	**RDX**	**PNUM**	**D7**	**D6**	**D5**	**D4**	**D3**	**D2**	**D1**	**D0**	**Hex**	**Function**
	1	↑	1		-	-	-	-	-	-	-	SSI_pos[8]		
	1	↑	1		SSI_pos[7:0]									
VGH_VGL CTRL	0	↑	1	6	1	0	1	1	1	1	1	1	(BFh)	
	1	↑	1		-	-	-	VGHP_DISP[4:0]						
	1	↑	1		-	-	-	VGHP_TP[4:0]						
	1	↑	1		-	-	-	VGHP_NOISE[4:0]						
	1	↑	1		-	-	-	-	VGLS_DP[3:0]					
	1	↑	1		-	-	-	-	VGLS_TP[3:0]					
	1	↑	1		-	-	-	-						

PWMCTRL (60h): PWM control

60H	PWMCTRL(PWM control)												
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
PWMCTRL	0	↑	1	-	0	1	1	0	0	0	0	0	(60h)
1st Parameter	1	↑	1	-	-	0	0	0	-	0	0	0	
2nd Parameter	1	↑	1	-	-	-	PWM_CLK_SEL[5:0]						
3rd Parameter	1	↑	1	-	-	-	-	-	0	0	0	0	

Description

PWM_CLK_SEL[5:0]: PWM clock select

CK_SEL	PWM Frequency(KHz)	CK_SEL	PWM Frequency(KHz)	CK_SEL	PWM Frequency(KHz)	CK_SEL	PWM Frequency(KHz)
0	58.824	16	6.920	32	3.565	48	2.401
1	58.824	17	6.536	33	3.460	49	2.353
2	39.216	18	6.192	34	3.361	50	2.307
3	29.412	19	5.882	35	3.268	51	2.262
4	23.529	20	5.602	36	3.180	52	2.220
5	19.608	21	5.348	37	3.096	53	2.179
6	16.807	22	5.115	38	3.017	54	2.139
7	14.706	23	4.902	39	2.941	55	2.101
8	13.072	24	4.706	40	2.869	56	2.064
9	11.765	25	4.525	41	2.801	57	2.028
10	10.695	26	4.357	42	2.736	58	1.994
11	9.804	27	4.202	43	2.674	59	1.961
12	9.050	28	4.057	44	2.614	60	1.929
13	8.403	29	3.922	45	2.558	61	1.898
14	7.843	30	3.795	46	2.503	62	1.867
15	7.353	31	3.676	47	2.451	63	1.838

'-': Don't care

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00h/00h/08h
S/W Reset	00h/00h/08h
H/W Reset	00h/00h/08h

LBCTRL (66h): Line Buffer control

66H	LBCCTRL(Line Buffer control)											
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
LBCTRL	0	↑	1	-	0	1	1	0	0	1	1	0	(66h)
1st Parameter	1	↑	1	-	-	-	-	-	-	-	SSI	-	
Description	**SSI:** For Source partial off function Resolution = 280 , this bit set to 0 Other resolution , this bit set to 1 & Set "CMD 0xBE ssi_en_start_pos[7:0]" '-': Don't care												
Register Availability	(see table below)												
Default	(see table below)												

Register Availability:

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default:

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

TIMINGCTRL (70h): Timing control

70H	TIMINGCTRL(Timing control)												
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
TIMINGCTRL	0	↑	1	-	1	1	1	0	0	0	0	0	(70h)
1st Parameter	1	↑	1	-	-	-	-	-	-	-	RTN[9:8]		
2nd Parameter	1	↑	1	-	RTN[7:0]								
3rd Parameter	1	↑	1	-	-	Y_Res[10:8]			-	-	X_Res[9:8]		
4th Parameter	1	↑	1	-	X_Res[7:0]								
5th Parameter	1	↑	1	-	Y_Res[7:0]								
6th Parameter	1	↑	1	-	-	-	-	VBP[8]	-	-	VFP[9:8]		
7th Parameter	1	↑	1	-	VBP[7:0]								
8th Parameter	1	↑	1	-	VFP[7:0]								
9th Parameter	1	↑	1	-	-	-	0	0	-	-	0	0	
10th Parameter	1	↑	1	-	0	0	0	0	0	0	0	0	
11th Parameter	1	↑	1	-	-	-	-	-	-	-	-	0	
12th Parameter	1	↑	1	-	0	0	0	1	1	0	1	0	

Description

RTN[9:0]: define frame rate. RTN min. = 2

$$\textbf{Frame Rate} = \frac{15{,}000{,}000}{(Y\_Res+VBP+VFP) \times RTN}$$

RTN[9:0]	FR(Hz)	RTN[9:0]	FR(Hz)	RTN[9:0]	FR(Hz)	RTN[9:0]	FR(Hz)
1BBh	70.84	251h	52.92	2E7h	42.24	37Dh	35.14
1C5h	69.27	25Bh	52.04	2F1h	41.67	387h	34.75
1CFh	67.78	265h	51.19	2FBh	41.13	391h	34.37
1D9h	66.34	26Fh	50.37	305h	40.60	39Bh	34.00
1E3h	64.97	279h	49.57	30Fh	40.08	3A5h	33.63
1Edh	63.65	283h	48.80	319h	39.57	3AFh	33.28
1F7h	62.39	28Dh	48.06	323h	39.08	3B9h	32.93
201h	61.17	297h	47.33	32Dh	38.60	3C3h	32.59
20Bh	60.00	2A1h	46.63	337h	38.13	3CDh	32.25
215h	58.88	2Abh	45.95	341h	37.67	3D7h	31.92
21Fh	57.79	2B5h	45.28	34Bh	37.23	3E1h	31.60
229h	56.75	2BFh	44.64	355h	36.79	3EBh	31.29
233h	55.74	2C9h	44.01	35Fh	36.36	3F5h	30.98
23Dh	54.77	2D3h	43.40	369h	35.95	3FFh	30.68
247h	53.83	2DDh	42.81	373h	35.54	-	

<table>
<tr><td></td><td>Note:
1. In this frame rate table , Y_Res = 400 , VFP = 60 , VBP = 18
2. The deviation of frame rate is +/- 5%
X_Res[9:0]: X resolution
Y_Res[10:0]: Y resolution
VBP[8:0]: Vertical back porch lines. VBP min =1
VFP[9:0]: Vertical front porch lines. VFP min =1
‘-’: Don’t care</td></tr>
<tr><td>Register Availability</td><td>
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table>
</td></tr>
<tr><td>Default</td><td>
<table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power On Sequence</td><td>00h/beh/51h/18h/00h/00h/04h/1ah/11h/33h/00h/3Fh</td></tr>
<tr><td>S/W Reset</td><td>00h/beh/51h/18h/00h/00h/04h/1ah/11h/33h/00h/3Fh</td></tr>
<tr><td>00h/beh/51h/18h/00h/00h/04h/1ah</td><td>00h/beh/51h/18h/00h/00h/04h/1ah/11h/33h/00h/3Fh</td></tr>
</table>
</td></tr>
</table>

DISPMODE (71h): Display mode

71H	DISPMODE(Display mode)											
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
LBCTRL	0	↑	1	-	0	1	1	1	0	0	0	1	(71h)
1st Parameter	1	↑	1	-	1	1	Nline[1:0]		0	-	Hsync_sel	Disp_mode	
Description	**Nline[1:0]:** For inversion in normal mode 0 = Column inversion , 1 = 1-Dot inversion , 2 = 2-Dot inversion **Hsync_sel:** 0 = internal h-sync , 1 = external h-sync (Valid @disp_mode=1) **Disp_mode:** 0 = internal mode , 1 = external mode (video with v-sync) '-': Don't care												

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	C3h
S/W Reset	C3h
H/W Reset	C3h

NVMID (B1h): NVM ID Setting

B1H	NVMID(NVM ID Setting)											
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
NVMID	0	↑	1	-	1	0	1	1	0	0	0	1	(B1h)
1st Parameter	1	↑	1	-	ID1_PROG_DATA[7:0]								
2nd Parameter	1	↑	1	-	ID2_PROG_DATA[7:0]								
3rd Parameter	1	↑	1	-	ID3_PROG_DATA[7:0]								
Description	**ID1/2/3_PROG_DATA[7:0]:** For NVM ID Program '-': Don't care												

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

GAMCTRP (B7h): Positive Voltage Gamma Control

B7H	GAMCTRP (Positive Voltage Gamma Control)												
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
GAMCTRP1	0	↑	1	-	1	0	1	1	0	0	0	0	(B7h)
1st Parameter	1	↑	1	-	-	-	-	-	VC0P[3:0]				
2nd Parameter	1	↑	1	-	-	-	VC1P[5:0]						
3rd Parameter	1	↑	1	-	-	-	VC2P[5:0]						
4th Parameter	1	↑	1	-	-	-	-	VC4P[4:0]					
5th Parameter	1	↑	1	-	-	-	-	VC6P[4:0]					
6th Parameter	1	↑	1	-	-	-	-	-	VC13P[3:0]				
7th Parameter	1	↑	1	-	-	VC20P[6:0]							
8th Parameter	1	↑	1	-	-	-	-	-	-	VC27P[2:0]			
9th Parameter	1	↑	1	-	-	-	-	-	-	VC36P[2:0]			
10th Parameter	1	↑	1	-	-	VC43P[6:0]							
11th Parameter	1	↑	1	-	-	-	-	-	VC50P[3:0]				
12th Parameter	1	↑	1	-	-	-	-	VC57P[4:0]					
13th Parameter	1	↑	1	-	-	-	-	VC59P[4:0]					
14th Parameter	1	↑	1	-	-	-	VC61P[5:0]						
15th Parameter	1	↑	1	-	-	-	VC62P[5:0]						
16th Parameter	1	↑	1	-	-	-	-	-	VC63P[3:0]				

Description

Adjust the gamma characteristics of the TFT panel.

Positive Gamma Control

'-': Don't care

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	Refer to description
S/W Reset	Refer to description
H/W Reset	Refer to description

GAMCTRN (B8h): Native Voltage Gamma Control

<table>
<tr><th>B8H</th><th colspan="13">GAMCTRN (Native Voltage Gamma Control)</th></tr>
<tr><td>Inst / Para</td><td>D/CX</td><td>WRX</td><td>RDX</td><td>-</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>GAMCTRN</td><td>0</td><td>↑</td><td>1</td><td>-</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>(B8h)</td></tr>
<tr><td>1st Parameter</td><td>1</td><td>↑</td><td>1</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td colspan="4">VC0N[3:0]</td><td></td></tr>
<tr><td>2nd Parameter</td><td>1</td><td>↑</td><td>1</td><td>-</td><td>-</td><td>-</td><td colspan="6">VC1N[5:0]</td><td></td></tr>
<tr><td>3rd Parameter</td><td>1</td><td>↑</td><td>1</td><td>-</td><td>-</td><td>-</td><td colspan="6">VC2N[5:0]</td><td></td></tr>
<tr><td>4th Parameter</td><td>1</td><td>↑</td><td>1</td><td>-</td><td>-</td><td>-</td><td>-</td><td colspan="5">VC4N[4:0]</td><td></td></tr>
<tr><td>5th Parameter</td><td>1</td><td>↑</td><td>1</td><td>-</td><td>-</td><td>-</td><td>-</td><td colspan="5">VC6N[4:0]</td><td></td></tr>
<tr><td>6th Parameter</td><td>1</td><td>↑</td><td>1</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td colspan="4">VC13N[3:0]</td><td></td></tr>
<tr><td>7th Parameter</td><td>1</td><td>↑</td><td>1</td><td>-</td><td>-</td><td colspan="7">VC20N[6:0]</td><td></td></tr>
<tr><td>8th Parameter</td><td>1</td><td>↑</td><td>1</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td colspan="3">VC27N[2:0]</td><td></td></tr>
<tr><td>9th Parameter</td><td>1</td><td>↑</td><td>1</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td colspan="3">VC36P[2:0]</td><td></td></tr>
<tr><td>10th Parameter</td><td>1</td><td>↑</td><td>1</td><td>-</td><td>-</td><td colspan="7">VC43P[6:0]</td><td></td></tr>
<tr><td>11th Parameter</td><td>1</td><td>↑</td><td>1</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td colspan="4">VC50P[3:0]</td><td></td></tr>
<tr><td>12th Parameter</td><td>1</td><td>↑</td><td>1</td><td>-</td><td>-</td><td>-</td><td>-</td><td colspan="5">VC57P[4:0]</td><td></td></tr>
<tr><td>13th Parameter</td><td>1</td><td>↑</td><td>1</td><td>-</td><td>-</td><td>-</td><td>-</td><td colspan="5">VC59P[4:0]</td><td></td></tr>
<tr><td>14th Parameter</td><td>1</td><td>↑</td><td>1</td><td>-</td><td>-</td><td>-</td><td colspan="6">VC61P[5:0]</td><td></td></tr>
<tr><td>15th Parameter</td><td>1</td><td>↑</td><td>1</td><td>-</td><td>-</td><td>-</td><td colspan="6">VC62P[5:0]</td><td></td></tr>
<tr><td>16th Parameter</td><td>1</td><td>↑</td><td>1</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td colspan="4">VC63P[3:0]</td><td></td></tr>
<tr><td>Description</td><td colspan="13">Adjust the gamma characteristics of the TFT panel.
Native Gamma Control
‘-’: Don’t care</td></tr>
<tr><td>Register Availability</td><td colspan="13"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="13"><table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>Refer to description</td></tr><tr><td>S/W Reset</td><td>Refer to description</td></tr><tr><td>H/W Reset</td><td>Refer to description</td></tr></table></td></tr>
</table>

GAMCTR (B9h): Gamma Control

B9H	GAMCTR (Gamma Control)											
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
GAMCTRP1	0	↑	1	-	1	0	1	1	1	0	0	1	(B9h)
1st Parameter	1	↑	1	-	-	AJ0P[2:0]			-	AJ1P[2:0]			
2nd Parameter	1	↑	1	-	-	AJ0N[2:0]			-	AJ1N[2:0]			
Description	Adjust the gamma characteristics of the TFT panel. '-': Don't care												
Register Availability													
Default													

Register Availability:

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default:

Status	Default Value
Power On Sequence	Refer to description
S/W Reset	Refer to description
H/W Reset	Refer to description

VCOM_SSI CTRL (BEh): VCOM_SSI Control

BEH	VCOM_SSI CTRL (VCOM_SSI Control)											
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
VCOM_SSI CTRL	0	↑	1	-	1	0	1	1	1	1	1	0	(BEh)
1st Parameter	1	↑	1	-	-	VCM[6:0]							
2nd Parameter	1	↑	1	-	-	-	-	-	-	-	-	SSI_pos[8]	
3rd Parameter	1	↑	1	-	SSI_pos[7:0]								

Description

VCM[6:0]: VCOMS Set.

VCM[6:0]	VCOMS (V)	VCM[6:0]	VCOMS (V)
00h	(-0.100)	30h	(-1.300)
01h	(-0.125)	31h	(-1.325)
02h	(-0.150)	32h	(-1.350)
03h	(-0.175)	33h	(-1.375)
04h	(-0.200)	34h	(-1.400)
05h	(-0.225)	35h	(-1.425)
06h	(-0.250)	36h	(-1.450)
07h	(-0.275)	37h	(-1.475)
08h	(-0.300)	38h	(-1.500)
09h	(-0.325)	39h	(-1.525)
0Ah	(-0.350)	3Ah	(-1.550)
0Bh	(-0.375)	3Bh	(-1.575)
0Ch	(-0.400)	3Ch	(-1.600)
0Dh	(-0.425)	3Dh	(-1.625)
0Eh	(-0.450)	3Eh	(-1.650)
0Fh	(-0.475)	3Fh	(-1.675)
10h	(-0.500)	40h	(-1.700)
11h	(-0.525)	41h	(-1.725)
12h	(-0.550)	42h	(-1.750)
13h	(-0.575)	43h	(-1.775)
14h	(-0.600)	44h	(-1.800)
15h	(-0.625)	45h	(-1.825)
16h	(-0.650)	46h	(-1.850)
17h	(-0.675)	47h	(-1.875)
18h	(-0.700)	48h	(-1.900)
19h	(-0.725)	49h	(-1.925)

1Ah	(-0.750)	4Ah	(-1.950)
1Bh	(-0.775)	4Bh	(-1.975)
1Ch	(-0.800)	4Ch	(-2.000)
1Dh	(-0.825)	4Dh	(-2.025)
1Eh	(-0.850)	4Eh	(-2.050)
1Fh	(-0.875)	4Fh	(-2.075)
20h	(-0.900)	50h	(-2.100)
21h	(-0.925)	51h	(-2.125)
22h	(-0.950)	52h	(-2.150)
23h	(-0.975)	53h	(-2.175)
24h	(-1.000)	54h	(-2.200)
25h	(-1.025)	55h~7Fh	-
26h	(-1.050)		
27h	(-1.075)		
28h	(-1.100)		
29h	(-1.125)		
2Ah	(-1.150)		
2Bh	(-1.175)		
2Ch	(-1.200)		
2Dh	(-1.225)		
2Eh	(-1.250)		
2Fh	(-1.275)		

Note:

1. VCOMS is used for feed through voltage compensation.
2. Setting limitation: VCOMS = (-0.1)V~(-2.2)V.

SSI_pos[8:0]: during "CMD 0x66 SSI" =1 .calculate based on panel X_resolution

SSI_pos = x_res/2 -4

'-': Don't care

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>00h/00h/00h</td></tr><tr><td>S/W Reset</td><td>00h/00h/00h</td></tr><tr><td>H/W Reset</td><td>00h/00h/00h</td></tr></table>

VGH_VGL CTRL (BFh): VGH VGL Control

BFH	VGH_VGL CTRL (VGH_VGL Control)												
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
VGH_VGL CTRL	0	↑	1	-	1	0	1	1	1	1	1	1	(BFh)
1st Parameter	1	↑	1	-	-	-	-	VGHP_DISP[4:0]					
2nd Parameter	1	↑	1	-	-	-	-	VGHP_TP[4:0]					
3rd Parameter	1	↑	1	-	-	-	-	VGHP_NOISE[4:0]					
4th Parameter	1	↑	1	-	-	-	-	-	VGLS_DISP[3:0]				
5th Parameter	1	↑	1	-	-	-	-	-	VGLS_TP[3:0]				
6th Parameter	1	↑	1	-	-	-	-	-	VGLS_NOISE[3:0]				

Description

VGHP_DISP/VGHP_TP/VGHP_NOISE[4:0]: VGH Set

VGHP[4:0]	VGH (V)	VGHP[4:0]	VGH (V)
00h	7.5	0Ch	13.5
01h	8.0	0Dh	14.0
02h	8.5	0Eh	14.5
03h	9.0	0Fh	15.0
04h	9.5	10h	15.5
05h	10.0	11h	16.0
06h	10.5	12h	16.5
07h	11.0	13h	17.0
08h	11.5	14h	17.5
09h	12.0	15h	18.0
0Ah	12.5	16h-1Fh	-
0Bh	13.0		

VGLS_DISP/VGLS_TP/VGLS_NOISE[3:0]: VGL Set

VGLS[3:0]	VGL (V)	VGLS[3:0]	VGH (V)
00h	6.46	08h	10.48
01h	7.02	09h	11.03
02h	7.51	0Ah	11.43

	03h	7.94	0Bh	12.07
	04h	8.54	0Ch	-
	05h	9.07	0Dh	-
	06h	9.51	0Eh	-
	07h	9.97	0Fh	-

Note:

1. VGHP - VGLS < 30v
2. VGHP_TP[4:0] / VGHP_NOISE[4:0] = VGHP_DISP[4:0] + 2v
3. Chip on Borad, VDDI=1.65 to 3.6V, VDD=2.65 to 3.6V, GND=0V,, Ta= -30℃~ 85℃

'-': Don't care

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00h/00h/00h/00h/00h/00h
S/W Reset	00h/00h/00h/00h/00h/00h
H/W Reset	00h/00h/00h/00h/00h/00h

12.4 Command Table 3

COMMAND Table 3														
Instruction	**D/CX**	**WRX**	**RDX**	**PNUM**	**D7**	**D6**	**D5**	**D4**	**D3**	**D2**	**D1**	**D0**	**Hex**	**Function**
VCOMOFFSET	0	↑	1	2	1	1	1	0	0	0	1	1	(E3h)	VCOM_ OFFSET
	1	↑	1		0	VMF[6:0]								
	1	↑	1		-	VMF_NEW[6:0]								
GAMMA_ANA	0	↑	1	5	0	1	1	1	0	0	1	1	(73h)	Gamma_ Analog Setting
	1	↑	1		-	-	-	-	-	1	0	0		
	1	↑	1		GVDD_AD[3:0]				GVEE_AD[3:0]					
	1	↑	1		-	-	0	1	0	-	1	0		
	1	↑	1		-	VRP[6:0]								
	1	↑	1		-	VRN[6:0]								
SVDD_SVCL_ SET	0	↑	1	2	0	1	1	1	1	0	1	0	(7Ah)	SVDD_ SVCL Setting
	1	↑	1		-	-	-	1	-	SVCL[2:0]				
	1	↑	1		-	-	-	-	-	1	SVDD[1:0]			
AVDD_AVCL_ SET	0	↑	1	2	0	1	1	1	1	0	1	1	(7Bh)	AVDD_ AVCL Setting
	1	↑	1		-	-	-	-	-	1	-	-		
	1	↑	1		-	AVCL[2:0]			-	AVDD[2:0]				

VCOMOFFSET (E3h): VCOM OFFSET SET

E3H	VCMOFSET (VCOM OFFSET SET)												
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
VCMOFSET	0	↑	1	-	1	1	0	1	1	1	0	1	(E3h)
1st Parameter	1	↑	1	-	0	VMF[6:0]							
2nd Parameter	1	↑	1	-	0	VMF_NEW[6:0]							

Description

VMF[6:0]/VMF_NEW[6:0]: finetune an offset to VCOM offset

VMF[6]	VMF[5:0]	VRP	VRN	VCOMS	VCOM
0	000000	VRP-64d	VRN+64d	VCM-64d	0
0	000001	VRP-63d	VRN+63d	VCM-63d	0
0	000010	VRP-62d	VRN+62d	VCM-62d	0
0	|	|	|	|	0
0	111110	VRP-2d	VRN+2d	VCM-2d	0
0	111111	VRP-1d	VRN+1d	VCM-1d	0
1	000000	VRP	VRN	VCM	0
1	000001	VRP+1d	VRN-1d	VCM+1d	0

1	000010	VRP+2d	VRN-2d	VCM+2d	0
1	|	|	|	|	0
1	111110	VRP+62d	VRN-62d	VCM+62d	0
1	111111	VRP+63d	VRN-63d	VCM+63d	0

'-': Don't care

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	40h/40h
S/W Reset	40h/40h
H/W Reset	40h/40h

GAMMA_ANA (73h): Gamma Analog Setting

73H	GAMMA_ANA (Gamma Analog Setting)											
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
GAMMA ANA	0	↑	1	-	0	1	1	1	0	0	1	1	(73h)
1st Parameter	1	↑	1	-	-	-	-	-	-	1	0	0	
2nd Parameter	1	↑	1	-	GVDD_AD[3:0]				GVEE_AD[3:0]				
3rd Parameter	1	↑	1	-	-	-	0	1	0	-	1	0	
4th Parameter	1	↑	1	-	-	VRP[6:0]							
5th Parameter	1	↑	1	-	-	VRN[6:0]							

Description

GVDD_AD[3:0]: positive gamma OP power set

GVDD_AD[3:0]	GVDD_AD (V)	GVDD_AD [3:0]	GVDD_AD (V)
00h	5.5	08h	6.3
01h	5.6	09h	6.4
02h	5.7	0Ah	6.5
03h	5.8	0Bh	6.6
04h	5.9	0Ch	6.7
05h	6.0	0Dh	6.8
06h	6.1	0Eh	-
07h	6.2	0Fh	-

GVEE_AD[3:0]: negative gamma OP power set

GVDD_AD[3:0]	GVDD_AD (V)	GVDD_AD [3:0]	GVDD_AD (V)
00h	3.4	08h	4.2
01h	3.5	09h	4.3
02h	3.6	0Ah	4.4
03h	3.7	0Bh	4.5
04h	3.8	0Ch	4.6
05h	3.9	0Dh	-
06h	4.0	0Eh	-
07h	4.1	0Fh	-

VRP[6:0]: VRP Set

VRP[6:0]	VRP (V)	VRP[6:0]	VRP (V)
00h	3.650 + (vcom offset)	40h	5.250 + (vcom offset)
01h	3.675 + (vcom offset)	41h	5.275 + (vcom offset)
02h	3.700 + (vcom offset)	42h	5.300 + (vcom offset)
03h	3.725 + (vcom offset)	43h	5.325 + (vcom offset)
04h	3.750 + (vcom offset)	44h	5.350 + (vcom offset)
05h	3.775 + (vcom offset)	45h	5.375 + (vcom offset)
06h	3.800 + (vcom offset)	46h	5.400 + (vcom offset)
07h	3.825 + (vcom offset)	47h	5.425 + (vcom offset)
08h	3.850 + (vcom offset)	48h	5.450 + (vcom offset)
09h	3.875 + (vcom offset)	49h	5.475 + (vcom offset)
0Ah	3.900 + (vcom offset)	4Ah	5.500 + (vcom offset)
0Bh	3.925 + (vcom offset)	4Bh	5.525 + (vcom offset)
0Ch	3.950 + (vcom offset)	4Ch	5.550 + (vcom offset)
0Dh	3.975 + (vcom offset)	4Dh	5.575 + (vcom offset)
0Eh	4.000 + (vcom offset)	4Eh	5.600 + (vcom offset)
0Fh	4.025 + (vcom offset)	4Fh	5.625 + (vcom offset)
10h	4.050 + (vcom offset)	50h	5.650 + (vcom offset)
11h	4.075 + (vcom offset)	51h	5.675 + (vcom offset)
12h	4.100 + (vcom offset)	52h	5.700 + (vcom offset)
13h	4.125 + (vcom offset)	53h	5.725 + (vcom offset)
14h	4.150 + (vcom offset)	54h	5.750 + (vcom offset)
15h	4.175 + (vcom offset)	55h	5.775 + (vcom offset)
16h	4.200 + (vcom offset)	56h	5.800 + (vcom offset)
17h	4.225 + (vcom offset)	57h	5.825 + (vcom offset)
18h	4.250 + (vcom offset)	58h	5.850 + (vcom offset)
19h	4.275 + (vcom offset)	59h	5.875 + (vcom offset)
1Ah	4.300 + (vcom offset)	5Ah	5.900 + (vcom offset)
1Bh	4.325 + (vcom offset)	5Bh	5.925 + (vcom offset)
1Ch	4.350 + (vcom offset)	5Ch	5.950 + (vcom offset)
1Dh	4.375 + (vcom offset)	5Dh	5.975 + (vcom offset)
1Eh	4.400 + (vcom offset)	5Eh	6.000 + (vcom offset)
1Fh	4.425 + (vcom offset)	5Fh	6.025 + (vcom offset)
20h	4.450 + (vcom offset)	60h	6.050 + (vcom offset)
21h	4.475 + (vcom offset)	61h	6.075 + (vcom offset)

22h	4.500 + (vcom offset)	62h	6.100 + (vcom offset)
23h	4.525 + (vcom offset)	63h	6.125 + (vcom offset)
24h	4.550 + (vcom offset)	64h	6.150 + (vcom offset)
25h	4.575 + (vcom offset)	65h	6.175 + (vcom offset)
26h	4.600 + (vcom offset)	66h	6.200 + (vcom offset)
27h	4.625 + (vcom offset)	67h	6.225 + (vcom offset)
28h	4.650 + (vcom offset)	68h	6.250 + (vcom offset)
29h	4.675 + (vcom offset)	69h	6.275 + (vcom offset)
2Ah	4.700 + (vcom offset)	6Ah	6.300 + (vcom offset)
2Bh	4.725 + (vcom offset)	6Bh	6.325 + (vcom offset)
2Ch	4.750 + (vcom offset)	6Ch	6.350 + (vcom offset)
2Dh	4.775 + (vcom offset)	6Dh	6.375 + (vcom offset)
2Eh	4.800 + (vcom offset)	6Eh	6.400 + (vcom offset)
2Fh	4.825 + (vcom offset)	6Fh	6.425 + (vcom offset)
30h	4.850 + (vcom offset)	70h	6.450 + (vcom offset)
31h	4.875 + (vcom offset)	71h	6.475 + (vcom offset)
32h	4.900 + (vcom offset)	72h	6.500 + (vcom offset)
33h	4.925 + (vcom offset)	73h	6.525 + (vcom offset)
34h	4.950 + (vcom offset)	74h	6.550 + (vcom offset)
35h	4.975 + (vcom offset)	75h	6.575 + (vcom offset)
36h	5.000 + (vcom offset)	76h	6.600 + (vcom offset)
37h	5.025 + (vcom offset)	77h	6.625 + (vcom offset)
38h	5.050 + (vcom offset)	78h	6.650 + (vcom offset)
39h	5.075 + (vcom offset)	79h	6.675 + (vcom offset)
3Ah	5.100 + (vcom offset)	7Ah	6.700 + (vcom offset)
3Bh	5.125 + (vcom offset)	7Bh	6.725 + (vcom offset)
3Ch	5.150 + (vcom offset)	7Ch	6.750 + (vcom offset)
3Dh	5.175 + (vcom offset)	7Dh	6.775 + (vcom offset)
3Eh	5.200 + (vcom offset)	7Eh	6.800 + (vcom offset)
3Fh	5.225 + (vcom offset)	7Fh	6.825 + (vcom offset)

VRN[6:0]: VRN Set

VRN[6:0]	VRN (V)	VRN[6:0]	VRN (V)
00h	-1.875 + (vcom offset)	40h	-3.475 + (vcom offset)
01h	-1.9 + (vcom offset)	41h	-3.5 + (vcom offset)

		02h	-1.925 + (vcom offset)	42h	-3.525 + (vcom offset)	
		03h	-1.95 + (vcom offset)	43h	-3.55 + (vcom offset)	
		04h	-1.975 + (vcom offset)	44h	-3.575 + (vcom offset)	
		05h	-2 + (vcom offset)	45h	-3.6 + (vcom offset)	
		06h	-2.025 + (vcom offset)	46h	-3.625 + (vcom offset)	
		07h	-2.05 + (vcom offset)	47h	-3.65 + (vcom offset)	
		08h	-2.075 + (vcom offset)	48h	-3.675 + (vcom offset)	
		09h	-2.1 + (vcom offset)	49h	-3.7 + (vcom offset)	
		0Ah	-2.125 + (vcom offset)	4Ah	-3.725 + (vcom offset)	
		0Bh	-2.15 + (vcom offset)	4Bh	-3.75 + (vcom offset)	
		0Ch	-2.175 + (vcom offset)	4Ch	-3.775 + (vcom offset)	
		0Dh	-2.2 + (vcom offset)	4Dh	-3.8 + (vcom offset)	
		0Eh	-2.225 + (vcom offset)	4Eh	-3.825 + (vcom offset)	
		0Fh	-2.25 + (vcom offset)	4Fh	-3.85 + (vcom offset)	
		10h	-2.275 + (vcom offset)	50h	-3.875 + (vcom offset)	
		11h	-2.3 + (vcom offset)	51h	-3.9 + (vcom offset)	
		12h	-2.325 + (vcom offset)	52h	-3.925 + (vcom offset)	
		13h	-2.35 + (vcom offset)	53h	-3.95 + (vcom offset)	
		14h	-2.375 + (vcom offset)	54h	-3.975 + (vcom offset)	
		15h	-2.4 + (vcom offset)	55h	-4 + (vcom offset)	
		16h	-2.425 + (vcom offset)	56h	-4.025 + (vcom offset)	
		17h	-2.45 + (vcom offset)	57h	-4.05 + (vcom offset)	
		18h	-2.475 + (vcom offset)	58h	-4.075 + (vcom offset)	
		19h	-2.5 + (vcom offset)	59h	-4.1 + (vcom offset)	
		1Ah	-2.525 + (vcom offset)	5Ah	-4.125 + (vcom offset)	
		1Bh	-2.55 + (vcom offset)	5Bh	-4.15 + (vcom offset)	
		1Ch	-2.575 + (vcom offset)	5Ch	-4.175 + (vcom offset)	
		1Dh	-2.6 + (vcom offset)	5Dh	-4.2 + (vcom offset)	
		1Eh	-2.625 + (vcom offset)	5Eh	-4.225 + (vcom offset)	
		1Fh	-2.65 + (vcom offset)	5Fh	-4.25 + (vcom offset)	
		20h	-2.675 + (vcom offset)	60h	-4.275 + (vcom offset)	
		21h	-2.7 + (vcom offset)	61h	-4.3 + (vcom offset)	
		22h	-2.725 + (vcom offset)	62h	-4.325 + (vcom offset)	
		23h	-2.75 + (vcom offset)	63h	-4.35 + (vcom offset)	
		24h	-2.775 + (vcom offset)	64h	-4.375 + (vcom offset)	
		25h	-2.8 + (vcom offset)	65h	-4.4 + (vcom offset)	
		26h	-2.825 + (vcom offset)	66h	-4.425 + (vcom offset)	

27h	-2.85 + (vcom offset)	67h	-4.45 + (vcom offset)
28h	-2.875 + (vcom offset)	68h	-4.475 + (vcom offset)
29h	-2.9 + (vcom offset)	69h	-4.5 + (vcom offset)
2Ah	-2.925 + (vcom offset)	6Ah	-4.525 + (vcom offset)
2Bh	-2.95 + (vcom offset)	6Bh	-4.55 + (vcom offset)
2Ch	-2.975 + (vcom offset)	6Ch	-4.575 + (vcom offset)
2Dh	-3 + (vcom offset)	6Dh	-4.6 + (vcom offset)
2Eh	-3.025 + (vcom offset)	6Eh	-4.625 + (vcom offset)
2Fh	-3.05 + (vcom offset)	6Fh	-4.65 + (vcom offset)
30h	-3.075 + (vcom offset)	70h	-4.675 + (vcom offset)
31h	-3.1 + (vcom offset)	71h	-4.7 + (vcom offset)
32h	-3.125 + (vcom offset)	72h	-4.725 + (vcom offset)
33h	-3.15 + (vcom offset)	73h	-4.75 + (vcom offset)
34h	-3.175 + (vcom offset)	74h	-4.775 + (vcom offset)
35h	-3.2 + (vcom offset)	75h	-4.8 + (vcom offset)
36h	-3.225 + (vcom offset)	76h	-
37h	-3.25 + (vcom offset)	77h	-
38h	-3.275 + (vcom offset)	78h	-
39h	-3.3 + (vcom offset)	79h	-
3Ah	-3.325 + (vcom offset)	7Ah	-
3Bh	-3.35 + (vcom offset)	7Bh	-
3Ch	-3.375 + (vcom offset)	7Ch	-
3Dh	-3.4 + (vcom offset)	7Dh	-
3Eh	-3.425 + (vcom offset)	7Eh	-
3Fh	-3.45 + (vcom offset)	7Fh	-

‘-’: Don’t care

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>04h/88h/1Ah/66h/4Dh</td></tr><tr><td>S/W Reset</td><td>04h/88h/1Ah/66h/4Dh</td></tr><tr><td>H/W Reset</td><td>04h/88h/1Ah/66h/4Dh</td></tr></table>

SVDD_SVCL_SET (7Ah): SVDD SVCL Setting

7AH	SVDD_SVCL_SET (SVDD SVCL Setting)												
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
SVDD_ SVCL SET	0	↑	1	-	0	1	1	1	1	0	1	0	(7Ah)
1st Parameter	1	↑	1	-	-	-	-	1	-	SVCL[2:0]			
2nd Parameter	1	↑	1	-	-	-	-	-	-	1	SVDD[1:0]		

Description:

SVCL[2:0]: SVCL set

SVCL[2:0]	SVCL (V)
00h	-3.35
01h	-3.5
02h	-3.66
03h	-3.84
04h	-4.22
05h	-4.4
06h	-4.59
07h	-4.79

SVDD[1:0]: SVDD set

SVDD[1:0]	SVDD (V)
00h	6.24
01h	6.42
02h	6.6
03h	6.79

	‘-’: Don’t care
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>14h/24h</td></tr><tr><td>S/W Reset</td><td>14h/24h</td></tr><tr><td>H/W Reset</td><td>14h/24h</td></tr></table>

AVDD_AVCL_SET (7Bh): AVDD AVCL Setting

7BH	AVDD_AVCL_SET (AVDD AVCL Setting)												
Inst / Para	D/CX	WRX	RDX	-	D7	D6	D5	D4	D3	D2	D1	D0	HEX
AVDD_ AVCL SET	0	↑	1	-	0	1	1	1	1	0	1	1	(7Bh)
1st Parameter	1	↑	1	-	-	-	-	-	-	1	-	-	
2nd Parameter	1	↑	1	-	-	AVCL[2:0]			-	AVDD[2:0]			
Description	**AVCL[2:0]:** AVCL set <table><tr><th>AVCL[2:0]</th><th>AVCL (V)</th></tr><tr><td>00h</td><td>-</td></tr><tr><td>01h</td><td>-</td></tr><tr><td>02h</td><td>-</td></tr><tr><td>03h</td><td>-4.05</td></tr><tr><td>04h</td><td>-4.4</td></tr><tr><td>05h</td><td>-4.58</td></tr><tr><td>06h</td><td>-4.78</td></tr><tr><td>07h</td><td>-5</td></tr></table> **AVDD[1:0]:** AVDD set <table><tr><th>AVDD[2:0]</th><th>AVDD (V)</th></tr><tr><td>00h</td><td>-</td></tr></table>												

	01h	-	
	02h	-	
	03h	6.25	
	04h	6.42	
	05h	6.6	
	06h	6.79	
	07h	6.99	

'-': Don't care

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	04h/45h
S/W Reset	04h/45h
H/W Reset	04h/45h

13 REVISION HISTORY

Version	Date	Description
V1.0	2024/04	First Release
V1.1	2024/07	Modify VDD & VDDI Supply Voltage
V1.2	2024/07	Modify Gate Resolution